Paediatrics AIIMS 2017

Question 1:

A 4-year-old child was brought to the hospital with a history of loose stools. No history of fever or blood in stools. The mother says he is irritable and eagerly drinks water if given. On examination, eyes are sunken and in the skin pinch test, the skin retracted within two seconds but not immediately. What treatment is advised for the child?

a) Give oral fluids and zinc supplementation only and ask mother to come back if danger sign develops.

b) Give oral fluids and ask mother to continue the same and visit again the next day.

c) Consider severe dehydration start IV fluids, IV antibiotics, and refer to higher centre.

d) Wait and observe.

Question 2:

A baby in the NICU is being evaluated for his features of microcephaly and hepatomegaly. CT scan revealed nodular periventricular calcifications. Congenital CMV infection is suspected. Which of the following samples is the best to detect the organism using PCR?

a) Liver biopsy

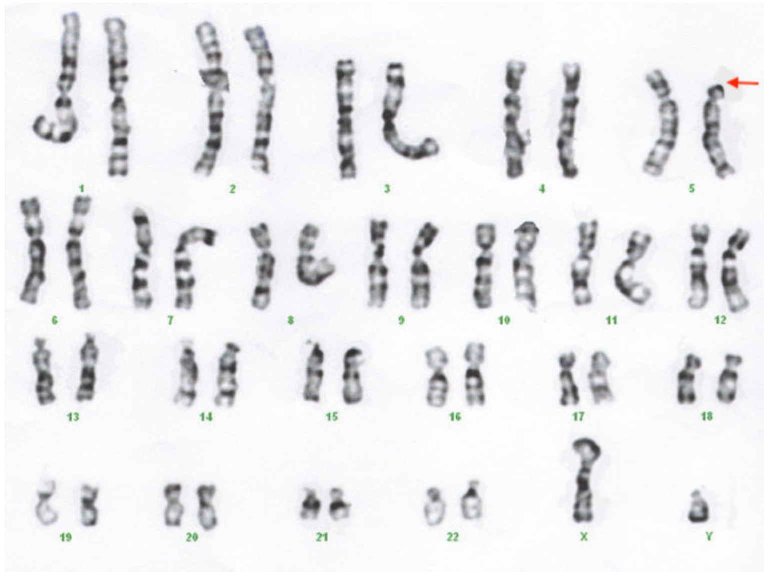
b) CSF

c) Blood

d) Urine

Question 3:

Identify the disease shown in the karyotype:



a) Cri-du–chat syndrome

b) Bloom syndrome

c) Angelman syndrome

d) Fragile X syndrome

Question 4:

Which of the following milestones would you generally expect a 3 year old child to display?

a) Hopping on one foot

b) Spoon handling

c) Copying a triangle

d) Repeats sentence of 10 syllables

Question 5:

Dried blood spot (DBS) testing is commonly done in infants for:

a) Inborn errors of metabolism

b) Blood grouping

c) Bilirubin testing

d) Coombs Test

Question 6:

A 4 day old baby is active and feeding well. He has a bilirubin level of 8 mg/dl. What is the next step in the management of the baby?

a) Exchange transfusion

b) Stop breastfeeding and start phototherapy

c) IV fluids and phototherapy

d) Continue breastfeeding

Question 7:

The most appropriate expansion of Apgar is:

a) Alert, pulse, grimace, activity, respiration

b) Assessment, pulse, grimace, alive, respiration

c) Appearance, pulse, grimace, activity, respiration

d) Appearance, pressure, grimace, activity, respiration

Question 8:

A 15-day-old infant had conjunctivitis, which later developed into respiratory distress and pneumonia. Chest Xray examination revealed bilateral lung infiltrates. Peripheral smear count showed 14,300/µL (N 43, L 54, E3). Which of the following is the most likely pathogenic organism?

a) Chlamydia trachomatis

b) Streptococcus pneumonia

c) Neisseria gonorrhoeae

d) Mycoplasma pneumonia

Question 9:

Of the following, all are findings of the condition in the below chest x-ray except:

a) Increased pulmonary blood flow

b) Narrow vascular pedicle

c) Pulmonary venous hypertension

d) Right atrial enlargement

Question 10:

In fetal alcohol syndrome, which of the following is not seen?

a) Thinning of corpus callosum

b) Microcephaly

c) Holoprosencephaly

d) Macrocephaly

Question 11:

A child presented with complaints of itching and discharges on flexure aspect of elbow and the lesions were red as shown below. What is your diagnosis?

a) Pemphigus vulgaris

b) Psoriasis

c) Post traumatic pigmentation

d) Atopic dermatitis

Question 12:

Which of the following fits into the criteria of severe variable deceleration, variations less than:

a) 90 beats per minute lasting for 60 seconds

b) 80 beats per minute lasting for 60 seconds

c) 100 beats per minute lasting for 60 seconds

d) 70 beats per minute lasting for 60 seconds

Question 13:

Which of the following constitutes Hutchinson's triad?

a) Interstitial keratitis, 8th nerve deafness, mulberry molars

b) Interstitial keratitis, 8th nerve deafness, Hutchinson molars

c) Interstitial keratitis, 8th nerve deafness, mulberry incisors

d) Interstitial keratitis, 8th nerve deafness, Hutchinson incisors

Answer Key

Question No.	Correct Option
1	a
2	d
3	a
4	b
5	a
6	d
7	c
8	a
9	d
10	d
11	d
12	d
13	d

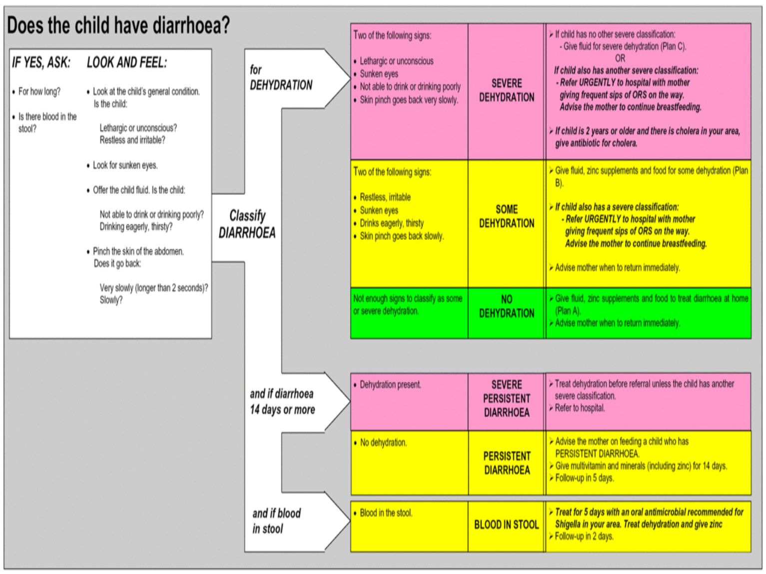
Detailed Explanations

Solution to Question 1:

In a child with diarrhea, thirst (eagerly drinks water), sunken eyes, restless and irritable behavior, and skin pinch showing slow retraction (<2 seconds) point towards some dehydration. Treatment includes oral rehydration therapy, zinc supplementation, and continued breastfeeding according to WHO integrated management of neonatal and childhood diseases (IMNCI) plan B.

The degree of dehydration is assessed and managed using the WHO IMNCI PLAN given below:

SIGNS	CLASSIFY AS	IDENTIFY TREATMENT (Urgent pre-referral treatments are in bold print)
Two of the following signs: • Lethargic or unconscious • Sunken eyes • Not able to drink or drinking poorly • Skin pinch goes back very slowly	**SEVERE DEHYDRATION**	► If child has no other severe classification - Give fluid for severe dehydration (Plan C). OR If child also has another severe classification: Refer URGENTLY to hospital with mother giving frequent sips of ORS on the way. Advise the mother to continue breastfeeding. ► If child is 2 years or older, and there is cholera in your area, give antibiotic for cholera.
Two of the following signs: • Restless, irritable • Sunken eyes • Drinks eagerly, thirsty • Skin pinch goes back slowly.	**SOME DEHYDRATION**	► Give fluid, Zinc supplements and food for some dehydration (Plan B) ► ***If Child also has a severe classification:*** ***Refer URGENTLY to hospital with mother giving frequent sips of ORS on the way. Advise the mother to continue breastfeeding.*** ► Advise mother when to return immediately. ► Follow-up in 5 days if not improving. ► If confirmed/symptomatic HIV, follow-up in 2 days if not improving.
• Not enough signs to classify as some or severe dehydration	**NO DEHYDRATION**	► Give fluid, Zinc supplements and food to treat diarrhoea at home (Plan A) ► Advise mother when to return immediately. ► *Follow-up in 5 days if not improving.* ► If confirmed/symptomatic HIV, follow-up in 2 days if not improving.



Solution to Question 2:

Among the following options, urine is the best sample to be sent for the diagnosis of congenital cytomegalovirus (CMV) infection.

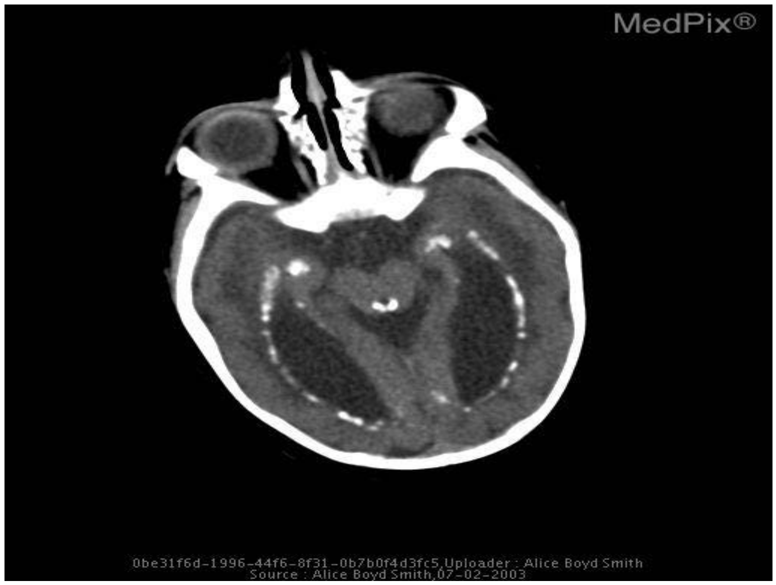
In congenital CMV infections, the investigation of choice is PCR or viral culture. It should be done within the first three weeks of life (after this period, it is difficult to distinguish between a congenital and an acquired infection). Since not all affected infants are viremic at birth, urine is a more reliable sample than blood.

Most CMV transmission occurs in the first trimester. The risk of transmission is high in the third trimester. 90% of the infected babies are asymptomatic and have lesser chances of having late sequelae.

The clinical manifestations include hepatosplenomegaly, petechial rashes, jaundice, chorioretinitis, and microcephaly. Neuroimaging shows intracranial periventricular calcifications. One of the long-term sequelae is sensorineural hearing loss.

There is no definitive treatment for congenital CMV infection. Treatment with ganciclovir is indicated only in patients with progressive neurologic disease and deafness.

The image below shows periventricular calcifications:



Solution to Question 3:

In the karyotype, there is a partial deletion of the short arm of chromosome 5, which leads to Cri-du-chat syndrome. Partial aneuploidy occurs as a result of the breakdown of the chromosomes.

Cri-du-chat syndrome is a rare genetic disorder. Features of the condition are hypotonia, short stature, characteristic cry called a cat's cry in the first few weeks of life, microcephaly with protruding metopic sutures, hypertelorism, bilateral epicanthic folds, high arched palate, flat nasal bridge, and intellectual disability.

Diagnosis is based on clinical findings and chromosomal analysis. Fluorescence in situ hybridization (FISH) is the specific test to confirm the diagnosis.

Given below is a picture of the facies in cri-du-chat syndrome:

Other options:

Option B: Bloom syndrome is a rare autosomal recessive disorder characterized by short stature, sun sensitivity, predeposition of development of cancer, and genomic instability.

Option C: Angelman syndrome is caused due to deletion in the maternal chromosome 15 or by paternal uniparental disomy. It primarily affects the nervous system. Characteristic features are delayed development, intellectual disability, severe speech impairment, and ataxia.

Option D: Fragile X syndrome is a genetic disorder also termed Martin Bell syndrome. It is the most common cause of inherited mental retardation (among the X-linked disorders), intellectual disability, and autism.

Solution to Question 4:

The ability to handle a spoon well and feed self is usually developed by 24 months (2 years) and would be expected in a 3 year old child.

Other options:

Option A: The ability to hop on one foot usually develops by 48 months (4 years).

Options C and D: The ability to draw a triangle from copy and repeating a sentence of 10 syllables usually develops at around 60 months (5 years) of age.

Some key milestones of children from 15 months of age are listed below:

Age	Motor milestones	Adaptive milestones	Language milestones	Social milestones
15 months	Walks alone, crawls up stairs.	Makes tower of 3 blocks, draws a line with crayon, inserts raisin or coin into bottle.	Jargon, follows simple commands, may name familiar objects, responds to own name.	Hugs parents, indicates some desires or needs by pointing.

Age	Motor milestones	Adaptive milestones	Language milestones	Social milestones
18 months	Runs stiffly, sits on a small chair, walks up stairs with 1 hand held, explores drawers.	Makes tower of 4 blocks, imitates scribbling.	10 words (average), names pictures, identifies 1 or more body parts.	Feeds self, seeks help when in trouble, may complain when wet or soiled.
24 months	Runs well, walks up and down stairs – 1 step at a time, opens doors, jumps.	Makes tower of 7 blocks, scribbles in circular pattern, fold paper once in imitation.	Puts 3 words together.	Handles spoon well, helps to undress, tells immediate stories, listen to stories when pictures are involved.
30 months	Goes up stairs alternating feet.	Makes tower of 9 blocks, makes vertical and horizontal strokes but not a cross.	Refers to self with pronoun 'I', knows full name.	Pretends during play, helps put things away.
36 months	Rides tricycle, stands momentarily on 1 foot.	Makes tower of 10 blocks, imitates construction of bridge with 3 blocks, copies circle, imitates cross.	Knows age and sex, counts 3 objects correctly, repeats 3 numbers or a sentence of 6 syllables.	Helps in dressing, washes hands, plays games (in parallel with other children).
48 months	Hops on 1 foot, throws ball overhand, uses scissors to cut pictures, climbs well.	Copies bridge from model, imitates construction of gate with 5 blocks, copies cross and square, identifies longer of 2 lines.	Counts 4 coins correctly, tells story.	Goes to toilet alone, plays with several children.
60 months	Skips	Draws triangle from copy, names the heavier of two weights.	Names 4 colors, speaks sentence of 10 syllables, counts 10 coins correctly.	Dresses and undresses, asks questions about meaning of words, engages in domestic role-playing.

Solution to Question 5:

Dried blood spots (DBS) taken from the heel are used in neonatal screening for inborn errors of metabolism (IEM) using tandem mass spectroscopy (TMS).

TMS employs two mass spectrometers in series to screen blood samples for their presence and to determine concentrations of various substances like amino acids, fatty acids, and other metabolites. A single sample can be used to screen multiple IEMs such as phenylketonuria, ethylmalonic encephalopathy, and glutaric acidemia.

Newborn screening using DBS is recommended for congenital hypothyroidism, G6PD deficiency, congenital adrenal hyperplasia, and galactosemia.

Universal screening is recommended for hearing loss, cyanotic congenital heart disease (pulse oximetry), hyperbilirubinemia, and congenital hip dysplasia (Ortolani and Barlow tests).

Hearing loss is screened with otoacoustic emission (OAE). Auditory Brainstem Response (ABR) is indicated in infants who fail OAE.

Screening for hypoglycemia should be performed in infants who are small or large for gestational age, born to mothers with diabetes, preterm, or symptomatic.

Solution to Question 6:

Since this child has a bilirubin level of only 8 mg/dl on the 4th day, it points to physiologic jaundice. The child is active and feeding well, the mother should be encouraged to continue breastfeeding frequently and exclusively.

All the other options would be possible treatment choices if the child had pathological jaundice.

Physiological jaundice/Icterus neonatorum is due to increased bilirubin production from the breakdown of fetal red blood cells combined with transient limitation in the conjugation of bilirubin by the immature neonatal liver. It first appears on the 2nd or 3rd days usually peaks on the 4th day at 5-6 mg/dL and decreases to <2 mg/dL between the 5th and 7th days after birth.

On the other hand, in pathological jaundice, appearance is within the first 24 hours after birth and may be present at or beyond the age of 2 weeks in term babies and 3 weeks in preterms babies. There may be an elevated direct (or conjugated) bilirubin level. Jaundice visible on the palms and soles, unresponsiveness to phototherapy, and a rapid rise in bilirubin levels during phototherapy indicate pathological jaundice.

Phototherapy is initiated in babies with pathological jaundice when the bilirubin levels exceed the 95th percentile on the total serum bilirubin (TSB) normogram. Babies are kept at a distance of 30-45 cm from the UV lamp (460-490 nm) with eyes and gonads covered. It is avoided in babies with conjugated hyperbilirubinemia due to the risk of bronze-baby syndrome. Dehydration and hypocalcemia are potential complications and breastfeeding is encouraged.

Exchange transfusion (ET) is done in cases of bilirubin encephalopathy, Rh incompatibility with significant hemolysis, and bilirubin values exceeding TSB cutoffs for ET.

Solution to Question 7:

The parameters to consider while calculating APGAR score in newborn babies are:

- Appearance
- Pulse rate
- Grimace
- Activity(muscle tone)
- Respiratory effort

The Apgar score is a systematic way of assessing the infant after birth at 1 and 5 minutes of life. It gives a maximum score of 10 which indicates the newborn is in the best possible condition.

Changes in the Apgar scores denote the response of the infant to resuscitation. Apgar scores should not be used to determine the need for resuscitation or to guide the steps of resuscitation. Apgar score alone should not be interpreted as evidence of asphyxia.

A score from 7-10 is considered normal. Scores between 0-3 require immediate resuscitation. If the score at 5 minutes remains less than 7, the infant should be assessed again every 5 minutes for the next 20 minutes.

Causes for false positive APGAR score are:

• Prematurity

• Drugs like analgesics, narcotics, sedatives, magnesium sulfate

• Acute cerebral trauma

• Precipitous delivery

• Airway obstruction (choanal atresia)

• Congenital pneumonia or sepsis

• Hemorrhage- hypovolemia

False-negative Apgar score is seen in maternal acidosis and high fetal catecholamine levels.

Solution to Question 8:

The organism most likely to cause conjunctivitis, progressing to respiratory distress, and then pneumonia is Chlamydia trachomatis.

It is not possible to differentiate between chlamydial pneumonia and other causes of atypical pneumonia, like Mycoplasma (option D) based solely on the clinical presentation. Here, the history of conjunctivitis makes it more likely to be a chlamydial infection.

Pneumonia caused by C. trachomatis can develop in infants born to women with an untreated chlamydial infection in 10-20% of cases. It has a very characteristic presentation in infancy, disease onset usually occurs around 1 and 3 months of age and is insidious, with a persistent cough, tachypnea, and absence of fever (atypical pneumonia). Auscultation reveals rales; wheezing is uncommon.

The absence of fever and wheezing helps to distinguish C. trachomatis pneumonia from respiratory syncytial virus pneumonia.

A distinctive laboratory finding is the presence of peripheral eosinophilia (>400 cells/µL). The most consistent finding on chest radiographs is hyperinflation accompanied by minimal interstitial or alveolar infiltrates.

Other options:

Option B: Pneumococcal pneumonia generally presents with features of lobar pneumonia. Fever, dyspnea, and productive cough are the primary symptoms.

Option C: Although Neisseria gonorrhoeae is an important cause of ophthalmia neonatorum, it does not cause pneumonia-like features.

Chlamydiae are obligate intracellular organisms that produce basophilic inclusion bodies. They are called energy parasites as they lack enzymes for ATP synthesis. They cannot grow on artificial media but can grow on cell lines (McCoy, HeLa, or the yolk sac of the chick embryo)

Chlamydiae occur in two forms, the elementary body (extracellular, infective form) and the reticulate body (intracellular, non-infective, and replicative form) which are found inside inclusion bodies.

Chlamydia trachomatis is the most common cause of ophthalmia neonatorum in developed countries. Neonatal inclusion conjunctivitis is caused by serotypes D to K. Chlamydia trachomatis inclusion conjunctivitis manifests relatively late, usually over 1 week after birth.

It is usually a venereal infection derived from the cervix or urethra of the mother during delivery. On examination, the conjunctiva is swollen and edematous. The discharge is initially watery, but later becomes mucopurulent or bloody.

Newborns with suspected chlamydia infection should have samples drawn from both the conjunctivae and oropharynx. Since it is an obligate intracellular organism, swabs should also include conjunctival epithelial cells if possible. Giemsa staining shows characteristic intracellular inclusion bodies. Diagnosis is usually established by PCR.

The treatment includes oral erythromycin for two weeks. Topical erythromycin to tetracycline ointment may also be used.

If left untreated, associated systemic diseases such as chlamydial otitis and pneumonia may develop later. In all cases, both parents must receive appropriate treatment for genital infection.

Given below is an image of ophthalmia neonatorum:

Solution to Question 9:

The given chest x-ray shows an egg-on-string appearance seen in the transposition of great arteries (TGA). Enlargement of the left atrium is seen in TGA, not the right atrium.

The reason for left atrial enlargement is due to increased pulmonary blood flow.

TGA is characterized by the aorta arising from the right ventricle and the pulmonary artery from the left ventricle. This abnormal configuration leads to a narrow vascular pedicle.

As pulmonary vascular resistance drops during the first few days of life, blood flow through the pulmonary vasculature is increased. There is also elevated pulmonary venous pressure seen in TGA, especially in infants with an intact interventricular septum.

The globular (egg) appearance of the heart due to abnormal convexity of the right atrial border and left atrial enlargement, and the narrow superior mediastinum (string) result in the classical egg-on-string appearance.

TGA is the most specific heart disease in infants of diabetic mothers.

Early management includes an immediate infusion of prostaglandin E_1 which keeps the ductus arteriosus patent for improving oxygenation. Rashkind balloon atrial septostomy is a palliative catheter-based intervention that can be done to improve the hypoxic state before definitive surgery. The procedure creates an opening in the atrial septum which allows better inter-circulatory mixing of the blood. Jatene arterial switch procedure is the definitive surgery.

Most infants with TGA do not survive till adulthood without corrective surgery.

Solution to Question 10:

Fetal alcohol syndrome (FAS) is associated with microcephaly rather than macrocephaly.

Alcohol exposure during pregnancy results in reduced brain volume in the frontal lobe, striatum, caudate nucleus, thalamus, and cerebellum. It also causes thinning of the corpus callosum and abnormal functioning of the amygdala.

Fetal Alcohol Spectrum Disorders (FASDs) are the leading preventable causes of developmental delay and intellectual disability. FASDs encompass different conditions such as FAS, partial FAS, alcohol-related neurodevelopmental disorder, alcohol-related birth defect, and neurobehavioral disorder associated with prenatal alcohol exposure.

Diagnostic criteria for FASDs include characteristic facial features, prenatal/postnatal growth deficiency, neurobehavioral impairment, and maternal alcohol consumption during pregnancy.

The facial features of FAS include short palpebral fissures, a thin upper lip vermilion border, a smooth philtrum, epicanthic folds, a flattened nasal bridge, lowset and unparallel ears, and retarded midfacial development.

CNS involvement leads to microcephaly, incomplete holoprosencephaly, and agenesis of the corpus callosum and the cerebellar vermis. The neurocognitive features include developmental delays, inattention, impulsivity, social impairments, and behavioral difficulties. Common comorbidities include attention-deficit/hyperactivity disorder (ADHD), self-regulation problems, and impulse control issues. Cardiac anomalies are also seen, with VSD being more common than ASD and tetralogy of Fallot.

Solution to Question 11:

The presence of diffuse, erythematous, itchy plaques on the flexural aspects of the elbow of a child is suggestive of atopic dermatitis.

Atopic dermatitis is the cutaneous manifestation of atopy, characterized by a family history of asthma, allergic rhinitis, or eczema.

There is a clear genetic predisposition. A characteristic defect is an impaired epidermal barrier. In many patients a mutation in the gene encoding filaggrin (a structural protein in the stratum granulosum) is responsible.

Patients may display many immunoregulatory abnormalities, including increased IgE synthesis, increased serum IgE levels, and delayed-type hypersensitivity reactions.

The disease progresses in three phases: Infantile, childhood, and adult.

In infants, the lesions of atopic eczema most frequently start on the face but may occur anywhere on the skin surface. Often, the napkin area is relatively spared. When the child begins to crawl, the exposed surfaces, especially the extensor aspect of the knees and elbows are most involved. The lesions consist of erythema and discrete or confluent oedematous papules. The papules are intensely itchy and may become exudative and crusted as a result of rubbing. Secondary infection and lymphadenopathy are common.

From 18 to 24 months onwards (childhood phase), the sites most characteristically involved are the elbow and knee flexures, a mixture of erythema, crusting, excoriation, hyper and hypopigmentation, and warty lichenification may be seen depending on the skin type on the sides of the neck, wrists, and ankles. The sides of the neck may show reticulate pigmentation, sometimes referred to as an atopic dirty neck. The erythematous and oedematous papules tend to be replaced by lichenification.

The picture in adults is essentially similar to that in later childhood, with lichenification, especially of the flexures and hands.

Solution to Question 12:

Severe variable decelerations are defined as: Less than 70 beats/minute lasting at least 60 seconds.

Variable deceleration is the abrupt decrease in FHR below the baseline. The decrease is 15 beats/minute or more, with a duration of 15 seconds or more but less than 2 minutes.

Mechanism: Umbilical cord compression, leads to occlusion of the umbilical vein, hence there is decreased venous return to maintain Cardiac Output.

Classification:

- Mild: more than 80 bpm irrespective of duration/ less than 30 seconds irrespective of HR/ 70-80 bpm lasting 60 seconds
- Moderate: more than 70 bpm lasting 30-60 seconds / 70-80 bpm lasting more than 60 seconds
- Severe: less than 70 bpm lasting at least 60 seconds

Solution to Question 13:

Hutchinson's triad refers to interstitial keratitis, 8th nerve deafness, and Hutchinson incisors. It is seen in late congenital syphilis.

The image below shows Hutchinson incisors:

Congenital syphilis is due to Treponema pallidum. Early features include hepato-splenomegaly, jaundice, diffuse lymphadenopathy, maculopapular or vesiculobullous rash involving hands and feet, persistent rhinitis (snuffles), and condylomatous lesions. Skeletal changes like saber shins, dental abnormalities like Hutchinson's teeth, and saddle nose (depression of the nasal root) are some late findings.

Other clinical features of congenital syphilis:

• Clutton joint - Due to synovitis, with sterile synovial fluid

• Juvenile tabes and paresis

• Rhagades - Linear scars radiating from mucocutaneous fissures of the mouth, anus, and genitalia

• Mulberry molars - Abnormal 1st lower molars with excessive cusps

• Sabre shin - Anterior bowing of the mid-portion of the tibia

• Olympian brow - Bony prominence of the forehead

Paediatrics AIIMS 2018

Question 1:

Which of the following is not an indicator of fetal lung maturity?

- **a)** Lecithin-sphingomyelin ratio > 2.5
- **b)** Positive shake test
- **c)** Increased phosphatidyl glycerol
- **d)** Nile blue test showing >50% of blue cells

Question 2:

How long should a child be isolated after being diagnosed with bacterial meningitis to prevent further transmission?

- **a)** Till 24 hours after starting antibiotics
- **b)** Till cultures become negative
- **c)** Till antibiotic course is complete
- **d)** Till 12 hours after admission

Question 3:

Choose the correct order of suctioning during neonatal resuscitation.

- **a)** Mouth-nose
- **b)** Nose-mouth
- **c)** Mouth-nose-oesophagus
- **d)** Trachea-nose-mouth

Question 4:

An adolescent girl complaints of dropping objects from hands precipitated during morning hours and exams. There is no history of loss of consciousness. One of her cousins has been diagnosed with epilepsy. EEG was suggestive of epileptic spikes. What is the most probable diagnosis?

a) Juvenile myoclonic epilepsy

b) Atypical absence

c) Choreo-athetoid epilepsy

d) Centrotemporal spikes

Question 5:

A newborn female presents with continuous dribbling of urine. The image below shows the pathology from which the patient is suffering. The most likely diagnosis is:

a) Gastroschisis

b) Bladder exstrophy

c) Superior vesical fissure

d) Anterior ectopic anus

Question 6:

The absence of which of the following milestones in a 3-year-old child is called as developmental delay?

a) Hopping on one leg

b) Drawing a square

c) Feeding by spoon

d) Passing a ball to someone

Question 7:

A 2-year-old child with a history of fall 1 year back with parietal bone fracture, has now presented with a painful and growing parietal swelling. What is the likely diagnosis?

a) Growing scalp hematoma

b) Growing fracture

c) Subdural hygroma

d) Chronic abscess

Question 8:

After the delivery of an infant of a diabetic mother, the blood glucose of the Infant was 60 mg/dl. Which other investigation would you do?

a) Serum potassium

b) Serum sodium

c) Serum chloride

d) Serum calcium

Question 9:

You have been called to declare the brain death of a 12-year-old child in the ICU. Which of the following is not seen in brain death?

a) Normal BP without pharmacological support

b) Positive spinal reflexes on stimulation

c) Sweating and tachycardia

d) Decorticate and decerebrate posturing

Question 10:

Craniopagus is defined as the fusion of ______.

a) Head and spine

b) Head only

c) Thorax and spine

d) Thorax only

Question 11:

Which of the following hormones is the major product of the fetal adrenal glands?

a) DHEA

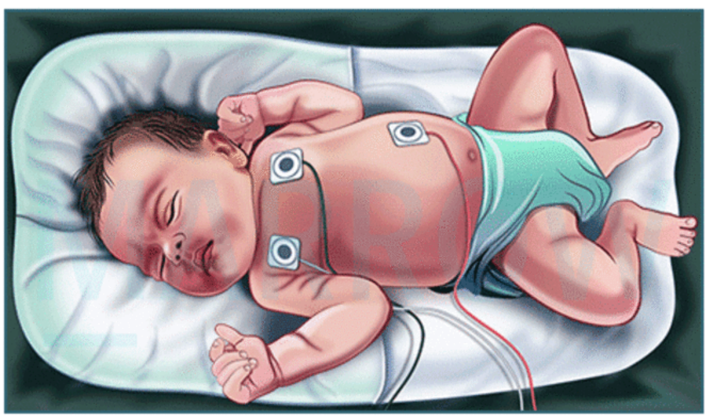
b) Cortisol

c) Corticosterone

d) Progesterone

Question 12:

A 2-day-old neonate was admitted to the NICU after he was observed to have movements as shown in the clip. His APGAR scores were 8 and 9 at 1 and 5 minutes, respectively. His blood sugar was within the normal range in the first 24 hours. What is the likely diagnosis?



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a) Focal tonic seizure

b) Focal clonic seizure

c) Spasms

d) Subtle seizure

Question 13:

A male child with coarse facies, macroglossia, thick lips with hepatosplenomegaly presents with copious mucous discharge from the nose. The most probable underlying diagnosis is

____

a) Hurler disease

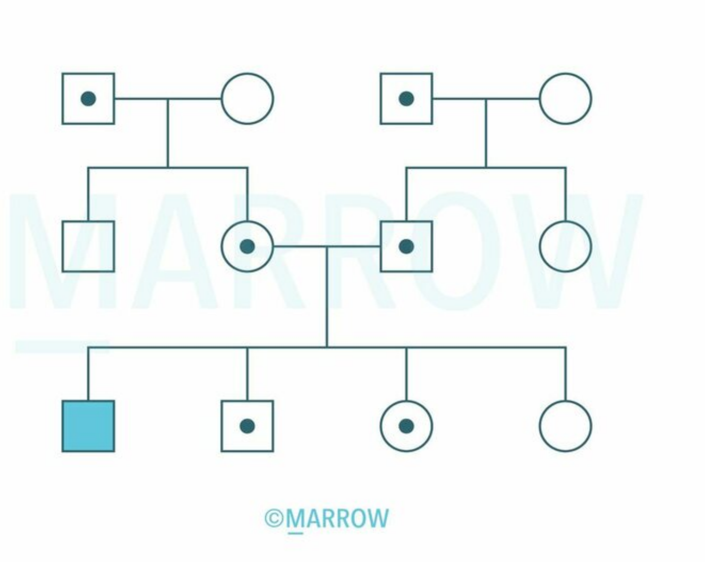
b) Beckwith-Wiedemann syndrome

c) Proteus Syndrome

d) Hypothyroidism

Question 14:

Which pattern of inheritance does the pedigree chart denote?



a) Autosomal-dominant disorder

b) X- linked dominant disorder

c) Autosomal-recessive disorder

d) X- linked recessive disorder

Question 15:

A preterm infant in the NICU was observed to have marked nasal flare, distinct xiphoid retractions, minimal intercostal retractions, and inspiratory lag with an audible grunt. What would be the Silverman Anderson score?

a) 10

b) 7

c) 8

d) 9

Question 16:

True regarding the difference between adult and pediatric resuscitation.

a) Ventricular dysrhythmias are common in children

b) The no flow phase precedes the period of cardiac arrest in children.

c) More ventilation to be given compared to chest compressions in children

d) Dysrhythmias are most often due to respiratory insufficiency in children

Answer Key

Question No.	Correct Option
1	d
2	a
3	a
4	a
5	b
6	c
7	b
8	d
9	d
10	b
11	a
12	b
13	a
14	c
15	c
16	d

Detailed Explanations

Solution to Question 1:

Nile blue test showing >50% orange cells is a sign of fetal lung maturity, not blue cells.

Tests for the assessment of fetal lung maturity include:

The Nile blue test involves staining desquamated fetal cells from amniotic fluid with 0.1% Nile blue sulfate. The presence of orange cells > 50% suggests lung maturity. Lecithin/sphingomyelin ratio at 31 to 32 weeks is 1, and at 35 weeks ratio is 2. The ratio ≥2 indicates lung maturity (option A).

The shake test or bubble test (Clement's test) (option B) involves mixing amniotic fluid with 95% ethanol, shaking it for 15 seconds, and observing it for 15 minutes. A complete ring of bubbles at the meniscus is a positive test and indicates lung maturity. This is a useful bedside test.

The presence of phosphatidylglycerol, (option C) saturated phosphatidylcholine and lamellar body count (>50,000/mcL) indicates lung maturity.

Glucocorticoids accelerate lung maturation and lower the incidence of respiratory distress and neonatal mortality in preterms if the birth was delayed for at least 24 hours after initiation of betamethasone. A single course of corticosteroids is recommended for women between 24 and 36 weeks who are at risk of delivery within 7 days. Intramuscular injections of either 2 doses of 12 mg betamethasone are given 24 hourly or 4 doses of 6 mg dexamethasone are given 12 hourly.

Solution to Question 2:

Isolation of the child is recommended until 24 hrs after the initiation of effective therapy after being diagnosed with acute bacterial meningitis to prevent further transmission.

Droplet precautions are recommended for patients with Neisseria meningitides and Haemophilus influenzae type b (Hib) meningitis until they have received 24 hours of effective therapy. Patients should be in private rooms, and hospital personnel should wear a face mask when they are within 3 feet (1 meter) of the patient.

The child with meningitis shows nonspecific symptoms like fever, anorexia, poor feeding, headache, myalgia, hypotension, petechia, purpura, or a rash. Meningial irritation is indicated by nuchal rigidity, back pain, and Kernig, and Brudzinski signs (usually not seen in infants). Seizures and altered mental status may be present.

The recommended empirical antibiotic regimen in a suspected case of meningitis outside the neonatal period is vancomycin combined with a third-generation cephalosporin. Patients allergic to penicillin or cephalosporin can be treated with meropenem, fluoroquinolones, or chloramphenicol.

Chemoprophylaxis with rifampicin/ceftriaxone/ciprofloxacin is recommended for all close contacts with meningococcal meningitis regardless of age or immunization status. Rifampicin prophylaxis should be given to all household contacts of patients with invasive disease caused by Haemophilus influenzae type b, if an immunocompromised child of any age resides in the household or close family member younger than 48 months has not been fully immunized.

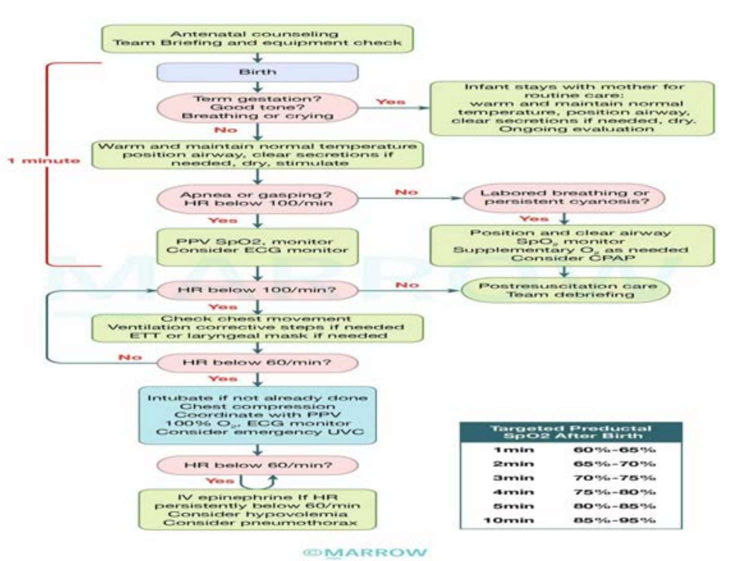
Solution to Question 3:

During neonatal resuscitation, the mouth is suctioned first and then the nares, in order to decrease the risk of aspiration. A 12 or 14 French suction catheter is used, with the suction pressure kept between 80-100mmHg.

Routine suctioning of the trachea during neonatal resuscitation is not recommended. The suctioning of the esophagus and stomach is avoided if not indicated specifically, as it can produce a vagal response due to stimulation in the posterior pharynx, resulting in apnea and/or bradycardia.

The steps in neonatal resuscitation follow the ABCs, Airway, Breathing, and Circulation.

The neonatal resuscitation algorithm is given below:



Chest compression should be initiated over the lower third of the sternum at a rate of 90/minutes. The ratio of chest compression to ventilation is 3: 1 (90 compressions: 30 breaths). The thumb technique (preferred) and 2 finger technique are the two different techniques used for performing chest compression. Umbilical vein cannulation is the preferred method for the administration of medications.

Solution to Question 4:

The given clinical scenario describes an adolescent girl with myoclonic jerks (a history of dropping objects) with preserved consciousness. This, along with abnormal EEG findings strongly suggests juvenile myoclonic epilepsy.

Juvenile myoclonic epilepsy (Janz syndrome) is the most common generalized epilepsy in young adults. It starts in early adolescence with one or more of the following features: bilateral myoclonic jerks in the morning, causing patients to drop things, generalized tonic-clonic or clonic-tonic-clonic seizures upon awakening, and juvenile absence seizure.

Consciousness is usually preserved. Precipitating factors include sleep deprivation, alcohol, and photic stimulation. EEG shows 4 to 5Hz polyspike and slow wave discharges. A family history of epilepsy is often present.

The treatment options for JME include sodium valproate, levetiracetam, and topiramate. Benzodiazepine, lamotrigine, perampanel, phenobarbital, primidone, and zonisamide can be considered as add-on drugs. Complete remission is uncommon and has a high risk of seizure recurrence, hence antiepileptics are continued lifelong in a majority of the patients.

Other options:

Option B: Atypical absence seizures have been associated with myoclonic components and head drop. They are precipitated by drowsiness and EEG shows 1 to 2 Hz spike and slow wave discharges.

Option D: In benign childhood epilepsy with centrotemporal spikes, the child typically wakes at night due to a focal seizure with preserved awareness. Buccal and throat tingling and tonic or clonic contractions on one side of the face are seen, with drooling and preserved consciousness. EEG shows typical wide-based centrotemporal spikes.

Solution to Question 5:

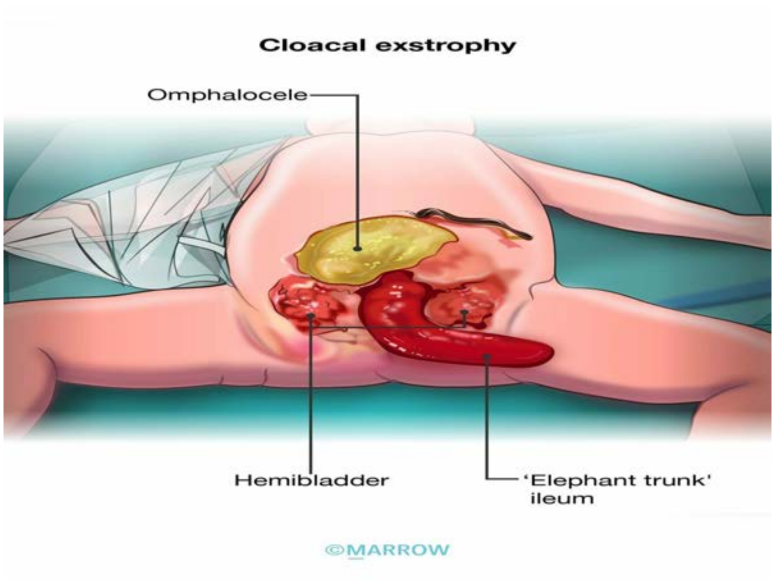
Neonate with continuous urine dribbling a protruding bladder with exposed mucosa suggests bladder exstrophy.

In bladder exstrophy, the bladder protrudes from the abdominal wall, and its mucosa is exposed. The umbilicus is displaced downwards, the pubic rami are widely separated and the rectus muscles are separated. Anomalies of the bladder occur when the mesoderm fails to invade the cephalad extension of the cloacal membrane, the extent of this failure determines the degree of the anomaly.

Males commonly have complete epispadias with dorsal chordee, undescended testes, short penis, and inguinal hernia. Epispadias with separation of the two halves of the clitoris and wide separation of labia is seen in females. Anus is displaced anteriorly in both sexes. Rectal prolapse may be seen.

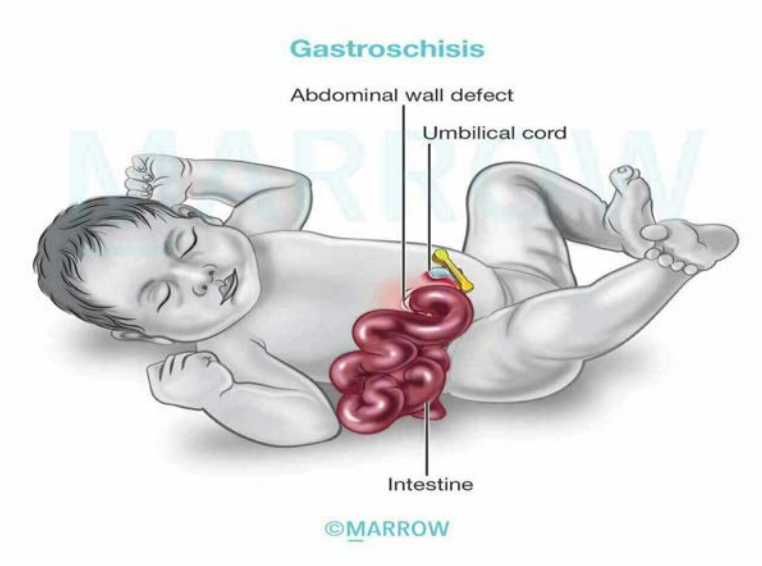
Management involves a staged surgical reconstruction of the bladder and genitalia. Untreated patients may develop total urinary incontinence and an increased incidence of bladder adenocarcinoma.

Other exstrophy anomalies are cloacal exstrophy and epispadias. Cloacal exstrophy (ileocaecal exstrophy) is a rare condition, in which there is a small exomphalos with an everted caecum and ileum separating halves of the bladder and in males, a split penis. The hindgut is truncated (giving tusks on the face of the elephant appearance).



Other options:

Option A: Gastroschisis is a condition where the intestinal loops herniate through the defect in the abdominal wall (shown below). The defect usually lies on one side of the umbilicus (right side commonly). The loops of the bowel are not covered by amnion as they herniate through the abdominal wall directly into the amniotic cavity. It is associated with elevated alpha-fetoprotein levels.



Option D: Mild forms of the imperforate anus are often called anal stenosis or an anterior ectopic anus.

Solution to Question 6:

Feeding by spoon is a social milestone that typically appears by 18 months. If this is absent, then the child should be termed as having a developmental delay. Hopping on one leg, drawing a square, and passing a ball to someone appear by 48 months of age.

Red flag signs in developmental screening and surveillance are indicators that suggest a serious disorder in a child's development, requiring referral to a pediatrician. These are given below:

Positive indicators: Presence of any of the following

• Loss of developmental skills at any age

• Concerns about vision or confirmed visual impairment

• Hearing loss at any age

• Persistently low muscle tone

• Lack of speech by 18 months without other means of communication

• Asymmetry of movements or features suggestive of cerebral palsy

• Persistent toe walking

• Complex disabilities

• Unusual head circumference compared to norms

• Clinician uncertainty about the developmental disorder.

Negative Indicators:

• Inability to sit unsupported by 12 months

• Inability to walk by 18 months in boys or 2 years in girls.

• Inability to walk normally

• Inability to run by 2.5 years

• Inability to hold objects in hand by 5 months or reach for them by 6 months.

• Inability to point at objects to share interest by 2 years

Solution to Question 7:

The clinical presentation of a child with a painful and growing parietal swelling, coupled with a history of a prior parietal bone fracture one year ago, strongly suggests the diagnosis of a growing fracture, also known as traumatic encephalocele or leptomeningeal cysts.

Skull fractures associated with a dural tear may fail to heal properly. The herniated brain, dilated ventricles, or a leptomeningeal cyst may cause an enlargement of this tear. These fractures are called growing skull fractures. They are usually seen in children < 3 years of age.

They present as a localized swelling or palpable skull defect that increases in size, most commonly seen in the parietal region. However, they may go undetected for years and present with delayed onset of neurologic symptoms (headache, seizure, or neurologic deficit).

Other options:

Option A: The extended duration of one year makes scalp hematoma unlikely, as it usually resolves within a few months.

Option C: Subdural hygromas are characterized by fluid accumulation in the subdural space. Most patients with subdural hygromas are asymptomatic, but uncommon symptoms may include headaches, changes in mental status, nausea and vomiting, focal neurological deficits, and seizures. They don't present with scalp swelling.

Option D: Chronic abscess may show inflammatory changes like erythema, pus, sinus, and systemic features like fever.

Solution to Question 8:

In an infant of a diabetic mother with normal blood glucose levels, the next investigation to consider is serum calcium due to the high risk of developing hypocalcemia. Hypocalcemia in these infants is attributed to the delayed response of the parathyroid hormone system.

Hypocalcemia is defined as a total serum calcium concentration of less than 8 mg/dL for a term baby (ionized calcium below 4mg/dL) and less than 7 mg/dL for a preterm baby. Asymptomatic neonates can be managed conservatively with early nutrition. Symptomatic neonates should receive iv or oral calcium replacement.

Maternal hyperglycemia causes fetal hyperglycemia, and the fetal pancreatic response leads to fetal hyperinsulinemia, which causes neonatal hypoglycemia. Neonatal hypoglycemia is defined as blood glucose <40 mg/dL (plasma glucose <45 mg/dL). The infant tends to be jittery, tremulous, hyperexcitable, hypotonic, lethargic, and poor sucking during the 1st three days after birth. These symptoms can also be related to hypocalcemia and hypomagnesemia which occur in the 1st 24 to 72 hours of life.

Respiratory distress syndrome, macrosomia, hyperbilirubinemia, polycythemia, renal vein thrombosis, visceromegaly, heart failure, and persistent pulmonary hypertension are other features seen in infants of diabetic mothers.

Congenital anomalies are seen in infants of mothers with pregestational diabetes. Neural tube defects (Encephalocele, meningomyelocele, anencephaly), transposition of great vessels (TGA), ventricular septal defect (VSD), atrial septal defect (ASD), hypoplastic left heart, aortic stenosis and coarctation of the aorta are commonly seen.

Caudal regression syndrome is rare, but a specific anomaly seen in infants of diabetic mothers. Small left colon syndrome is a rare anomaly that develops in the second and third trimesters.

Solution to Question 9:

Decerebrate or decorticate posturing is not seen in brain death as they are motor responses to noxious stimuli that originate from the midbrain and cortex.

Brain death, also known as death by neurologic criteria refers to the irreversible and total cessation of all functions of the entire brain, including the brainstem.

The three components for diagnosing brain death are a demonstration of coexisting irreversible coma with a known cause, absence of brain stem reflexes, and apnea. Coma requires the patient to be unresponsive, even to noxious stimuli. The diagnosis of brain death can still be made despite positive spinal reflexes. (option B)

Brainstem reflexes (pupillary responses, oculocephalic reflex, oculovestibular responses, corneal reflex, jaw reflex, facial grimacing, pharyngeal gag reflex, coughing) are absent. Apnea is the absence of respiratory drive at $PaCO_2$ of ≥60mmHg and > 20mmHg above baseline, which is consistent with brain death.

The following manifestations are occasionally seen and should not be misinterpreted as evidence of brainstem function:

• Normal systolic blood pressure - SBP >100 mmHg (option A)

• Movements originating from the spinal cord or peripheral nerve may occur

• Sweating, flushing, tachycardia (option C)

• Deep tendon reflexes; superficial abdominal reflexes; triple flexion response

• Babinski reflex

• Spontaneous movements of limbs other than pathologic flexion or extension response

• Respiratory-like movements (shoulder elevation and adduction, back-arching, intercostal expansion without significant tidal volumes)

Solution to Question 10:

Craniopagus is a type of conjoint twin where there is a fusion of the head or cranium.

Foramen magnum or the skull base never fuses in the case of craniopagus. They usually will have fused skulls and meninges. A shared cerebral cortex may occasionally be found.

Craniopagus has an incidence of 5%, and separation of the twins has a poor prognosis.

Dizygotic/fraternal twins result from the fertilization of two separate ova. Monozygotic/identical twins arise from the division of a single fertilized ovum. The outcome of the monozygotic twinning process depends on when division occurs.

Types of conjoined twins:

• Thoracopagus: united at the thorax

• Craniopagus: united at the head

• Cephalothoracopagus: fusion of both head and thorax

• Omphalopagous or xiphipagous: united anteriorly from the xiphoid to the umbilicus

• Parapagus: lateral fusion with two heads, sharing a single thorax, abdomen, and pelvis

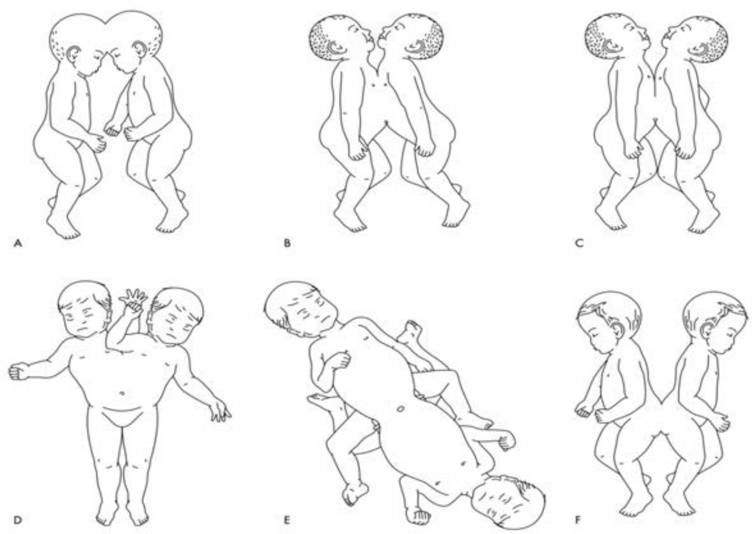
• Ischiopagus: united at the pelvis

• Pygopagus: united in the sacral region

• Heteropagous: asymmetric forms

The image below shows the types of conjoined twins:

Monozygotic types	Time of zygote division
Dichorionic diamniotic	<72 hours
Monochorionic diamniotic(more common)	4 to 8 days
Monochorionic monoamniotic	8 to 12 days
Conjoined twins	>12 days



Common forms of conjoined twins: (A) craniopagus, (B) thoracopagus, (C) omphalopagus, (D) parapagus, (E) ischiopagus, and (F) pyopagus.

Solution to Question 11:

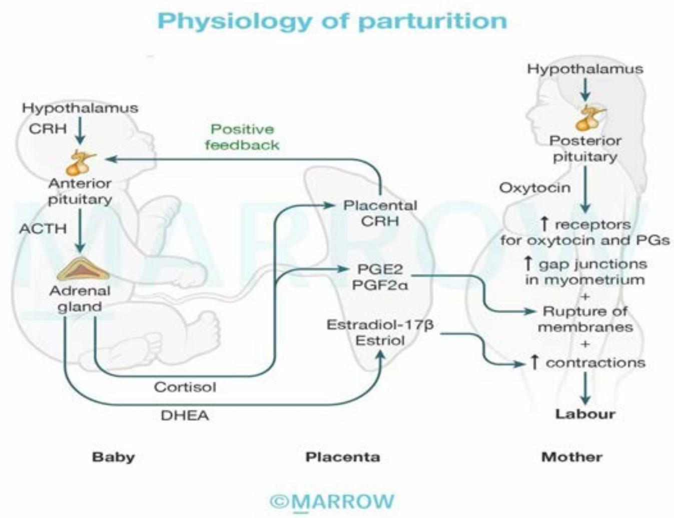
The major steroid products of the fetal adrenal gland are DHEA (dehydroepiandrosterone) and DHEA-S.

They serve as important substrates for estrogen synthesis in the placenta.

The fetal-adrenal-placental axis is vital for normal childbirth. Placental corticotropin-releasing hormone (CRH) stimulates the fetal adrenal gland to produce DHEA-S and cortisol. Cortisol further stimulates the production of placental CRH, creating a positive feedback loop that enhances adrenal steroid hormone production.

The steroid products released by the fetal adrenal gland have effects on the placenta and membranes. These effects contribute to transforming the myometrium from a quiescent to a contractile state.

Congenital adrenal hyperplasia is an autosomal recessive condition in which cortisol synthesis in the adrenal gland is affected. Cortisol deficiency increases the secretion of ACTH, which leads to adrenocortical hyperplasia and overproduction of intermediate metabolites.



Solution to Question 12:

The video clip of a 2-day-old neonate exhibiting repetitive, rhythmic movement in unilateral upper and lower limbs suggests a neonatal seizure of focal clonic type.

A focal clonic seizure is characterized by multiple, repetitive, rhythmic contractions of muscle groups of the limbs, face, or trunk, maybe unifocal or multifocal. It may occur synchronously or asynchronously in muscle groups on one side of the body. It may occur simultaneously but asynchronously on both sides. It cannot be suppressed by restraint.

Other options:

Option A: Focal tonic seizures are characterized by sustained posturing of a single limb, sustained asymmetrical posturing of the trunk, and sustained eye deviation. They cannot be provoked by stimulation or suppressed by restraint.

Option C: Spasms are sudden generalized jerks lasting 1-2 sec and are usually associated with a single, very brief, generalized discharge. They may be flexor, extensor, or mixed extensor/flexor and can occur in clusters. They too cannot be provoked by stimulation or suppressed by restraint.

Option D: Subtle seizures include transient eye deviations, nystagmus, blinking, mouthing, abnormal extremity movements (rowing, swimming, cycling, pedalling), fluctuations in heart rate, hypertension episodes, and apnea. They are more common in preterm than in term babies.

Myoclonic seizures are random, single, rapid muscle contractions of the limbs, face, or trunk. They are not repetitive or may recur at a slow rate. They can be generalized, focal, or fragmentary. They can be provoked by stimulation.

Clinically apparent seizures in a neonate should be treated if they last >3 minutes or are brief serial seizures. In all neonates with seizures, hypoglycemia, CNS infection, and hypocalcemia should be ruled out before starting anti-epileptic drug treatment.

Causes of neonatal seizure according to the age of presentation:

1 to 4 days	4 to 14 days	2 to 8 weeks
Hypoxic ischemic encephalopathyMaternal use of narcotics or barbituratesDrug toxicity: lidocaine, penicillinIntraventricular hemorrhageHypocalcemia, sepsis, hypoglycemia, hypomagnesemia, hypo or hypernatremiaGalactosemia, urea cycle disorders	Meningitis (bacterial)Encephalitis (enteroviral, herpes simplex)Hypocalcemia, hypoglycemiaGalactosemia, fructosemia,Hyperinsulinism, hyperammonemia syndrome	Herpes simplex or enteroviral encephalitis, bacterial meningitisSubdural hematoma, child abuseAminoaciduria, urea cycle defectsMalformations of cortical developmentTuberous sclerosisSturge-Weber syndrome

Solution to Question 13:

The given clinical scenario of a child with coarse facies, macroglossia, thick lips, hepatosplenomegaly, and copious discharge from the nose is suggestive of Hurler's disease.

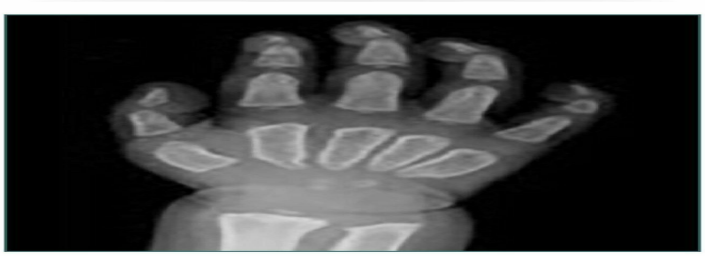
It is the severe form of mucopolysaccharidoses type 1 (MPS 1), an autosomal recessive disorder caused by a defect in α-L iduronidase, leading to the accumulation of heparan sulfate and dermatan sulfate. The child usually presents between 6 and 24 months with hepatosplenomegaly, coarse facial features (shown below), corneal clouding, large tongue, prominent forehead, joint stiffness, short stature, and skeletal dysplasia.

Most patients have recurrent upper respiratory tract and ear infections, noisy breathing, and persistent copious nasal discharge. Radiograph shows skeletal dysplasia, called dysostosis multiplex - thick ribs, ovoid vertebral bodies, enlarged and trabeculated diaphyses of the long bones with irregular metaphysis and epiphysis.

Macrocephaly, enlarged J-shaped sella, premature closure of lambdoid sagittal sutures, and shallow orbits are seen with disease progression. Obstructive airway disease, cardiac complications, and respiratory infections are the common causes of death.

The image below shows the radiograph shows enlarged diaphyses of the metacarpal and phalanges.

A



B

Other options:

Option B: Beckwith-Wiedemann syndrome is an overgrowth syndrome characterized by macrosomia, macroglossia, hemihypertrophy, omphalocele, and renal anomalies. It is associated with an increased risk of Wilms tumor, hepatoblastoma, rhabdomyosarcoma, neuroblastoma, and adrenal cortical carcinoma.

Option C: Proteus syndrome is a disturbance of cellular growth involving ectodermal and mesodermal tissues, caused by mutation in the AKT1 gene. They present with hemimegalencephaly, asymmetric overgrowth of the extremities, verrucous cutaneous lesions, thickening of the bone, and excessive growth of muscles.

Option D: A child with untreated congenital hypothyroidism may have stunted growth, short extremities, narrow palpebral fissures, swollen eyelids, depressed nasal bridge, thick broad protruded tongue, and short and thick neck, dry and scaly skin.

Solution to Question 14:

The above pedigree chart denotes an autosomal recessive pattern. In the above pedigree chart, both males and females are carriers (equally affected) which indicates an autosomal pattern. Parents are unaffected (both are carriers/heterozygotes), and no affected members in other generations. 25% of the children are normal, 50% are carriers, and only 25% are affected. This indicates a a recessive pattern.

Pedigree charts in various modes of inheritance:

Solution to Question 15:

The Silverman-Anderson score of the child would be 8.

Calculation of the Silverman-Anderson score of this child:

Total score: 8

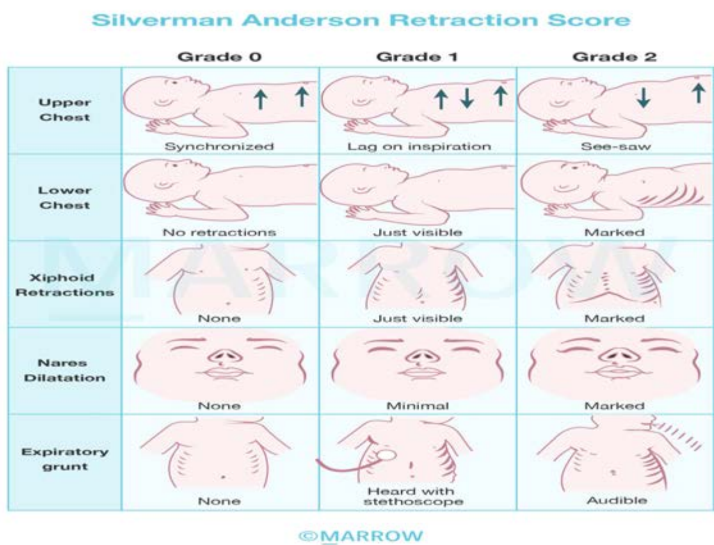
A score of more than 7 indicates respiratory failure.

The Silverman-Anderson score is a commonly used score to assess respiratory distress. It helps to assess the progression of distress over time and initiate treatment. It is more suited for preterms with respiratory distress syndrome. Scoring is to be done every half an hour and charting is carried out to monitor progress.

The image below shows Silverman Anderson retraction score:

Criteria	Finding	Score
Expiratory grunt	Audible with the naked ear	2
Nasal flaring	Marked	2

Criteria	Finding	Score
Xiphoid retraction	Marked	2
Intercostal retraction (lower chest)	Minimal	1
Chest movement (upper chest)	Lag	1



Downes' Score is also used for assessing the severity of respiratory distress. It is more comprehensive and can be applied to any gestational age and condition. Downe's score of >7 means impending respiratory failure.

Score	0	1	2
Respiratory rate	<60/min	60–80/min	>80/min
Air entry	Normal	Mild decrease	Marked decrease
Cyanosis	Nil	In-room air	On >40% O2
Grunting	None	Only on auscultation	Audible with the naked ear
Retraction	Nil	Mild	Moderate

Solution to Question 16:

Dysrhythmias in the pediatric age group are most often the result of respiratory insufficiency or hypoxia and not of primary cardiac causes, as in adults.

Hypoxia, hypercapnia, and acidosis lead to bradycardia, hypotension, and secondary cardiac arrest in children.

Other options:

Option A: Supraventricular tachycardia is the most common tachydysrhythmia that necessitates treatment in pediatric patients. Ventricular dysrhythmia is an uncommon presenting rhythm in infants and children.

Option B: The pre-arrest phase is the period that precedes the period of cardiac arrest.

Four phases of cardiac arrest have been described as follows:

• The prearrest phase- a period that precedes the period of cardiac arrest.

• The no-flow phase- reflects untreated cardiac arrest before it is recognized by a bystander in the community or by a medical provider in the hospital

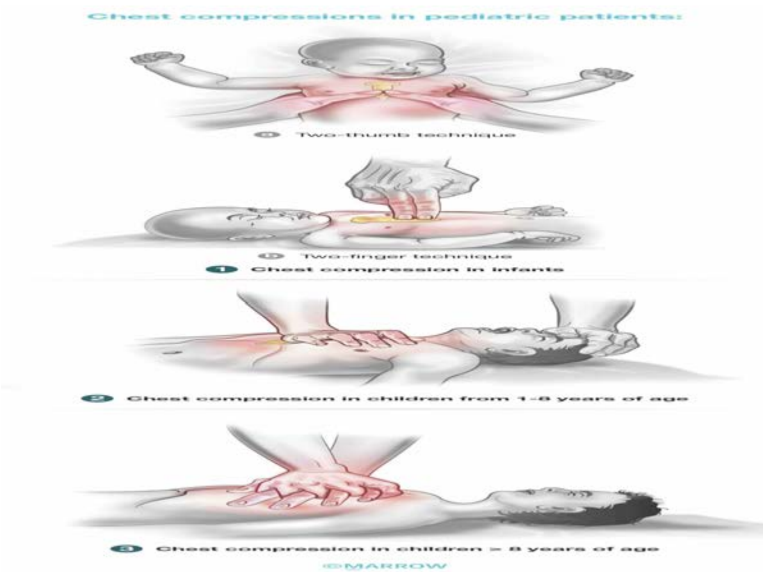
• The low-flow phase- begins with the onset of cardiopulmonary resuscitation (CPR)

• The post-resuscitation phase - The post-resuscitation phase begins with the return of spontaneous circulation (ROSC).

Option C: The chest compression to ventilation ratio should be 30:2 or 15:2. Excessive ventilation results in decreased venous return into the chest, decreased coronary and cerebral blood flow, diminished cardiac output, and ultimately a decreased likelihood of return of spontaneous circulation.

Guidelines for pediatric basic life support:

Maneuver	Neonate	Infant <1 year	Child 1 year to puberty
Rescue breath	30-60/minute (every 1 to 2 seconds)	12-20/minute (every 3 sec onds)	12-20/minute (every 3 sec onds)
Pulse check site	Umbilical	Brachial	Carotid/femoral
Chest compression site	one finger below the inte rmammary line	One finger below the inte rmammary line or lower half of the sternum	The lower half of the ster num
Compression method	Two fingers or two thum b	Two fingers or two thum b	The heel of one hand or t wo hands
Compression depth	One-third of the chest	One-third to one-half of t he chest	One-third to one-half of t he chest
Compression rate	120/minute	100/minute	100/minute
Compression to ventilati on ratio	3: 1	15: 2 (single rescuer - 30: 2)	15: 2 (single rescuer - 30: 2)


Chest compressions in pediatric patients:
Two-thumb technique
Two-finger technique
1 Chest compression in infants
2 Chest compression in children from 1-8 years of age
3 Chest compression in children > 8 years of age
©MARROW

Paediatrics AIIMS 2019 (May & Nov)

Question 1:

Which of the following is a component of the neuromuscular criteria for maturity according to the expanded New Ballard score?

- **a)** Heel-knee test
- **b)** Finger-nose test
- **c)** Scarf sign
- **d)** Cubital angle

Question 2:

An 18-month-old baby was brought to the hospital with complaints of poor feeding and abnormal twitching of his lower limb. On examination, he has fever with fontanelle bulging and the child was lethargic. The most probable cause is:

- **a)** Febrile seizures
- **b)** Meningitis
- **c)** Pseudotumor cerebri
- **d)** Intracranial haemorrhage

Question 3:

A child presents with brown-colored urine and oliguria for the last 3 days. He has mild facial and pedal edema. His blood pressure is 126/90. He has 2+ proteinuria with >100 red cells/HPF and a few granular casts. His creatinine is 0.9, urea is 56. What is the likely diagnosis?

- **a)** PSGN
- **b)** Nephrolithiasis
- **c)** Posterior Urethral Valve
- **d)** Minimal change disease

Question 4:

Sign of effective neonatal resuscitation is:

a) Color change

b) Bilateral Air entry

c) Increased heart rate

d) None of the above

Question 5:

Gene involved in juvenile myoclonic epilepsy:

a) CHYNN-1

b) GABRA-1

c) FMR-1

d) All of the above

Question 6:

A 3-week neonate presents with ambiguous genitalia with Na+ - 127; K+ - 6 meq/L with a BP of 50/22 mm Hg. Along with IV fluids, what specific management is done?

a) Spironolactone

b) Potassium binding resins

c) Hydrocortisone

d) Broad spectrum antibiotics

Question 7:

A 6 years old child with developmental delay, can ride a tricycle, can climb upstairs with alternate feet, can say their name, and knows own sex, but cannot narrate a story. What is the developmental age?

a) 3 years

b) 4 years

c) 5 years

d) 2 years

Question 8:

The procedure displayed below is:

a) Intubation

b) Nasogastric tube insertion

c) Orogastric tube Insertion

d) Oropharyngeal airway insertion

Question 9:

An infant in the NICU was observed to have the features as seen in the video below. In addition to these features, there is an audible grunt. What would the Silverman Anderson score be?

a) 8

b) 6

c) 5

d) 4

Question 10:

Which of the following denotes an "infant at risk"?

a) Infant of a working mother

b) Infant of mother who has not taken folic acid during pregnancy

c) Infant with history of premature birth at <32 weeks gestation

d) Infant of mother with pre-eclampsia

Question 11:

A 7-year-old child presents with leg pain. She had fever and cough 3 days prior. Currently, there is pain on palpation of the calf muscles. Reflexes and other clinical examinations are normal. Lab test reveals CK levels of 2000 IU. What is the likely diagnosis?

a) Viral Myositis

b) Duchenne

c) GBS

d) Dermatomyositis

Question 12:

An 8-year-old boy with sensorineural hearing loss and eye defects presents to the OPD with hematuria. His renal biopsy on light microscopy was normal. Family history reveals the maternal uncle had Gr 5 CKD an is undergoing dialysis. Which of the following would be the probable diagnosis?

- **a)** Alport syndrome
- **b)** Goodpasture syndrome
- **c)** IgA nephropathy
- **d)** PSGN

Question 13:

The following device shown attached to the baby is used in all except :

- **a)** In CPR
- **b)** To know about apnea
- **c)** In neonatal resuscitation
- **d)** Measure central cyanosis

Answer Key

Question No.	Correct Option
1	c
2	b
3	a
4	c
5	b
6	c
7	a
8	c
9	a
10	c
11	a
12	a
13	d

Detailed Explanations

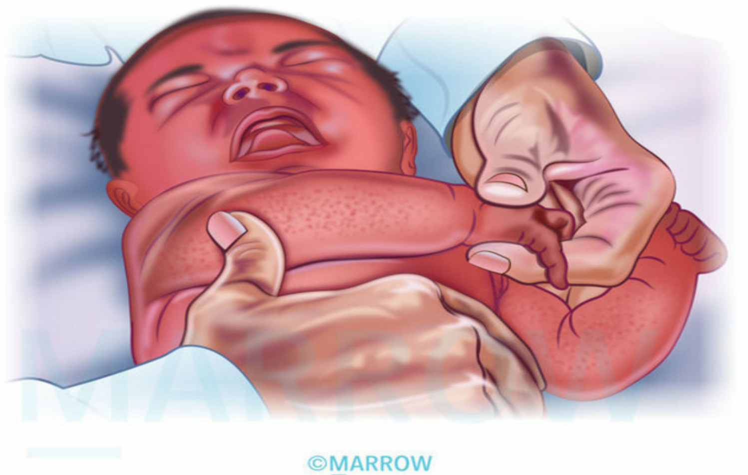
Solution to Question 1:

Scarf sign is a component of the neuromuscular criteria for maturity according to the expanded new Ballard score.

Scarf sign: If the infant's arm is pulled medially across the chest (as shown in the image below),

• In a hypotonic infant, the elbow will cross the midline.

• In a term infant with a normal tone, the elbow will not reach the midline.



Neuromuscular criteria for maturity (according to the expanded New Ballard score) include:

• Posture

• Square window (wrist)

• Arm recoil

• Popliteal angle

• Scarf sign

• Heel to ear

Physical criteria for maturity (according to the expanded New Ballard score) include:

• Skin

• Lanugo

• Eye/ear

• Plantar crease

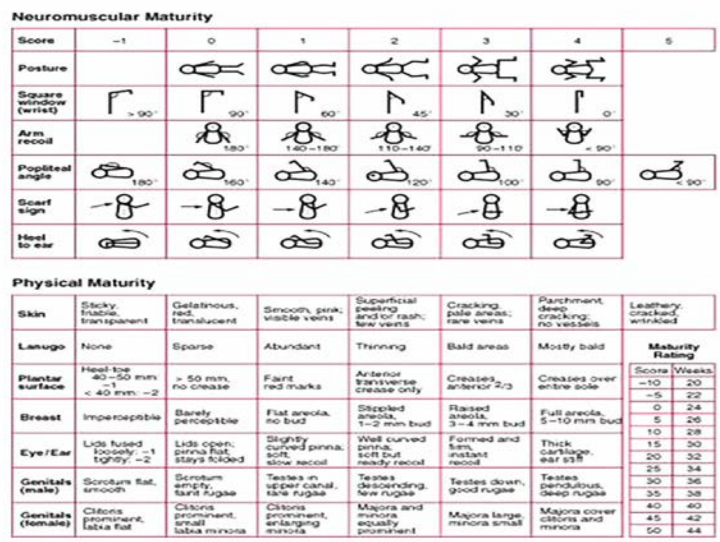
• Breast

• Genitals

The expanded new Ballard score is used to assess the physical and neuromuscular maturity of the newborn by scoring six criteria and thereby calculating the gestational age. It is accurate between 20 to 44 weeks.

The lowest score on the expanded New Ballard Score is -10, which corresponds to 20 weeks gestational age, and the highest score is 50, which corresponds to 44 weeks gestational age. An increase in the score by 5 points corresponds to an increase of 2 weeks gestational age

Given in the image below is the scoring system for the physical and neuromuscular criteria:



Solution to Question 2:

The clinical features of poor feeding, fever, seizures (abnormal lower limb twitching), bulging fontanelle (due to raised ICP), and lethargy are all suggestive of meningitis.

Febrile seizures (Option A) are diagnosed in the absence of any other features suggestive of CNS infection.

Pseudotumor cerebri (Option C) is rarely seen under the age of 10 years. There is an association with obesity, female gender, pregnancy, and drugs such as tetracycline, vitamin A, and retinoid derivatives. It has a chronic course of illness, with papilledema being present in a majority of cases.

Intracranial hemorrhage (Option D) will present with seizures, focal neurological deficits, altered tone, and altered consciousness.

Bacterial meningitis may be caused due to Neisseria meningitidis, Streptococcus pneumoniae, or Haemophilus influenzae type b.

Non-specific symptoms and signs include fever, rash, lethargy, refusal to feed, vomiting, etc. Specific symptoms and signs include photophobia, nuchal rigidity, bulging of the fontanelle, and clinical signs of meningeal irritation (Kernig and Brudzinski signs), but this is not usually seen in infants. Seizures and focal neurological signs may be present.

The diagnosis of meningitis is confirmed by CSF analysis via lumbar puncture - which typically shows micro-organisms on gram stain and culture, neutrophilic pleocytosis, increased protein, and decreased glucose concentration. Blood cultures should be done in all cases of suspected meningitis.

Acute CNS complications include seizures, increased ICP, cranial nerve palsies, stroke, cerebral or cerebellar herniation, and thrombosis of the dural venous sinuses. Subdural effusions and non-communicating hydrocephalus may develop.

Solution to Question 3:

A child presenting with acute onset (3 days) gross hematuria, hypertension, edema, oliguria, and mild to moderate proteinuria is suggestive of a nephritic syndrome picture. The only disease which causes this out of the given options is post-streptococcal glomerulonephritis (PSGN).

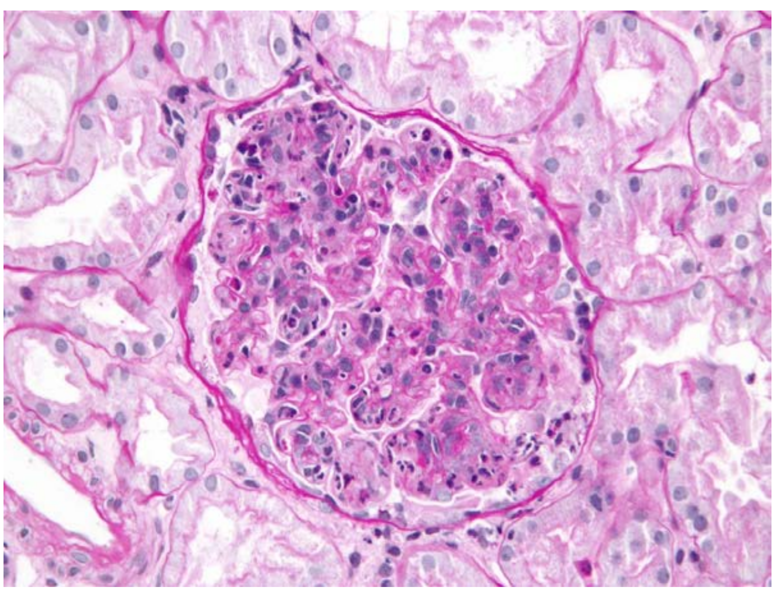
PSGN is primarily caused by Group A β-hemolytic streptococcal infections in children aged 5 to 12 years. It is caused by immune complexes containing streptococcal antigens (M protein of cell wall) and specific antibodies. PSGN typically manifests as an acute nephritic syndrome, appearing either 1 to 2 weeks after streptococcal pharyngitis or 3 to 6 weeks after streptococcal pyoderma. The severity of kidney involvement can vary, ranging from asymptomatic microscopic hematuria with normal renal function to gross hematuria with acute renal failure. Edema, hypertension, and oliguria may be present depending on the extent of renal involvement.

Laboratory findings commonly include a significant decrease in serum C3 levels during the acute phase, with values returning to normal within 6 to 8 weeks. To confirm the diagnosis, evidence of a preceding streptococcal infection is necessary. The antistreptolysin O (ASO) titer often increases following pharyngitis, while the anti-deoxyribonuclease B level is a reliable indicator of cutaneous streptococcal infection.

Light microscopic examination reveals enlarged, hypercellular glomeruli with diffuse mesangial and endothelial proliferation, as well as leukocyte infiltration. Immunofluorescence analysis demonstrates granular deposits of IgG and complement C3 along the glomerular basement membrane (GBM) and in the mesangium. Electron microscopy primarily shows subepithelial humps, with subendothelial deposits observed in early stages of the disease.

Managing the acute effects of renal insufficiency (sodium restriction and diuretics) and hypertension is the mainstay. Antibiotic therapy with a 10-day course of penicillin is recommended to limit the spread of nephritogenic pathogens. But antibiotics do not alter the natural course of PSGN. Complete resolution of the hematuria and proteinuria in the majority of children occurs within 3–6 weeks of the onset of nephritis but 3–10% of children may have persistent microscopic hematuria, non-nephrotic proteinuria, or hypertension.

Below is the image showing light microscopy findings of post-streptococcal glomerulonephritis:



Solution to Question 4:

An increase in the heart rate is the best sign of effective neonatal resuscitation. The response to neonatal resuscitation at various steps is assessed based on the heart rate.

As soon as a baby is born, the tone, cry, breathing efforts, and respiratory efforts are assessed. If the tone is poor, or if the baby is not breathing, the following measures are done immediately - warming to maintain the normal temperature, proper positioning of the baby's airway, i.e. sniffing position with the head slightly extended, clearing of airway secretions, and tactile stimulation.

After the initial steps of neonatal resuscitation, the heart rate of the baby must be assessed. If the heart rate is < 100 beats per minute (bpm), positive pressure ventilation is provided along with SPO2 monitoring to ensure adequate ventilation.

If the heart rate remains at < 100 bpm, ventilation corrective steps and/or endotracheal intubation have to be considered. The ventilation corrective steps are as follows and can be remembered with the mnemonic MRSOPA.

• Mask has to be reapplied

• Reposition head

• Suction the secretions

• Open mouth slightly and ventilate

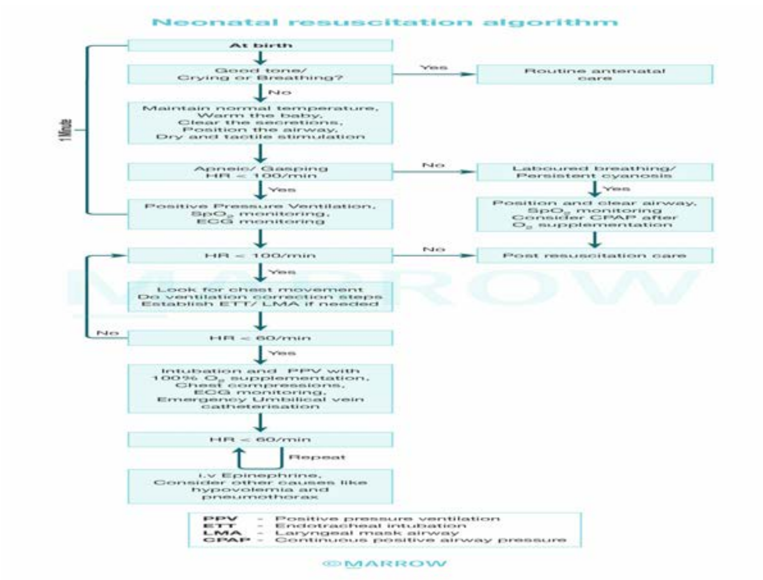
• Pressure is increased slightly

• Airway may be changed

If the heart rate is < 60 bpm, chest compressions are initiated over the lower third of the sternum at a rate of 90 compressions/min. The ratio of compressions to ventilation is 3: 1 (90 compressions: 30 breaths).

If the heart rate still remains <60 bpm, despite effective compressions and ventilation, then epinephrine (1:10,000 solution at 0.1–0.3 mL/kg intravenously or 0.5–1 mL/kg intratracheally) should be administered. The dosage can be repeated every 3–5 minutes. Persistent bradycardia may signify inadequate ventilation or severe cardiac compromise.

The neonatal resuscitation algorithm is given below:



Solution to Question 5:

Mutation in GABRA-1 is associated with juvenile myoclonic epilepsy.

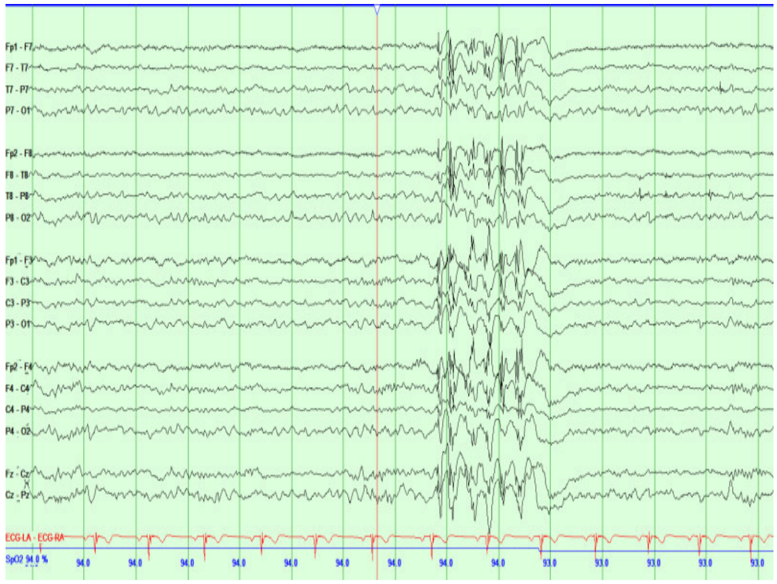
Juvenile myoclonic epilepsy (JME), also known as Janz syndrome or impulsive petit mal, is the most common generalized epilepsy in young adults. It may be caused due to mutation in many genes including CACNB4, CLNC2, GABRD, GABRA1, and Myoclonin1/ EFHC1

It typically begins between the ages of 12 and 18 (adolescence). The defining seizure type in JME is generalized myoclonic seizures, which occur in all patients. These seizures are typically mild and primarily affect the upper extremities in the morning causing the patient to drop things. However, they often go unnoticed and are only brought to medical attention after a generalized tonic-clonic seizure, which tends to occur after sleep deprivation or alcohol consumption. Some cases of childhood absence epilepsy may evolve into JME.

Diagnosis of JME relies on the patient's clinical history and electroencephalogram (EEG) findings. EEG typically shows bursts of generalized irregular 4- to 6-Hz spike-and-wave activity, particularly after awakening. JME is generally a lifelong condition.

Valproate is considered the most effective medication. Lamotrigine, levetiracetam, and topiramate are also approved for JME.

The characteristic EEG pattern is shown in the image below:



Solution to Question 6:

The given clinical case of a neonate presenting with ambiguous genitalia, hyponatremia, hyperkalemia, and hypotension points towards a diagnosis of congenital adrenal hyperplasia (CAH). Hydrocortisone must be administered along with IV fluids for the management of this baby.

The most common enzyme deficiency in CAH is 21-hydroxylase deficiency. Babies with 21-hydroxylase deficiency typically present around 10-14 days of age with the subset of symptoms due to aldosterone deficiency, cortisol deficiency, and androgen excess.

Aldosterone deficiency leads to salt-wasting crises marked by hyponatremia, hyperkalemia, hypotension, and metabolic acidosis.

Cortisol deficiency causes manifestations like hypoglycemia, hyperpigmentation, weight loss, weakness, anorexia, vomiting, and shock.

Androgen excess in 21-hydroxylase deficiency results in virilization of females, leading to female pseudohermaphroditism, and precocious puberty in males. Virilization is usually apparent at birth in females and within 2 to 3 years of life in males.

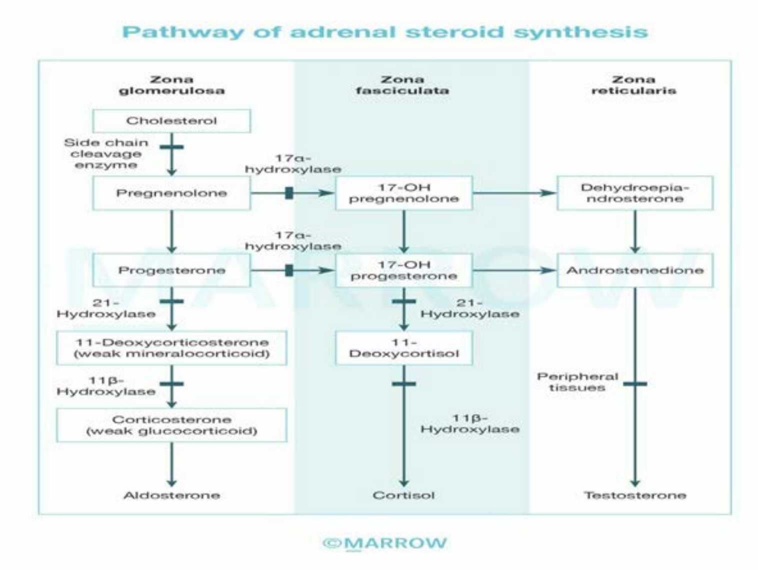
Without treatment, shock, cardiac arrhythmias, and death may occur within days or weeks.

Diagnosis is confirmed by elevated 17-hydroxyprogesterone levels or by genetic testing.

Glucocorticoid replacement therapy is the primary treatment for cortisol deficiency, aiming to suppress androgen production and prevent complications associated with androgen excess. Fludrocortisone is given to control mineralocorticoid deficiency.

If the condition is diagnosed prenatally, dexamethasone may be administered to the mother.

The image below summarizes the pathway involved in steroid hormonal production:



Solution to Question 7:

The developmental age of a child who can ride a tricycle, can climb upstairs with alternate feet, can say their name, and knows own sex is 3 years. Telling stories is a language milestone expected to be attained at 4 years of age.

The developmental milestones at 3 years of age are as follows:

Domain	Milestones
Motor	Rides tricycleStands momentarily on one footWalks upstairs with alternate foot
Fine Motor	Builds tower of 9 blocksCopies circle
Language	Knows name and sexAsks questions
Social	Shares toysKnows full name and gender

Solution to Question 8:

The procedure depicted is the insertion of an orogastric tube.

It is indicated in conditions like choanal atresia and cleft palate - where a nasogastric tube cannot be passed, or may cause inadvertant injury.

An orogastric tube is used to facilitate enteral rehydration and administer medications in children with dehydration or who are critically ill. It can also be used for gastric decompression in cases of distention due to small intestinal atresia or stenosis.

Solution to Question 9:

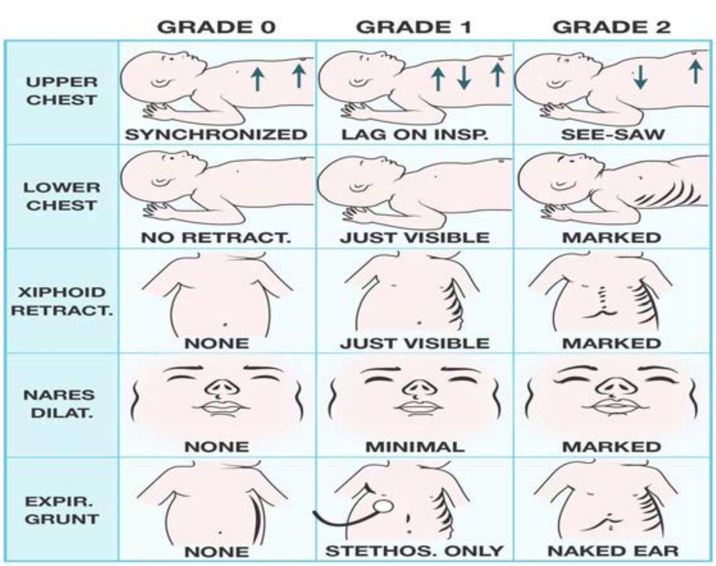
In the given case, the baby shows a see-saw chest movement (2), minimal intercostal retraction (1), visible xiphoid retraction (1), marked nasal flaring (2) and there is an audible respiratory grunt (2). The Silverman-Anderson score of the baby would be 8 (2+1+1+2+2).

Given below is the Silverman-Anderson scoring system (as per 2022 IAP guidelines) to assess the severity of respiratory distress in a child:

Interpretation:

Score	Upper chest retraction(Chest movement)	Lower chest retraction(Intercostal retraction)	Xiphoid retraction	Nasal dilatation	Grunt
0	Synchronized	none	none	none	none
1	Lag on inspiration	Just visible	Just visible	Minimal	Stethoscope
2	See-saw	Marked	marked	marked	naked ear

Score	Inference
0	No distress
1-4	Mild respiratory distress
5-7	Moderate respiratory failure
>7	Severe or impending respiratory distress



Silverman Anderson retraction score is more suited for preterms with hyaline membrane disease.

Downes' Score is a more comprehensive scoring system and can be applied to any gestational age and condition. Scoring is to be done every half an hour and charting is carried out to monitor progress.

Downe's score of >6 means impending respiratory failure.

Score	0	1	2
Respiratory rate	<60/min	60–80/min	>80/min
Air entry	Normal	Mild decrease	Marked decrease
Cyanosis	Nil	In-room air	On >40% O2
Grunting	None	Only on auscultation	Audible with the naked ear
Retraction	Nil	Mild	Moderate

Solution to Question 10:

An infant with a history of premature birth at <32 weeks gestation will be considered a high-risk infant according to AIIMS protocols on Neonatology.

The inclusion criteria for high-risk neonates according to AIIMS protocol are:

1. Birth weight <1500 grams

2. Gestation age <32 weeks

3. Infants with a birth weight of ≥ 1500gm or gestation age ≥ 32 weeks with any of the following risk factors:

• Intrauterine growth centile <3rd centile

• Meningitis

• Received mechanical ventilation for 48 hours or more

• Hypoxic ischemic encephalopathy stage 2 or higher

• Major malformation

• Inborn error of metabolism/ chromosomal or genetic disorders/ intrauterine infections

• Symptomatic hypoglycemia

• Symptomatic polycythemia

• Retrovirus positive mother

• Hyperbilirubinemia requiring exchange transfusion OR Rh isoimmunization/cholestasis

• Abnormal neurological examination at discharge/ seizures

• Major morbidities such as chronic lung diseases, IVH grade III or more and periventricular leukomalacia

Note: Although Park's Textbook of Preventive and Social Medicine mentions infants of working mothers (Option A) as high-risk infants, premature infant with < 32 weeks gestation is chosen over it, as AIIMS protocols are used as the preferred reference for AIIMS exams.

Solution to Question 11:

The clinical picture showing pain in the calves occuring within a week of an upper respiratory tract infection (URI), with elevated creatine kinase levels suggests the diagnosis of viral myositis. The specific history of URI suggests influenza myositis.

Infectious myositis has a male predominance and is typically seen in young adults. Childhood form presents with fever, malaise, and rhinorrhea and is usually followed by severe pain, especially in the calves. The muscle pain is usually worse with movement and the symptoms last for about a week. In children, toe walking and wide-based gait may be seen because of the involvement of the gastrocnemius-soleus muscles.

Other viral causes of myositis include HIV-1 (one of the most common causes of myositis), group B coxsackievirus (epidemic myalgia), HTLV-1 (Human T-lymphotropic virus), and cytomegalovirus (CMV).

Lab features of influenza myositis include elevated CK which may be as high as 500 times normal the value. Urine myoglobin is usually positive. Detection of the virus with PCR testing of nasopharyngeal specimens can be carried out but is not of much clinical significance. The histopathological examination would reveal muscle fiber necrosis without much inflammatory change.

Treatment includes bed rest, IV fluids, and symptomatic management with analgesics and antipyretics. Amantadine and other antiviral agents may be considered in adults. Complications include myocarditis and respiratory dysfunction. Rhabdomyolysis is an uncommon complication.

Solution to Question 12:

The clinical presentation of a child with hematuria, sensorineural hearing loss, and eye defects with normal renal biopsy is consistent with Alport Syndrome. It is a hereditary nephritis, which explains the family history of chronic kidney disease.

Alport syndrome is a genetic disease caused by mutations in the COL4A5 gene, which codes for the alpha chain of type IV collagen. In approximately 85% of cases, the syndrome is X-linked.

In the early years (around 5 years), patients are often asymptomatic but may have persistent microscopic hematuria. Between 5 and 10 years of age, proteinuria, and hypertension may develop, which can be managed with angiotensin-converting enzyme (ACE) inhibitors. By 10 years of age, serum creatinine levels start to rise. Around 15 years, patients may experience ear, nose, and throat (ENT) manifestations, with high-frequency sensory neural hearing loss being a common feature in males. Ocular manifestations, such as anterior lenticonus (pathognomonic) and dot and fleck retinopathy (most common), typically appear around the age of 20. Anterior lenticonus is particularly correlated with the development of end-stage renal disease (ESRD). By 25 to 30 years of age, most individuals with Alport's syndrome progress to ESRD.

Gross hematuria following upper respiratory tract infections is not typically seen in Alport's syndrome and should prompt consideration of other conditions, such as IgA nephropathy.

Skin immunofluorescence to demonstrate the lack of the $\alpha 5$(IV) collagen chain and genetic testing can be used for the diagnosis. Light microscopy and immunofluorescence (IF) on renal biopsy may be normal. Electron microscopy reveals irregular areas of thinning and thickening of the glomerular basement membrane (GBM), along with splitting of the lamina densa, giving it a basket-weave appearance.

The mainstay of management is the control of hypertension and ACE inhibitors to slow down the progression of renal damage. Although transplant recipients develop anti-GBM antibodies, graft survival is good.

Thin GBM disease is an autosomal dominant disease characterized by persistent microscopic hematuria but no progression to ESRD or extrarenal manifestations. EM can be used to confirm the diagnosis, showing GBM thickness of less than 200 nm.

Other options:

Option B: Goodpasture syndrome is an immune-mediated disease with antibodies against the alpha-3 chain of GBM type IV collagen; it is not hereditary and the light microscopy would reveal extra capillary proliferation with crescents & necrosis.

Option C: In IgA Nephropathy, light microscopy would reveal focal mesangial proliferative glomerulonephritis & mesangial widening.

Option D: In post-streptococcal glomerulonephritis (PSGN), there will be diffuse endocapillary proliferation and accompanying leukocytic infiltration seen in light microscopy.

Solution to Question 13:

The image shows a pulse oximeter. It cannot be reliably used to measure central cyanosis.

A pulse oximeter employs spectrophotometry to measure the percentage saturation of oxygen at a peripheral site. In cases of markedly deranged hemoglobin (Hb) levels and abnormal Hb, pulse oximetry doesn't correlate with central cyanosis, which depends on the absolute value of reduced Hb.

A pulse oximeter can be used for assessing response to resuscitation, titrating oxygen therapy in newborns, monitoring apnea (bradycardia and desaturation), confirming endotracheal tube placement, evaluating the effectiveness of bag and mask ventilation, and monitoring for hypoxia during suction and laryngoscopy.

Cyanosis is characterized by a bluish discoloration of the skin and mucous membranes due to elevated levels of reduced Hb (more than 4 g/dL) or Hb derivatives. It is examined in the nail beds and mucous membranes, earlobes, lips, and fingers. It can be classified into central and peripheral types.

Central cyanosis can be due to reduced oxygen saturation or the presence of abnormal Hb derivatives. Causes include high altitudes, impaired pulmonary function, anatomic shunts, and abnormal Hb. Both the mucous membranes and skin will be affected.

The absolute amount of reduced Hb rather than its relative quantity is significant in cyanosis. In severe anemia, even with marked arterial desaturation, cyanosis may not be present. Conversely, polycythemia patients are prone to cyanosis even at higher oxygen saturation levels.

The presence of abnormal Hb species may lead to errors in SpO_2 readings. In a patient with methemoglobinemia, the SpO_2 will be 80% to 85%, irrespective of arterial saturation (SaO_2). A patient with carbon monoxide poisoning will have a falsely elevated SpO_2 level. A co-oximeter is a better device to measure abnormal hemoglobins.

Carbon monoxide poisoning does not lead to true cyanosis. It causes cherry red discoloration.

Peripheral cyanosis occurs due to decreased blood flow and excessive oxygen extraction from normally saturated arterial blood. The mucous membranes of the oral cavity and sublingual mucosa are spared. The causes include reduced cardiac output, cold exposure, and arterial and venous obstruction. Here, the degree of cyanosis will generally correspond to SpO_2 readings. However, the SpO_2 levels may be falsely low despite normal SaO_2.

Paediatrics AIIMS 2020 (May and Nov INI-CET)

Question 1:

Which of the following is least likely regarding Rh incompatibility in a newborn?

a) Increased unconjugated bilirubin

b) Increased fecal urobilinogen

c) Increased urine bilirubin

d) Increased stercobilinogen

Question 2:

A 3-year-old toxic child presents with tachypnea, high-grade fever, inspiratory stridor, and developed difficulty in swallowing with drooling of saliva within 4-6 hours of fever. Along with airway management which of the following needs to be given?

a) Nebulisation with racemic adrenaline

b) IV ceftriaxone

c) Diphtheria antitoxin

d) Start steroids

Question 3:

A child transfers an object from one hand to another. What does it signify?

a) Visual motor co-ordination

b) Ability to compare objects

c) Ability to handle small objects

d) Voluntary release

Question 4:

Surgical correction of VSD is performed in an infant. This will help in relieving which of the following features?

a) Respiratory alkalosis

b) Failure to thrive

c) Arrythmia

d) Heart block

Question 5:

All the following PFT findings in a child are suggestive of asthma except?

a) Increase in FEV1> 12% after salbutamol inhalation

b) FEV1/FVC< 80%

c) Day- night variation of FEV1 > 15%

d) Decrease in FEV1>15% after exercise

Question 6:

Which of the following is not a symptom of severe dehydration?

a) Drowsy child

b) Skin pinch retracts slowly

c) Thirsty child

d) Sunken fontanelle

Question 7:

Which of the following is not a criteria for severe acute malnutrition?

a) Weight for height less than -3 SD

b) MUAC < 115mm

c) B/L Pedal Edema

d) Weight for age less than -3 SD

Question 8:

Phototherapy converts bilirubin to _________

a) Stercobilin

b) Lumirubin

c) Biliverdin

d) Urobilin

Question 9:

Identify the vitamin which when deficient can lead to neonatal seizures.

a) Pyridoxine

b) Thiamine

c) Riboflavin

d) Pantothenic acid

Question 10:

A baby assessed 5 minutes after birth is found to be pink, with blue extremities, and irregular respiration. Heart rate is 80 beats per minute. There is grimace and some flexion. What is the Apgar score for this baby?

a) 2

b) 3

c) 4

d) 5

Question 11:

A child suffering from steroid-dependent nephrotic syndrome is brought to the OPD for vaccination. Which of the following is true about vaccination in this child?

a) All live vaccines under NIS can be given as per schedule

b) All killed vaccines under NIS can be given

c) Only pneumococcal vaccine should be given to the child in acute phase

d) Live vaccines are contraindicated in the child's sibling

Question 12:

A young girl presents with the complaints of continuous dribbling of urine. However, she also voids normally. Her parents tell you that she had never been totally dry since her birth. What is the most likely diagnosis?

a) Ectopic ureter

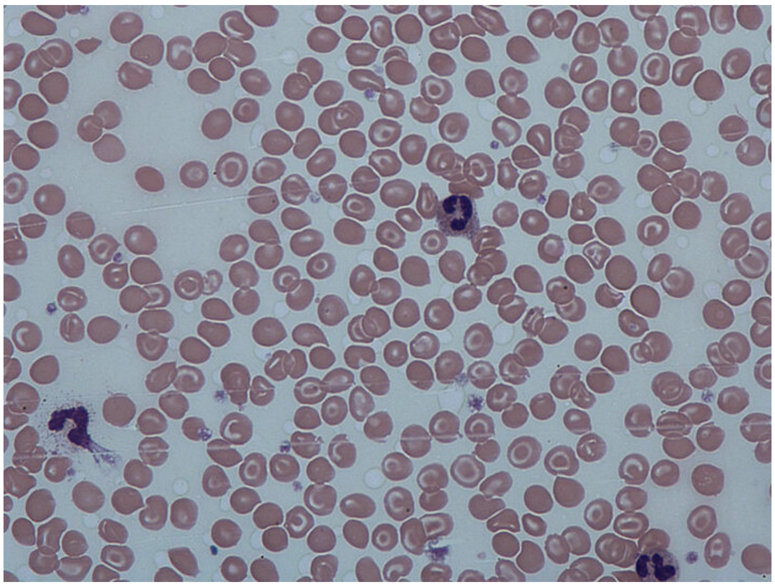
b) Ureterocele

c) Congenital megaureter

d) Vesicoureteric reflux

Question 13:

A 14-month-old child with Hb 6.3 was undergoing repeated blood transfusions. The last transfusion was given in the 9th month. Which of the following investigation is used for a definitive diagnosis?



a) High performance liquid chromatography

b) Ferritin levels estimation

c) Bone marrow aspiration

d) Globin gene sequencing

Answer Key

Question No.	Correct Option
1	c

2	b
3	b
4	b
5	c
6	c
7	d
8	b
9	a
10	d
11	b
12	a
13	d

Detailed Explanations

Solution to Question 1:

Amongst the given options, increased urine bilirubin is least likely with Rh incompatibility in newborns.

Rh incompatibility leads to hemolysis in the newborn, causing the release of large amounts of hemoglobin. This hemoglobin is converted to biliverdin and then to unconjugated bilirubin.

The newborn liver has a limited capacity to convert this large amount of unconjugated bilirubin to its soluble form through glucuronide conjugation. As a result, levels of unconjugated bilirubin increase (option A). Unconjugated bilirubin, being water-insoluble and bound to albumin, is not excreted in urine.

Intestinal bacteria convert unconjugated bilirubin to urobilinogen and stercobilinogen, which are excreted in urine and stool (options B and D).

Solution to Question 2:

The clinical presentation of a toxic-looking child with high-grade fever, tachypnea, inspiratory stridor, dysphagia, and salivary drooling strongly suggests acute epiglottitis. Once the airway is secured, empirical antimicrobial therapy with intravenous ceftriaxone is recommended in the management of acute epiglottitis.

The most common causative agents of epiglottitis in vaccinated children are Streptococcus pyogenes, Streptococcus pneumoniae, non-typeable Haemophilus influenzae, and Staphylococcus aureus. The incidence of Haemophilus influenzae type B has decreased significantly due to vaccination.

The child presents with a high fever, sore throat, dyspnea, dysphagia, and progressive respiratory obstruction. The child may exhibit a toxic appearance, with salivary drooling and assuming a tripod position. The presence of stridor indicates near-complete obstruction of the airway.

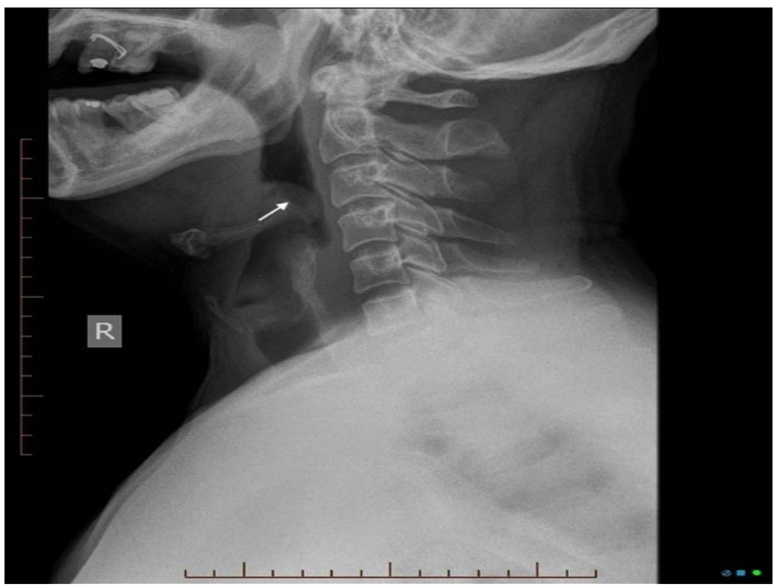
Laryngoscopy should be promptly conducted in a controlled setting, such as an operating room or intensive care unit when the diagnosis of epiglottitis is certain or likely based on clinical evaluation.

Acute epiglottitis is a medical emergency due to unpredictable clinical course and sudden airway obstruction. Immediate airway management through endotracheal or nasotracheal intubation or tracheostomy is crucial, regardless of the severity of respiratory distress. Empiric treatment with antibiotics like ceftriaxone, cefepime, or meropenem is recommended until culture and susceptibility results are available. Dexamethasone may also be administered to reduce pharyngeal edema.

Other options:

Option A: Nebulisation with racemic adrenaline is given in moderate to severe croup.

The image below shows the thumb sign in epiglottitis:



Solution to Question 3:

The milestone of transfer of objects from one hand to another in a child signifies the ability of the child to compare objects. The transfer of objects is a motor milestone occurring at 6 months of age.

Other options:

Option A: When the child reaches for objects it implies the onset of visuomotor coordination. It is attained at 4 months

Option C: Thumb finger grasp implies the ability to handle small objects. It is attained at 8 months.

Option D: The voluntary release/ Release phenomenon is used to denote the disappearance of the palmar grasp, which is seen at 4 months.

Solution to Question 4:

Surgery for ventricular septal defect (VSD) mainly helps to relieve congestive cardiac failure which manifests as a failure to thrive in an infant. The main aim of performing surgery in VSD is to get the symptoms of heart failure in control and to prevent the development of pulmonary vascular disease.

Ventricular septal defect (VSD) is the most common congenital heart disease. The most common subtype is the membranous type.

Small VSDs with trivial left-to-right shunts and normal pulmonary artery pressure (PAP) are the most common. These patients are asymptomatic. A loud, harsh, or blowing holosystolic murmur is present and heard best over the lower left sternal border, and it is accompanied by a thrill.

Small VSDs close spontaneously, especially within the first two years of life. Muscular VSDs have a higher closure rate (up to 80%) compared to membranous VSDs.

Large VSDs with excessive pulmonary blood flow and pulmonary hypertension can lead to signs of congestive heart failure in early infancy. These signs may include dyspnea, feeding difficulties, poor weight gain, profuse perspiration, irritability, weak crying, and recurrent pulmonary infections. The holosystolic murmur associated with a large VSD is less harsh and more blowing compared to that of a small VSD. This is due to the absence of a significant pressure gradient across the defect.

The indications of surgery in VSD are:

• Large defects in patients of any age with uncontrolled symptoms or failure to thrive.

• Moderate to large defects associated with pulmonary hypertension in infants between 6 and 12 months, even if symptoms are controlled by medication.

• Patients older than 24 months with a Qp: Qs ratio greater than 2:1 (ratio of total pulmonary blood flow-Qp to total systemic blood flow-Qs).

• Supracristal VSDs of any size, due to the higher risk of developing aortic valve regurgitation.

Severe pulmonary vascular disease nonresponsive to pulmonary vasodilators is a contraindication to VSD closure.

Other options:

Options A and C: Respiratory alkalosis and arrhythmia are other rare complications of VSD.

Options D: Heart block may occur perioperatively in VSD.

Solution to Question 5:

Day-night variation of FEV1 > 15% is not suggestive of asthma, variation ≥20% is suggestive of asthma.

Lung Function Abnormalities in Childhood Asthma

• Spirometry showing airflow limitation: Low FEV1 (relative to the percentage of predicted norms) and FEV1:FVC ratio <0.80 (option B)

• Bronchodilator response (to inhaled β-agonist): Improvement in FEV1 ≥12% and ≥200 mL after inhalation of a short-acting beta agonist (option A).

• Exercise challenge: Worsening in FEV1 ≥15% (option D)

• Daily peak expiratory flow (PEF) or FEV1 monitoring: day to day and/or A.M.-to-P.M. variation ≥20%

• Exhaled nitric oxide (FeNO): A value of >20ppb (parts per billion) supports the clinical diagnosis of asthma in children.

FEV1: Forced expiratory volume in 1 sec; FVC: Forced vital capacity.

Childhood asthma can be classified into two types: recurrent wheezing in early childhood, often triggered by respiratory viral infections and resolves during preschool/lower school years, and chronic asthma associated with allergies that persist into later childhood and adulthood.

Based on the severity, acute asthma is classified as mild, moderate, and severe.

Clinical parameter	Mild	Moderate	Severe
Color	Normal	Normal	Pale
Sensorium	Normal	Anxious	Agitated
Respiratory rate	Increased	Increased	Increased
Dyspnea	Absent	Moderate	Severe
Use of accessory muscles	Nil or minimal	Chest indrawing	Indrawing, nasal flare
Rhonchi	Expiratory and/or inspiratory	Expiratory and/or inspiratory	Expiratory or absent.
Oxygen saturation	>95%	90-95%	<90%
Speech	Sentence	Phrases	Difficulty in speech

Solution to Question 6:

A thirsty child signifies some dehydration, not severe dehydration.

Clinical features of severe dehydration in a child include:

• Drowsy, lethargic, or unconscious, with deeply sunken eyes and absent tears

• Sunken fontanelle

• Drinks poorly or not able to drink

• Deep breathing

• Skin pinch goes back very slowly (recoil in >2 seconds)

• Tachycardia, with bradycardia in most severe cases

• Weak, thready, or impalpable pulse

• Prolonged capillary refill

• Cold, mottled, and cyanotic extremities

• Minimal urine output

Solution to Question 7:

Severe acute malnutrition (SAM) according to the WHO criteria is the weight for height (not age) less than minus -3 SD.

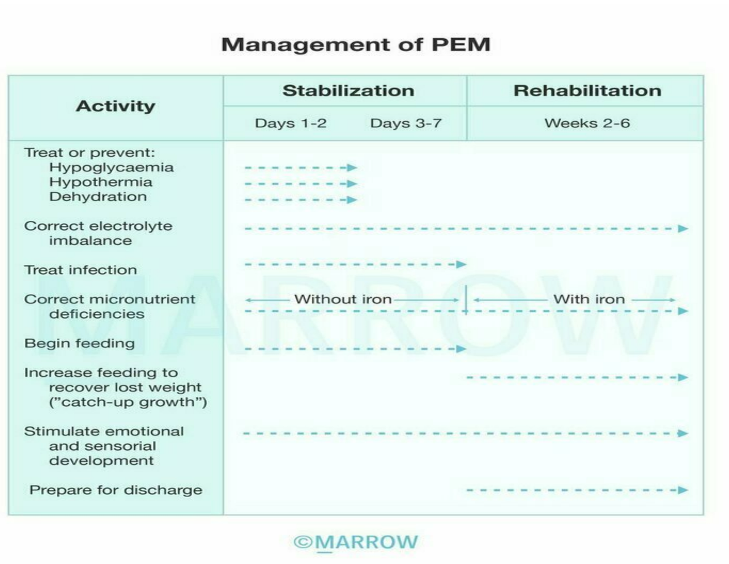
Severe acute malnutrition (SAM) in children of ages 6 to 59 months is defined by the WHO as any of the following:

• Weight-for-length (or height) below –3 SD of the WHO child growth standards

• Presence of bipedal edema

• Mid-upper arm circumference (MUAC) <11.5 cm.

Management of SAM is given below:



Solution to Question 8:

Phototherapy converts bilirubin to lumirubin. It is used to manage clinical jaundice and indirect hyperbilirubinemia.

It involves exposing the patient to high-intensity light in the visible spectrum. Bilirubin absorbs light most effectively in the blue range (420 to 470nm).

Phototherapy converts toxic bilirubin into two forms: lumirubin, formed by irreversible structural isomerization that can be excreted by the kidneys, and a reversible isomer (4Z,15E-bilirubin), formed by photoisomerization that can be excreted in bile without conjugation.

The effectiveness of phototherapy depends on factors such as the appropriate wavelength range, distance to the infant, skin surface area exposure, and the infant's bilirubin metabolism. The infant's eyes should be covered to prevent corneal damage. Continuous conventional phototherapy is typically used, with frequent repositioning of the infant.

Phototherapy can be discontinued once two TSB (total serum bilirubin) values 12 hours apart fall below current age-specific cut-offs.

Complications of phototherapy may include loose stools, erythematous macular rash, purpuric rash (linked to transient porphyrinemia), overheating, dehydration (due to increased insensible water loss and diarrhea), and hypothermia from exposure.

Bronze baby syndrome is a dark grayish-brown skin discoloration seen in infants undergoing phototherapy in the presence of direct hyperbilirubinemia. Porphyria is a contraindication for phototherapy because photosensitivity and blistering can be severe in these infants.

Solution to Question 9:

Pyridoxine deficiency can cause refractory seizures in the neonatal period and infancy.

Pyridoxal 5'-phosphate (P5P) is the active form of pyridoxine and functions as a cofactor for enzymes involved in neurotransmitter synthesis, amino acid metabolism, and glycogen metabolism. Intracellular P5P deficiency in the brain can lead to a seizure disorder that does not respond to typical anticonvulsant medications but shows a response to pyridoxine.

Antiquitin deficiency is the most common cause of pyridoxine-dependent epilepsy (due to mutations in the ALDH7A1 alleles)

They primarily present with generalized seizures in the early days of life. The seizures can be of various types, recurrent focal motor seizures, generalized tonic seizures, and myoclonus. Additional manifestations may include dystonia, respiratory distress, abdominal distention with vomiting, hepatomegaly, hypoglycemia, and hypothermia.

Increased α-aminoadipic semialdehyde and pipecolic acid levels are observed in the CSF, plasma, and urine. Treatment with vitamin B6 (50-100 mg/day) shows significant improvement in seizures. Life-long therapy is needed.

Other options:

Option B: Thiamine (vitamin B1) deficiency causes beriberi.

Option C: Riboflavin (vitamin B2) deficiency causes glossitis, cheilosis, keratitis, and corneal vascularization.

Option D: Pantothenic acid (vitamin B5) deficiency causes burning feet syndrome.

Solution to Question 10:

The Apgar score for this baby will be 5.

• Color; body pink, extremities blue: score-1

- Heart rate; 80 beats per minute: score 1
- Grimace: Score-1
- Some flexion: Score-1
- Irregular respiration: Score-1

The Apgar score is a systematic way of assessing the infant after birth at 1 and 5 minutes of life. It gives a maximum score of 10 which indicates the newborn is in the best possible condition. Scores between 0-3 require immediate resuscitation.

Changes in the Apgar scores denote the response of the infant to resuscitation. If the score at 5 minutes remains less than 7, the infant should be assessed again every 5 minutes for the next 20 minutes.

Apgar Scoring System

Indicator		0 Points	1 Point	2 Points
A	Appearance (skin colour)	Blue/Pale	Pink body, blue extremeties	Pink
P	Pulse	Absent	< 100 BPM	> 100 BPM
G	Grimace (reflex irritability)	No response	Minimal response to stimulation	Cough/sneeze when stimulated
A	Activity (muscle tone)	Floppy	Some flexion	Well flexed
R	Respiratory efforts	Absent	Slow and irregular	Strong cry

Causes for false positive Apgar score are:

- Prematurity
- Drugs like analgesics, narcotics, sedatives, magnesium sulfate
- Acute cerebral trauma
- Precipitous delivery
- Airway obstruction (choanal atresia)
- Congenital pneumonia or sepsis
- Hemorrhage- hypovolemia

False-negative Apgar score is seen in maternal acidosis and high fetal catecholamine levels.

Solution to Question 11:

In a child with steroid-dependent nephrotic syndrome, all killed vaccines can be given.

Option A: Live virus vaccines are administered with caution in children receiving corticosteroid therapy. They can be given to patients in whom the prednisone dose is below 1mg/kg daily or 2mg/kg on alternate days. If the patient is receiving corticosteroid-sparing agents (cyclophosphamide, cyclosporine), live virus vaccines are contraindicated.

Option C: Only live vaccines are contraindicated in this child. Full pneumococcal vaccination (with the 13-valent conjugate vaccine and 23-valent polysaccharide vaccine) and influenza vaccination can be given annually to the child and their household contacts.

Option D: Immunization of siblings/healthy household contacts with live vaccines is necessary to reduce the risk of infection transmission to the immunosuppressed child. Direct exposure to the gastrointestinal/respiratory secretions of the household contacts is avoided for about 3-6 weeks after vaccination.

Minimal change nephrotic syndrome is the most common cause of nephrotic syndrome. It is characterized by heavy proteinuria, edema, hypoalbuminemia, and hyperlipidemia. The diagnosis is confirmed by performing a urinalysis and urine protein: creatinine ratio. Nephrotic range proteinuria is proteinuria greater than 3.5 g per 24 hours (3+ or 4+) or a urine protein : creatinine ratio exceeding 2.

The recommended initial treatment for nephrotic syndrome is a daily dose of 2mg/kg (max 60mg) of prednisolone for 6 weeks followed by an alternate dose of 1.5mg/kg (max 40mg) for the next 6-8 weeks.

Definitions regarding the course of nephrotic syndrome:

• Frequently relapsing nephrotic syndrome: ≥2 relapses in 6 months or 3 or more relapses in 12 months.

• Steroid-dependent nephrotic syndrome: Relapse during steroid tapering or within 2 weeks of discontinuation.

• Steroid resistance: Failure to achieve remission despite therapy with daily prednisolone for 6 weeks

Alternative drugs used in steroid-dependent patients, frequent relapsers, steroid-resistant patients, and those with severe corticosteroid toxicity are cyclophosphamide, calcineurin inhibitors: cyclosporine and Tacrolimus, mycophenolate, levamisole, and rituximab (a chimeric monoclonal antibody against CD20).

Solution to Question 12:

The clinical presentation of a young girl with continuous dribbling of urine, despite normal voiding, strongly suggests the diagnosis of an ectopic ureter.

An ectopic ureter refers to a condition where the ureter drains outside the bladder. It is more common in females and opens either into the urethra below the sphincter or into the vagina. Hence, they present with constant urinary dripping all day, even though the child voids regularly (paradoxical incontinence).

The terminal part of the ureter is frequently narrowed, leading to hydroureteronephrosis. Urinary tract infections (UTIs) are common due to urinary stasis.

In males, ectopic ureters usually enter the posterior urethra above the external sphincter, hence they are usually continent and most patients present with UTI or epididymitis.

Investigations include renal ultrasound (USG), voiding cystourethrogram (VCUG), and renal scan to assess function. USG shows a hydronephrotic kidney or dilated upper pole and ureter. If there is a satisfactory function, the recommended treatment options are ureteral reimplantation into the bladder or ureteroureterostomy. The poor function may require partial or total nephrectomy.

Other options:

Option B: A ureterocele is a cystic dilation of the terminal ureter, typically obstructive due to a small ureteral opening. In childhood, they commonly present with infections. Larger ureteroceles can cause obstruction at the bladder neck.

Option C: Congenital megaureter (dilated ureter) presents UTI, urinary stones, hematuria, and flank pain because of urinary stasis.

Option D: Vesicoureteral reflux (VUR) is characterized by the retrograde flow of urine from the bladder to the ureter and kidney. They usually present with recurrent attacks of UTI.

Solution to Question 13:

The given clinical scenario of repeated transfusion, low hemoglobin level, and target cells in the smear are suggestive of thalassemia. Globin gene sequencing is used for its definitive diagnosis.

High-performance liquid chromatography (HPLC) is used as a screening test for thalassemia (option A).

Thalassemia is a hemoglobinopathy that arises due to deficient synthesis of the globin chains. Depending on the chain affected, it is classified into α-thalassemia and β-thalassemia.

β-thalassemia is caused by point mutations of the β-thalassemia gene on chromosome 11, which diminish the synthesis of β-globin chains. Splicing mutations are the most common cause of (β+) thalassemia.

There is a deficit in the synthesis of HbA with an elevation of HbA2 and HbF levels in β-thalassemia. Children with severe thalassemia present with distinctive facial features such as maxilla hyperplasia, a flat nasal bridge, and frontal bossing. Other manifestations include pathologic bone fractures, significant hepatosplenomegaly, and cachexia.

Blood smears show marked anisocytosis, poikilocytosis, and microcytic hypochromic anemia. Target cells, basophilic stippling, and fragmented red cells also are common. A radiographic image of the skull shows the classic 'crew cut' appearance.

Hb Electrophoresis and high-performance liquid chromatography (HPLC) are used as screening tests for thalassemia. DNA diagnosis to detect β-thalassemia mutations is used for definitive diagnosis. The Mentzer index is the ratio of MCV to RBC count. It is used to differentiate between iron-deficiency anemia and β-thalassemia trait. The Mentzer index is <13 in thalassemia minor syndrome.

Transfusions are administered every 3-4 weeks to maintain a pretransfusion hemoglobin level of 9.5-10.5 g/dL. Transfusion-induced hemosiderosis is a significant complication in transfusion-dependent thalassemia. Iron deposition occurs in the liver, endocrine organs, and heart, resulting in complications including hypothyroidism, hypogonadotropic hypogonadism, growth hormone deficiency, hypoparathyroidism, diabetes mellitus, heart failure, and arrhythmias.

The image below shows thalassemic facies and crew cut appearance on X-ray:

Paediatrics NEET 2018

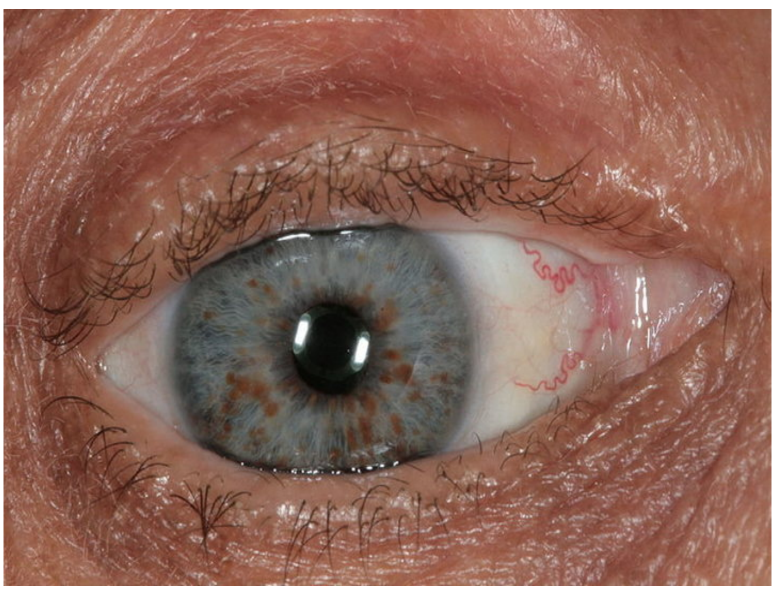
Question 1:

A fetus with intrauterine growth restriction was born prematurely with jaundice, hepatosplenomegaly, microcephaly, and diffuse petechiae at birth. A brain CT was done which revealed periventricular calcifications. What is the best method for the diagnosis of the etiological agent?

- **a)** Urine examination
- **b)** Liver biopsy
- **c)** Blood examinaton
- **d)** CSF examination

Question 2:

Which of the following is false regarding the given image?



- **a)** It represents melanocytic hamartoma
- **b)** It causes no visual disturbance
- **c)** It is associated with facial angiofibroma
- **d)** They are diagnostic of a neurocutaneous syndrome

Question 3:

Congenital adrenal hyperplasia most commonly presents as _______.

a) Male pseudohermaphroditism

b) Female pseudohermaphroditism

c) True hermaphroditism

d) 46, XY intersex

Question 4:

A newborn presented with a tuft of hair over the lumbosacral region. What is the likely diagnosis?

a) Spina bifida occulta

b) Spinal lipoma

c) Dermal sinus

d) Any of the above

Question 5:

What is the most common cause of ventriculomegaly in newborns?

a) Arnold-Chiari malformation

b) Dandy-Walker syndrome

c) Arachnoid villi malformation

d) Aqueductal stenosis

Question 6:

When do you consider administering epinephrine in a neonate during resuscitation?

a) Heart rate remains at < 60 beats/minute despite effective compressions and ventilations.

b) Heart rate remains at < 100 beats/minute despite effective compressions and ventilations.

c) Heart rate does not improve after 30 seconds with bag and mask ventilation.

d) Infants with severe respiratory depression fail to respond to positive-pressure ventilation via bag and mask.

Question 7:

Which of the following is not true regarding ataxia-telangiectasia?

a) Mutations in 11q gene is implicated

b) Follows autosomal dominant mode of inheritance

c) Humoral and cellular immunodeficiency

d) Linked to adenocarcinomas

Question 8:

Warthin Finkeldey cells are seen in:

a) Rubella

b) Rubeola

c) Rabies

d) Typhoid

Question 9:

Which of the following is an X-linked disorder?

a) Color blindness

b) Thalassemia

c) Sickle cell anemia

d) Cystic fibrosis

Question 10:

The following parameter in ALL indicate a poor prognosis ________.

a) Age >10 years

b) Leukocyte count <50,000/mm3

c) Hyperdiploidy

d) Trisomy of chromosomes 4, 10, and 17

Question 11:

Viral infection causing transient aplastic anemia is ______

a) HIV

b) Polio

c) Parvovirus B-19

d) HHV-8

Question 12:

What is true about ovotesticular disorders of sex development?

a) Karyotype is 46 XY.

b) Ovotestis is seen in the gonadal biopsy

c) The response of testosterone to hCG is increased.

d) Uterus is usually absent.

Question 13:

Which one of the following is not included in the criteria for neurofibromatosis 1?

a) Two or more iris hamartomas

b) Acoustic neuromas

c) Dysplasia of the sphenoidal and tibial bone

d) Cafe-au-lait spots

Answer Key

Question No.	Correct Option
1	a
2	c
3	b
4	d
5	d
6	a
7	b

8	b
9	a
10	a
11	c
12	b
13	b

Detailed Explanations

Solution to Question 1:

Periventricular calcifications and microcephaly in a baby with suspected congenital infection point toward cytomegalovirus (CMV). The best diagnostic method for CMV is PCR or viral culture on urine.

Since not all affected infants are viremic at birth, urine is a more reliable sample than blood.

Intrauterine growth restriction (IUGR), prematurity, and hepatosplenomegaly raise the suspicion of congenital infection. Hepatosplenomegaly, jaundice, pneumonitis, petechiae, and meningoencephalitis are common to toxoplasmosis, rubella, CMV, herpes simplex virus (HSV), enteroviruses and syphilis. Hydrocephalus is seen in CMV, toxoplasmosis, HSV, and rubella.

The characteristic clinical features of various congenital infections are described in the table below:

The triad of congenital CMV infection includes intracranial calcifications (periventricular), microcephaly, and chorioretinitis.

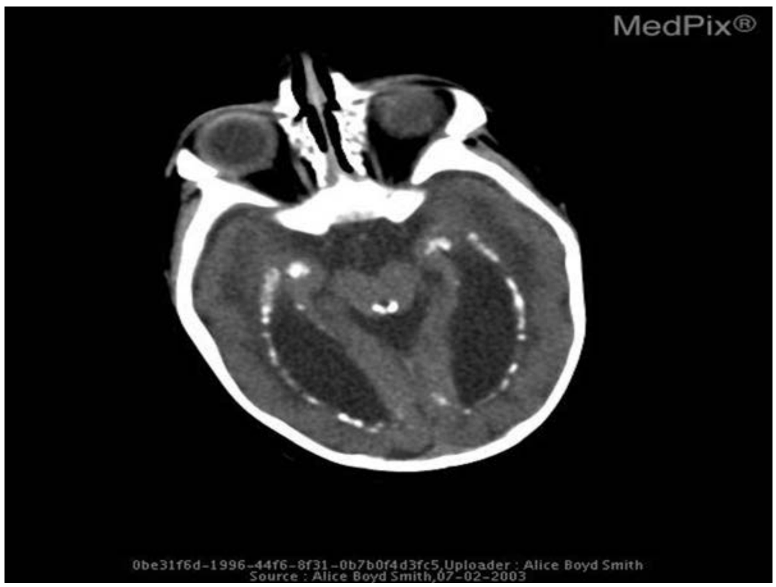
Most CMV transmission occurs in the first trimester. The risk of transmission is high in the third trimester. 90% of the infected babies are asymptomatic and have lesser chances of having late sequelae.

In symptomatic individuals, ganciclovir can limit hearing loss and improve the developmental outcome, while there is no definitive management for congenital CMV infection.

The image below shows periventricular calcification in the head CT seen in CMV infection:

Congenital infections	Characteristicfeatures
Cytomegalovirus (CMV)	Periventricular calcifications Microcephaly
Toxoplasmosis	ChorioretinitisDiffuse intracranial calcificationsHydrocephalus
Rubella	Sensorineural hearing lossSalt-and-pepperretinopathyCataractGlaucomaCongenital defects likepatent ductus arteriosus

Congenital infections	Characteristicfeatures
Herpes simplex virus (HSV)	VesiclesMicrocephaly
Syphilis	Bone lesions (osteochondritis, periostitis)Maculopapular ra shes
Enterovirus (polio, rhinoviru s, hepatitis A)	MyocarditisParalysis
Varicella zoster virus	Limb hypoplasiaCicatricial sk in lesionsCortical atrophy
Zika virus	Subcortical calcificationsMicr ocephalyCongenital contractu res (arthrogryposis)
HIV	ThrushFailure to thriveRecur rent bacterial infectionsCalcif ication of basal ganglia



The image below shows a blueberry muffin rash, seen in TORCH complex infections and neuroblastoma.

Solution to Question 2:

The given image shows discrete tan-colored nodules over the iris called Lisch nodules (pigmented melanocytic hamartomas). They are seen in neurofibromatosis-1 (NF-1). Facial angiofibroma (adenoma sebaceum) is a feature of tuberous sclerosis (TS) and not neurofibromatosis.

The image below shows facial angiofibroma seen in tuberous sclerosis (Bourneville disease).

Lisch nodules are specifically associated with NF-1 and are usually multiple and bilateral. They are best visualized with a slit lamp examination and don’t cause any visual disturbances.

Other important ocular manifestations in NF-1 include eyelid plexiform neurofibroma, which causes S-shaped deformity of the upper lid, with a 'bag of worms' like texture, optic nerve glioma, and spheno-orbital encephalocele due to dysplasia of greater wing of the sphenoid.

NF-1 (von Recklinghausen's disease) is an autosomal dominant disorder caused by a loss-of-function mutation of the NF-1 gene located on chromosome 17q11.2. NF-1 produces neurofibromin, which acts as a tumor suppressor and a negative regulator of RAS-MAP kinase signaling.

Clinical features of NF-1 include café-au-lait macules, axillary freckling, Lisch nodules, neurofibromas, and optic gliomas.

NF-1 is more common than NF2 and typically diagnosed in childhood. NF-2 is rarer but often presents with bilateral acoustic neuroma, juvenile cataract and schwannoma.

The various neurocutaneous syndromes and their ocular manifestations are as follows:

Neurocutaneous syndrome	Ocularmanifestations
Neurofibromatosistype 1	Eyelid plexiform neurofibromaOptic nerve gliomaBilateral Lisch nodulesCongenital ectropion uveaeProminent corneal nervesCongenital glaucomaSpheno-orbital encephalocele
Neurofibromatosistype 2	Posterior subcapsular cataractHamartomas of theretinal pigment epithelium (RPE) and retina
Sturge-Weber syndrome	Choroidal hemangiomaIris heterochromiaIpsilateral glaucoma
Tuberous sclerosis	Atypical iris colobomaIris hypopigmentationRetinal astrocytoma
von Hippel-Lindausyndrome	Retinal hemangiomas

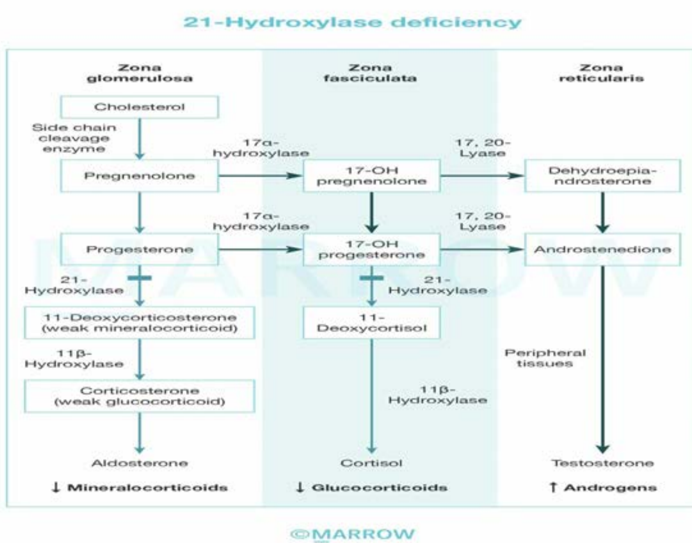
Solution to Question 3:

Congenital adrenal hyperplasia (CAH) most commonly presents as female pseudohermaphroditism as the most common enzyme deficiency is 21-hydroxylase deficiency.

A female pseudohermaphrodite is a genotypically normal female (46, XX), whose external genitalia resembles a male's.

The image below shows ambiguous genitalia in a female with CAH:

The mechanism underlying the endocrine defects in 21-hydroxylase deficiency is depicted in the image below:



Babies with 21-hydroxylase deficiency typically present around 10-14 days of age with the subset of symptoms due to aldosterone deficiency, cortisol deficiency and androgen excess.

Aldosterone deficiency leads to salt-wasting crises marked by hyponatremia, hyperkalemia, hypotension, and metabolic acidosis.

Cortisol deficiency causes manifestations like hypoglycemia, hyperpigmentation, weight loss, weakness, anorexia, vomiting, and shock.

Androgen excess in 21-hydroxylase deficiency results in virilization of females, leading to female pseudohermaphroditism, and precocious puberty in males. Virilization is usually apparent at birth in females and within 2 to 3 years of life in males.

Without treatment, shock, cardiac arrhythmias, and death may occur within days or weeks.

Other options:

Option A: Male pseudohermaphroditism is seen in CAH due to 17α-hydroxylase deficiency resulting in incomplete virilization. 17α-hydroxylase deficiency is 2nd most common enzyme affected in CAH.

Option C: True hermaphroditism is presently referred to as an ovotesticular disorder of sex development(DSD). Here, both ovarian and testicular tissue is present. It is not due to CAH.

Option D: 46 XY intersex (now called 46 XY DSD) has under virilization of genitalia. Adrenal causes include lipoid adrenal hyperplasia, 3β-hydroxysteroid dehydrogenase deficiency, 17-hydroxylase/17,20-lyase deficiency, and not 21 hydroxylase deficiency.

Solution to Question 4:

A tuft of hair over the lumbosacral region in a newborn is suggestive of occult spinal dysraphism or neural tube defect. It comprises a range of conditions, including spina bifida occulta, intradural lipoma, and dorsal dermal sinus.

Spinal dysraphism is due to the defective fusion of the midline embryonic neural, vertebral, and mesenchymal structures. Associated clinical features are high-arched feet, muscle size and strength differences between legs, and abnormal walking patterns. In newborns and young infants, the neurological examination is usually normal, but older children may experience a lack of sensation in the perineal area and back pain.

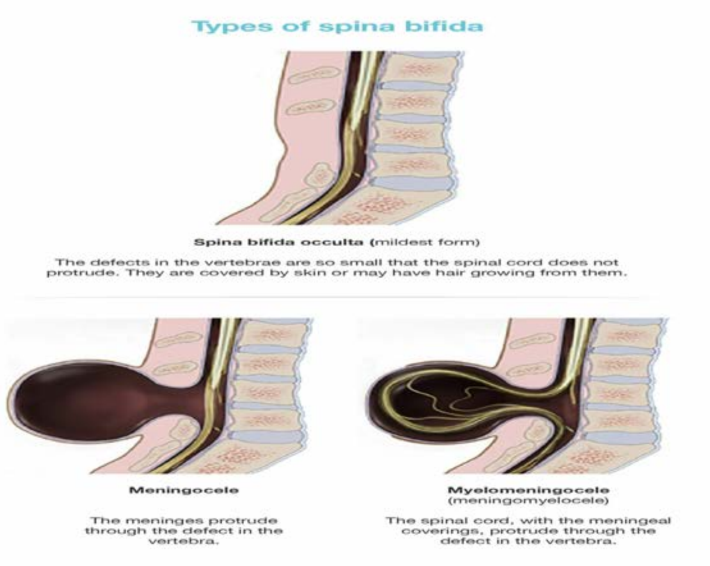
Most of the patients will have a cutaneous abnormality overlying the lower spine, including a small dimple, a tuft of hair, or a subcutaneous lipoma.

The image shows a tuft of hair in a case of spinal dysraphism:

Spinal dysraphism also includes meningocele, myelomeningocele, lipomeningocele, diastematomyelia (split spinal cord), thickened filum terminale, and cauda equina tumour.

The initial investigation is usually ultrasound in the neonate and the best investigation is MRI.

Spina bifida is the most common type of spinal dysraphism, in which the spinal column does not close completely, leaving the spinal cord and its protective covering exposed.



Myelomeningocele is the most severe form of spina bifida, where the spinal cord and nerve roots are exposed and may be damaged. A sac-like protrusion, containing spinal cord tissue, meninges, and cerebrospinal fluid, is visible outside the body, often leading to significant neurological impairments and disabilities.

Solution to Question 5:

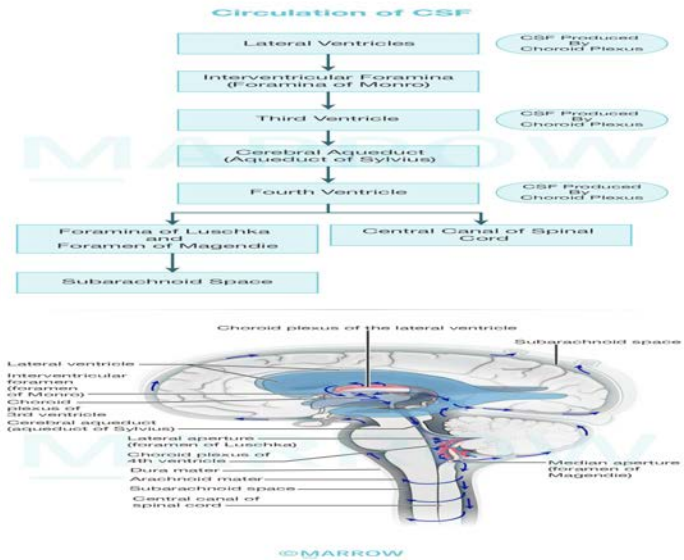
Aqueductal stenosis is the most common cause of ventriculomegaly and hydrocephalus in newborns.

Ventriculomegaly is the dilation of the ventricles. When ventriculomegaly occurs in association with raised cerebrospinal fluid (CSF) pressure, it is called hydrocephalus.

Pathologic etiologies can be congenital or acquired. Congenital causes include structural abnormalities such as aqueductal stenosis, agenesis of the corpus callosum, Dandy-Walker malformations, Chiari malformations, and neural tube defects. Acquired causes are intracranial infections(cytomegalovirus, toxoplasmosis), intracranial hemorrhage, ischemic injury, and brain parenchymal tumors.

Aqueductal stenosis is the narrowing or obstruction of the cerebral aqueduct, a narrow channel that connects the third and fourth ventricles in the brain. As a result, the fluid accumulates within the ventricles, causing them to enlarge.

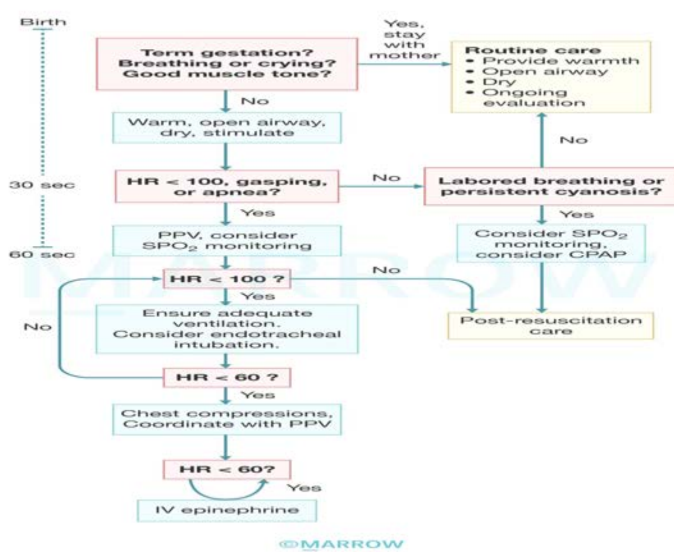
Treatment for aqueductal stenosis includes ventriculostomy, diuretics, serial lumbar punctures in specific cases, and CSF diversion. In severe cases, CSF diversion procedures (shunting), may be necessary to redirect the flow of CSF and relieve pressure on the brain.



Solution to Question 6:

As per the newborn resuscitation protocol (NRP), a heart rate < 60 beats/minute despite effective compressions and ventilation is the indication to administer epinephrine.

The NRP algorithm is shown below:



The response to neonatal resuscitation at various steps is assessed based on the heart rate, as a rise in heart rate is the most effective indicator of successful ventilatory effort.

If the heart rate is less than 100, positive pressure ventilation (PPV) is provided with a bag and mask.

In case of no response even after doing ventilation corrective steps (Options C and D), endotracheal intubation is done.

The ventilation corrective steps are as follows and can be remembered with the mnemonic MRSOPA.

• Mask has to be reapplied

• Reposition head

• Suction the secretions

• Open mouth slightly and ventilate

• Pressure is increased slightly

• Airway may be changed

If the heart rate is below 60, then chest compressions are initiated over the lower third of the sternum at a rate of 90 compressions/min. The ratio of compressions to ventilation is 3:1.

If the heart rate continues to remain <60 despite effective compressions and ventilation, then IV epinephrine is given.

Solution to Question 7:

Ataxia telangiectasia (AT) or Louis–Bar syndrome is an autosomal recessive disorder.

The defect in AT is in the ATM gene localized to chromosome 11q22-23. The ATM gene is involved in the repair of double-strand DNA breaks. This defect in DNA repair leads to combined immunodeficiency (impaired immunoglobin isotype switching and association with thymic hypoplasia) and predisposition to an increased risk of lymphoreticular tumors such as lymphoma or leukemia.

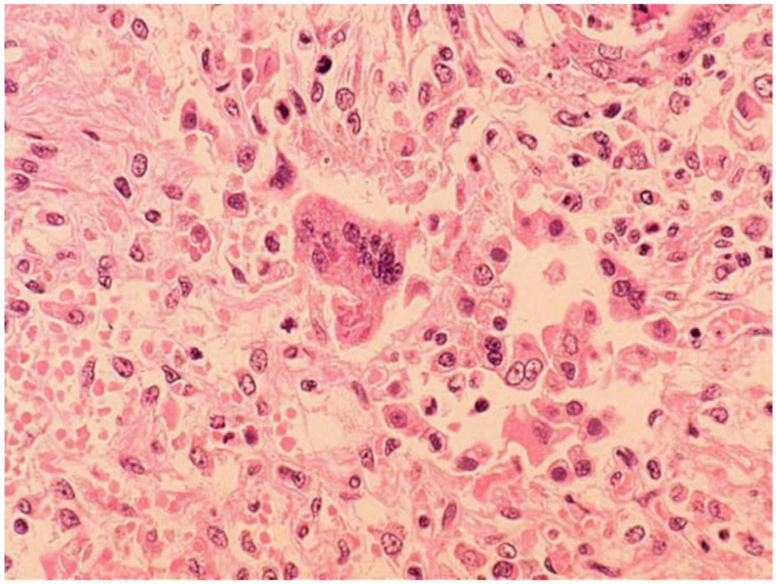
AT is characterized by progressive ataxia (beginning during infancy), telangiectasia (initially on bulbar conjunctiva), sinopulmonary infections, hypogonadism, sensitivity to ionizing radiation, insulin resistance, and raised alpha-fetoprotein levels.

Solution to Question 8:

Warthin-Finkeldey cells are multinucleated giant cells that are present in the prodromal stage of measles (rubeola), HIV lymphadenitis, and various benign and malignant lymphoid conditions.

The cytopathic effect of measles consists of multinucleate syncytium formation with numerous acidophilic nuclear and cytoplasmic inclusions.

The image given below shows the Warthin-Finkeldey giant cells:



Measles is an infection caused by Paramyxovirus which is a single-stranded enveloped RNA virus. It spreads 4 days before and 5 days after the appearance of the rash through body secretions.

Clinical features include mild fever, conjunctivitis, upper respiratory tract infection, and pathognomic Koplik spots. The Koplik spots are red lesions with bluish-white spots seen in the inner cheeks at the level of premolars. The red maculopapular rashes begin on the forehead, behind the ears, and on the upper neck on the 4th day. It then spreads towards the torso, and extremities and reaches the palms and soles.

The diagnosis is mainly made clinically but a serological demonstration of IgM antibodies and IgG antibodies is also used. Complications include pneumonia, croup, and tracheitis. Subacute sclerosing panencephalitis is a fatal complication of chronic measles diagnosed by elevated IgG levels in cerebrospinal fluid.

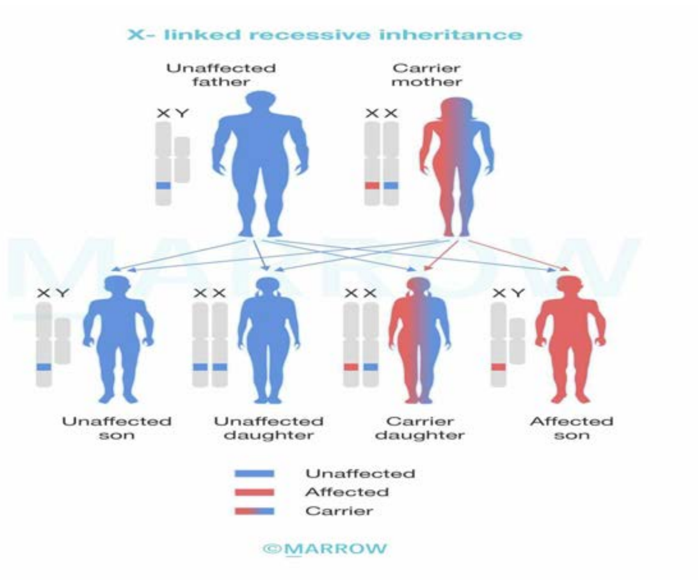
Solution to Question 9:

Color blindness is primarily inherited as an X-linked recessive trait. The genes for color vision are located on the X chromosome.

Males have a single X chromosome inherited from mothers. If the X chromosome inherited carries a mutation, males exhibit the disease. However, females become carriers of the color blindness gene and have a 50% chance of passing it on to their children.

Female carriers usually are protected because of the random inactivation of one X chromosome. But there is a possibility that the mutant X chromosome remains active if the normal alleles are inactivated in some of the cells so the heterozygous female can express the disorder partially.

The image shows the pattern of X-linked inheritance:



In color blindness, the mechanism to appreciate one or more primary colors is either defective (anomalous) or absent (anopia). It may be congenital or acquired. Congenital color blindness can be dyschromatopsia (defective color vision of any one of the RGB colors) or achromatopsia (complete color blindness of all three RGB colors). RGB colors are red, green, and blue.

Tests for color blindness include the Ishihara test, Hardy–Rand–Rittler test, and Farnsworth–Munsell 100-hue test.

Solution to Question 10:

Age >10 years indicates a poor prognosis in ALL.

Leukocyte count <50,000/mm3, hyperdiploidy, and trisomy of chromosomes 4,10,17 indicate a good prognosis.

Solution to Question 11:

Parvovirus B19

• Common, most prevalent are the rash illness of erythema infectiosum and transient aplastic crisis.

• Transmission by respiratory route but also transmissible in blood and blood products.

• Shows erythrotropism – it specifically targets the erythroid cell line, especially the pronormoblast stage. Viral infection produces cell lysis, leading to depletion of erythroid precursors and a transient arrest of erythropoiesis. This may lead to a precipitous fall in haemoglobin which may even require blood transfusion.

Clinical manifestations of Parvovirus B19 infection

• Erythema infectiosum/Fifth disease (slapped cheek appearance)

• Transient aplastic anemia

• Petechial/ purpuric skin eruptions: Papular purpuric "gloves and socks" syndrome

• Arthritis and arthralgia

• Chronic anemia in immunocompromised persons

• Primary maternal infection is associated with non-immune fetal hydrops and intrauterine fetal demise.

Solution to Question 12:

Ovotesticular disorder of sex development (DSD) is characterized by the presence of both ovarian and testicular tissue in the same individual. An ovotestis is present in approximately 2/3 of affected individuals.

Most affected individuals have a 46, XX chromosomal karyotype, which normally results in female sexual development. In about 10% of patients, testicular tissue in an individual with a 46, XX karyotype is present as a result of a translocation of the SRY gene on the Y chromosome to the X chromosome or another chromosome. There is a defect in R-spondin1 protein encoded by the RSPO1 gene, described in 46, XX ovotesticular DSD.

An abnormal vagina and an underdeveloped uterus are usually present. The testes are usually undescended, and the penis may present with hypospadias. Upon reaching puberty, breast development, feminization, and menstruation may occur. Most affected individuals are infertile, but ovulation or spermatogenesis is possible. The response of testosterone to hCG is normal or reduced.

Solution to Question 13:

In neurofibromatosis 1 (NF1), two or more of the following clinical manifestations are present:

• Six or more café-au-lait macules

• Axillary or inguinal freckling

• Two or more iris Lisch nodules (hamartomas of iris)

• Two or more neurofibromas or 1 plexiform neurofibroma

• A distinctive osseous lesion such as sphenoidal and tibial bone dysplasia

• Optic gliomas

• A 1st-degree relative with NF-1

Paediatrics NEET 2019

Question 1:

Unconjugated hyperbilirubinemia, which did not subside even after 3 weeks of birth, was observed in a neonate. On investigating, liver enzymes, PT/INR and albumin levels were normal. No hemolysis was seen on a peripheral blood smear. A drop in bilirubin level was observed within a week after treating with phenobarbital. What is the most likely diagnosis?

a) Rotor syndrome

b) Crigler Najjar type 2

c) Dubin Johnson syndrome

d) Crigler Najjar type 1

Question 2:

Which one of the following is not a feature in cerebral palsy?

a) Hypotonia

b) Microcephaly

c) Ataxia

d) Flaccid paralysis

Question 3:

Which of the following is not seen in a child with cystic fibrosis?

a) Sweat chloride test chloride conc of 70 mEq/L

b) Increase immunoreactive trypsinogen level

c) Hyperkalemia

d) Contraction alkalosis

Question 4:

A 9-year-old boy has swelling of knee joints and non-blanching rashes as seen in the picture. Urine exam reveals haematuria +++ and proteinuria +. Platelet levels are normal. What is the probable diagnosis?

a) Systemic lupus erythematosus (SLE)

b) Henoch Schonlein purpura (HSP)

c) Immune thrombocytopenic purpura (ITP)

d) Polyarteritis Nodosa (PAN)

Question 5:

A child in your unit is diagnosed with severe acute malnutrition (SAM). According to the WHO criteria, SAM is defined best by:

a) Weight for age less than - 2SD

b) Weight for height less than + 2SD

c) Weight for age less than + 3SD

d) Weight for height less than - 3SD

Question 6:

The most common manifestation of congenital toxoplasmosis is ______

a) Hydrocephalus

b) Chorioretinitis

c) Hepatosplenomegaly

d) Thrombocytopenia

Question 7:

Molecular defect causing liver disease in Dubin-Johnson syndrome is__________

a) ATP7A

b) ATP7B

c) ABCC2

d) SERPINA 1

Question 8:

Where would you place the pulse oximeter to measure pre-ductal oxygen saturation in an infant who was born 3 minutes ago?

a) Left upper limb

b) Left lower limb

c) Right upper limb

d) Right lower limb

Question 9:

All among the following are clinical features of necrotizing enterocolitis except _________

a) Vomiting

b) Abdominal mass

c) Erythema of the abdominal wall

d) Metabolic alkalosis

Question 10:

The most common cause of HIV infection in a newborn is _________

a) Perinatal transmission

b) Breast milk

c) Transplacental

d) Exchange transfusion with infected blood

Question 11:

Which of the following is the main factor for the ductal closure postnatally?

- **a)** Increase in partial pressure of oxygen (paO2)
- **b)** Increase in systemic vascular resistance
- **c)** Increase in circulating prostaglandin levels
- **d)** Decrease in pulmonary venous resistance

Question 12:

Cardiac edema and neuropathy are seen in which of the following vitamin deficiencies?

- **a)** Biotin
- **b)** Thiamine
- **c)** Pyridoxine
- **d)** Riboflavin

Question 13:

The commonest indication for liver transplantation in children is ______

- **a)** Alagille syndrome
- **b)** Biliary atresia
- **c)** Caroli disease
- **d)** Hepatcellular carcinoma

Question 14:

While you are evaluating a baby, you show him a bright pink teddy bear, that he reaches out to with both hands. What is the earliest age by which this milestone is typically achieved?

- **a)** 4 months
- **b)** 5 months
- **c)** 6 months
- **d)** 7 months

Question 15:

A baby assessed 5 minutes after birth, is found to be cyanosed with irregular gasping respiration. Heart rate is 60 beats/min, and shows minimal response to stimulation with some flexion of extremities. The Apgar score for this newborn is ______

- **a)** 2
- **b)** 5
- **c)** 3
- **d)** 4

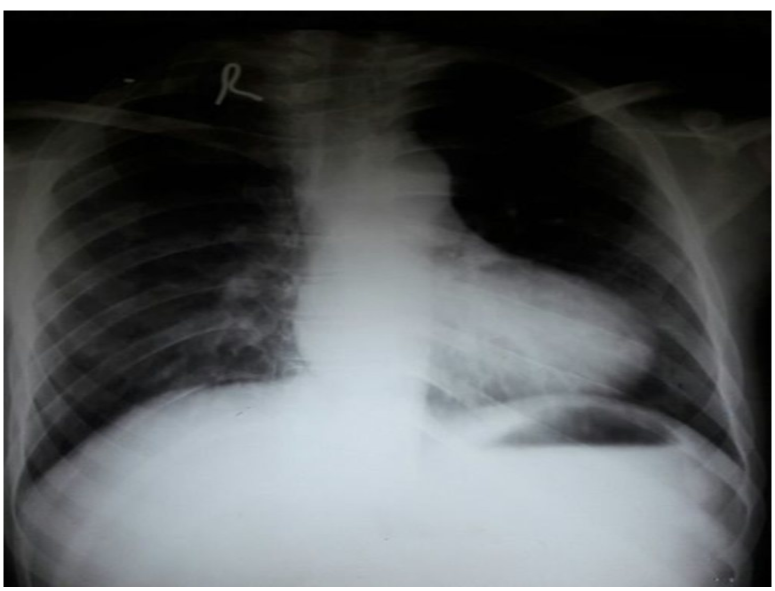
Question 16:

Most common heart defect in congenital rubella syndrome is:

- **a)** ASD
- **b)** VSD
- **c)** PDA
- **d)** PS

Question 17:

Which among the following options describes the chest x-ray image?



a) Egg on string appearance

b) Coeur en sabot

c) Scimitar sign

d) Snow man sign

Answer Key

Question No.	Correct Option
1	b
2	d
3	c
4	b
5	d
6	b
7	c
8	c
9	d
10	a
11	a
12	b
13	b
14	a
15	d
16	c
17	b

Detailed Explanations

Solution to Question 1:

Jaundice persisiting 3 weeks after birth is suggestive of pathological jaundice. Among the given options, Crigler Najjar type 2 presents with unconjugated hyperbilirubinemia which responds to phenobarbital therapy.

Dubin Johnson (option C) and Rotor (option A) syndromes lead to conjugated hyperbilirubinemia.

Crigler Najjar type 2 is an autosomal recessive disease with non-hemolytic, unconjugated hyperbilirubinemia caused by homozygous missense mutations in the enzyme UDP UGT1A1 resulting in reduced (partial) enzymatic activity.

The differences between Crigler-Najjar type 1 and type 2 are described in the table below:

Crigler Najjar Type 1		Crigler Najjar Type 2
Autosomal recessive		Autosomal recessive
UDPGT isabsent		UDPGT isreduced
Severe unconjugated hyperbil irubinemia		Less severe
Pale stools		Normal-coloured stools
May proceed tokernicterusan ddoes notrespond to phenob arbitone		Does not proceed to kernicter us andis responsive to pheno barbitone

Solution to Question 2:

Flaccid paralysis is not a feature of cerebral palsy (CP).

CP is an upper motor neuron disease characterized by rigidity and scissoring gait. Flaccid paralysis would be seen in lower motor neuron disease.

Other options:

Option A: Hypotonia is a characteristic feature of athetoid CP. These infants generally have poor head control.

Option B: Microcephaly can be seen in CP due to perinatal insults to the developing brain.

Option C: Ataxia is a common feature of extrapyramidal CP.

The subtypes of cerebral palsy are given in the table below:

UL - upper limb; LL - lower limb; GDD - global developmental delay

Types	Etiology	Presentation	MRI findings	
Spastic(m/c type)	Diplegia(second m/ c in India)	PrematurityIschemi aInfection	Bilateral spasticity (LL>UL)Scissoring g aitCommando craw lLearning disabiliti es	Periventricular leuk omalacia
Hemiplegia	Vascular insultInfec tionGenetic	Early hand prefere nce(UL>LL)Walkin g delayed up to 18-24mEquinovarusde formity of footSeizu res (33%)	Porencephalic cystF ocal infarct changes	

Types	Etiology	Presentation	MRI findings	
Quadriplegia(most severe)(m/c type in India)	IschemiaInfection	Marked motor impairmentFlexioncontracturesStrong association withseizures Severeintellectual impairment andGDDDifficulty swallowing (supranuclear palsy)	Multi-cystic encephalomalaciaSevere periventricular leukomalacia	
Dyskinetic(15-20%)	ExtrapyramidalAthetoidChorioathetoid	AsphyxiaKernicterusMitochondrialGenetic	Rigidity/hypotonia, dystonia, athetosis, choreaCognitive disabilityHip displacementSeizures	Basal ganglia scarringThalamic scarring

Solution to Question 3:

Patients with cystic fibrosis (CF) can present with hypokalemia and not hyperkalemia.

Excess production of sweat in CF leads to a significant loss of chloride ions from the extracellular fluid (ECF). The depletion of chloride from the ECF causes an intracellular shift of potassium and activation of renin-aldosterone system, leading to hypokalemia.

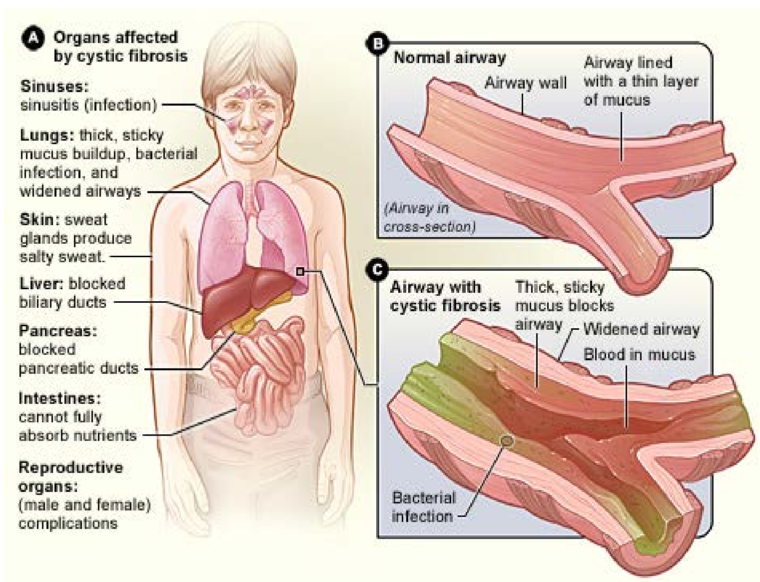
The excessive depletion of sodium chloride in CF also affects the acid-base balance. It leads to increased reabsorption of bicarbonate by the kidneys through the chloride-bicarbonate exchanger on the intercalated cells. This process is known as contraction alkalosis (option D) and results in hyponatremic hypochloremic alkalosis.

Cystic fibrosis, also known as mucoviscidosis, is an autosomal recessive disorder caused by a mutation in the cystic fibrosis transmembrane regulator (CFTR) gene on chromosome 7q. This gene is responsible for encoding ATP-gated chloride channels in the apical membranes of epithelial cells, which regulate the flow of ions across various epithelial membranes in the body, including sweat glands, respiratory airways, and pancreatic ducts. The defect in CFTR leads to elevated salt concentrations in sweat and the production of thick, viscous secretions in the respiratory and gastrointestinal tracts.

Diagnostic tests for CF include the sweat chloride test (option A), where an increased chloride concentration (>60 mEq/L) is considered diagnostic. Immunoreactive trypsinogen levels (option B) are also measured for newborn screening, as they tend to be elevated in CF. Genetic testing for CFTR gene mutations is considered the gold standard and should be performed for all suspected cases. In equivocal cases, nasal transepithelial potential difference can be estimated as an additional diagnostic test.

The criteria for diagnosis is the presence of clinical symptoms affecting at least one organ system (respiratory, gastrointestinal, or genitourinary) or a positive newborn screening or a sibling with cystic fibrosis, along with evidence of dysfunction of the CFTR gene (two positive sweat chloride test results obtained on separate days or identification of two CFTR mutations by genetic testing or an abnormal measurement of nasal potential difference).

The image below shows the organs involved in cystic fibrosis:



Solution to Question 4:

The given clinical scenario showing swelling of knee joints (arthritis), non-blanching rash in the lower limbs (palpable purpura), along with hematuria and proteinuria (renal involvement) in a child is suggestive of Henoch Schonlein purpura (HSP).

HSP (IgA vasculitis) is a small vessel vasculitis most commonly seen in the age group of 3-8 years. It generally follows an upper respiratory tract infection and there is a history of atopy in one-third of the cases.

The pathology of HSP involves leukocytoclastic vasculitis and the deposition of IgA in the affected blood vessels. IgA is deposited in the glomerular mesangium in a distribution similar to that of IgA nephropathy.

Clinical manifestations include palpable purpura, the hallmark feature of HSP, which characteristically involves the extensor surfaces of arms and legs. Musculoskeletal symptoms such as arthritis and arthralgia are common. Gastrointestinal symptoms can include abdominal pain, vomiting, diarrhea, and bleeding. Renal involvement may lead to gross or microscopic hematuria, nephritic syndrome, or nephrotic syndrome. Renal manifestations are more severe in adults. A rapidly progressive form of glomerulonephritis is seen in some cases. Nephritis usually follows the appearance of the rash.

Diagnosis of HSP is primarily based on clinical findings, including the palpable purpura, along with the presence of other symptoms. Laboratory tests are not specific to HSP but may show nonspecific markers of inflammation and sometimes elevated levels of IgA. Imaging studies and biopsies of affected tissues can provide additional diagnostic information

Treatment of HSP is generally supportive, focusing on hydration, pain relief, and nutrition. Corticosteroids may be used in severe cases or for significant gastrointestinal involvement. Intravenous immune globulin (IVIG) and plasma exchange are occasionally employed for severe disease. Nephritis due to IgA vasculitis should be treated with immunosuppressants like prednisolone and azathioprine. Complications of HSP can include serious gastrointestinal problems, kidney disease, and recurrences, but overall, the prognosis for childhood HSP is favorable. Regular monitoring of blood pressure and urine is recommended to detect and manage potential complications.

Solution to Question 5:

Severe acute malnutrition (SAM) is best defined as weight for height less than median minus -3SD

Severe acute malnutrition (SAM) in children of ages 6 to 59 months is defined by the WHO as any of the following:

• Weight-for-length (or height) below –3 SD of the median WHO child growth standards.

• Presence of bipedal edema

• Mid-upper arm circumference <11.5 cm

Once the child is diagnosed with severe acute malnutrition, the doctor must assess for severe edema (+++); low appetite (failed appetite test); medical complications (severe anemia, pneumonia, diarrhea, dehydration, cerebral palsy, tuberculosis, HIV, heart disease, etc.) and one or more danger signs as per IMNCI

If any of the complications are present, the child is diagnosed with complicated SAM and is managed inpatient in the facility.

If the above-mentioned signs are absent, the child is diagnosed with uncomplicated SAM and supervised management at home is given.

Solution to Question 6:

Chorioretinitis is the most common manifestation of congenital toxoplasmosis.

The characteristic triad of congenital toxoplasmosis is chorioretinitis, hydrocephalus, and diffuse intracranial calcifications. Clinical manifestations apart from the triad include hydrops fetalis, intrauterine growth retardation, prematurity, peripheral retinal scars, persistent jaundice, and mild thrombocytopenia.

Congenital toxoplasmosis occurs when a pregnant woman acquires the infection during gestation. The organism may cross to the fetus transplacentally or at the time of vaginal delivery. The severity of the disease in the neonate is greater if the woman gets infected early in her pregnancy. The causative agent is Toxoplasma gondii, an obligate intracellular protozoan.

The diagnosis is made by the detection of IgM anti-toxoplasma antibodies. IgG can cross the placenta is hence, cannot confirm infection in the infant. IgM immunosorbent agglutination assay (ISAGA) and IgA ELISA are the most useful diagnostic tests.

Spiramycin can be administered for cases of acute infection in early pregnancy to reduce vertical transmission. It does not cross the placenta and therefore cannot be used to treat an infected fetus. Pyrimethamine-sulfonamide (to be avoided in the first trimester) is used when a fetal infection is confirmed or in maternal infections after 18 weeks.

Solution to Question 7:

The molecular defect causing liver disease in Dubin–Johnson syndrome is ABCC2.

Dubin-Johnson syndrome is an autosomal recessive disorder due to a defect in ABCC2. ABCC2 codes for canalicular protein with ATP-binding cassette. A defect in this gene leads to the mutation of the multiple drug-resistant protein 2 (MRP2) receptor. This in turn leads to a defective transfer of conjugated bilirubin from the liver to bile.

It presents with abdominal pain, mild icterus, and conjugated hyperbilirubinemia. Biopsy shows black liver due to an accumulation of epinephrine metabolite.

Conditions causing conjugated hyperbilirubinemia are summarised in the table below:

Important diseases and their molecular defects:

Dubin-Johnson syndrome	Rotor syndrome
Autosomal recessive	Autosomal recessive
The absent function ofmultiple drug-resistant proteins (MRP-2)	Defects in anion transport
Black liver on biopsy (due to accumulation of epinephrine metabolite)	No abnormal liver pigmentation.
Normal levels of coproporphyrin in urine	Increased levels of coproporphyrin in urine
The gall bladder is not visualized on oral cholecystography	The gall bladder is visualized on oral cholecystography

ATP7A	Menke’s disease
ATP7B	Wilson disease
CFTR	Cystic fibrosis
SERPINA 1	α-1 antitrypsin deficiency
ABCC2 (MRP2 protein defect)	Dubin Johnson syndrome

Solution to Question 8:

The right upper limb is the best area to check pre-ductal oxygen saturation in a newborn infant as it receives blood supply primarily from the right brachial artery, which arises from the aorta before the ductus arteriosus.

In routine neonatal screening for critical congenital heart diseases (CCHD), saturation probes are placed on the right upper limb (pre-ductal) and the right lower limb (post-ductal) respectively.

If the difference between pre-ductal and post-ductal saturation is more than five percent, it confirms the presence of hypoxemia. Any reading showing an SpO2 <90% or greater than a three percent difference between the right hand and right foot for three successive readings separated by an hour each would suggest the presence of a congenital cardiac disease.

Screening is done to identify critical conditions before they present symptomatically, and before the neonate leaves the hospital post-delivery. This would allow for early and effective treatment of an underlying disease.

Critical congenital heart diseases include coarctation of the aorta, double outlet right ventricle, Ebstein anomaly, hypoplastic left heart syndrome, interrupted aortic arch, pulmonary atresia, single ventricle, tetralogy of Fallot (TOF), total anomalous pulmonary venous return (TAPVC), d-transposition of the great arteries (TGA), tricuspid atresia, and truncus arteriosus.

Solution to Question 9:

Necrotizing enterocolitis (NEC) can cause metabolic or respiratory acidosis (not alkalosis).

Acidosis can be attributed to sepsis, lactic acidosis due to decreased perfusion of the abdominal wall, decreased reabsorption of bicarbonate from the intestine, and reduced respiratory compensation due to apnea.

NEC is the most common life-threatening gastrointestinal disorder in the newborn period. It is characterized by various degrees of mucosal or transmural inflammation and necrosis of the intestinal tract. It most commonly affects the distal ileum and proximal colon.

Three major risk factors implicated are prematurity (greatest risk factor), formula feeding, and bacterial colonization of the intestine. Incidence and mortality increase with decreasing birth weight and gestational age.

It usually presents within the 2nd or 3rd week of life with abdominal distension and tenderness, feed intolerance, and occult/gross blood in the stool. Systemic features include hypotension, bradycardia, lethargy, and temperature instability. Other signs and symptoms include vomiting, changes in stool pattern, abdominal mass, apnea, respiratory distress, hyponatremia, shock, and disseminated intravascular coagulation.

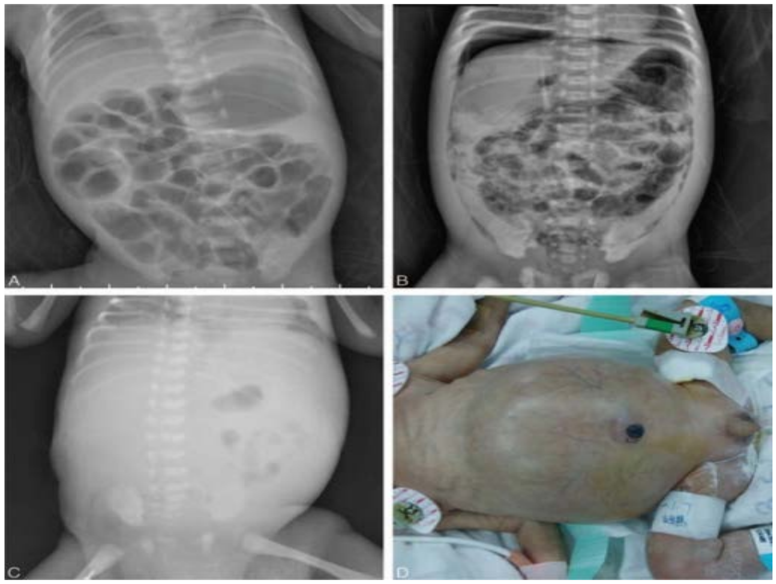
Diagnosis can be made with a plain abdominal X-ray. Pneumatosis intestinalis (air in the bowel wall) is diagnostic. Portal vein gas is a sign of severe disease. Pneumoperitoneum indicates perforation of the intestinal wall.

Staging is done using the modified Bell's staging criteria:

Management of NEC includes supportive measures - nil per oral, nasogastric decompression, intravenous fluids, broad-spectrum antibiotics, and inotropic support if indicated. Perforation is an absolute indication for surgery. The most effective preventive strategy is the use of human milk. Prebiotics and probiotics can be used with variable efficacy.

The images below show pneumatosis intestinalis (A), pneumoperitoneum (B), gasless abdomen (C), and distention of the abdomen with periumbilical erythema (D):

Stage	Features	Radiological signs	Management	Treatment
ISuspected NEC	Mildabdominal distention, vomitingTemperature instability, apnea, bradycardiaI A:Occult blood in stoolsI B:Gross bloodin stools	Normal or intestinal dilatation, mild ileus	MedicalNil per oral Total parenteral nutrition - IV fluids containing dextroseIV antibiotics -ampicillin, gentamycin, metronidazole	Antibiotics for 3 days
II AMildly ill	Absentbowel sounds	Pneumatosis intestinalis	Antibiotics for 14 days	
II BModerately ill	II A + abdominal wall edemaThrombocytopeniaHyponatremiaMetabolic acidosis	II A +portal venous gas, with or without ascites		
III ASeverely ill, bowel intact	PeritonitisDefinite ascites	II B +definite ascites		
III BSeverely ill, bowelperforated	Intestinal perforation	Pneumoperitoneum	Surgical(due to bowel perforation)	Stable- Laparotomy (resection of necrosed part, followed by anastomoses)Unstable-Primary peritoneal drainage, and once stabilized definitive surgery



Solution to Question 10:

The most common cause of HIV infection in a newborn is perinatal transmission. Around 70 to 80% of affected children acquire the virus intrapartum.

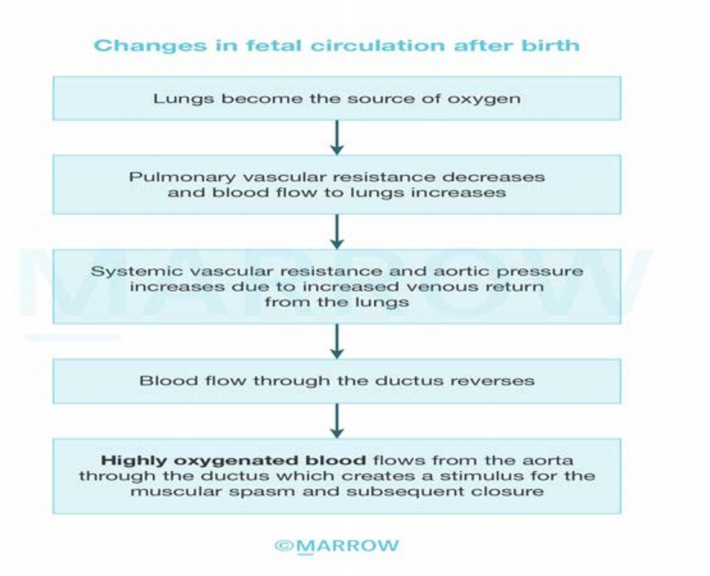
The mechanism of perinatal transmission has been postulated to be mucosal exposure to infected blood and cervicovaginal secretions in the birth canal. The intrauterine contractions during active labor/delivery could also increase the risk of late micro transfusions.

Vertical transmission of HIV can occur in utero (during pregnancy), during delivery (perinatal), and after birth by breastfeeding (least common).

The risk factors for vertical transmission are high maternal viremia, preterm delivery (<34 weeks), vitamin A deficiency, prolonged rupture of membranes (>4 hours) low birth weight, and low maternal CD4 count.

Solution to Question 11:

Oxygen is the most important factor controlling ductal closure.



Gestational age also appears to play an important role; the ductus of a premature infant is less responsive to oxygen, even though its musculature is developed. Hence, patent ductus arteriosus (PDA) is more common in premature infants.

The fetal circulatory adjustments after birth are given below:

The patency of ductus arteriosus may either provide an alternative route for blood to bypass congenital defects such as pulmonary atresia or coarctation of the aorta.

Therapeutic medications can be used to either maintain the patency (prostaglandin E1) or promote closure (indomethacin). Drugs like indomethacin are not recommended during the third trimester of pregnancy.

	Functional closure	Anatomical closure
Ductus venosus	After removal of the placenta	7 days
Foramen ovale	At birth - after breathing is initiated	Months to years
Ductus arteriosus	10–15 hours	10–21 days

Solution to Question 12:

Cardiac edema and neuropathy are associated with the deficiency of thiamine (vitamin B1).

Early symptoms of thiamine deficiency include peripheral neuropathy, irritability, loss of deep tendon reflexes, and proprioception. As the disease progresses, increased intracranial pressure, meningismus, and coma develop.

Wet beriberi (cardiac) is associated with tachycardia, cardiomegaly, cardiac failure, and edema. It is commonly seen in the refugee camps where people consume polished rice based monotonous diets. Diagnosis of thiamine deficiency can be established by erythrocyte transketolase activity assessment.

Dry beriberi is associated with neurological deficits such as optic atrophy, ataxia, and irritability.

Complications include lactic acidosis and Wernicke encephalopathy.

Solution to Question 13:

Biliary atresia is the most common indication for liver transplantation in children.

Extrahepatic biliary atresia (EHBA) is defined as a partial or total absence of permeable bile duct between the porta hepatis and the duodenum. It is due to inflammatory fibrosis of the biliary tree. It leads to obstruction to bile flow thereby causing the leaking of conjugated bilirubin into the circulation.

Infants have symptoms of jaundice, pale stools, and dark urine right after birth. In advanced disease, infants present with failure to thrive, hepatomegaly, and ascites due to liver cirrhosis.

HIDA scan is the investigation of choice for extrahepatic biliary atresia. An intraoperative cholangiogram is a gold standard investigation for diagnosis.

Children with biliary atresia, who do not achieve successful drainage after hepatoportoenterostomy (Kasai's procedure), require liver transplantation within the first year of life.

Management of extrahepatic biliary atresia		
Type I	Atresia restricted to the common bile duct	Roux-en-Y hepaticojejunostomy
Type II	Atresia of the common hepatic duct	Kasai procedure(Roux-en-Y hepatoportoenterostomy)
Type III	Atresia of the right and left hepatic duct	

Solution to Question 14:

The earliest age of appearance of bidextrous reach is 4 months.

Fine motor reflexes begin to appear only after the palmar grasp reflex disappears at 3-4 months. The development of unidextrous reach occurs by 6 months of age.

Solution to Question 15:

The Apgar score in this baby is 4. It is calculated as:

Appearance: blue - score 0

Pulse: <100 beats/min - score 1

Grimace: with minimal response to stimulation - score 1

Activity: some flexion of the extremities - score 1

Respiratory effort: irregular - score 1

The total score is 4.

Apgar score is not used to guide resuscitation, and resuscitation must be initiated before the 1-minute score. The change of score at sequential time points following birth reflects how well the baby is responding to resuscitative efforts.

Causes for low Apgar score are:

• Fetal distress

• Prematurity

• Drugs like analgesics, narcotics, sedatives, magnesium sulfate

• Acute cerebral trauma

• Precipitous delivery

• Airway obstruction (choanal atresia)

• Congenital pneumonia or sepsis

• Haemorrhage- hypovolemia

False-negative Apgar score is seen in maternal acidosis and high fetal catecholamine levels.

A total score of 10 indicates that the infant is in the best possible condition. An infant with a score of 0-3 requires immediate resuscitation. If the score at 5 minutes remains less than 7, the infant should be assessed again every 5 minutes for the next 20 minutes.

Solution to Question 16:

Patent ductus arteriosus (PDA) is the most common congenital heart disease associated with rubella, followed by pulmonary artery stenosis (PS).

Cardiac anomalies have been observed in 50% of children infected during the first 8 weeks of gestation. The incidence of heart disease in congenital rubella syndrome is: PDA > PS (right >left) > ventricular septal defect > atrial septal defect.

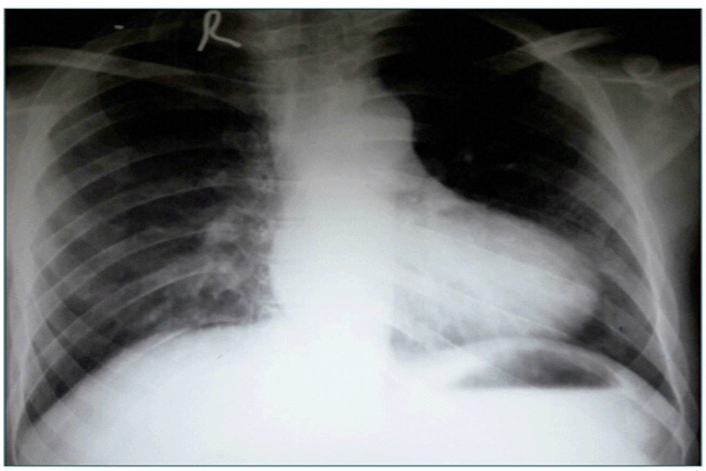
Solution to Question 17:

The given chest x-ray shows 'coeur-en-sabot' or boot-shaped heart, which is seen in tetralogy of Fallot (TOF).

The characteristic silhouette is due to cardiac apex elevation as a result of right ventricular hypertrophy. The hilar areas and lung fields are relatively clear because of diminished pulmonary blood flow or the small size of the pulmonary arteries and the aorta is usually enlarged.

The image below describes the likeness of the cardiac silhouette to a boot:

Tetralogy of Fallot - boot-shaped heart



TOF is cyanotic congenital heart disease with the following four components:

- Obstruction to right ventricular outflow (pulmonary stenosis)
- Malalignment type of ventricular septal defect (VSD)
- Dextroposition of the aorta so that it overrides the ventricular septum (overriding of the aorta)
- Right ventricular hypertrophy (RVH)

The severity of symptoms depends on the degree of the right ventricular outflow obstruction. In mild degrees, the symptoms become pronounced as the baby grows due to RV infundibular hypertrophy and may present with heart failure. In severe degrees, pulmonary blood flow predominantly depends on the ductus arteriosus. As soon as the duct closes, the baby presents with cyanosis and circulatory collapse in the neonatal period.

Clinical features include significant hypercyanotic spells in infants which are characterised by hyperpnea, cyanosis, and gasping followed by syncope. The spells occur most frequently in the morning on initially awakening or after episodes of vigorous crying. These spells are due to a prolonged reduction of pulmonary blood flow resulting in severe systemic hypoxia and metabolic acidosis.

Older children present with exertional dyspnea and assume a squatting position to relieve the dyspneic episode after which the child is able to resume the activity.

Paediatrics NEET 2020

Question 1:

Two girls in the same class are diagnosed with meningococcal meningitis. Their 12-year old friend from the same school is afraid of contracting the disease. What advice should be given to the exposed students?

- **a)** Two doses of polysaccharide vaccine
- **b)** Antibiotic prophylaxis
- **c)** Two doses of conjugate vaccine
- **d)** Single dose of meningococcal vaccine

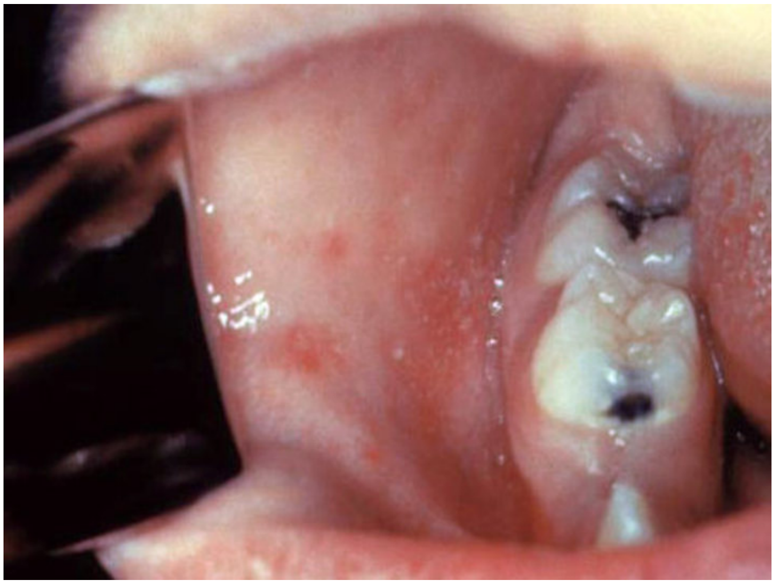
Question 2:

A laborer's second child is brought to the OPD with a swollen belly and dull face. He has been fed rice water (rice milk) in his diet mostly. On investigations, the child is found to have low serum protein and low albumin. What is the probable diagnosis?

- **a)** Kwashiorkor
- **b)** Marasmus
- **c)** Indian childhood cirrhosis
- **d)** Kawasaki disease

Question 3:

A 3-year-old male has the following buccal mucosa finding. After three days, he presents to your OPD with a rash. What is your diagnosis?



a) Measles

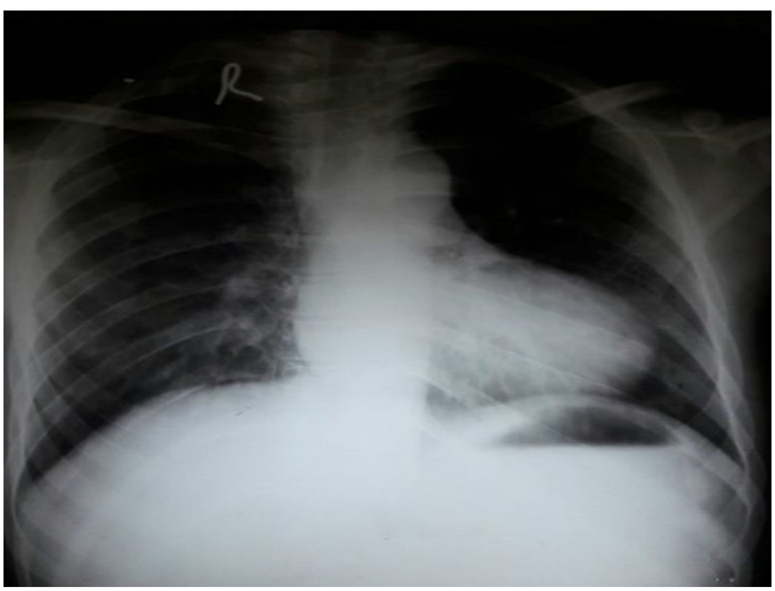
b) Leukoplakia

c) Scarlet fever

d) Kawasaki disease

Question 4:

The given chest X-ray is suggestive of:



a) Tetralogy of Fallot

b) Atrial septal defect

c) Ventricular septal defect

d) Ebstein's anomaly

Question 5:

The greenish-black color of the first stool in the newborn is due to :

a) Bilirubin

b) Biliverdin

c) Meconium

d) Gut flora of neonate

Question 6:

Maternal antibodies that do not provide significant immunity in a neonate:

a) Tetanus

b) Polio

c) Diptheria

d) Measles

Question 7:

Anuj has been■ diagnosed with diphtheria and is being treated in the ward. His 3-year-old brother, Ajay who lives in the same house has received vaccination against diphtheria about 16 months back. What is the next best step for the management of the younger sibling, Ajay?

a) One booster dose

b) Nothing as the child is already exposed

c) Erythromycin + diphtheria toxoid

d) Erythromycin only

Question 8:

According to current recommended dietary guidelines for children, energy from saturated fats should be __________ of total energy intake.

a) < 10%

b) < 5%

c) < 15%

d) < 30%

Question 9:

A 3-year-old girl was brought by her mother with complaints of enlarged clitoris. Karyotype revealed 46XX. What is the likely cause?

a) 21 hydroxylase deficiency

b) 11 hydroxylase deficiency

c) 3 beta dehydrogenase deficiency

d) 2, 3 lyase deficiency

Question 10:

A newborn is brought with evisceration of bowel loops from a defect in the abdominal wall to the right of the umbilicus. The most probable diagnosis is

a) Omphalocele

b) Umbilical hernia

c) Gastroschisis

d) Ectopia vesicae

Answer Key

Question No.	Correct Option
1	b
2	a
3	a
4	a
5	b
6	b
7	d
8	a

9	a
10	c

Detailed Explanations

Solution to Question 1:

Antibiotic prophylaxis is provided to the close contacts of persons with confirmed meningococcal disease.

Close contacts of individuals with confirmed meningococcal disease are at an increased risk of developing the illness. Prophylaxis is also offered to individuals with direct and prolonged contact with a case of meningococcal disease. Antibiotics are effective in preventing additional cases by eradicating the carriage of the invasive strain.

Treatment is provided to the contacts within 24 hours of identification of the index case. The antibiotics effective include rifampicin, ciprofloxacin, ceftriaxone, and azithromycin.

Prophylaxis is not generally recommended for medical personnel, except in cases where there is exposure to aerosols of respiratory secretions, such as during mouth-to-mouth resuscitation, intubation, or suctioning. This recommendation applies within 24 hours before or after the initiation of antibiotic therapy in the index case.

For the treatment of cases, penicillin is the preferred drug, but ceftriaxone or other third-generation cephalosporins are recommended for patients with penicillin allergy. Droplet infection control precautions should be followed for hospitalized patients for 24 hours after effective therapy has been initiated.

Offering quadrivalent meningococcal A, C, W, Y conjugate vaccines in addition to antimicrobial prophylaxis can further reduce the risk of disease among close contacts of cases caused by vaccine capsular groups.

Meningococcal vaccines are available in two forms: polysaccharide vaccines and polysaccharide-protein conjugate vaccines. Polysaccharide vaccines can be given as a single dose to individuals aged 2 years and older. They are offered in bivalent (A, C), trivalent (A, C, W135), and quadrivalent (A, C, W135, Y) formulations.

Meningococcal conjugate vaccines are available as monovalent (A or C) or quadrivalent (A, C, W135, Y) formulations. Conjugate vaccines, which offer increased immunogenicity and potential for herd protection, are preferred over polysaccharide vaccines, especially in children under 2 years of age. Both types of vaccines are safe to use during pregnancy.

Solution to Question 2:

The given clinical scenario of a child with a swollen belly, dull face with hypoproteinemia along with poor diet history suggests the diagnosis of kwashiorkor.

Kwashiorkor is a form of severe acute malnutrition (SAM) characterized by edema and apathy. In edematous malnutrition, the initial appearance of edema is commonly observed in the feet and lower legs, progressing into generalized edema affecting the hands, arms, and face. The skin over the swollen areas may show dark, crackled, peeling patches (flaky paint dermatosis) with pale skin underneath, making it susceptible to infections.

There may be sparse and easily pluckable hair. Hepatomegaly with fatty infiltration is a common finding. Associated symptoms of misery, apathy, and loss of appetite are seen in children with edema.

Other options:

Option B: Marasmus is a severe form of wasting characterized by significant loss of fat and muscle tissue as the body utilizes these resources for energy. It is often triggered by acute starvation or illness in individuals with borderline nutritional status.

Option C: Indian childhood cirrhosis (ICC) is a chronic liver disease that affects infants and young children. Common symptoms include jaundice, pruritus, lethargy, and hepatosplenomegaly. The disease progresses rapidly, leading to the development of cirrhosis.

Option D: Kawasaki disease (KD) is a systemic inflammatory disorder with arteritis of medium to large-sized vessels. It presents with fever, rash, conjunctivitis, and cervical lymph node enlargement.

Solution to Question 3:

The given image shows small red lesions with bluish-white spots in the center on the inner aspects of the cheek opposite to the lower premolar, indicating the presence of Koplik spots. The appearance of these spots, along with a rash appearing three days later, strongly suggests a diagnosis of measles.

The measles virus is a single-stranded RNA virus. Measles infection has 4 phases: the incubation period, prodromal illness, exanthematous phase (characterized by the rash), and recovery phase.

After an incubation period of 8-12 days, the prodromal phase of measles starts with mild fever and is followed by conjunctivitis, photophobia, coryza, cough, and increasing fever. Koplik spots, appearing 1-4 days prior to the rash, are a characteristic sign of measles.

Symptoms gradually intensify over a period of 2-4 days until the first day of the rash. The rash typically begins on the forehead (around the hairline), behind the ears, and on the upper neck as a red maculopapular eruption. It then spreads downward to the torso and extremities, often becoming confluent on the face and upper trunk.

Complications include acute otitis media (most common), pneumonia (the most common cause of death), croup, bronchiolitis, reactivation of tuberculosis, diarrhea, appendicitis, febrile seizures, and encephalitis. Subacute sclerosing panencephalitis is a rare chronic complication of measles.

Other options:

Option B: Leukoplakia appears as a white patch in the oral cavity that cannot be wiped off.

Option C: Scarlet fever is Group A Streptococcus (GAS) pharyngitis and is characterized by a distinctive sandpaper rash. The rash appears as a diffuse, finely papular, erythematous eruption that turns the skin bright red and blanches on pressure. It starts around the neck and spreads to the trunk and extremities.

Option D: Kawasaki disease (KD) is a systemic inflammatory disorder with arteritis of medium to large-sized vessels. It presents with fever, rash (maculopapular, erythema multiforme urticarial, or micropustular), conjunctivitis, and cervical lymph node enlargement.

Solution to Question 4:

The given chest X-ray shows a boot-shaped cardiac silhouette which is suggestive of the Tetralogy of Fallot.

The boot-shaped (coeur en sabot) cardiac silhouette is seen due to the right ventricular hypertrophy and upward shift of the cardiac apex. The hilar areas and lung fields appear relatively clear due to reduced pulmonary blood flow or the small size of the pulmonary arteries, or both.

In Ebsteins's anomaly (Option D) box-shaped heart is seen.

Solution to Question 5:

The greenish-black color of the first stool in the newborn is due to biliverdin.

Meconium is a dark, viscous material that is normally present in the intestine of the fetus and passed within 1st 48hour of life. Meconium appears after the 20th week and at term, it is uniformly distributed throughout the gut up to the rectum indicating the presence of intestinal peristalsis.

It is composed of various products of secretion, such as desquamated fetal cells, glycerophospholipids from the lungs, lanugo, scalp hair, vernix, and undigested products from swallowed amniotic fluid. The greenish-black color of meconium is due to the presence of bile pigments excreted by the biliary tract, especially biliverdin (pigment from the oxidation of bilirubin).

During intrauterine hypoxia, the relaxation of the anal sphincter due to vagal stimulation can lead to the passage of meconium into the amniotic fluid. Additionally, the release of arginine vasopressin (AVP) from the fetal pituitary gland due to intrauterine hypoxia can stimulate the colon muscles to contract, resulting in intraamniotic defecation. Meconium aspiration syndrome can occur when meconium is inhaled, as it is toxic to the lungs.

Other options:

Option A: Bilirubin is responsible for the yellow color of the stools.

Option D: Neonates do not harbor gut flora at birth.

Solution to Question 6:

Maternal antibodies do not provide significant immunity to polio infection in a neonate.

Administering an oral polio vaccine (OPV) dose at birth can provide early mucosal protection, safeguarding against enteric pathogens before they can hinder the intestinal mucosal immune response. Furthermore, administering OPV during the period when infants are protected by maternal antibodies may help prevent vaccine-associated paralytic polio (VAPP).

All pregnant women should receive a tetanus toxoid, reduced diphtheria toxoid, and acellular pertussis (Tdap) vaccine during each pregnancy, as early in the 27–36-week-of-gestation window as possible. The maternal antibodies formed from Tdap will provide immunity against tetanus, diphtheria, and pertussis in the newborn. Administering the Tdap vaccine earlier, between the 27th and 36th week of gestation, maximizes the transfer of passive antibodies to the infant (options A and C).

Infants less than 6 months are generally protected from measles by maternal antibodies (option D).

Solution to Question 7:

In the given clinical scenario, a 3-year-old child is likely to have received the 4th dose (1st booster) of the diphtheria vaccine between 16–24 months of age according to National Immunization Schedule (i.e., 16 months ago). Since he has received a booster dose within 5 years, only erythromycin is indicated in case of close contact with the diphtheria patient.

All household contacts with diphtheria patients should be closely monitored for 7 days. Culture should be taken from both the throat and nose. All contacts should receive antibiotic prophylaxis (benzathine penicillin G or erythromycin), regardless of immunization status.

Children who have not received a booster dose of the diphtheria toxoid vaccine within 5 years should be given the vaccine in an age-appropriate form. Children who have not yet received their 4th dose should be vaccinated accordingly.

If fewer than three doses of diphtheria toxoid have been given, or vaccination history is unknown, an immediate dose of diphtheria toxoid should be given and the primary series completed according to the current schedule.

Asymptomatic carriers of diphtheria are treated with antimicrobial prophylaxis for a period of 10-14 days. If an individual has not received a booster dose of diphtheria toxoid within 1 year, an age-appropriate preparation of the vaccine should be administered promptly.

Droplet precautions (respiratory colonization) or contact precautions (cutaneous colonization) of cases and carriers are maintained until two consecutive negative cultures (collected 24 hours apart) are obtained after cessation of therapy.

Solution to Question 8:

According to current dietary guidelines for children, energy from saturated fats should be <10% of total energy intake. Diets that are low in saturated fat and cholesterol (<300mg/day) without transfat are preferred.

Acceptable macronutrient distribution ranges (AMDA % of energy) are given below:

Macronutrient	Age1 to 3years	Age4 to 18years
Fats	30 to 40	25 to 35
Protein	5 to 20	10 to 30
Carbohydrate	45 to 65	45 to 65

Solution to Question 9:

The given clinical scenario of a girl (46XX) with an enlarged clitoris is highly suggestive of congenital adrenal hyperplasia (CAH) due to 21-hydroxylase deficiency.

Congenital adrenal hyperplasia/adrenogenital syndrome is a family of autosomal-recessive disorders caused by the deficiency of enzymes involved in the synthesis of steroid hormones in the adrenal cortex. The majority of cases of CAH are caused by 21-hydroxylase deficiency. This deficiency leads to a lack of synthesis of cortisol and aldosterone, resulting in the salt-wasting form of the disease, which is the most severe form.

Severe 21-hydroxylase deficiency presents with symptoms including weight loss, dehydration, weakness, hypotension, and electrolyte imbalances. Accumulated steroid precursors like 17-hydroxyprogesterone, are shunted into the pathway of androgen synthesis leading to elevated levels of androstenedione and testosterone. This can result in abnormal genital development in affected female fetuses during early gestation.

Females may have masculinized external genitalia. This can include enlargement of the clitoris, partial or complete labial fusion, and a shared opening of the vagina and urethra (urogenital sinus).

Untreated individuals develop signs of androgen excess after birth. These signs include accelerated skeletal maturation, premature epiphyseal closure leading to stunted adult stature, excessive muscular development, the appearance of pubic and axillary hair, acne, and deepening of the voice. In males, enlargement of the penis, scrotum, and prostate. In females, the clitoris may become further enlarged.

Diagnosis of confirmed by elevated 17-hydroxyprogesterone levels. Glucocorticoid replacement with hydrocortisone is done. In a salt-wasting crisis, fludrocortisone supplementation is given. Vaginoplasty and clitoral recession are done for females with ambiguous genitalia.

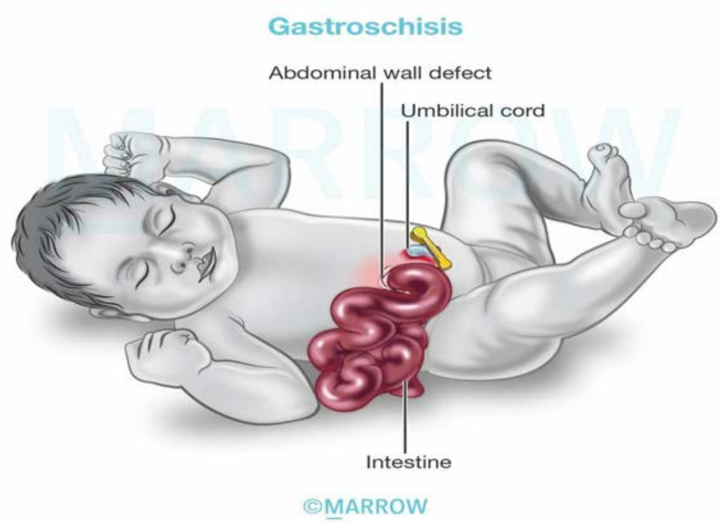
Solution to Question 10:

The most probable diagnosis for the evisceration of bowel loops, coming out of a defect in the abdominal wall to the right of the umbilicus is gastroschisis. It occurs when there is a failure in the closure of the body wall in the abdominal region.

Gastroschisis is an abdominal wall defect located on one side of the umbilicus (right side commonly) allowing protrusion of the intestines, stomach, bladder, and sometimes ovaries or undescended testes. The loops of the bowel are not covered by amnion as they herniate through the abdominal wall directly into the amniotic cavity. The affected bowel loops can be damaged by exposure to amniotic fluid.

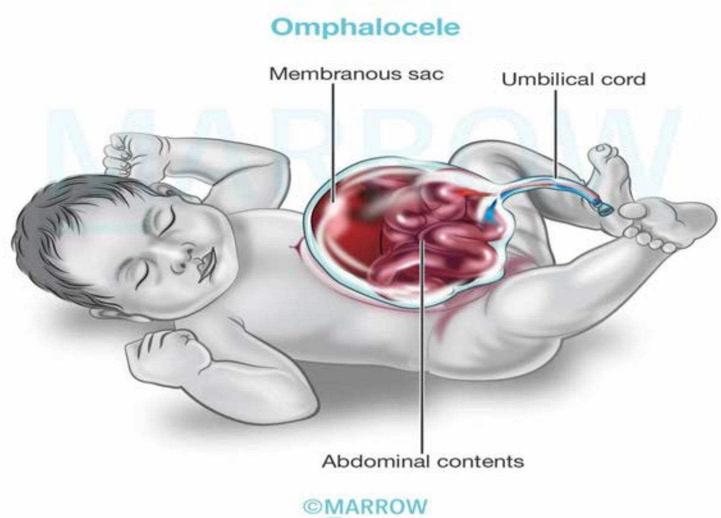
Risk factors include teenage pregnancy, recreational drug use, smoking, and genitourinary infection during pregnancy. After birth, careful management is required to prevent twisting or kinking of the blood supply to the intestines. The abdomen is wrapped with transparent plastic wrap, a nasogastric tube is inserted, and fluid resuscitation is initiated.

Depending on the severity, primary closure under general anesthesia or gradual reduction using a preformed silo may be performed. Parenteral nutrition is often needed for several weeks to support intestinal recovery.



Other options:

Option A: In omphalocele bowel loops fail to return from the physiologic umbilical hernia. Since the umbilicus is covered by a reflection of the amnion, the omphalocele is also covered by the amnion.



Option B: Congenital umbilical hernia occurs due to the incomplete closure or weakness of the muscular umbilical ring. It presents as a soft swelling covered by skin that protrudes during activities like crying, coughing, or straining, and it can be easily pushed back through the fibrous ring at the umbilicus. The hernia contains the omentum or segments of the small intestine.

The image below shows an umbilical hernia:

Option D: Ectopia vesicae/bladder exstrophy is a congenital abnormality that occurs when the lower abdominal wall does not form properly and the bladder protrudes from the abdominal wall, with its mucosa exposed.

The image below shows ectopia vesicae/bladder exstrophy

Paediatrics INI-CET 2021

Question 1:

Which of the following is a true statement about congenital CMV infection?

- **a)** About 20-30% of the infections are symptomatic
- **b)** Triad of SNHL, periventricular calcification and enamel hypoplasia
- **c)** Most children asymptomatic at birth can develop conductive hearing loss later in life
- **d)** If mother is IgG positive for CMV, the child is unlikely to develop infection

Question 2:

Which of the following are components of kangaroo mother care?

- **a)** Early discharge, skin to skin contact, prevent infection
- **b)** Early discharge, skin to skin contact, exclusive breastfeeding
- **c)** Skin to skin contact, exclusive breastfeeding, prevent infection
- **d)** Early discharge, exclusive breastfeeding and prevent infection

Question 3:

Which of the following is given as a part of the therapy for a child with severe COVID?

- **a)** Corticosteroids
- **b)** Ivermectin
- **c)** Remdesivir
- **d)** All of the above

Question 4:

Which of the following is seen in physiological jaundice?

- **a)** Increased indirect bilirubin, increased urobilinogen, absent bilirubin in urine
- **b)** Increased indirect bilirubin, absent urobilinogen, absent bilirubin in urine
- **c)** Increased direct bilirubin, increased urobilinogen, absent bilirubin in urine

d) Increased direct bilirubin, absent urobilinogen, absent bilirubin in urine

Question 5:

A child presents with fever and vesicular lesions on the upper limb and the lower limb. Neck stiffness was present. Similar lesions were present on the palms, soles, and oral cavity. CSF analysis revealed normal glucose levels and elevated lymphocytes and protein. What is the most likely diagnosis?

a) Bacterial meningitis with sepsis

b) Coxsackie viral meningitis and Hand Foot Mouth disease

c) Herpes simplex gingivostomatitis and meningoencephalitis

d) Tuberculous meningitis

Question 6:

Match the following developmental milestones:

a) 1-a, 2-b, 3-c, 4-d

b) 1-b, 2-a, 3-d, 4-c

c) 1-c, 2-b, 3-a, 4-d

d) 1-d, 2-c, 3-b, 4-a

Question 7:

Which of the following congenital heart disease has equal saturation in all the chambers of the heart?

a) Tetralogy of Fallot

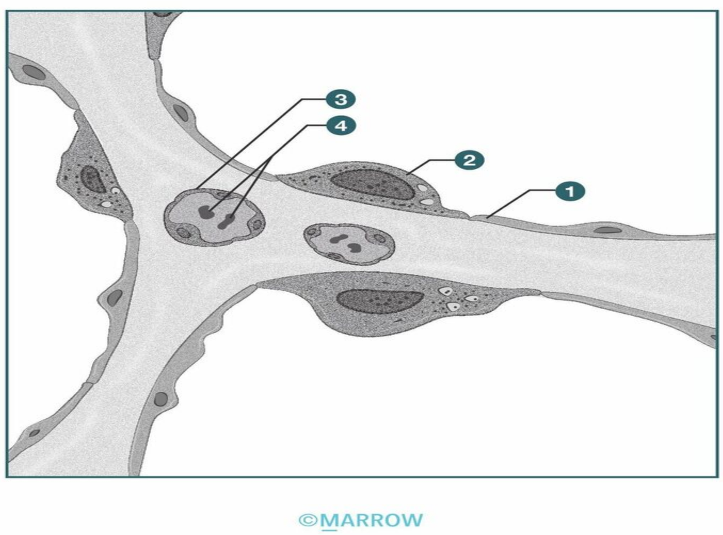
b) Total anomalous pulmonary venous circulation

c) Tricuspid atresia

d) Transposition of great arteries

Question 8:

Which of the following marked structure is damaged in respiratory distress syndrome?



a) 1

b) 2

c) 3

d) 4

Question 9:

Which of the following are causes of neonatal seizures ?

a) 1, 2, 3 only

b) 1, 2, 3, and 4

c) 1, 3, 4 only

d) 2, 4 only

Question 10:

A neonate is found to have increased irritability, poor feeding, and 2 episodes of seizures. Lumbar puncture was done and the findings were suggestive of meningitis. What is the most likely causative organism?

- **a)** Escherichia coli
- **b)** Streptococcus agalactiae
- **c)** Neisseria meningitidis
- **d)** Listeria monocytogenes

Question 11:

At what age does a child attain half-height of the adult height?

- **a)** 12 - 18 months
- **b)** 20 - 24 months
- **c)** 32 - 36 months
- **d)** 40 - 48 months

Answer Key

Question No.	Correct Option
1	d
2	b
3	a
4	a
5	b
6	b
7	b
8	b
9	b
10	a
11	b

Detailed Explanations

Solution to Question 1:

If the mother is IgG positive for CMV, then the child is unlikely to develop the infection.

In one-third of mothers who develop a primary infection during pregnancy, the virus gets transmitted to the fetus. In previously infected mothers with IgG antibodies, only 1-3% would transmit the virus to the fetus.

Most CMV transmission occurs in the first trimester. The risk of transmission is high in the third trimester. Only 10% of infants with congenital CMV infection are symptomatic (option A).

Hepatosplenomegaly, jaundice, and petechiae constitute the clinical triad (option B) in congenital CMV. Neuroimaging shows intracranial periventricular calcifications. Other manifestations include IUGR, hydrops, microcephaly, pneumonitis, chorioretinitis, seizures, sensorineural hearing loss, abnormal dentition, and hypocalcified enamel.

About half of the symptomatic infants develop long-term sequelae which include sensorineural hearing loss (SNHL), vision loss, seizure disorders, mental retardation, and developmental delay. But in asymptomatic infants, only about 13.5% develop long-term sequelae which is commonly SNHL (option C).

The investigation of choice is PCR or viral culture. It should be done within the first two to three weeks of life. After this period, it is difficult to distinguish between a congenital and an acquired infection. Since not all affected infants are viremic at birth, urine and/or saliva are more reliable samples than blood.

There is no definitive treatment for congenital CMV infection. Treatment with ganciclovir is indicated only in patients with progressive neurologic disease and deafness.

Solution to Question 2:

The four components of kangaroo mother care (KMC) are skin-to-skin positioning of the baby on the mother's (or caretaker's) chest, exclusive breastfeeding, early discharge and ambulatory care, and support for the mother and her family for taking care of the baby.

Eligibility criteria for KMC include all hemodynamically stable low birthweight (LBW) babies. All mothers without any serious illness can provide KMC. Short sessions of KMC can be initiated in babies on medical treatment like IV fluids or oxygen therapy.

The initiation of KMC can differ depending on the weight of the newborn. Babies weighing less than 1200 g at birth may require several days to weeks before KMC can begin. For babies weighing between 1200 g and 1800 g, the initiation of KMC may only take a few days. KMC can be started immediately after birth for babies weighing over 1800 g.

The baby should be placed between the mother's breasts in an upright position, with the face turned to one side, The hips are abducted and flexed in a 'frog position' and the arms are also flexed. The abdomen of the baby should be at the level of the mother's epigastrium. In this position, the mother's breathing would stimulate the baby and prevent the development of apnea.

KMC is not necessary once the baby attains 2500 g of weight or 37 weeks of gestation.

The image below shows proper KMC positioning:

The duration of KMC should be gradually increased from short sessions of more than one hour to 24 hours a day.

KMC is comparable to incubators in terms of safety and thermal protection as evidenced by mortality rates. Additionally, it has been shown to reduce severe morbidity by facilitating breastfeeding.

KMC promotes the bonding between mother and child, fostering a strong emotional connection during the early stages of life.

Note: Incubatory care, feeding via nasal catheter, and prevention of infection are part of intensive care for low birthweight babies.

Solution to Question 3:

Among the given options, only corticosteroids are given for children with severe COVID-19.

Severe COVID-19 infection is defined as SpO2 <90% on room air with any of the following findings: acute respiratory distress syndrome (ARDS), signs of severe pneumonia, septic shock, multi-organ dysfunction syndrome (MODS) or pneumonia with cyanosis, grunting, severe chest retraction, lethargy, somnolence, seizures.

It is important to note that CT chest is not indicated in the diagnosis or management of COVID-19 in children and is only considered when there is no improvement in the respiratory status.

Regarding the management of severe COVID-19 infection, the Ministry of Health and Family Welfare (MoHFW) guidelines mention that, in mild or asymptomatic COVID-19 infection, steroids are avoided and are found to be harmful. However, in rapidly progressive moderate or severe cases, corticosteroids are used in the treatment.

This can be administered in two ways: Dexamethasone at a dose of 0.15 mg/kg, with a maximum dose of 6 mg, once a day, or Methylprednisolone at a dose of 0.75 mg/kg, with a maximum dose of 30 mg, once a day.

Steroids should be given for 5-7 days, with a gradual tapering off over 10-14 days, and are avoided within the first 3-5 days of symptom onset, as they prolong viral shedding.

The use of antivirals (option C) or monoclonal antibodies is not recommended for children less than 18 years of age, irrespective of the severity of the infection.

Antimicrobials should not be prescribed unless there is clinical suspicion of a superadded infection or the patient is in septic shock.

Anticoagulants are not indicated routinely, except in patients having a strong personal or family history of thromboembolism or more than four high-risk factors for developing thrombosis.

Solution to Question 4:

In physiological jaundice, there is increased indirect bilirubin, increased urobilinogen, and absent bilirubin in urine.

Physiological jaundice (icterus neonatorum) is due to increased bilirubin production from the breakdown of fetal red blood cells combined with transient limitation in the conjugation of bilirubin by the immature neonatal liver. It first appears on the 2nd or 3rd days usually peaks on the 4th day at 5-6 mg/dL and decreases to <2 mg/dL between the 5th and 7th days after birth.

Physiological jaundice would lead to an increase in indirect (unconjugated) bilirubin. Increased levels of unconjugated bilirubin in the blood lead to increased production of urobilinogen which would be present in the urine. Urine bilirubin is not usually found in this condition.

Risk factors for physiological jaundice include maternal age, maternal diabetes, prematurity, drugs (vitamin K3, novobiocin), polycythemia, male sex, trisomy 21, cutaneous bruising, blood extravasation as in cephalhematoma, breastfeeding, weight loss, and a family history of physiological jaundice.

On the other hand, in pathological jaundice, the appearance of jaundice is within the first 24 hours after birth and may be present at or beyond the age of 2 weeks in term babies and 3 weeks in preterms babies. Other signs of pathological jaundice include an increase in serum bilirubin exceeding 5mg/dL within a 24-hour period, a serum bilirubin level greater than 12mg/dL in a full-term infant or between 10-14mg/dL in a preterm infant, and a direct bilirubin level more than 2mg/dL at any given time. Jaundice visible on the palms and soles, unresponsiveness to phototherapy, and a rapid rise in bilirubin levels during phototherapy indicate pathological jaundice.

When the concentration of unconjugated bilirubin exceeds 20-25mg/dL, bilirubin would penetrate the blood-brain barrier and lead to kernicterus.

Phototherapy is initiated in babies with pathological jaundice when
the bilirubin levels exceed the 95th percentile on the total serum bilirubin (TSB) normogram. Babies are kept at a distance of 30-45 cm from the UV lamp (460-490 nm) with eyes and gonads covered. It is avoided in babies with conjugated hyperbilirubinemia due to the risk of bronze-baby syndrome. Dehydration and hypocalcemia are potential complications and breastfeeding is encouraged.

Exchange transfusion (ET) is done in cases of bilirubin encephalopathy, Rh incompatibility with significant hemolysis, and bilirubin values exceeding TSB cutoffs for ET.

Solution to Question 5:

The presence of characteristic vesicular lesions on the palms, soles, and oral cavity is suggestive of Hand-foot-and-mouth disease (HFMD) caused by the coxsackie virus. The presence of neck stiffness and CSF showing elevated lymphocytes with normal glucose and elevated protein levels are suggestive of viral meningitis. Coxsackie A viruses can lead to aseptic meningitis.

HFMD is a viral infection commonly caused by coxsackievirus A16, which belongs to the subgroup of enteroviruses. In larger outbreaks, other viruses such as enterovirus 71, coxsackie A viruses 5, 6, 7, 9, 10, and coxsackie B viruses 2 and 5 can also cause the disease.

HFMD typically presents as a mild illness with vesicular lesions on the tongue, posterior pharynx, buccal mucosa, palate, and lips. Lesions may also appear on the hands, fingers, buttocks, and groin, with a higher prevalence on the hands. These vesicles usually resolve within a week.

Infections with enterovirus 71 can be more severe with neurological and cardiopulmonary manifestations. Coxsackievirus A16 infections occasionally lead to complications such as encephalitis, acute flaccid paralysis, myocarditis, pericarditis, and shock. Coxsackievirus A6 can cause a severe and atypical form of HFMD.

Diagnosing HFMD typically relies on clinical examination of the characteristic lesions. Confirmatory testing can involve viral culture using samples such as blood, cerebrospinal fluid (CSF), urine, and mucosal swabs. However, RT-PCR testing is more sensitive and specific for detecting the viral genome even when cultures are negative.

Treating HFMD involves mainly supportive care as there is no specific proven antiviral agent for enteroviruses. In cases of enterovirus 71-related cardiopulmonary disease, milrinone is suggested as an effective treatment. Immunoglobulins have been utilized to manage severe cases, although outcomes vary.

Apart from HFMD, coxsackie A viruses can cause herpangina characterized by febrile vesicular pharyngitis with ulcerating small vesicles on the oral fauces, palate, uvula, and posterior pharyngeal wall.

On the other hand, coxsackie B viruses can cause pleurodynia or Bornholm disease, myocarditis, and pericarditis in newborns, as well as orchitis and juvenile diabetes.

Other options:

Option C: Herpes simplex virus (HSV) presents similarly with vesicular lesions but lacks typical herpetic signs. The absence of atypical symptoms like mouth pain, drooling, and characteristic herpetic lesions rule out HSV gingivostomatitis. HSV-induced encephalitis causes anosmia, focal seizures and behavioural changes due to cortex or limbic system damage.

The image shows a typical HSV vesicular lesion below:

Option A: Bacterial meningitis with sepsis is unlikely given the vesicular lesions' distribution pattern (bacteria are less likely to cause vesicles in the oral cavity). Moreover, CSF findings in bacterial meningitis usually show decreased glucose, elevated neutrophils, and elevated protein levels.

Option D: Tuberculous meningitis is less likely as there is no history of TB exposure provided. In most cases, CSF findings in tuberculous meningitis demonstrate elevated lymphocytes and protein levels, along with decreased glucose.

Solution to Question 6:

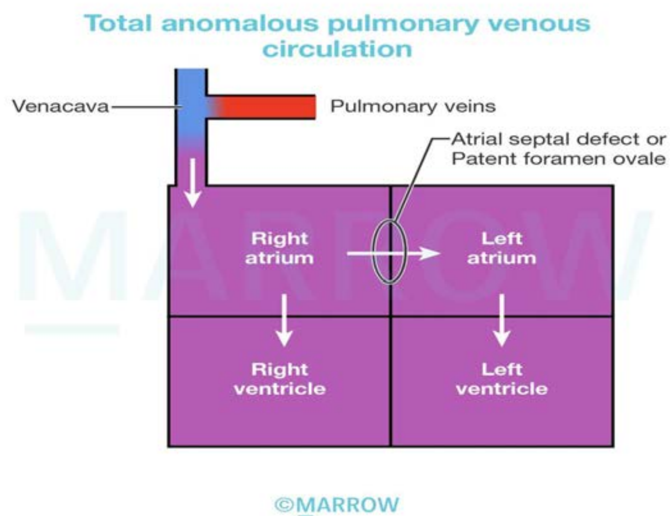
The correctly matched developmental milestones are as follows:

1.Transfers objects	b. 6-7 months
2.Social smile	a. 1-2 months
3. Pincer grasp	d. 9-12 months
4. Walks 1-2 steps	c. 12-15 months

Solution to Question 7:

In total anomalous pulmonary venous circulation (TAPVC), there is equal saturation in all chambers of the heart.

In TAPVC, the oxygenated blood from the pulmonary veins reaches the right atrium (having deoxygenated blood) instead of the left atrium. It occurs due to the draining of the pulmonary veins to the superior vena cava or to the inferior vena cava or to both. This is a total mixing lesion. The blood from this right atrium (RA) passes to the right ventricle and pulmonary artery. This blood also reaches the left atrium through the atrial septal defect or the patent foramen ovale, and subsequently, to the left ventricle The left atrium is not receiving any blood because all of the pulmonary veins are draining to the right atrium. Therefore, this shunt will be the only source of blood to it. Since blood from right atrium reaches all the other chambers, the oxygen saturation of the blood in all four chambers is the same.



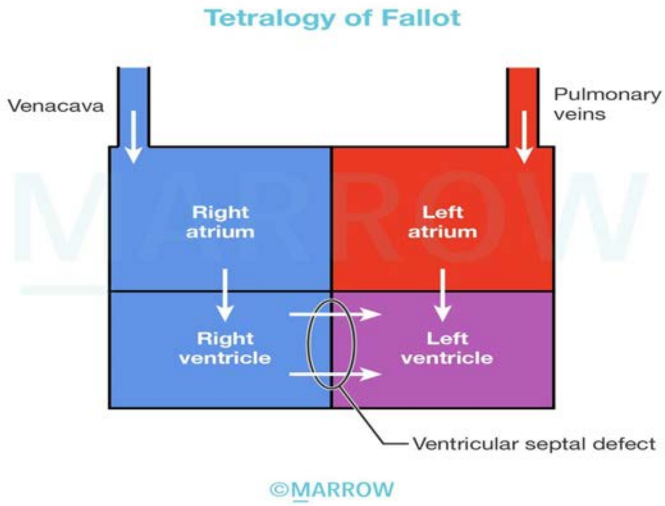
In (TAPVC), there are three main types: supra-cardiac, cardiac, and infra-cardiac. In supra-cardiac TAPVC, the pulmonary veins merge to form a single vein that drains into either the superior vena cava or the left innominate vein. In cardiac TAPVC, the pulmonary veins directly drain into the right atrium or coronary sinus. The most dangerous type is infra-cardiac TAPVC, which is always obstructive. In this type, the pulmonary veins drain into either the portal vein or the inferior vena cava.

Diagnosis of TAPVC is the demonstration of any vein with doppler flow away from the heart because the normal venous flow is usually toward the heart.

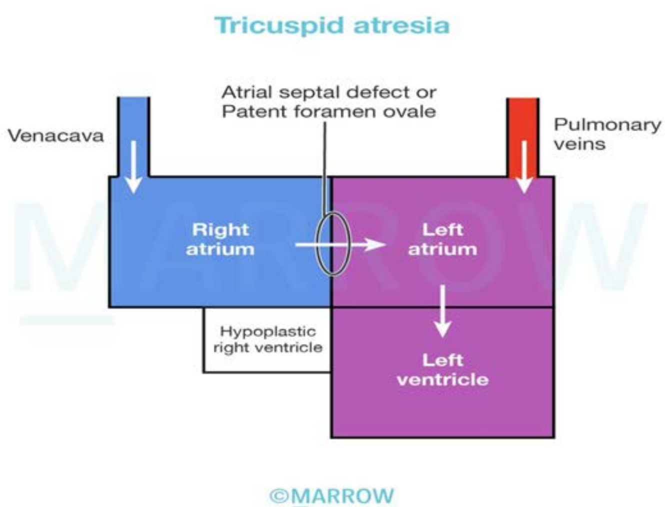
Surgical correction of TAPVR is done in infancy. Pulmonary venous confluence is anastomosed directly to the left atrium, the ASD is closed, and any connection to the systemic venous circuit is interrupted.

Other options

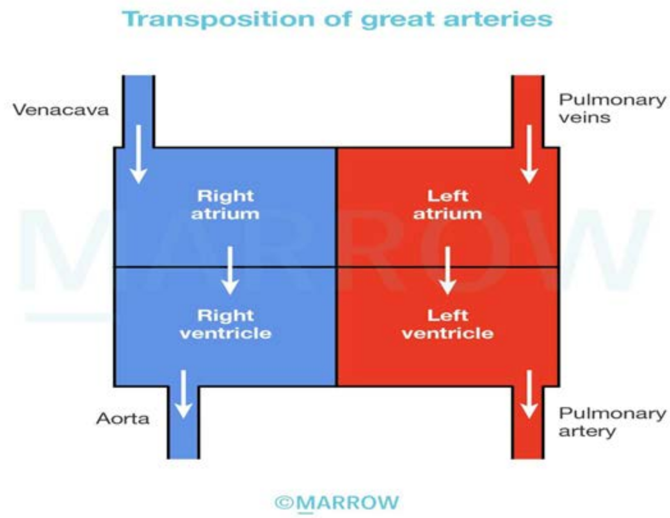
Option A - Tetralogy of Fallot (TOF) comprises right ventricular outflow tract obstruction, ventricular septal defect (VSD), overriding of the aorta over the ventricular septum, and right ventricular hypertrophy. Right ventricular outflow tract obstruction increases the pressure in the right ventricle. This causes right to left shunt through the ventricular septal defect. The deoxygenated blood from the right ventricle mixes with the oxygenated blood in the left ventricle. Therefore, in this condition, the left atrium has a higher oxygen saturation compared to the left ventricle due to mixing. The right atria and ventricles have the least oxygen saturation.



Option C - Tricuspid atresia is the congenital absence of the tricuspid valve. Therefore, the blood in the right atrium is shunted to the left atrium through the patent foramen ovale or the atrial septal defect. In the left atrium, the blood from the right atrium mixes with the oxygenated blood and the oxygen saturation decreases. However, the oxygen saturation in the left atrium and left ventricle is still greater compared to that of the hypoplastic right ventricle.



Option D - Transposition of great arteries (TGA) is characterized by the aorta arising from the right ventricle and the pulmonary artery from the left ventricle. Because of this, the pulmonary and systemic circulations remain separate. The right side of the heart has deoxygenated blood and the left side of the heart has oxygenated blood. The pulmonary artery saturation is always higher than the aortic saturation.



Solution to Question 8:

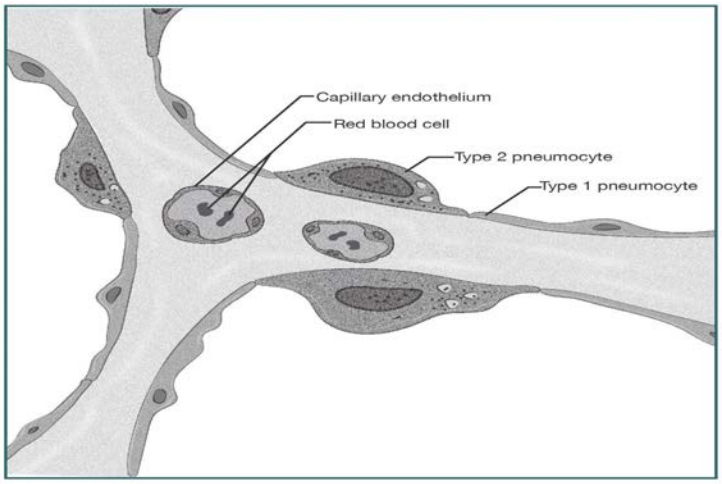
Type II pneumocytes, which are marked as 2 in the above image, are damaged in respiratory distress syndrome.

The cells forming the lining epithelium of alveoli (also called pneumocytes) are mainly of the following types:

• Type I pneumocytes - large flattened cells which present a very thin diffusion barrier for gases. These cells have large cytoplasmic extensions and form the primary lining cells of the alveoli. They are connected to each other by tight junctions.

• Type II pneumocytes (granular pneumocytes) - These are rounded secretory cells bearing microvilli on their free surfaces. These cells secrete the surfactant which decreases the surface tension between the thin alveolar walls and thus prevents alveoli from collapsing during respiration. Type II cells are important in alveolar repair as well.

The below image shows the pneumocytes lining the alveoli:



The alveoli are surrounded by pulmonary capillaries. The alveoli also contain other specialized cells, including pulmonary alveolar macrophages, lymphocytes, plasma cells, neuroendocrine cells, and mast cells.

Respiratory distress syndrome (RDS) is seen in premature infants, due to the decreased production and secretion of the lung surfactant. The surfactant reduces the alveolar surface tension. This phospholipid is synthesized in type II pneumocytes (also called granular pneumocytes) in the endoplasmic reticulum.

The surfactant appears in the amniotic fluid between 28 and 32 weeks of gestation. Mature levels of the pulmonary surfactant are present usually after 35 weeks of gestation. In a premature infant, there is a decrease in both the quantity and quality of surfactant resulting in respiratory distress syndrome.

Administration of antenatal glucocorticoids to mothers who deliver at 24 to 34 weeks can increase surfactant production and reduce the risk of RDS. It includes either giving 2 doses of betamethasone 12mg/dose with 24 hours between each dose, or 4 doses of dexamethasone 6mg/dose, with 12 hours between each dose.

Mild to moderate RDS can be managed with continuous positive airway pressure (CPAP). CPAP is a noninvasive modality of support where a continuous distending pressure (5-7 cm H20) is applied at the nostril level to keep the alveoli open in a spontaneously breathing baby. Severe cases will require mechanical ventilation.

Surfactant administration is reserved for severe cases and moderate cases not improving even with fiO2> 40% in CPAP.

With treatment, it rarely leads to respiratory failure and has a good prognosis.

Solution to Question 9:

All of the above-given options, hyponatremia, hypomagnesemia, hypocalcemia, and hypernatremia are causes of neonatal seizures.

The causes of neonatal seizure according to the age of presentation are summarised in the table below:

Hypoxic-ischemic encephalopathy is the most common cause. Metabolic disorders cause neonatal seizures commonly. Hypoglycemia is the most common metabolic cause of neonatal seizures.

Neonatal seizures may be of different types - subtle, tonic, clonic, spasms, and myoclonic.

Subtle seizures are the most common type of neonatal seizures. They involve movements like transient eye deviations, nystagmus, blinking, mouthing, cycling, and apnea, and are more common in premature infants.

Tonic seizures are characterized by increased muscle tension, either focal (sustained posturing of a single limb) or generalized (bilateral tonic limb extensions or flexions of the upper limbs with tonic extension of the lower limbs and trunk). They can not be provoked by stimulation or suppressed by restraint.

Clonic seizures involve repetitive contractions of muscle groups. Spasms are sudden generalized jerks associated with brief discharges.

Myoclonic seizures are the least common type of neonatal seizures and are rapid jerks (<50 msec) without rhythmicity.

Electroencephalogram (EEG) is the preferred diagnostic investigation for seizures, while MRI is the recommended neuroimaging modality for etiological evaluation. Phenobarbital is the drug of choice for treatment. In term neonates, hypoxic-ischemic encephalopathy (HIE) is the most common cause of seizures, while intraventricular hemorrhage (IVH) is more prevalent in preterm neonates.

1 to 4 days	4 to 14 days	2 to 8 weeks
Hypoxic ischemic encephalopathy	Bacterial meningitisEncephalitis (enteroviral, herpes simplex)	Encephalitis (enteroviral, herpes simplex)Bacterial meningitis
Drug withdrawal, maternal drug use	HypocalcemiaHypoglycemia	Subdural hematomaChild abuse
Drug toxicity: lidocaine, penicillin	GalactosemiaFructosemia	AminoaciduriaUrea cycle defects
Intraventricular hemorrhage		Malformations of cortical development
HypocalcemiaSepsisHypoglycemiaHypomagnesemiaHypo or hypernatremia		Tuberous sclerosis
GalactosemiaUrea cycle disorders		Sturge-Weber syndrome

Solution to Question 10:

The given clinical scenario describes neonatal meningitis. E. coli > Streptococcus agalactiae, part of group B streptococci, are the most likely causative organism of neonatal meningitis.

Early features of meningitis include temperature instability (hypo or hyperthermia), lethargy, bradycardia, and feed intolerance. Late features include seizures, bulging fontanelle, and nuchal rigidity.

The most common cause of bacterial meningitis in a 0 to 2-month baby is group B streptococci (Option B) worldwide and gram-ve bacteria like E.coli and Klebsiella (option A) in India. In babies aged 2 to 24 months, pneumococci are the primary causative pathogens, followed by H. influenzae. For children older than 2 years, pneumococci remain a prominent cause of meningitis, with Neisseria meningitidis being the second most common pathogen.

The most common etiological agents of neonatal meningitis and sepsis are listed in the table below:

Neonates have immature cellular and humoral immunity. This makes them prone to meningitis and sepsis. Thus, an infant with a fever who is 28 days or younger, should be evaluated for sepsis.

Option C: Meningococcal infections are usually asymptomatic. If invasive disease occurs, then the clinical spectrum varies widely from meningococcal meningitis (highest proportion) to bacteremia without sepsis, meningococcal septicemia with or without meningitis, pneumonia, chronic meningococcemia, and occult bacteremia.

Option D: Meningitis caused by Listeria monocytogenes is rare but clinically significant. Infants can acquire the infection in utero or through an infected birth canal. Neonatal listeriosis can manifest as early-onset disease, typically within 1-2 days of birth, primarily as a septicemic form. Alternatively, it can present as a late-onset disease in infants older than 5 days, predominantly as a meningitic form.

	India	Worldwide
Neonatal meningitis	E. coli, Klebsiella	Group BStreptococci
Neonatal sepsis	Klebsiella,S.aures,E. coli	Group BStreptococci

Solution to Question 11:

A child attains half of the adult height by 20 -24 months.

The normal length at birth is 50 cm, increasing to around 75 cm at 1 year of age. By the end of the second year, the length ranges from 85 to 90 cm.

From age 2 to around 12 years, there is a steady growth of about 6 cm per year. During pubertal growth spurts after age 12, females grow an average of 8 cm per year, while males grow around 10 cm per year.

By around 2 years of age, a normal child reaches half of their adult height (around 80-85 cm). Height doubles by age 4 and triples by age 12 compared to the birth length.

Between 18-24 months, a child experiences significant growth and development.

Physically, they improve their balance, learn to run and climb stairs, and reach about half of their adult height. Head growth slows slightly, with 85% of adult head circumference achieved by the age of 2 years.

Cognitive development includes the establishment of object permanence and problem-solving skills (e.g., using a stick to obtain a toy that is out of reach).

Emotionally, they display self-recognition and recognize when toys are broken. When toddlers see anything abnormal in their noses while gazing into a mirror for the first time, they tend to touch their own faces rather than reaching out toward the reflected image.

Linguistically, their vocabulary expands from 10-15 words to 50-100 words, and they start combining words to form simple sentences while understanding two-step commands (eg., give me the ball and then get your shoes).

Paediatrics NEET 2021

Question 1:

A child with infantile hypertrophic pyloric stenosis has multiple episodes of non-bilious vomiting. Which of the following metabolic abnormality is likely to be seen in this condition?

a) Hypochloremic, hypokalemic acidosis

b) Hypochloremic, hypokalemic alkalosis

c) Hypokalemic hyperchloremic alkalosis

d) Hyperchloremic hypokalemic acidosis

Question 2:

A 6-year-old girl born out of nonconsanguineous marriage presented with rachitic changes in limbs since 1½ years of age for which she has received multiple courses of vitamin D. Investigations showed the following. What is the most likely diagnosis?

a) Vitamin D dependent rickets

b) Hypoparathyroidism

c) Chronic renal failure

d) Hypophosphatemic rickets

Question 3:

During the anthropometric evaluation, the height for age of a child is < - 2SD. This is most likely due to?

a) Chronic malnutrition

b) Acute malnutrition

c) Recent infection

d) No malnutrition

Question 4:

A child presents with generalised edema and was found to have hypercholesterolemia. Urine analysis shows protein 3+ and fat bodies. What is the likely diagnosis?

a) Nephrotic syndrome

b) Nephritic syndrome

c) Goodpasture's disease

d) IgA nephropathy

Question 5:

A young child with congenital hydrocephalus was treated with a successful ventriculoperitoneal (VP) shunt. Four months post surgical repair, the child presented with fever, nuchal rigidity, and irritability. You suspect meningitis. What is the next best step?

a) Wait and watch

b) Check shunt patency by nuclear study

c) Blood culture and take CSF sample from shunt tap

d) Blood culture and take CSF sample by lumbar puncture

Question 6:

In fetal circulation, which vessel carries deoxygenated blood back to the placenta?

a) Umbilical Vein

b) Umbilical artery

c) Descending aorta

d) Pulmonary Artery

Question 7:

A 10-year-old boy presented with seizures. His past history is significant for an episode of fever with rash at 1 year of age which resolved spontaneously. What is the most helpful investigation to diagnose his condition?

a) IgG measles in CSF

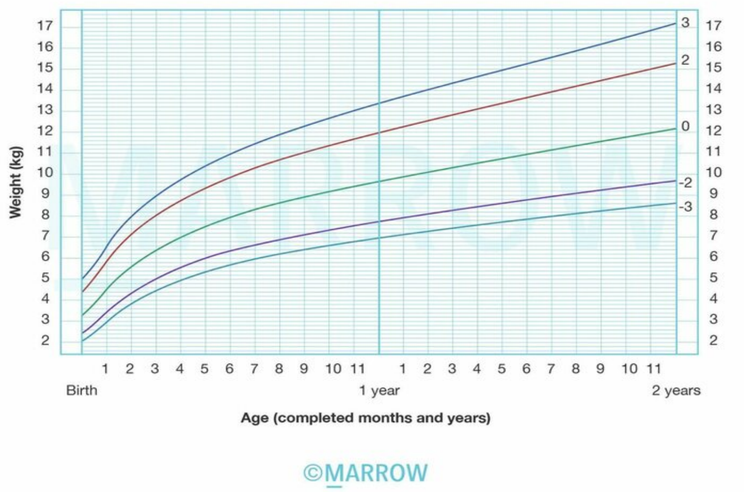
b) MRI mesial temporal lobe sclerosis

c) IgM measles in CSF

d) C1Q4 antibodies in the CSF

Question 8:

A 16-month-old boy is found to weigh 8kg. A WHO growth chart(weight-for-age) is provided below. Plot the growth chart for this child. What is the most appropriate advice based on the plotted graph?



a) No malnutrition - Assure the mother

b) Mild malnutrition - home treatment

c) Severe malnutrition - refer to nutrition rehabilitation center

d) Moderate malnutrition - teach the mother on how to feed

Answer Key

Question No.	Correct Option
1	b
2	d
3	a
4	a
5	c
6	b
7	a

8	d

Detailed Explanations

Solution to Question 1:

Multiple episodes of non-bilious vomiting in infantile hypertrophic pyloric stenosis (IHPS) lead to hypochloremic hypokalemic alkalosis.

Emesis causes loss of gastric fluid containing H+ ions, leading to the release of bicarbonate into the bloodstream. Without normal neutralization in the small intestine, metabolic alkalosis develops. This is the generation phase of alkalosis.

Volume depletion during vomiting hinders the urinary excretion of bicarbonate. This is due to reduced glomerular filtration, increased proximal tubule resorption of sodium and bicarbonate, and elevated aldosterone levels promoting bicarbonate resorption and hydrogen ion secretion in the collecting duct. This is the maintenance phase.

Loss of chloride ions in the gastric acid leads to hypochloremia.

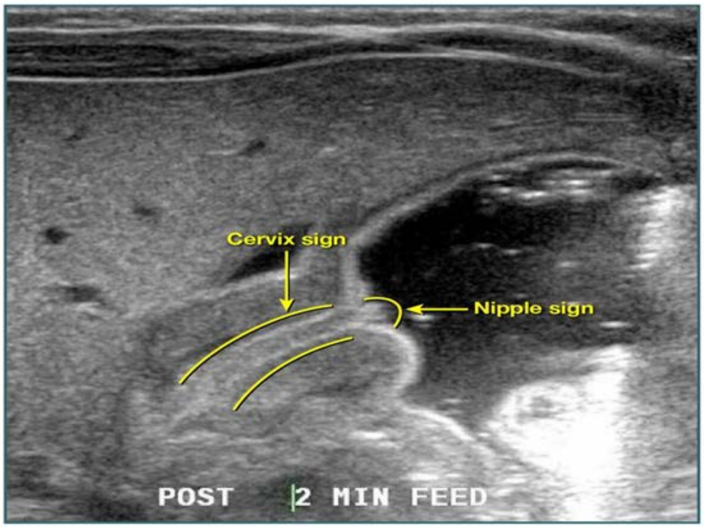
Hydrogen ions exit from the cells in exchange for potassium leading to hypokalemia. Loss of fluid due to excessive vomiting activates the RAAS pathway. This worsens hypokalemia as the aldosterone causes active reabsorption of sodium in exchange for the secretion of K+ and H+ ions into the urine. The loss of H+ ions in urine lowers the urinary pH, resulting in paradoxical aciduria.

Infantile hypertrophic pyloric stenosis (IHPS) is associated with hypertrophy of the circular muscle of the pylorus resulting in constriction and obstruction of the gastric outlet. It is common between 3 and 6 weeks of age. First-born male infants are at the highest risk.

Infants usually present with projectile, non-bilious vomiting. Clinically, an olive-shaped mass is palpable in the right upper quadrant or epigastrium. When the infant is given a test feed, visible peristalsis is usually observed moving from left to right (not from right to left), followed by projectile vomiting. It is associated with hypokalemic hypochloremic metabolic alkalosis with paradoxical aciduria.

Ultrasound examination is the investigation of choice for this condition. The diagnostic criteria for pyloric stenosis are a muscle thickness of ≥4 mm and a pyloric length of ≥16 mm. The nipple sign is produced by the double-layered hypertrophic mucosa that protrudes into the stomach. The sonographic appearance of the hypertrophied pylorus resembles that of the uterine cervix and is therefore called the cervix sign.

The image below shows the cervix and nipple sign seen in IHPS:



Management includes the correction of metabolic abnormality prior to surgery. Administering 5% dextrose and 0.45% saline with added potassium at a rate of 150-175 mL/kg per 24 hours can correct the fluid deficit in infants. Monitoring urine output (>2 cc/kg/h) is important to ensure successful rehydration.

The hypertrophic pylorus is surgically corrected by Ramstedt's pyloromyotomy (open or laparoscopic approach). It involves splitting the pyloric muscle while preserving the underlying submucosa.

Solution to Question 2:

A history of resistant rickets with normal levels of calcium (9 to 11 mg/dL), parathyroid hormone (PTH), arterial blood gas (ABG), with decreased phosphate (normal - 2.5 to 4.3 mg/dL), and elevated levels of alkaline phosphatase (ALP) (normal, 33 to 96 U/L) in a child is suggestive of hypophosphatemic rickets.

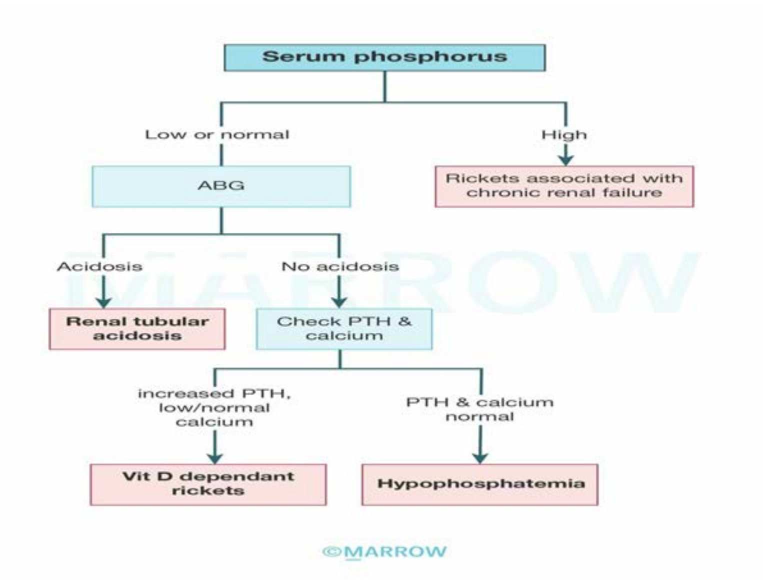
Hypophosphatemic rickets:

X-linked dominant hypophosphatemic rickets (XLH) is the most common genetic cause of rickets while the most common, overall is nutritional vitamin D deficiency. Mutations in the PHEX gene (phosphate-regulating endopeptidase on x chromosome) leads to increased FGF-23 (phosphatonin), which inhibits phosphate reabsorption in the proximal tubule levels and hence, causes increased excretion of phosphate. It also inhibits renal 1-alpha hydroxylase leading to reduced 1,25 dihydroxy vitamin D.

Causes of hypophosphatemic rickets include familial vitamin D resistance, autosomal dominant hypophosphatemic rickets, autosomal recessive hypophosphatemic rickets, tumor-induced rickets, McCune Albright syndrome, epidermal nevus syndrome, and neurofibromatosis.

In X-linked dominant hypophosphatemic rickets lower extremity abnormalities, short stature, poor growth, delayed dentition, and tooth abscesses are common.

The chart attached below summarizes the evaluation of hypophosphatemic rickets.



Patients with hypophosphatemic rickets respond well to oral phosphorus and 1,25-D (calcitriol). Phosphorus supplementation is usually given in 1-3 g elemental phosphorus divided into 4–5 doses. Frequent dosing reduces diarrhea, which is a complication of high-dose oral phosphorus.

Calcitriol is administered 30–70 ng/kg/day divided into 2 doses. Burosumab, a monoclonal antibody to FGF-23 can be used in children > 1 year of age with XLH rickets.

Further, Joulie's solution is also useful in the treatment of familial hypophosphatemic rickets. Joulie's solution includes contains 30.4 mg of phosphate/mL. The most common side effect is diarrhea.

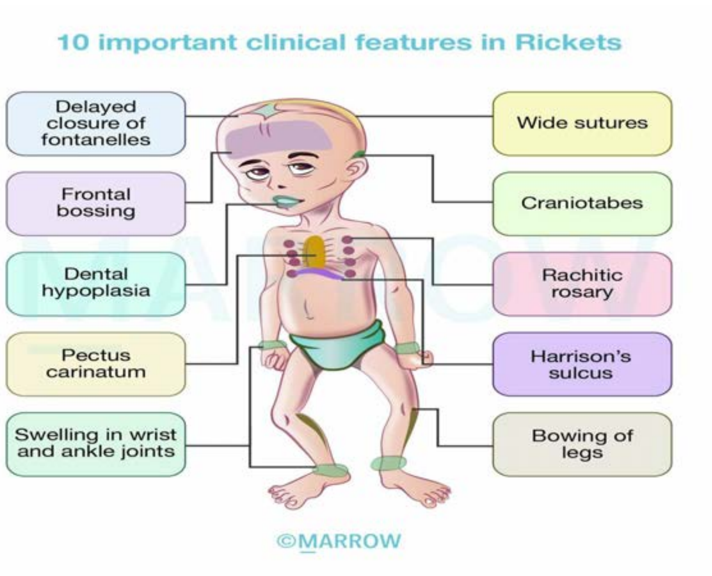
Other options:

Option A: Vitamin D-dependent rickets type 1A is due to mutation in the gene coding for 1α-hydroxylase leading to reduced calcitriol levels (1,25-D). Type 1B is due to a mutation in the gene for 25-hydroxylase leading to reduced calcidiol (25-D). Type 2A have mutations in the gene coding for vitamin D receptor. Type 2B is due to the over-expression of a hormone response element-binding protein that interferes with the actions of 1,25-D. All of them have normal to low calcium levels, low phosphorus, elevated PTH and ALP levels.

Option B: Hypoparathyroidism will present with features of hypocalcemia. There will be a decrease in serum PTH levels and calcium levels.

Option C: In chronic renal failure, 1α-hydroxylase activity is reduced leading to reduced 1,25-D. They have normal to low calcium levels and elevated PTH and ALP levels. They have hyperphosphatemia due to decreased renal excretion of phosphate.

The image below summarizes the common features of rickets in children.



Solution to Question 3:

Low height for the age (height for age < -2SD) in children indicates chronic malnutrition and stunting.

Classification of nutritional status depending on the anthropometry (height/length for age):

Nutritional status is often assessed in terms of anthropometry:

• Low weight for height, or wasting, indicates acute malnutrition.

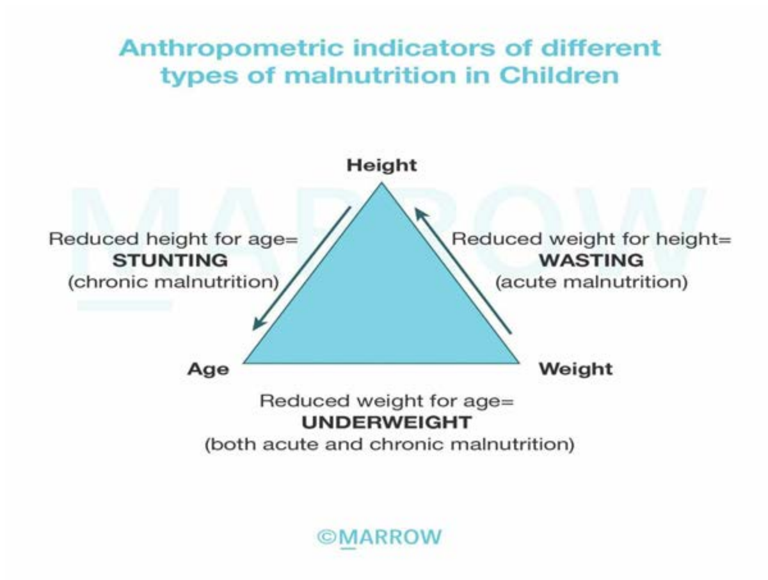
• High weight-for-height indicates overweight.

• Low height-for-age, or stunting, indicates chronic malnutrition. This is the best indicator of long-term nutritional status in the pediatric population.

• Low weight-for-age indicates acute or chronic malnutrition. It does not differentiate between wasting and stunting. Hence, it has limited clinical significance.

Mid-upper arm circumference (MUAC) is an age-independent parameter used for the screening of wasting in children. In children aged 6 months to 5 years, MUAC <11.5cm is classified as severe malnutrition.

Classification	Criteria
Moderatelystunted (or) mode rate chronic malnutrition	Height/length-for-age< (–2 S D)and> (–3 SD)
Severelystunted (or) severe c hronic malnutrition	Height/length-for-age< (–3 S D)



Solution to Question 4:

The clinical scenario showing a child with generalized edema, hyperlipidemia, massive proteinuria, and fat bodies in urine is highly suggestive of nephrotic syndrome.

Nephrotic syndrome is characterized by nephrotic range proteinuria/heavy proteinuria >3.5 g/24 hr or urine protein: creatinine ratio > 2 and is associated with a triad of hypoalbuminemia (≤2.5 g/dL), edema, and hyperlipidemia (cholesterol > 200 mg/dL).

Nephrotic syndrome in children is primarily idiopathic, with minimal change disease being the most common glomerular lesion. Podocyte injury or genetic mutations in podocyte proteins can lead to nephrotic-range proteinuria.

Edema is the most common presenting symptom. They have elevated levels of cholesterol, triglycerides, low-density lipoproteins (LDL), and very low-density lipoproteins (VLDL). It is caused by both increased lipid synthesis and decreased lipid catabolism.

They are prone to infections especially by encapsulated bacteria due to factors such as urinary losses of immunoglobulin G (IgG) and complement factors (C3, C5, factors B and D). Nephrotic syndrome is a hypercoagulable state due to vascular stasis, increased platelet, increased hepatic production of fibrinogen, and urinary losses of antithrombotic factors such as antithrombin III and protein S.

The diagnosis is confirmed by performing a urinalysis and urine protein: creatinine ratio. The recommended initial treatment for nephrotic syndrome is a daily dose of 2mg/kg (max 60mg) of prednisolone for 6 weeks followed by an alternate dose of 1.5mg/kg (max 40mg) for the next 6-8 weeks.

Definitions regarding the course of nephrotic syndrome:

• Remission is characterized by a urine protein: creatinine ratio of <0.2 or <1+ protein on urine dipstick testing for 3 consecutive days

• Response in nephrotic syndrome is defined as achieving remission within the initial 4 weeks of corticosteroid therapy

• Relapse is indicated by an increase in urine protein: creatinine ratio >0.2 or a reading of 2+ and higher for 3 consecutive days.

• Frequently relapsing: ≥2 relapses in 6 months or 4 relapses in 12 months.

• Steroid-dependent nephrotic syndrome: Relapse during steroid tapering or within 2 weeks of discontinuation.

• Steroid resistance: Failure to achieve remission despite therapy with daily prednisolone for 4 weeks and alternate days for the next 4 weeks.

Other options :

Option B: Nephritic syndrome classically presents with tea- or cola-colored urine (hematuria), edema, hypertension, and oliguria.

Option C: Goodpasture disease is an autoimmune condition characterized by pulmonary hemorrhage, rapidly progressive glomerulonephritis, and elevated anti-glomerular basement membrane antibody levels.

Option D: IgA nephropathy is an immune complex-mediated disease, with a predominance of IgA within glomerular mesangial deposits. They present with gross hematuria and proteinuria.

Solution to Question 5:

In a child with suspected shunt-associated meningitis, the next best step useful in confirming the diagnosis is by taking a blood culture and a CSF sample from a shunt tap is preferable to a lumbar puncture (LP) (option D).

Shunt aspiration allows for better detection of microbes and avoids the need for a separate procedure, reducing the risk of contamination. VP shunt fluid is obtained from a subcutaneous reservoir using a 25-gauge needle.

Bacterial meningitis may be caused due to meningococcus, streptococcus pneumoniae, or Haemophilus influenzae type b. The clinical presentation of meningitis in children with VP shunts is similar to acute bacterial meningitis in other children. These include fever, rash, vomiting, and refusal to feed. Clinical signs of meningeal irritation (Kernig and Brudzinski signs) are not usually seen in infants.

Blood cultures should be done in all cases of suspected meningitis. Treatment is aggressive with systemic antibiotics. Third-generation cephalosporins along with vancomycin can be used. Wait and watch policy is not followed (option A).

A ventriculoperitoneal (VP) shunt is used to drain excess cerebrospinal fluid (CSF) due to an obstruction in the normal outflow or there is a decreased absorption of the fluid. A silicone rubber device with a proximal portion inserted into the ventricle, a unidirectional valve, and a distal segment is commonly used for shunt placement.

Complications of the shunting procedure include infection due to skin flora entering the shunt (Staphylococcus epidermidis), intracerebral or intraventricular hemorrhage, shunt over drainage, shunt nephritis, shunt disconnection, obstruction, inguinal hernia, and seizures.

Shunt patency studies are performed to determine whether shunt revision surgery is needed in the malfunctioned ventriculoperitoneal shunt. This is done using radionucleotides (option B)

Hydrocephalus represents a group of conditions that result from impaired circulation or absorption of CSF.

Causes of communicating hydrocephalus (increased production or reduced absorption) includes choroid plexus papilloma, meningeal malignancy, and meningitis. Causes of noncommunicating (obstructive) hydrocephalus include aqueductal stenosis, infectious (toxoplasmosis, mumps), Chiari malformation, Dandy-Walker malformation, and mass lesions.

In infants, signs include rapid head enlargement, bulging fontanel, and downward eye deviation. Long-tract signs and dilated scalp veins may also be observed. Older children may experience irritability, lethargy, poor appetite, vomiting, and headaches. Diagnosis can be done using MRI, ultrasound, and measurement of ICP.

Medical management with medications like acetazolamide and furosemide can provide temporary relief, but long-term outcomes are often unsatisfactory. Most cases require surgical intervention, such as ventriculoperitoneal shunting or endoscopic third ventriculostomy.

Solution to Question 6:

Umbilical arteries carry deoxygenated blood back to the placenta during the fetal period. The umbilical vein carries oxygenated blood away from the placenta.

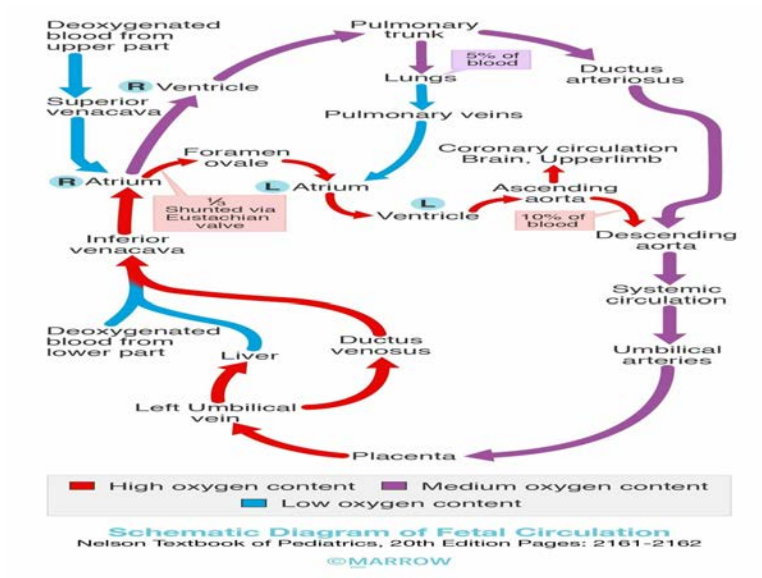
Arteries carry oxygenated blood away from the heart, except for the pulmonary and umbilical arteries which carry deoxygenated blood.

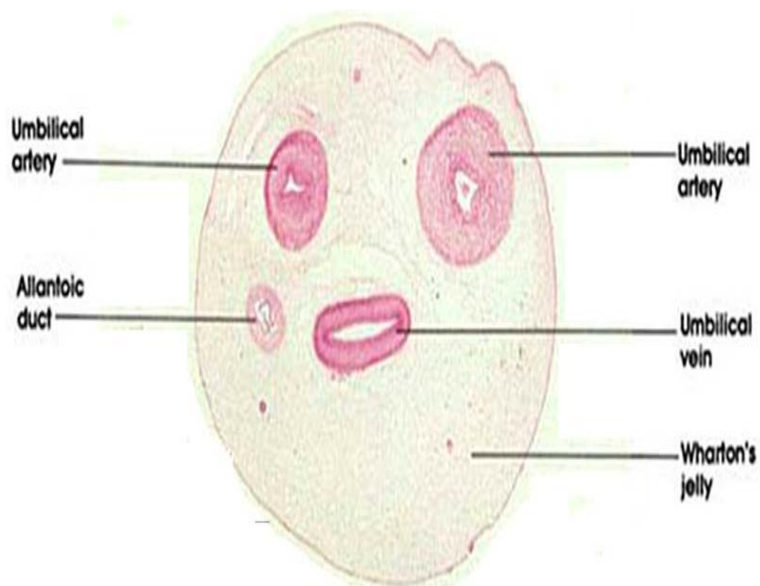
The blood gets oxygenated in the placenta and this passes to the umbilical vein (pO_2 30 to 35mmHg). From the umbilical vein, 50% of the blood enters the hepatic circulation and 50% bypasses the liver to enter the inferior vena cava (IVC) (pO_2: 26 - 28mmHg) through ductus venosus.

The right atrium receives the oxygenated blood from the IVC and superior vena cava (less oxygenated). The right atrial pressure is more than the left atrial pressure resulting in the major flow of blood from the right atrium to the left atrium via the foramen ovale. The blood from the left atrium flows to the left ventricle. The upper part of the fetal body (coronaries, cerebral arteries, and upper extremities) is exclusively perfused by the left ventricle via the ascending aorta.

The major portion of this blood bypasses the lung and flows from the pulmonary artery into the descending aorta via the ductus arteriosus. Finally, some of this blood from the aorta and the lower parts of the body is pumped back to the placenta via umbilical arteries.

A schematic diagram of fetal circulation is given below:





The concept of fetal circulation and the various embryological remnants of the vascular entities of fetal circulation is a high-yield frequently tested topic in NEET-PG examination. The concepts are summarized in the pearl attached below.

Solution to Question 7:

The clinical scenario of a child with a previous history of fever with rash at 1 year, now presenting with seizures, is highly indicative of subacute sclerosing panencephalitis (SSPE), a rare chronic complication of measles. Anti-measles antibodies are elevated in cerebrospinal fluid (CSF) (IgG >IgM); hence, it is the most helpful investigation to diagnose this condition.

SSPE is a fatal progressive demyelinating disease caused by chronic CNS infection with the genetically altered variant of the measles virus. It occurs several years after primary measles infection and is more common in males.

Clinical manifestations progress through four stages:

• Stage I: involves mild symptoms like irritability and poor school performance.

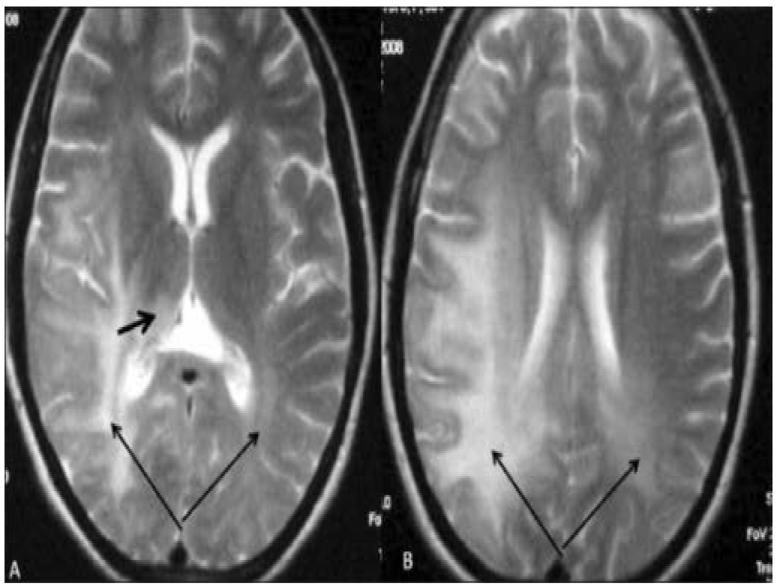
• Stage II: is characterized by myoclonic seizures.

• Stage III:choreoathetosis, immobility, dystonia, lead-pipe rigidity, and cognitive decline are seen.

• Stage IV: leads to coma and death.

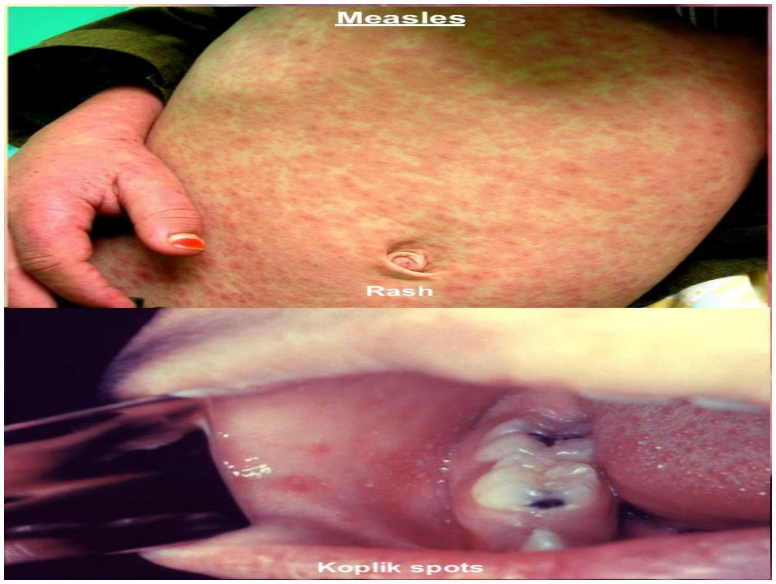
Diagnosis of SSPE is established by documentation of a compatible clinical course along with one of the following supporting findings: elevated measles antibody (IgG >IgM) in CSF analysis or characteristic electroencephalographic (EEG) patterns, suppression-burst episodes (high voltage sharp slow waves) or typical histological findings or isolation of virus from brain tissue.

CT and MRI show evidence of multifocal white matter lesions, cortical atrophy, and ventricular enlargement. Increased T2 signaling in the white matter of the brain is seen as the disease progresses (shown below):



Treatment is mainly supportive. Isoprinosine along with interferons is said to have additional effects. Carbamazepine is found to be useful in controlling myoclonic jerks in the early stages of the illness.

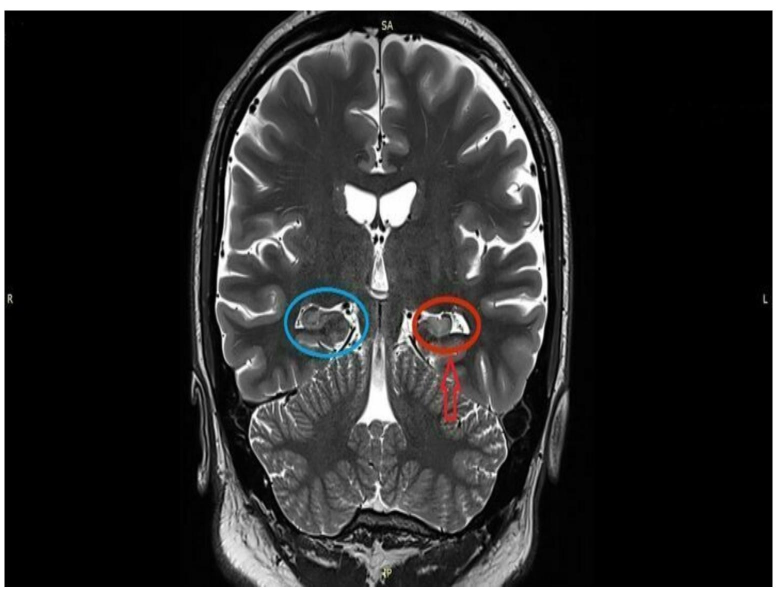
In measles, after an incubation period of 8-12 days, the prodromal phase starts with mild fever and is followed by conjunctivitis, photophobia, coryza, cough, and increasing fever. Koplik spots, appearing 1-4 days prior to the rash, are a characteristic sign of measles. The rash (shown below) appears on the 4th day of fever and typically begins on the forehead (around the hairline), and behind the ears, as a red maculopapular eruption.



The most common complication of measles is acute otitis media. The most common cause of death in measles is pneumonia. Low serum retinol levels in children with measles are associated with higher morbidity and mortality and hence Vitamin A therapy is indicated for all patients with measles.

Other options:

Option B: A common cause of temporal lobe epilepsy is mesial (medial) temporal sclerosis. It is the most common cause of surgically remediable partial epilepsy in adolescents and adults. It is often preceded by febrile seizures. Pathologically, these patients have atrophy and gliosis of the hippocampus. MRI shows hippocampal sclerosis (shown below).



Option D: C1Q4 antibodies in the CSF are seen in neuromyelitis optica. Neuromyelitis optica, also known as Devic's disease, is an aggressive inflammatory demyelinating disorder characterized by recurrent attacks of optic neuritis and myelitis.

Solution to Question 8:

The weight for age of the child falls in between -2SD and -3SD as shown in the growth chart below, which is suggestive of moderate malnutrition and the mother should be taught to feed the baby properly.

The primary treatment in mild and moderate malnutrition involves providing sufficient protein and energy intake, with a recommended minimum of 150 kcal/kg/day. Nutritious home food is advised, and frequent feeding (up to 7 times a day) may be required to achieve the necessary energy intake. High-fat foods are often used to increase energy in therapeutic diets.

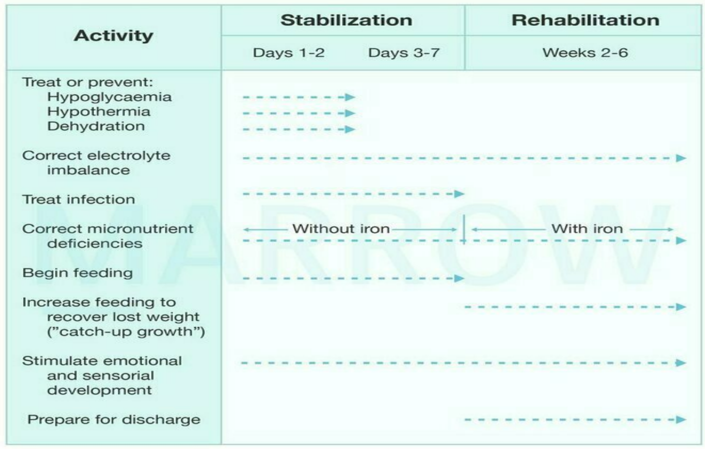
Animal-source foods, such as milk and eggs, are recommended by WHO as they provide necessary amino acids and nutrients. Plant-source foods like legumes and cereals can also be good protein sources. A slight increase in protein intake (3g/kg/day) is sufficient for catch-up growth with high energy intake. Milk and vegetable protein mixtures can be used as a protein source and adequate minerals and vitamins are also added.

Weight gain is the best measure of treatment efficacy for mild and moderate malnutrition.

Severe acute malnutrition (SAM) should be referred to a hospital for proper management. Uncomplicated SAM can be managed at home under supervision. Complicated SAM should be treated as an inpatient.

Management of SAM is summarized in the image below.

Management of PEM



The World Health Organization(WHO) has established international standards for normal child growth from birth to 5 years of age. The normalization of the anthropometric measures is done in terms of z-scores (standard deviation scores). The anthropometric indicators are marked in a growth chart which is compared with a reference curve.

The anthropometric measures used are weight for age, height for age, weight for height, BMI, and mid-upper arm circumference.

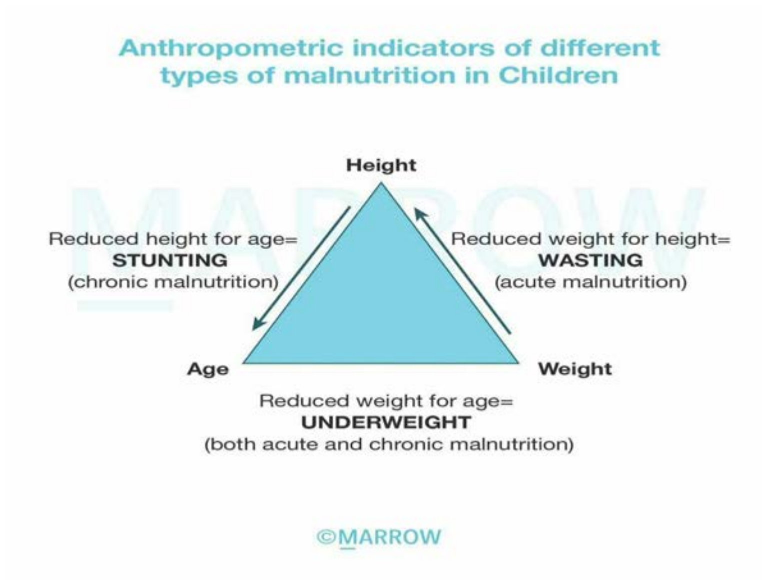
• Low weight for height, or wasting, indicates acute malnutrition.

• High weight-for-height indicates overweight.

• Low height-for-age, or stunting, indicates chronic malnutrition. This is the best indicator of long-term nutritional status in the pediatric population.

• Low weight for the age indicates acute or chronic malnutrition. It does not differentiate between wasting and stunting. Hence, it has limited clinical significance.

Mid-upper arm circumference (MUAC) is used for the screening of wasting in children between 6 months to 5 years. In children aged 6 months to 5 years, MUAC <11.5cm is considered as severe malnutrition.


Anthropometric indicators of different types of malnutrition in Children
Height
Reduced height for age=
STUNTING
(chronic malnutrition)
Reduced weight for height=
WASTING
(acute malnutrition)
Age
Weight
Reduced weight for age=
UNDERWEIGHT
(both acute and chronic malnutrition)
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Paediatrics INI-CET 2022

Question 1:

In neonatal resuscitation, which of the following is the most effective indicator of successful ventilatory effort?

- **a)** Colour change
- **b)** Rise in heart rate
- **c)** Air entry
- **d)** Chest rise

Question 2:

Which can be used as an alternative to mother's milk in a low birth weight (LBW) baby?

- **a)** Cow's milk
- **b)** Preterm formula
- **c)** Term formula
- **d)** Human donor milk

Question 3:

A 4-year-old male child with a body weight of 15kg and height of 100cm is admitted with renal failure. His blood urea was 100 mg/dL and serum creatinine was 1 mg/dL. What is the closest calculated eGFR in the patient?

- **a)** 33 ml/min/1.73 m2 BSA
- **b)** 40 ml/min/1.73 m2 BSA
- **c)** 55 ml/min/1.73 m2 BSA
- **d)** 80 ml/min/1.73 m2 BSA

Question 4:

Adrenocorticotropic hormone is the drug of choice in which of the following conditions?

- **a)** Juvenile myoclonic epilepsy

b) Rolandic epilepsy

c) West syndrome

d) Lennox Gastaut syndrome

Question 5:

Regarding urinary tract infection (UTI) in children, which of the following is true?

a) 1,3,4

b) 2,4

c) 2,3,4

d) 1,4

Question 6:

Handedness is seen at the age of ______.

a) 1 year

b) 2 years

c) 3 years

d) 4 years

Question 7:

Which of the following cyanotic heart diseases causes increased pulmonary blood flow?

a) 1,4

b) 1,2

c) 3,4

d) 2,4

Question 8:

An 11-year-old boy was brought to the outpatient clinic with intentional tremors, and poor scholastic performance. His sister has similar complaints. On examination, hepatomegaly is seen. The eye finding is shown in the image. What is the probable diagnosis?

a) Wilson's disease

b) Huntington's chorea

c) Glutaric aciduria

d) Hepatitis A

Question 9:

Which alternative method can be used for the procedure shown in the image below?

a) Head circumference

b) Arm span

c) Crown-rump length

d) Knee height

Question 10:

A child presents with acute exacerbation of asthma. Which of the following would you do?

a) 3,4

b) 1,4

c) 1,3,4

d) 2,3,4

Question 11:

A young man is brought to the hospital with high-grade fever and altered consciousness. On examination, the patient had neck rigidity and pain when bending the neck. A lumbar puncture was performed and it showed a WBC count of 1500 cells/µL, majority being neutrophils, elevated pressure with protein of 120 mg/dL, and a glucose level of 70 mg/dL (blood glucose level - 300 mg/dL). How will you manage the index case?

a) Piperacillin+ tazobactum

b) Amphotericin + flucytosine

c) Vancomycin+ ceftriaxone

d) Antitubercular therapy

Question 12:

Which of the following increase the risk of recurrence of febrile seizures?

a) 1,2,4

b) 1,4

c) 1,2,3

d) 1,3,4

Question 13:

Choose the most appropriate answer.

a) Both assertion and reason are independently false statements.

b) The assertion is independently a true statement, but the reason is independently a false statement.

c) Both assertion and reason are independently true statements, but the reason is not the correct explanation for the assertion.

d) Both assertion and reason are independently true statements, and the reason is the correct explanation for the assertion.

Question 14:

A child is brought to the casualty with a history of choking. On examination, there is decreased air entry in the left lung. What would be the correct management?

a) 1, 2, and 4

b) 1 and 2

c) 3 and 4

d) 1 and 4

Question 15:

A 9-month-old boy presents with progressive pallor and abdominal distension. The Mentzer index was found to be less than 13. What are the investigations to be done in this patient?

a) 1, and 2

b) 1, 3, and 4

c) 1 and 4

d) 1, 2, 3, and 4

Question 16:

A child presents with complaints of fever, rash, body ache, and throat ache. He has a history of thorn prick injury a week back. What antibiotics would you give empirically to this child?

a) Ceftriaxone

b) Amoxicillin+ clavulanate

c) Vancomycin

d) Meropenem

Question 17:

A child with thalassemia is undergoing multiple blood transfusions. What is the best method to detect iron overload?

a) Liver iron concentration

b) MRI T2 myocardium

c) Serum ferritin

d) NTBI

Question 18:

How would you manage a child with status epilepticus?

a) 1, 2, 3, 4

b) 2, 1, 4, 3

c) 1, 3, 4, 2

d) 1, 2, 4, 3

Question 19:

A child presents with cold feet and a history of wearing socks even in the summers. On examination, lower limb pulses are diminished as compared to the radial pulse. Prominent radio femoral delay is present. What is the underlying condition?

a) There is hypertrophy of ductus arteriosus.

b) Coarctation distal to the origin of the left subclavian artery

c) Supply is from posterior to anterior thoracic artery.

d) Collaterals develop to supply upper limb.

Question 20:

An unconsious child is brought to the casualty. What is the correct sequence of the management?

a) 3-1-2-4-5

b) 1-2-3-4-5

c) 3-1-2-5-4

d) 1-2-4-3-5

Question 21:

Which of the following statements are true regarding cytomegalovirus (CMV) infection?

a) 3, 4
b) 1, 2
c) 2, 4
d) 1, 3

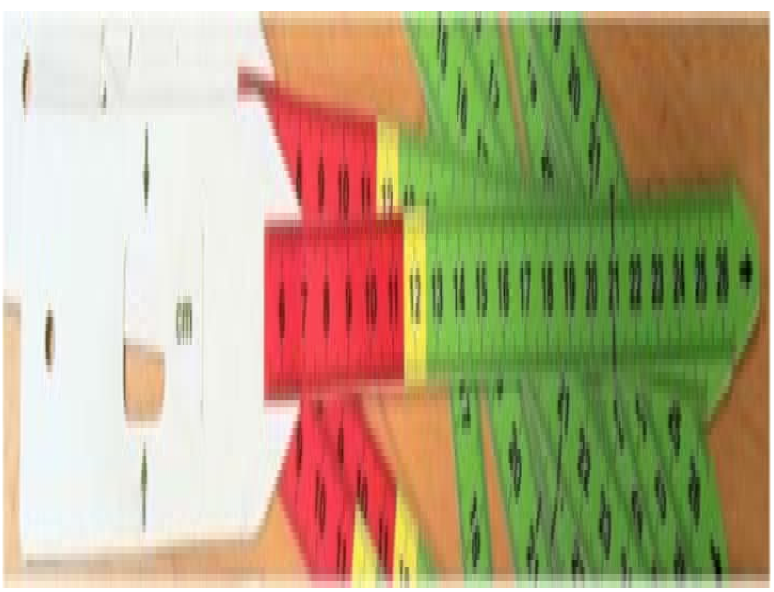
Question 22:

The hormone that does not play a role in the growth of the fetus is:

a) Insulin
b) Growth hormone
c) Thyroxine
d) Glucocorticoids

Question 23:

Which of the following statements is true regarding the tape shown in the image?



a) 1, 3

b) 2, 4

c) 2, 3

d) 1, 4

Answer Key

Question No.	Correct Option
1	b
2	d
3	b
4	c
5	b
6	c
7	c
8	a
9	b
10	d
11	c
12	c
13	c
14	d
15	c
16	b
17	a
18	d
19	b
20	c
21	d
22	b
23	b

Detailed Explanations

Solution to Question 1:

An increase in the heart rate is the most effective indicator of successful ventilatory effort. The response to neonatal resuscitation at various steps is assessed based on the heart rate.

As soon as a baby is born, the tone, cry, breathing efforts, and respiratory efforts are assessed. If the tone is poor, or if the baby is not breathing, the following measures are done immediately - warming to maintain the normal temperature, proper positioning of the baby's airway, i.e. sniffing position with the head slightly extended, clearing of airway secretions, and tactile stimulation.

After the initial steps of neonatal resuscitation, the heart rate of the baby must be assessed. If the heart rate is < 100 beats per minute (bpm), positive pressure ventilation is provided along with SPO2 monitoring to ensure adequate ventilation.

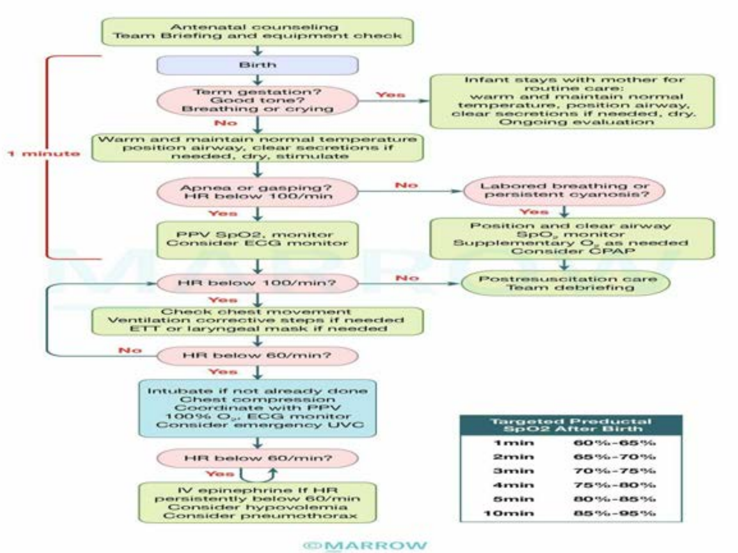
If the heart rate remains at < 100 bpm, ventilation corrective steps and/or endotracheal intubation have to be considered. The ventilation corrective steps are as follows and can be remembered with the mnemonic MRSOPA.

- Mask has to be reapplied
- Reposition head
- Suction the secretions
- Open mouth slightly and ventilate
- Pressure is increased slightly
- Airway may be changed

If the heart rate is < 60 bpm, chest compressions are initiated over the lower third of the sternum at a rate of 90 compressions/min. The ratio of compressions to ventilation is 3: 1 (90 compressions: 30 breaths).

If the heart rate still remains <60 bpm, despite effective compressions and ventilation, then epinephrine (1:10,000 solution at 0.1–0.3 mL/kg intravenously or 0.5–1 mL/kg intratracheally) should be administered. The dosage can be repeated every 3–5 minutes. Persistent bradycardia may signify inadequate ventilation or severe cardiac compromise.

The neonatal resuscitation algorithm is given below:



Solution to Question 2:

Human donor milk is the best alternative to mother's milk for low birth weight (LBW) babies.

It is recommended that all LBW babies are exclusively fed breast milk regardless of the feeding method. This can be achieved through the use of expressed breast milk (EBM) from the mother or human donor milk.

Donor milk offers advantages over preterm formula, such as reduced risk of necrotizing enterocolitis (NEC) and feeding intolerance. However, due to lower protein and micronutrient levels in donor milk, human milk fortifiers should be added to meet the nutritional needs of infants.

In cases where breastfeeding is not feasible or the mother is ill, formula feeds can be utilized. Term formula feed (option C) is suitable for infants weighing over 1500 grams at birth, while preterm formula feed (option B) is recommended for very low birth weight infants.

Cow milk (option A) is not preferred as it lacks essential nutrients such as vitamin C, iron, and essential fatty acids. Additionally, its high protein and lactose content can lead to digestive issues and allergic reactions in infants.

Once the mother's condition stabilizes or breastfeeding contraindications are resolved, exclusive breastfeeding should be resumed.

The mode of nutrition of LBW depends on their gestational age. Babies born before 28 weeks of gestation receive total parenteral nutrition (TPN). Enteral feeding is not initiated at this stage as the gastrointestinal tract is not fully developed.

Babies born between 28 and 32 weeks of gestation may exhibit suckling bursts but their coordination of suckling, swallowing, and breathing is not fully developed. Hence, EBM given orally may result in aspiration. EBM is administered through a nasogastric tube or orogastric tube.

As gestational age advances to 32-34 weeks, the coordination of feeding improves, and EBM can be given with a paladai or Katori spoon.

For babies born after 34 weeks of gestation, with fully developed coordination, direct breastfeeding is possible.

Solution to Question 3:

eGFR in the 4-year-old child calculated by the modified Schwartz formula is 40 ml/min/1.73 m2 body surface area (BSA).

The modified Schwartz formula is the most commonly used formula in the pediatric age group to estimate creatinine clearance in acute settings. It is based on serum creatinine, patient height, and an empirical constant k. The k value in the modified Schwartz formula is 0.413 for ages 1-17 years. This formula has higher accuracy and precision compared to the original Schwartz formula.

eGFR = 0.413 x height (cm) / serum creatinine (mg/dL)

In the given scenario, the height of the patient is 100 cm and serum creatinine is 1mg/dL. Therefore, the closest eGFR is 40 mL/min/1.73 m2 BSA.

eGFR in adults is calculated using the Modification of Diet in Renal Disease (MDRD) equation or the Chronic Kidney Disease Epidemiology Collaboration (CKD-EPI) equation. The calculation is based on age and serum creatinine with specific modifications in females and African Americans.

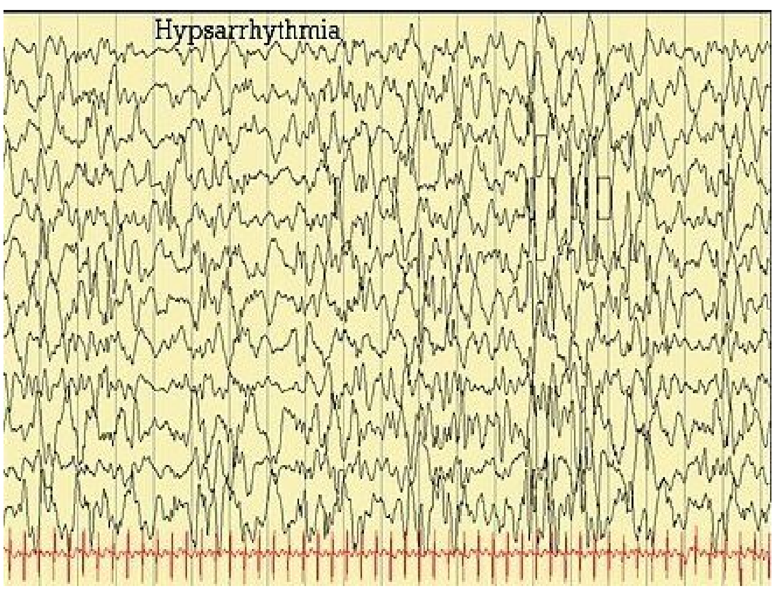
Glomerular filtration rate (GFR) is a measure of kidney function. It can be estimated accurately by quantitating the clearance of a substance filtered across the capillaries but the substance should not be absorbed nor secreted by tubules.

GFR can be estimated by intravenous injection of radioisotope technetium diethylene triamine pentaacetate (Tc-99m DTPA), but it is expensive and time-consuming. Therefore eGFR is estimated by the clearance of creatinine.

Solution to Question 4:

Adrenocorticotropic hormone (ACTH) is the drug of choice in West syndrome.

West syndrome is a severe generalized epilepsy that begins between the ages of 2 and 12 months. It is characterized by a triad of infantile spasms that occur in clusters, developmental regression, and hypsarrhythmia on electroencephalogram (EEG). Hypsarrhythmia refers to a high voltage, slow, chaotic background with multifocal spikes (shown below).



Cryptogenic cases (unknown etiology) have normal development prior to onset, while symptomatic West syndrome is associated with preceding developmental delay owing to perinatal encephalopathies, metabolic disorders, or malformations.

Hormonal therapy, such as ACTH injections or oral steroids, is the preferred treatment for West syndrome. The patient's response is monitored through awake and asleep EEGs conducted at 1, 2, and 4th weeks of starting hormonal therapy. Side effects may include hypertension, electrolyte imbalance, infections, hyperglycemia and/or glycosuria, and gastric ulcers.

Vigabatrin is a first-line treatment option for infantile spasms in patients with tuberous sclerosis and is also considered a second-line choice if hormonal therapy has been unsuccessful in other cases of infantile spasms.

The ketogenic diet is considered a third-line therapy for infantile spasms. Other alternative treatment options for spasms include valproate, benzodiazepines (such as nitrazepam and clonazepam), topiramate, lamotrigine, zonisamide, pyridoxine, and intravenous gamma globulin (IVIG).

Other options:

Option A: Juvenile myoclonic epilepsy (JME)/ Janz syndrome starts in early adolescence and is characterized by myoclonic jerks in the morning. EEG shows generalized irregular 4- to 6-Hz spike-and-wave activity, particularly after awakening. Valproate is considered the most effective medication. Lamotrigine, levetiracetam, and topiramate are also approved for JME.

Option B: Rolandic epilepsy is benign epilepsy seen in children. They present at night or while awakening. They are partial seizures with no loss of consciousness. They resolve in adulthood. Treatment with carbamazepine is useful.

Option D: Lennox-Gastaut syndrome is seen in 2-10 years and is characterized by a triad of developmental delay, multiple seizures (atypical absences, myoclonic, astatic, and tonic seizures), and specific EEG changes (1 to 2 Hz spike and slow waves, polyspike bursts). Treatment with topiramate, clobazam, felbamate, lamotrigine, and valproate is useful.

Solution to Question 5:

The true statements regarding UTI in children are bowel and bladder dysfunction increases the risk of recurrence, and micturating cystourethrogram (MCU) is done in children with recurrence of UTI.

The chief cause of UTI in children is Escherichia coli, others include Klebsiella, Enterobacter, and Staphylococcus saprophyticus. Risk factors for recurrent UTI are age <6 months, obstructive uropathy, female sex, vesicoureteric reflux (VUR), bladder dysfunction, constipation, and repeated catheterization.

Clinical features in neonates include fever, vomiting, diarrhea, poor weight gain, and lethargy. If the child cries during voiding or has a weak urine stream with a palpable bladder, it implies urinary obstruction. High fever (>39ºC), persistent vomiting, dehydration, renal angle tenderness, or raised creatinine is considered a complicated UTI.

A colony count of >105/mL of a clean catch sample is considered significant bacteriuria. Any colonies on suprapubic aspiration and >50,000/mL on urethral catheterization are considered significant. Patients with recurrent UTIs need to be evaluated with ultrasonography, DMSA scan (dimercaptosuccinic acid), and MCU.

Evaluation following the first episode of UTI is given below:

Third-generation cephalosporins or aminoglycosides are preferred empirical antibiotics for urinary tract infections (UTIs). Treatment duration is 7-10 days for simple UTIs and 10-14 days for complicated UTIs. Asymptomatic bacteriuria doesn't require treatment. Infants <3 months and those with complicated UTIs should receive parenteral antibiotics.

Cotrimoxazole, nitrofurantoin, cephalexin, and cefadroxil are given as prophylaxis to prevent recurrent UTIs. Cotrimoxazole and nitrofurantoin can be given in infants >3 months of age.

Age	Evaluation
<1 year	USG, MCU, DMSA renal scan
1 to 5 years	USG, DMSA scan. MCU is done if USG is abnormal
>5 years	USG. If USG is abnormal, MCU and DMSA are done.

Solution to Question 6:

Handedness is seen at the age of 3 years. The child may become frustrated if attempts are made in changing the hand preference.

Solution to Question 7:

The cyanotic congenital heart diseases with increased pulmonary blood flow are transposition of the great arteries (TGA) and total anomalous pulmonary venous communication (TAPVC).

Classification of cyanotic congenital heart diseases based on pulmonary blood flow is as follows:

Atrial and ventricular septal defects are also associated with increased pulmonary blood flow. However, they are acyanotic.

Decreased pulmonary blood flow	Increased pulmonary blood flow
Tetralogy of FallotPulmonary atresia with an intact ventricular septumEbstein anomalyIsolated right ventricular hypoplasia.	Transposition of the great vessels.Total anomalous pulmonary venous return (TAPVR) Truncus arteriosus

Solution to Question 8:

The probable diagnosis in the given case, based on the given clinical scenario and the image which shows icterus in the eyes, is Wilson's disease.

Wilson's disease is an autosomal-recessive condition caused by mutations in the ATP7B gene, which encodes for the ATPase that removes copper, preventing its accumulation in the liver and brain.

Clinical features are jaundice, ascites, hepatomegaly, dysarthria, dystonia, wing-beating tremor, a decline in school performance, and personality change. Deposition of copper in the lens results in sunflower cataracts and Kayser-Fleischer (KF) ring is seen due to the deposition of copper in Descemet's membrane of the cornea.

Diagnosis is made by the presence of a KF ring and decreased levels of serum copper and ceruloplasmin. Increased 24-hour urine copper excretion is seen. Measurement of hepatic copper levels in liver biopsy is the gold standard.

Treatment is the use of intramuscular British anti-lewisite or oral D-penicillamine. These drugs increase the excretion of copper in the urine. Pyridoxine is given concomitantly. Trientine hydrochloride is a suitable alternative to those intolerant to penicillamine.

Zinc acetate helps in copper absorption. It is used in presymptomatic patients, pregnant women, and as maintenance therapy. Liver transplantation is considered in advanced stages.

Other options:

Option B: Huntington's chorea is a trinucleotide (CAG) repeat disorder with an autosomal dominant mode of inheritance. It is characterized by involuntary movements known as chorea.

Option C: Glutaric aciduria is an autosomal recessive disease presenting during the first 6 months of life. Clinical features are hypotonia and jitteriness at birth, progressive hyperkinetic movements (dystonia, choreoathetosis), and may have seizures

Option D: Hepatitis A is a viral infection transmitted by the fecal-oral route, which is the commonest, parenteral route, and also by sexual transmission. The onset of jaundice is preceded by symptoms like nausea, vomiting, anorexia, and mild fever. Hepatitis A resolves completely but a relapse may occur in a few cases.

Solution to Question 9:

The procedure shown in the image is the measurement of the height of a child using a stadiometer. The alternative method that can be used for the procedure is arm span. The proportion of arm span changes with age but is close to height.

For children older than 2 years of age, a stadiometer is used to measure their height. It is measured with the head in the Frankfurt plane, and the back of the head, thorax, buttocks, and heels all on the same axis as that of the stadiometer.

Arm span provides an assessment of proportionality. It is measured by asking the child to stand, the back against the wall with arms outstretched, and the distance between the tip of the middle fingers are measured.

The body proportion in a child is assessed by the ratio of the upper segment (US) and lower segment (LS) and by comparing the arm span with height.

Upper segment: Lower segment ratio

A higher US:LS ratio is associated with bone disorders and conditions involving short-limb dwarfism, such as Turner syndrome, whereas a lower ratio is associated with hypogonadism and Marfan syndrome.

Other options:

Option-A: Head circumference is measured as the maximal circumference along the supraorbital ridge in the front and the occipital prominence at the back of the head. It is measured using a flexible tape. It reflects the brain growth in a child.

Option-C: Crown-rump length is used for the assessment of fetal growth as early as 6–8 weeks of gestation. This length is used to determine the gestational age in the first trimester.

Option-D: Knee height is the distance from the sole of the foot to the anterior surface of the femoral condyle, with the ankle and knee flexed to right angles. It is used when standing height cannot be measured.

Age	US: LS ratio
Birth	1.7
3 years	1.3
7 years	1

Solution to Question 10:

A child coming with acute exacerbation of asthma will be administered oxygen, salbutamol nebulization three times in 60 minutes, and oral corticosteroids.

Exacerbation of asthma is known as an increase in symptoms such as cough, wheezing, and/or breathlessness. If the child has cyanosis, silent chest, poor respiratory effort, altered sensorium, and oxygen saturation of <90%, the exacerbation is considered life-threatening.

Based on the severity, acute asthma is classified as mild, moderate, and severe.

Home treatment for acute exacerbation involves parents administering short-acting, beta-2-agonists by metered-dose inhaler with/without a spacer and facemask. One puff for every 30–60 seconds up to a maximum of 10 puffs with monitoring. If symptoms are relieved after inhalation, the child can continue salbutamol inhalation every 4–6 hours. If there is no improvement or partial improvement, the child should be transferred to a hospital.

In a child presenting to the hospital with severe symptoms, start inhalation of oxygen, short-acting beta-2-agonists such as salbutamol or albuterol that can be used every 15–20 minutes for the first hour and administer oral prednisolone (1–2 mg/kg). Oxygen inhalation is stopped if an oxygen saturation of >95% cannot be maintained.

Clinical parameter	Mild	Moderate	Severe
Color	Normal	Normal	Pale
Sensorium	Normal	Anxious	Agitated
Respiratory rate	Increased	Increased	Increased
Dyspnea	Absent	Moderate	Severe
Use of accessory muscles	Nil or minimal	Chest indrawing	Indrawing, nasal flare

Clinical parameter	Mild	Moderate	Severe
Rhonchi	Expiratory and/or inspiratory	Expiratory and/or inspiratory	Expiratory or absent.
Oxygen saturation	>95%	90-95%	<90%

Solution to Question 11:

The given clinical scenario with high fever, altered consciousness, neck rigidity, and cerebrospinal fluid (CSF) showing neutrophilic pleocytosis with low sugar point toward a diagnosis of bacterial meningitis. The initial treatment involves the administration of 3rd-generation cephalosporin like ceftriaxone along with vancomycin for 10–14 days.

Hyperglycemia due to the pathological stress from the infection can mask the reduced CSF glucose. It can be correctly inferred from CSF/blood glucose ratio. A ratio <0.4 is highly significant for bacterial meningitis.

The most common cause of meningitis in adults over 20 years of age is Streptococcus pneumoniae.

The classic clinical triad of meningitis includes fever, headache, and nuchal rigidity. Other common symptoms include decreased consciousness, nausea, vomiting, photophobia, and seizures. Cutaneous signs such as petechiae, purpura, or an erythematous macular rash may be present.

Kernig's sign and Brudzinski's sign can indicate meningeal irritation. Kernig's sign is positive when there is resistance and pain in extending the knee with the hip flexed at 90 degrees. Brudzinski's sign is positive when there is involuntary flexion of the hips and knees in response to passive neck flexion. However, these signs may not be reliable in certain patient groups, such as infants, elderly individuals, or those with altered mental status.

When bacterial meningitis is suspected, blood cultures should be obtained, and empirical antimicrobial therapy should be initiated without delay. The diagnosis is confirmed by examining cerebrospinal fluid (CSF). CSF analysis shows increased polymorphonuclear leukocytes, decreased glucose concentration, increased protein concentration, and increased opening pressure. CSF bacterial cultures and Gram stain are used to identify the causative organism.

Prompt initiation of empirical antimicrobial therapy is crucial in suspected cases of bacterial meningitis. Commonly used antibiotics include third- or fourth-generation cephalosporins (such as ceftriaxone, cefotaxime, or cefepime) that cover Streptococcus pneumoniae, Neisseria meningitidis, and Haemophilus influenzae and Vancomycin if there is a high suspicion of methicillin-resistant Staphylococcus aureus (MRSA) infection. Ampicillin with metronidazole is added when Listeria monocytogenes is suspected, particularly in neonates, pregnant women, individuals over 50 years, and immunocompromised individuals.

Solution to Question 12:

Age <1 year, temperature of 38–39°C (100.4-102.2ºF), and fever duration of <24 h (statement 1,2,3) are the major risk factors for the recurrence of febrile seizures.

Minor risk factors include a family history of epilepsy, male gender, daycare, complex febrile seizure, family history of febrile seizures, and lower serum sodium at the time of presentation.

Febrile seizures occur in children of the age of 6–60 months (peak at 12 to 18 months) with a temperature of 38ºC (100.4°F) or higher, without underlying intracranial infections or prior unprovoked seizures.

Types of febrile seizures:

Lumbar puncture should be performed in the first episode of seizures with fever, for all infants <6 months of age (to exclude meningitis). Electroencephalogram and neuroimaging are advised for children with suspected epilepsy or neurological abnormalities.

Treatment with lorazepam, diazepam, and midazolam for seizures lasting more than 5 minutes. Rectal diazepam is prescribed to use at home as rescue medication if febrile seizures last more than 5 minutes. Buccal or intranasal midazolam may be used alternatively.

For frequently recurring febrile seizures, intermittent oral clonazepam or diazepam can be given during febrile illnesses to reduce the risk of recurrence. Continuous prophylaxis is no longer recommended.

Antipyretics can be given but it does not decrease the risk of recurrence. Chronic antiepileptic therapy may be considered for children at high risk of developing epilepsy. Screening for and treating iron deficiency is recommended as it is associated with an increased risk of febrile seizures.

Simple febrile seizures	Complex (atypical) febrile sei zures
Occurs within 24 hours of th e onset of fever	Can occur at any time during a fever
Generalized seizures	Focal seizures
Duration is <15 min	Prolonged
No recurrence within 24 hrs of seizures	Recurrent
The chances of going into epi lepsy are 1%	The chances of going into epi lepsy are 6%

Solution to Question 13:

The correct answer is option C.

The mousy or musty odor of urine in phenylketonuria (PKU) is due to the accumulation of phenylalanine metabolites including phenylpyruvate, phenylacetate, and phenyl lactate.

PKU can lead to low levels of tyrosine, which is synthesized from phenylalanine. Tyrosine gives rise to catecholamines like dopamine, norepinephrine, and epinephrine. Thus, low levels of tyrosine may lead to a deficiency of dopamine, norepinephrine, and epinephrine. However, this is not the reason for the mousy odor of urine.

PKU is caused by a deficiency of the enzyme phenylalanine hydroxylase (classic PKU) or defects in the enzymes involved in producing tetrahydrobiopterin. These deficiencies impair the conversion of phenylalanine to tyrosine, leading to elevated levels of phenylalanine and low levels of tyrosine in the body. The excess phenylalanine in the blood is metabolized to phenyl ketones (phenylpyruvate and phenylacetate) which are excreted in the urine.

Affected infants initially appear normal at birth. If left untreated, they gradually develop profound intellectual disability. Older untreated children may exhibit hyperactivity, autistic behaviors, and neurologic signs such as seizures, spasticity, and tremors. Abnormalities like microcephaly, prominent maxillae, and growth retardation are seen. They are also prone to low bone mineral density and osteopenia.

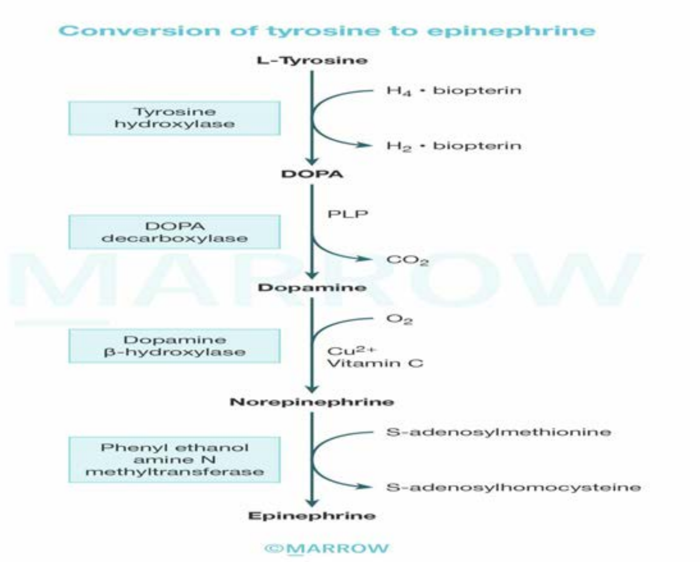
Untreated PKU patients may exhibit hypopigmentation (fair hair, light skin color, and blue eyes) due to low levels of tyrosine, which is necessary for the formation of melanin pigment.

If a woman with PKU is not following a low-phenylalanine diet during pregnancy, it can lead to maternal PKU syndrome in the offspring. This condition is associated with microcephaly and congenital heart disease.

The preferred screening method is tandem mass spectrometry. If screening is positive, the diagnosis is confirmed by quantitatively measuring phenylalanine levels.

The primary treatment for PKU is a low-phenylalanine diet, which should be initiated as soon as the diagnosis is established. Lifelong restriction of dietary phenylalanine is recommended. Tyrosine becomes an essential amino acid in PKU and it needs to be obtained from the diet.

The series of reactions involving the production of catecholamines is shown in the image below:



Solution to Question 14:

A history of choking and reduced breath sounds over the left lung in a child are suggestive of a foreign body lodged in the left bronchus. The next step in the management of this child is to obtain a chest x-ray, followed by bronchoscopic removal of the foreign body.

Food items (nuts, seeds hard candy, gum, bones raw fruits, and vegetables) account for the majority of choking cases in children. The most common site of foreign body lodging is the bronchus. Foreign body aspiration into the airway has three stages: immediate symptoms of coughing and choking, an asymptomatic interval, and later complications such as fever, atelectasis, and pneumonia.

When a child presents with a history of choking, detailed history taking and physical examination of the nose, oral cavity, pharynx, neck, and lungs are necessary.

The next step for suspected foreign body aspiration is to obtain a chest X-ray, including the abdomen, despite most foreign bodies being radiolucent. Secondary findings on the X-ray can indicate foreign body involvement, such as air trapping, atelectasis, obstructive emphysema, asymmetric hyperinflation, mediastinal shift, or consolidation. While a CT scan has high accuracy, it is not typically performed due to radiation risks.

The treatment of choice for a foreign body in the airway is its endoscopic removal with a rigid instrument, usually a bronchoscope, by a specialist otolaryngologist or pulmonologist. Bronchoscopy must be performed even if there is a high suspicion of an airway foreign body in spite of negative or inconclusive imaging.

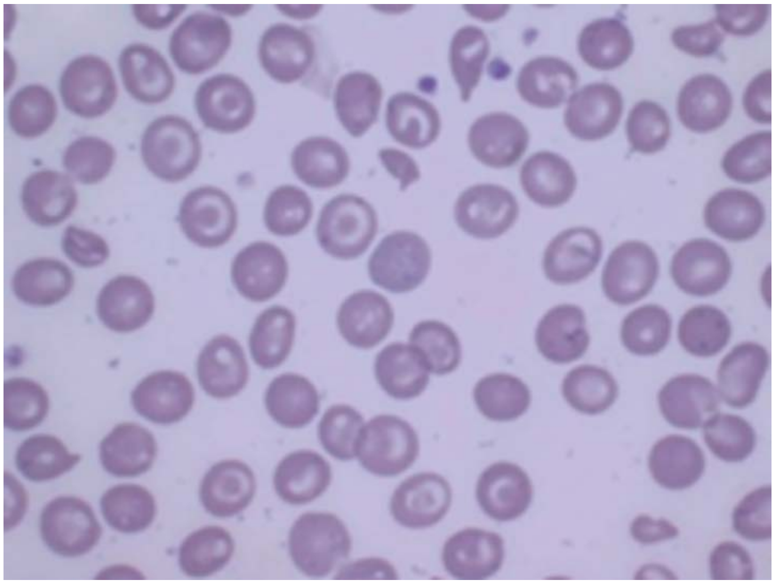
Solution to Question 15:

The investigations to be done in this patient are HPLC (high-performance liquid chromatography) and peripheral smear

The clinical history of a child with pallor, Mentzer's index <13, and abdominal distention (suggestive of hepatosplenomegaly) is suggestive of thalassemia.

Peripheral smear shows hypochromic microcytic anemia, target cells, basophilic stippling, fragmented red cells, and nucleated RBCs. HPLC showing decreased levels of HbA and an increase in HbA2 and HbF levels is sufficient to make a diagnosis.

The Mentzer index is the ratio of MCV (mean corpuscular volume) to RBC count. It is used to differentiate between iron-deficiency anemia and β-thalassemia trait. MCV and MCH (mean corpuscular hemoglobin) are low in both IDA and thalassemia. RBC count may be normal in thalassemia, whereas it is reduced in IDA. The Mentzer index is <13 in thalassemia minor syndrome.



Bone marrow examination (option 3) and coagulation studies (option 2) have no role in the diagnosis of thalassemia.

Thalassemia is a hemoglobinopathy that arises due to deficient synthesis of the globin chains. Depending on the chain affected, it is classified into α-thalassemia and β-thalassemia.

There is a deficit in the synthesis of HbA with an elevation of HbA2 and HbF levels in β-thalassemia. Children with severe thalassemia present with facial features (shown below) such as maxilla hyperplasia, a flat nasal bridge, and frontal bossing. Other manifestations include pathologic bone fractures, significant hepatosplenomegaly, and cachexia.

Hb Electrophoresis and HPLC are used as screening tests for thalassemia. DNA diagnosis to detect β-thalassemia mutations is used for definitive diagnosis. A radiographic image of the skull shows the classic 'crew cut' appearance.

Blood transfusions are administered every 3-4 weeks to maintain a pretransfusion hemoglobin level of 9.5-10.5 g/dL. Oral iron chelation to prevent iron toxicity is also recommended.

Hematopoietic stem cell transplantation is the only definitive therapy.

Antenatal counseling and heterozygote screening can be used in the prevention of the condition.

α-thalassemia is caused by deficient synthesis of α-chains. Deletion or malfunction of three α-globin chains can result in HbH disease. Hb Bart's hydrops fetalis is due to the absence of any normally functioning α-globin genes.

Solution to Question 16:

A child with a history of fever with rash and throat ache (sore throat) following a thorn prick injury could be having scarlet fever caused by group A Streptococcus (GAS). This child could be treated empirically with amoxicillin-clavulanate.

GAS commonly presents in children with pharyngitis and rash. It commonly spreads through droplets but can also be inoculated by a wound (thorn prick in this case) or burn infected with GAS.

The rash typically begins as diffuse, erythematous, papular eruptions. Sandpaper rash and the strawberry tongue are characteristic lesions that develop in the course of the disease. Pyrogenic exotoxin (erythrogenic exotoxin) produced by Group A Streptococcus is responsible for scarlet fever.

Streptococcal rapid antigen detection test or throat culture should be done before starting treatment with antibiotics. First-line treatment is by penicillin or amoxicillin for 10 days. Clavulanate can be added to combat any beta-lactamases produced by the bacteria.

Solution to Question 17:

The best method to assess iron overload is quantitative liver iron using T2 MRI. It eliminates the need for invasive procedures like liver biopsy.

Excessive iron stores resulting from transfusions are a major concern in the management of β-thalassemia major. Accurate assessment of iron overload is crucial for optimal treatment.

Quantitative measurement of cardiac iron by T2 MRI (option B) is initiated at the age of 10. However, there can be a discrepancy between liver iron and cardiac iron levels due to variations in tissue loading and unloading rates.

While serial serum ferritin levels (option C) can be used as a screening tool, they may not accurately reflect quantitative iron stores. Relying solely on serum ferritin levels can lead to undertreatment or overtreatment of iron overload.

Normally, iron in the plasma is bound to transferrin. But in iron-overload conditions, iron exceeds the iron binding capacity leading to the appearance of non-transferrin-bound iron (option D). It can act as an indicator of iron overload, but it is influenced by plasma factors like citrate, uric acid, and albumin. Hence, it cannot be used as a tool to measure quantitative iron overload that can help guide in planning chelation therapies.

Other tests that help calculate iron overload include echocardiography, bone mass densitometry, and testing for endocrinopathy. These tests are done at the time of diagnosis and done yearly to monitor iron overload. Serum ferritin tests can be done every 3 months.

Hypothyroidism, hypogonadotropic hypogonadism, growth hormone deficiency, hypoparathyroidism, and diabetes mellitus are seen due to iron deposition in endocrine organs. Iron deposition in the heart can lead to heart failure, arrhythmias, and cardiac-related mortality.

Iron chelation therapy is recommended when serum ferritin levels exceed 1,000 ng/mL and/or liver iron concentration is above 5,000 μg/g dry weight. It is not approved for children under 2 years of age. Three iron chelators are available: deferoxamine, deferasirox, and deferiprone. Deferoxamine is administered subcutaneously or intravenously due to its poor oral bioavailability. Deferasirox is a once-daily oral iron chelator. Deferiprone is an oral iron chelator approved as a second-line agent.

Solution to Question 18:

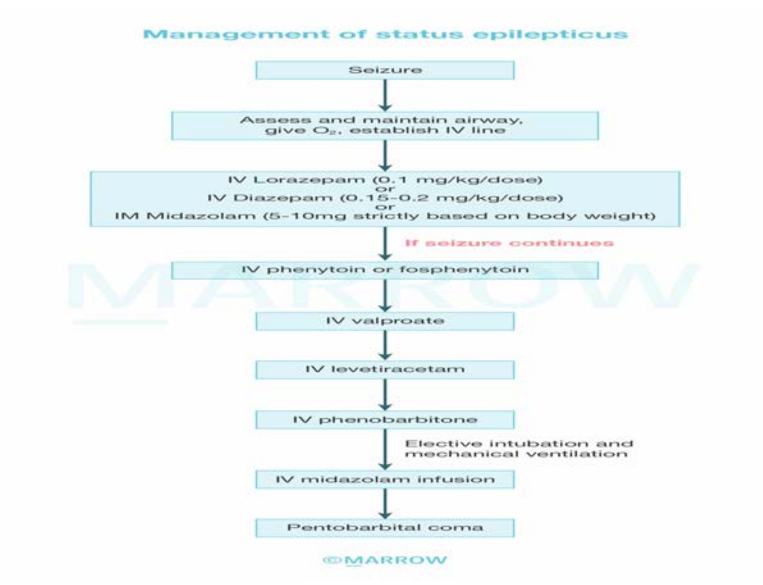
The order of management of a child with status epilepticus is airway and breathing→ iv lorazepam→IV Fosphenytoin→Midazolam, Propofol, Thiopentone (1, 2, 4, 3)

For generalized tonic-clonic seizures (GTCS), status epilepticus (SE) is defined as continuous convulsive activity or recurrent generalized convulsive seizure activity without regaining consciousness.

t1 is the time at which treatment should be initiated. t2 is the time at which continuous seizure activity leads to long-term sequelae such as neuronal injury.

The image given below depicts the management of status epilepticus in a flowchart:

Type of seizure	t1	t2
GTCS	5 min	30 min
Focal seizures with impaired awareness	10 min	30 min
Absence seizures	10-15 min	unknown



Refractory status epilepticus (SE) refers to SE that does not respond to therapy involving at least two medications (such as benzodiazepine and another medication). Super refractory status epilepticus is SE that does not resolve or recur within 24 hours or more despite treatment that involves continuous infusions of medications like midazolam or pentobarbital.

Solution to Question 19:

The given clinical scenario of a child with cold feet with diminished lower limb pulses and radio femoral delay is suggestive of coarctation of the aorta. It is caused by coarctation distal to the origin of the left subclavian artery.

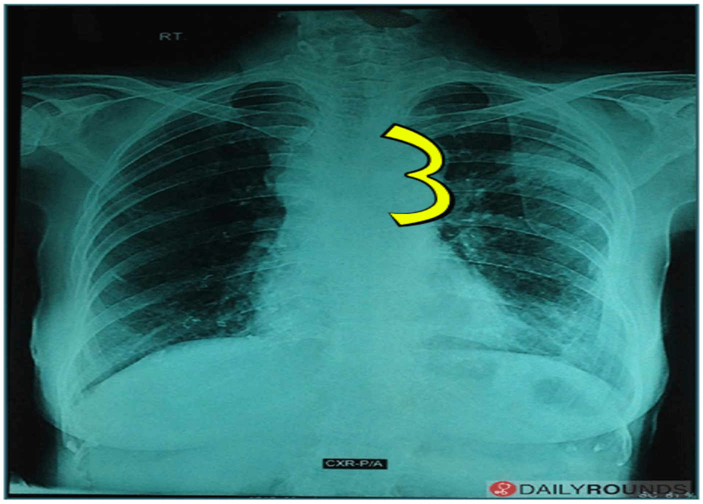
Coarctation of the aorta is a congenital heart disease, which involves constriction of the aorta at any point from the transverse arch to the iliac bifurcation. In 98% of cases, there is a narrowing of the descending aorta, usually distal to the origin of the left subclavian artery at the origin of the ductus arteriosus.

Coarctation of the aorta can be classified into two types based on the location of constriction: preductal (infantile type) and postductal (adult type). Preductal coarctation occurs proximal to the ductus arteriosus, while postductal coarctation occurs distal to it.

Clinical features include differences in blood pressure and pulse in the upper and lower extremities, radio femoral delay, and prominent pulsations in the upper limb. Cardiac enlargement and pulmonary congestion are noted in infants with severe coarctation. During childhood, the findings are present after the 1st decade. The heart is mildly or moderately enlarged because of left ventricular prominence. The enlarged left subclavian artery typically produces a prominent shadow in the left superior mediastinum. A continuous murmur over the infra-scapular region or the scapula can be auscultated.

The condition is associated with the bicuspid aortic valve, systemic hypertension, an aortic aneurysm, left ventricular hypertrophy, left heart failure, and premature atherosclerosis.

The figure of 3 sign refers to the indentation of the aorta at the coarctation site. There is pre and post-stenotic dilatation along with the left para-mediastinal shadow. Chest radiograph shows notching of ribs and the figure of 3 sign:



Unless surgically corrected in infancy, coarctation of the aorta results in the development of collateral circulation, mainly from branches of the subclavian, superior intercostal, and internal mammary arteries, to create channels for arterial blood to bypass the area of coarctation.

Hypertrophy of the ductus arteriosus (Option A) is not typically associated with coarctation of the aorta as it closes shortly after birth and is not directly involved in coarctation. The blood supply to the upper limb is higher as the coarctation is distal to the origin of the subclavian artery, therefore collaterals do not develop to supply the upper limb (Option D).

Solution to Question 20:

The correct sequence to provide basic life support is to assess responsiveness, check for breathing and pulse, start bag and mask ventilation commence and then cardiopulmonary resuscitation (CPR).

In an unresponsive patient, ensure scene safety first, then check for responsiveness, and activate the emergency response. In the presence of two or more rescuers, one remains with the victim while the other activates the emergency response system. Evaluate breathing and pulse within 10 seconds.

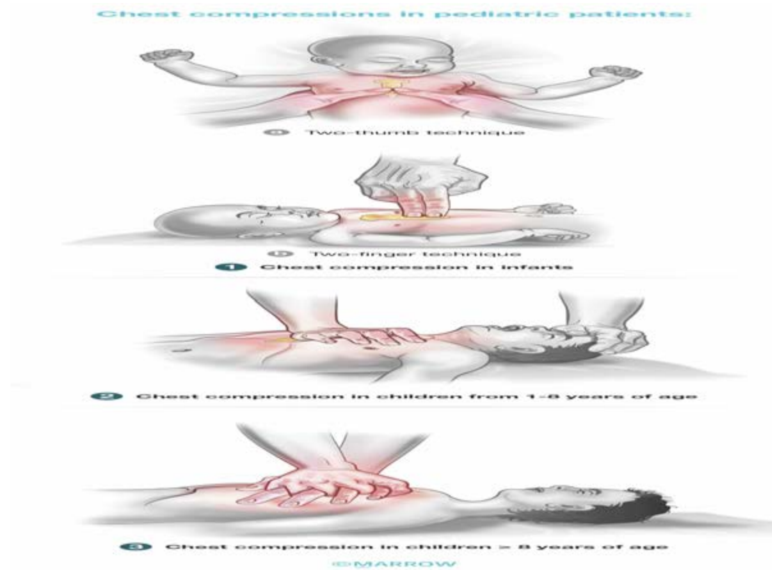
If the person does not have a pulse and is not breathing and there is only one rescuer, start CPR immediately with cycles of 30 compressions and 2 breaths if it is not a witnessed collapse. If it is a witnessed collapse with one rescuer, activate the emergency response system (ERS) if not done already, and retrieve the defibrillator.

After about 2 minutes, if still alone, activate the emergency response system and retrieve the automated external defibrillator (AED). If there are 2 or more rescuers, then one should start CPR while the other retrieves AED. When the second rescuer arrives change the cycle to 2 breaths for every 15 compressions. Breaths can be provided with the help of a bag and mask.

If the patient has a pulse but no normal breathing or only gasping, provide 1 breath every 3 to 5 seconds or about 12 to 20 breaths/min. Add compressions if pulse remains ≤60/min with signs of perfusion. Activate the emergency response system after 2 minutes. Continue rescue breathing, and check pulse every 2 minutes. If no pulse, start CPR.

When AED arrives, analyze the rhythm. If it is a shockable rhythm, give 1 shock. Resume CPR immediately for about 2 minutes. If it is a non-shockable rhythm, resume CPR immediately for about 2 minutes.

The image below shows chest compressions in pediatric patients.



Solution to Question 21:

The true statements regarding CMV infection are statements 1 and 3.

Most CMV transmission occurs in the first trimester. The risk of transmission is high in the third trimester.

Only 10% (statement 2) of the infected babies are symptomatic. Asymptomatic neonates have lesser chances of having late sequelae (statement 1).

Congenital CMV infection can result from either primary or reactivation infection of the mother. The rate of transmission of infection to the fetus in a mother who has a primary infection is about 30% (statement 3) while in a mother with a non-primary infection, it is 1-2%. However, the total number of infants who have acquired an infection from a mother with a non-primary infection is 3-4 times more than infants who have acquired an infection from mothers with a primary infection.

The clinical manifestations include
hepatosplenomegaly, petechial rashes, jaundice, chorioretinitis, and microcephaly. Neuroimaging shows intracranial periventricular calcifications. One of the long-term sequelae is sensorineural hearing loss.

The investigation of choice is PCR or viral culture. It should be done within the first three weeks of life (statement 4). After this period, it is difficult to distinguish between a congenital and an acquired infection. Since not all affected infants are viremic at birth, urine and/or saliva are more reliable samples than blood.

There is no definitive treatment for congenital CMV infection. Treatment with ganciclovir is indicated only in patients with progressive neurologic disease and deafness.

Below is the image showing the blueberry muffin rash in CMV affected baby:

Solution to Question 22:

Growth hormone, despite being present at higher levels in the fetus, does not play a significant role in fetal growth.

The human fetus produces thyroxine from the 12th week of gestation. Thyroxine and insulin have an important role in regulating tissue growth and differentiation of the fetus, particularly during late gestation. Glucocorticoids play an important role towards the end of gestation in the prepartum maturation of organs such as the liver, lungs, and gastrointestinal tract.

Insulin-like growth factor (IGF)-1 and IGF-II also play an important role in fetal growth.

Solution to Question 23:

The tape shown in the image is Shakir's tape. It is used to assess severe acute malnutrition (SAM). It is used by frontline field workers for nutritional screening of malnourished children and those at risk, aged 6-59 months.

Shakir's tape is used for the measurement of mid-upper arm circumference (MUAC). To locate the midpoint of the left arm, the arm is flexed at the right angle, and the mid-point is marked halfway between the acromion and olecranon process. Color coding is as follows:

• Red is less than 11.5 cm, which indicates severe acute malnutrition

• Yellow is from 11.5 to 12.5 cm, which indicates moderate acute malnutrition

• Green is from 12.5 to 26.5 cm, indicating no acute malnutrition

The presence of any one of the following confers severe acute malnutrition in children 6 to 59 months of age:

• Weight-for-height (length) below -3 standard deviation (<-3SD) on the WHO growth standard

• Presence of bipedal edema

• Mid-upper arm circumference (MUAC) below 11.5 cm

Paediatrics NEET 2022

Question 1:

A 1-day-old neonate has not passed urine since birth. What is the next step in management?

a) Continue breast feeding and observe

b) Admit to NICU

c) Start artificial feeding

d) Start intravenous fluids

Question 2:

A 7-year-old boy presented with abdominal pain, vomiting, oliguria, and periorbital puffiness following chemotherapy. Investigations reveal hyperuricemia, raised creatinine levels, and hyperkalemia. What is the next best step in the management of this condition?

a) Hydration

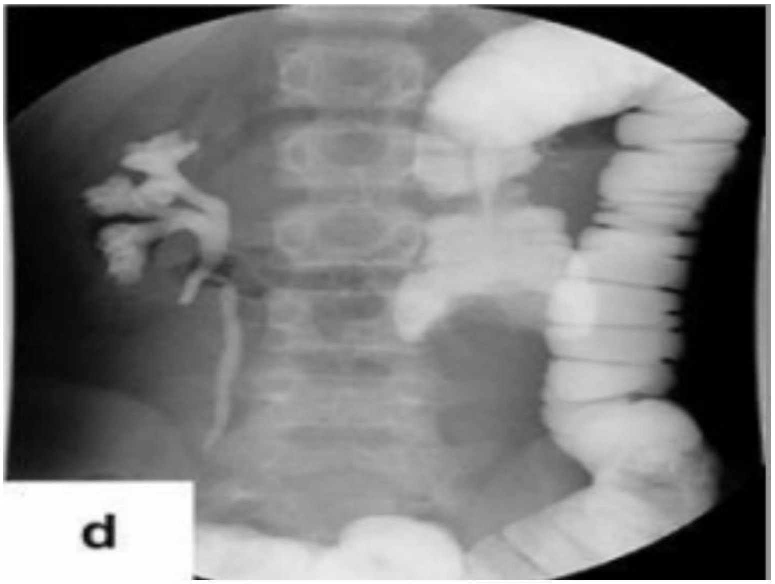
b) Probenecid

c) Allopurinol

d) Rasburicase

Question 3:

A baby presented with abdominal pain. On examination, a mass is palpated in the right lumbar region. A barium enema is done, and the image is given below. What is the likely diagnosis?



a) Intussusception

b) Volvulus

c) Duodenal atresia

d) Intestinal obstruction

Question 4:

An 8-day-old newborn was found to have thyroid-stimulating hormone level of more than 100 mIU/L. Which of the following will be the next best investigation?

a) Urine iodine excretion

b) Serum thyroid receptor antibody

c) Radiotracer uptake with technetium

d) Perchlorate secretion test

Question 5:

A 10-month-old infant was brought with complaints of jerking movement of limbs towards the body. On examination, there is a regression of developmental milestones. Electroencephalogram shows hypsarrythmia. Which of the following is the drug of choice in this condition?

a) Phenytoin

b) Adrenocorticotropic hormone (ACTH)

c) Levetiracetam

d) Phenobarbitone

Question 6:

A 2-month-old infant born to an HIV-positive mother presents with recurrent diarrhea. What is the next best step?

a) Test stool for giardia and give antibiotics

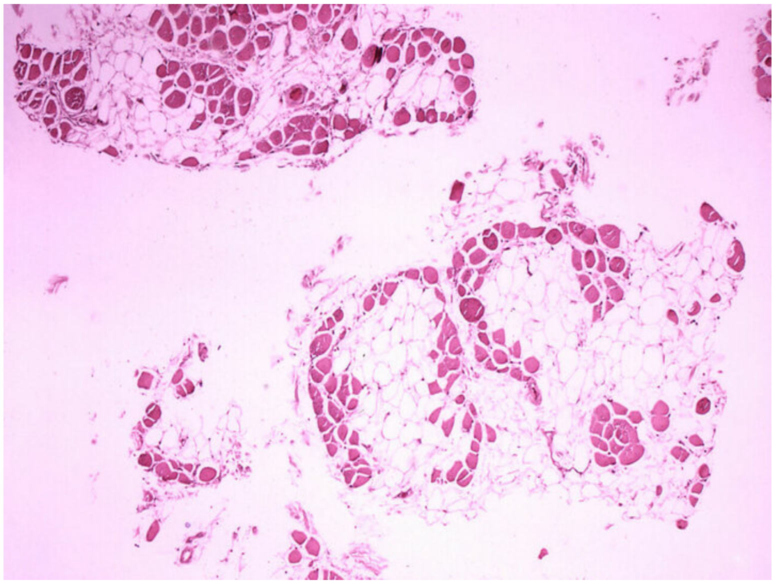
b) Dried spot sample for HIV DNA PCR

c) Antibody tests for HIV

d) Aerobic culture

Question 7:

An 8-year-old child has difficulty walking and getting up from a squatting position. A muscle biopsy was done and is as shown in the image. Which of the following is true about this condition?



a) Death occurs in 3rd decade

b) Previous history of viral prodrome

c) It is a mitochondrial storage disorder

d) Early treatment has excellent prognosis

Question 8:

A child presented with a history of loose stools with an increase in frequency for 4 days. On examination, he is drowsy, unable to feed, and skin on pinching goes back very slowly. According to the integrated management of neonatal and childhood illness (IMNCI), this child will be classified as having:

a) Mild dehydration

b) Some dehydration

c) Severe dehydration

d) Moderate dehydration

Question 9:

Which of the following children are considered at-risk babies?

a) 2,3

b) 1,2,3,4

c) 4,5

d) 1,4

Question 10:

Identify the condition.

a) Bladder exstrophy

b) Omphalocele

c) Persistent vitellointestinal duct

d) Gastroschisis

Question 11:

A previously healthy child presented with acute-onset dyspnea. A chest X-ray shows unilateral hyperinflation of the lungs. What is true for this patient?

a) Focal area of decreased air entry will be suggestive of foreign body

b) Flexible bronchoscopy used for removal

c) In complete obstruction, ball and valve mechanism causes hyperinflation

d) The child has developed acute laryngotracheobronchitis

Question 12:

A male child presented with arthralgia and abdominal pain. On examination, there was palpable purpura over the lower limbs. There is past history of upper respiratory tract infection prior to the onset of presenting symptoms. Which of the following is the treatment for this condition?

a) Azathioprine

b) Methotrexate

c) Cyclosporine

d) Glucocorticoids

Question 13:

A 10-year-old presents with edema and anasarca. A diagnosis of minimal change disease is made. Which of the following is true about this condition?

a) Light microscopy shows effacement of podocytes

b) Good response to steroids

c) Most common in adults

d) Non-selective proteinuria

Question 14:

A 10-year-old child presents with diarrhea and weight loss. On examination, the height and weight are lesser than expected. Laboratory investigations were positive for class II HLA-DQ2. Which of the following will you advise the child?

a) Fat-free diet

b) Lactose-free diet

c) Low carbohydrate diet

d) Gluten-free diet

Answer Key

Question No.	Correct Option
1	a
2	a
3	a
4	c
5	b
6	b
7	a
8	c
9	a
10	a
11	a
12	d
13	b
14	d

Detailed Explanations

Solution to Question 1:

In this clinical scenario, a 1 day-old neonate has not passed urine since birth. Most healthy neonates void within 24 hours of life. Delay beyond the first 24 hours of life is generally benign but may result in parental anxiety. Continuing breastfeeding and observing is the next best step in management.

The first urination should occur by 12-24 hours of age and the first passage of meconium should occur by 48 hours of age.

Delay in the passage of urine is seen in conditions causing blockage to urine outflow or birth defects that affect the spinal cord. Some commonly seen conditions are vesicoureteric reflux, ureteropelvic junction obstruction, posterior urethral valve, ureterocele, and spina bifida.

Delay in the passage of the first stool in term neonates may be associated with conditions causing a lower intestinal obstruction, which include meconium plug syndrome, Hirschsprung disease, and imperforate anus. More generalized problems, such as sepsis or hypothyroidism, should be considered.

Solution to Question 2:

The clinical scenario and the laboratory investigations are suggestive of tumor lysis syndrome. Aggressive hydration is the next best step in the management of this condition.

Tumor lysis syndrome (TLS) is a condition that can occur spontaneously after the initiation of chemotherapy.

During chemotherapy, the treatment kills the malignant cells, which leads to increased serum uric acid levels from the turnover of nucleic acids causing hyperuricemia. Tumor lysis results in the release of intracellular potassium, and phosphate pools, which causes hyperkalemia, and hyperphosphatemia. Increased phosphate levels lead to reciprocal depression of serum calcium/hypocalcemia. Deposition of uric acid crystals, calcium phosphate, and hyperphosphatemia leads to renal failure.

Signs and symptoms associated with TLS can develop within 12-48 hours after the initiation of chemotherapy and include:

• Abdominal pain and distension

• Dysuria, oliguria, flank pain and hematuria

• Anorexia, vomiting, cramps, seizures, spasms, altered mental status, and tetany due to hypocalcemia

• Weakness and paralysis due to hyperkalemia

• Hemodialysis in patients refractory to medical treatment

The Cairo-Bishop criteria are used to assess the severity of TLS.

Management of TLS includes:

• Before starting therapy, the electrolytes should be measured and aggressive hydration should be done.

• Allopurinol is a xanthine oxidase inhibitor that decreases the synthesis of uric acid. It should be started in TLS to prevent further accumulation of uric acid.

• Febuxostat is a selective non-purine xanthine oxidase inhibitor that can be used to treat hyperuricemia. It does not require dose calculations and can be used in patients with mildly decreased renal function.

• Rasburicase is an enzyme that degrades uric acid. It is preferred in TLS over allopurinol to decrease the accumulation of uric acid.

Option B: Probenecid inhibits the organic anion transporter in the nephron. It thereby inhibits the reabsorption of uric acid. It is contraindicated in TLS as it increases the risk of uric acid stones due to increased uric acid levels.

Solution to Question 3:

The clinical scenario of pain abdomen with a palpable mass in the right lumbar region along with the barium enema showing claw sign is suggestive of intussusception.

Intussusception occurs when a portion of the bowel is telescoped into an adjacent segment. Intussuscipiens is the one that receives and intussusceptum are the tubes that advance. The apex of the intussusceptum is more prone to ischemia which leads to gangrene.

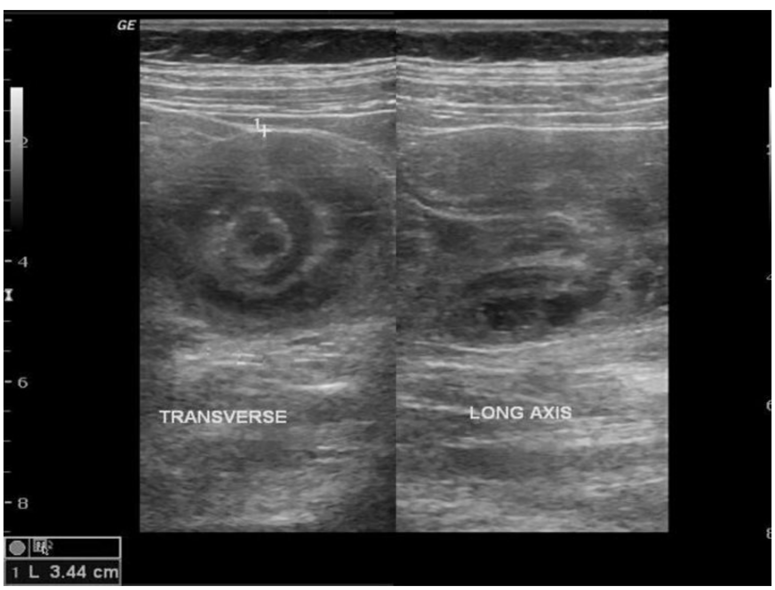
The most common location of intussusception is ileocolic.

It can be seen in infants during the weaning period following the introduction of new food, vaccination, or upper respiratory tract infection, and the currently approved rotavirus vaccines are associated with very minimal risk of intussusception. An area of enlarged submucosal Payer's patch acts as a lead point for intussusception.

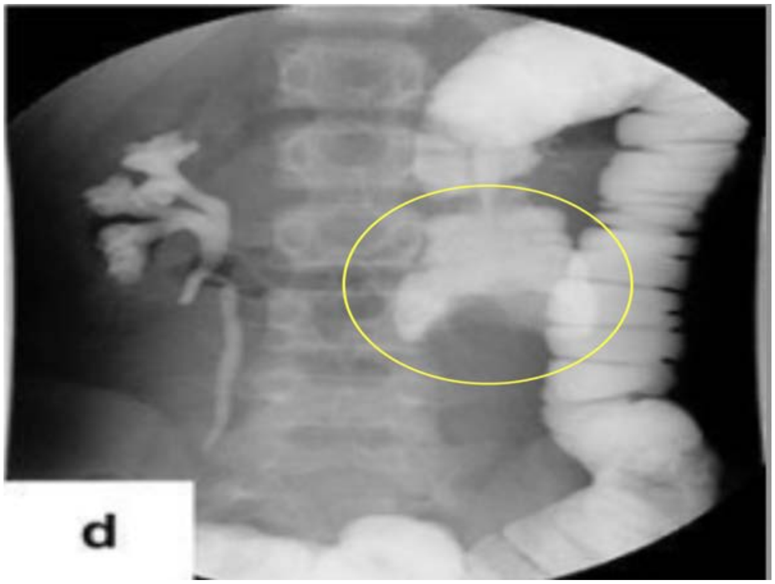
Some lead points in children are Meckel's diverticulum, intestinal polyp, inflamed appendix, submucosal hemorrhage seen in Henoch-Schonlein purpura, and intestinal duplication cysts.

Patients usually present with cramping abdominal pain, palpable sausage-shaped abdominal mass, vomiting, and passage of red currant jelly stools.

Diagnosis is confirmed by abdominal ultrasound which shows target sign/bull's eye sign (shown below) and pseudokidney sign (seen longitudinally).



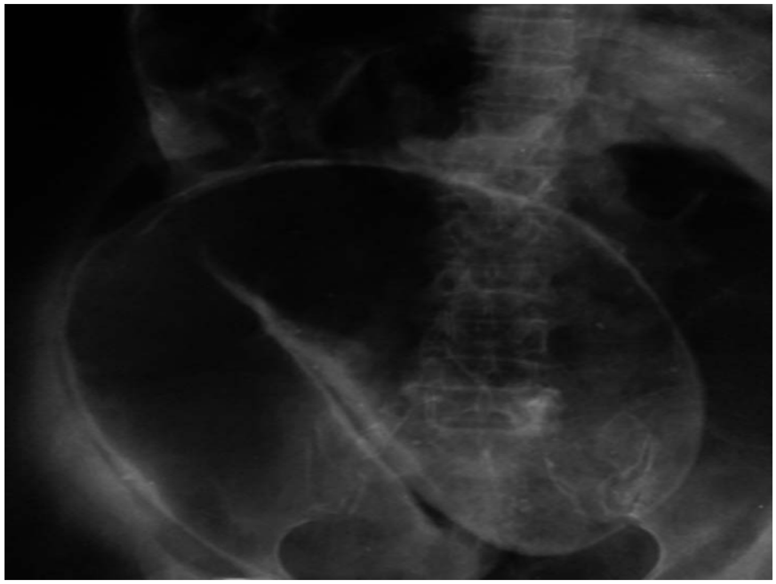
Barium enema shows a typical claw sign (shown below). This appearance is due to the convex filling defect caused by the apex of the intussusceptum.



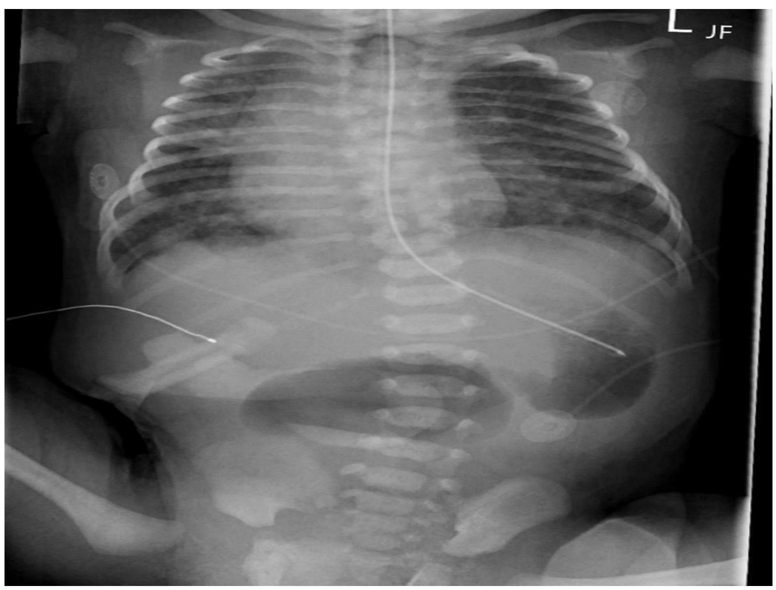
The image below shows the coiled spring appearance seen in intussusception. This is due to the barium in the lumen of the intussusceptum and in the intraluminal space.

Treatment of intussusception includes hydrostatic reduction by enema using contrast material or air. Surgical reduction is only done in the presence of refractory shock, suspected bowel necrosis or perforation, peritonitis, and multiple recurrences after hydrostatic reduction. An appendicectomy is an essential component of surgical management.

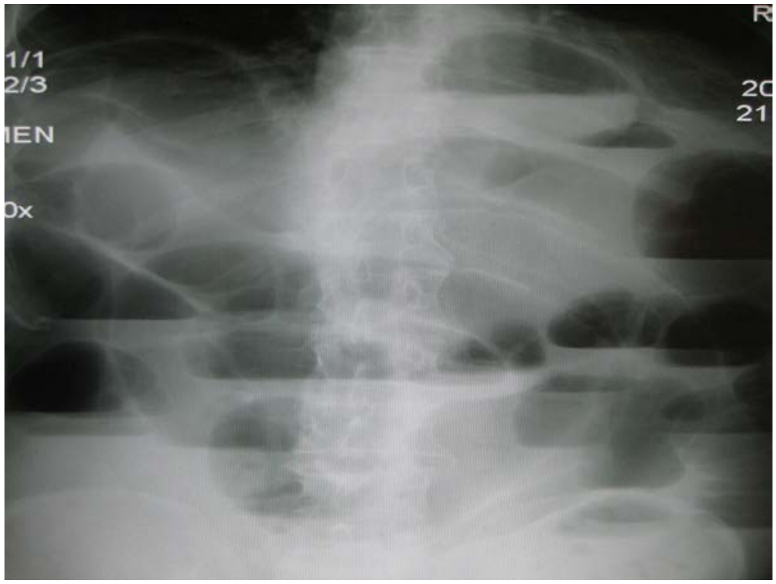
Option B: Sigmoid volvulus presents with abdominal pain and distension with constipation. A plain x-ray of the abdomen shows an omega sign, coffee-bean sign (shown below), or bent inner tube sign.



Option C: Duodenal atresia is commonly seen in newborns. Babies usually present with bilious/non-bilious vomiting immediately after birth. A plain X-ray abdomen reveals two foci of gas, one in the stomach and the other in the duodenum called the double bubble sign (shown below).



Option D: Intestinal obstruction presents with abdominal pain and distension with constipation. A plain x-ray of the abdomen shows multiple air-fluid levels (shown below). Normally, 3 air-fluid levels can be seen in the plain x-ray at the fundus of the stomach, at the duodenum, and at the cecum. A barium meal is usually contraindicated in acute intestinal obstruction.



Solution to Question 4:

The given laboratory finding of increased thyroid-stimulating hormone level in an 8-day-old baby is suggestive of congenital hypothyroidism. Radiotracer uptake with technetium will be the next investigation in this condition to confirm the presence of the thyroid gland.

Congenital hypothyroidism is the most common preventable cause of mental retardation. Causes of congenital hypothyroidism can be classified based on the presence of goiter as:

Clinical features of congenital hypothyroidism:

• Prolongation of physiologic hyperbilirubinemia- early sign (it is due to delayed maturation of glucuronide conjugation)

• Constipation- usually does not respond to treatment

• Wide open anterior and posterior fontanelles (>0.5 cm) and cranial sutures

• Developmental delay, delayed bone growth, delayed dentition

• Cardiovascular- slow pulse rate, cardiomegaly, asymptomatic pericardial effusion, edema

• Respiratory- macroglossia, feeding difficulties, choking spells, respiratory difficulties

• Hematological- macrocytic anemia refractory to treatment with hematinics

• Nervous system- hypothermia, hypotonia, slow relaxation of deep tendon reflexes, sluggishness, and somnolence.

The classical presentation is described as the 6 P's of congenital hypothyroidism: pot-bellied, pale, puffy-faced child with protruding umbilicus, protuberant tongue, and poor brain development.

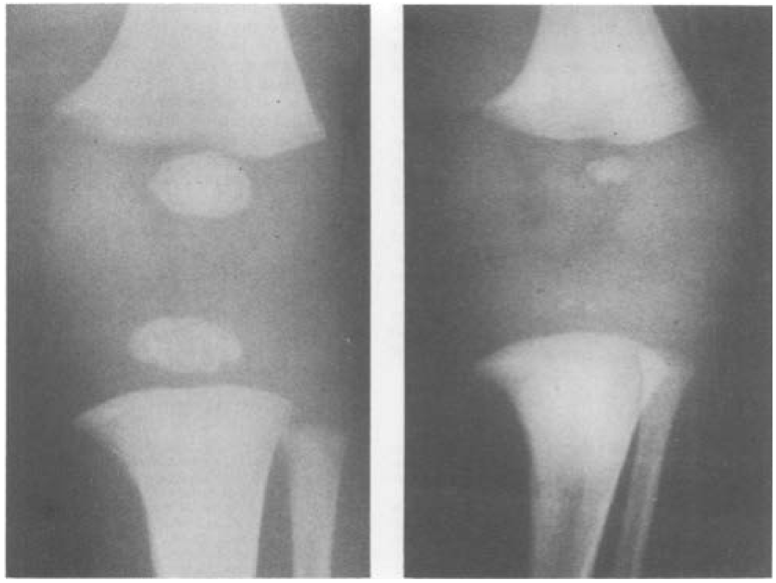
The image given below shows a pot belly with an umbilical hernia seen in congenital hypothyroidism.

Goitrous(Palpable goiter)	Non-goitrous(No palpable goi ter)
Thyroid dyshormonogenesis (30%)Iodine deficiencyPlace ntal passage of antithyroid dr ugs	Thyroid dysgenesis(65%)Thy roid receptor blocking antibo dies (5%)Central hypothyroid ism

Laboratory investigations in congenital hypothyroidism would demonstrate low T4 levels and increased TSH levels. Screening is usually done on days 1 and 5 days of life, dried blood samples obtained from a heel prick are used for this. Initial investigations in a child with high TSH levels include evaluation of radiotracer uptake study with radioactive iodine or technetium and ultrasound of the thyroid to confirm the presence of a thyroid gland.

An X-ray shows a delay in bone development. The distal femoral and proximal tibial epiphyses normally present at birth are often absent.

In the image below, the left side shows a normal neonate with both knee epiphyses present and the right side shows a neonate with congenital hypothyroidism in whom both epiphyses are small.



Treatment with levothyroxine should be started immediately after diagnosis. Thyroxine (T4) of dose 10-15 μg/kg/day should be started and adjusted to achieve the normal free T4 range. In central hypothyroidism, cortisol replacement should be started before thyroid replacement.

Neonatal hypothyroidism is a sensitive indicator of environmental iodine deficiency and can also be an effective indicator for monitoring the impact of a program.

Other options:

Option A: Urinary iodine excretion is used for surveillance and epidemiological studies for classifying countries according to the level of iodine deficiency by the WHO, and is an effective indicator of the program against iodine deficiency.

Option B: Serum thyroid receptor antibody is used to diagnose transient congenital hypothyroidism caused due to transplacental passage of TSH receptor blocking antibodies. It is usually seen in babies born to mothers suffering from autoimmune hypothyroidism taking thyroxine.

Option D: A perchlorate secretion test can be used to diagnose thyroid peroxidase deficiency.

Solution to Question 5:

The clinical scenario along with the electroencephalogram showing hypsarrythmia is suggestive of the West syndrome. Adreno-corticotropic hormone (ACTH) is the drug of choice for this condition.

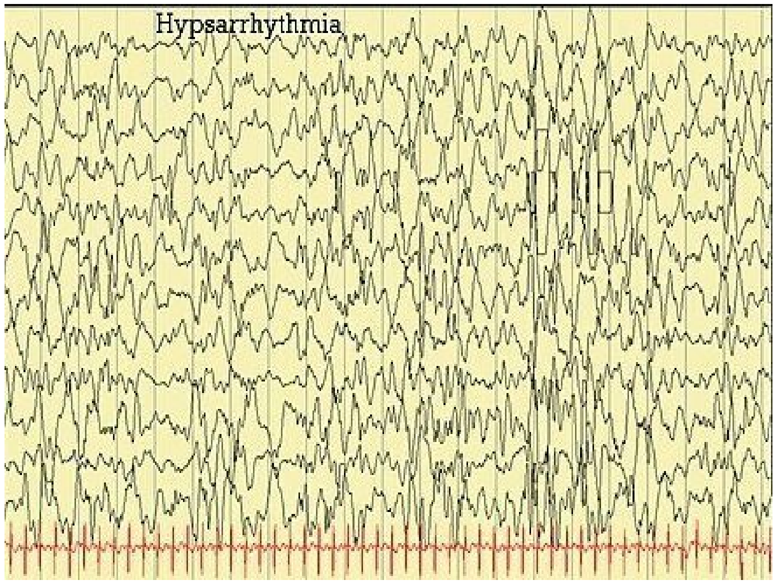
West syndrome is severe generalized epilepsy that begins between the ages of 2 and 12 months. It is characterized by a triad of:

• Infantile epileptic spasms

• Developmental regression

• Hypsarrhythmia on electroencephalogram (EEG)

Hypsarrhythmia refers to a high voltage, slow, chaotic background with multifocal spikes as shown below.



West syndrome has been known by many names like infantile spasms, salaam attacks, salaam convulsions infantile spastic epilepsy, and jackknife convulsion syndrome. During an attack babies typically attain complete heaving of the head forward towards their knees, and then immediately relax into the upright position.

The image given below illustrates infantile spasms in a child.

Diagnosis is confirmed by EEG which typically shows hypsarrythmia.

The treatment of choice in West syndrome is either adreno-corticotropic hormone (ACTH) or corticosteroids. An antiseizure medication, mainly vigabatrin can be used when associated with tuberous sclerosis.

Note: The table given below shows the EEG features seen in various seizure disorders.

Options A and C: Phenytoin and levetiracetam are used in focal seizures and status epilepticus.

Option D: Phenobarbitone is used as an antiepileptic in neonatal seizures and as an enzyme inducer in Crigler-Najjar syndrome type II.

Disease	Typical EEG feature
Lennox-Gastaut syndrome	1.5-2.5 Hz spike and polyspike bursts in sleep
Absence seizures	3 Hz spike and slow-wave (spike and dome pattern)
Juvenile myoclonic epilepsy	4-5 Hz spike
West syndrome	Hypsarrhythmia (high-voltage waves with chaotic background)

Solution to Question 6:

In the above scenario, the diagnosis of HIV infection should be established/ruled out before evaluating for secondary infections. Hence, a dried spot sample for HIV DNA PCR is the next best step.

AIDS enteropathy, a syndrome of malabsorption with partial villous atrophy not associated with a specific pathogen, has been postulated to be a

result of direct HIV infection of the gut.

The diagnosis of HIV infection in babies born to HIV-infected mothers is done by HIV proviral DNA or RNA PCR, HIV culture, or HIV p24 antigen. The national program (Early Infant Diagnosis) now uses HIV total nucleic acid (RNA + proviral DNA) PCR test on dried blood spot samples. All positive tests need to be confirmed with an HIV PCR test on another sample.

An infant born to an HIV-positive mother should be breastfed exclusively or by replacement feeds exclusively for 6 months of age. Infants should receive ARV prophylaxis with syrup nevirapine or zidovudine.

Involvement of the gastrointestinal tract is common in HIV-infected babies. Common pathogens identified are bacteria (Salmonella, Campylobacter, Shigella, Mycobacterium avium complex), protozoa (Giardia, Cryptosporidium, Isospora, microsporidia), viruses (CMV, HSV, rotavirus), and fungi (Candida). MAC and protozoal infections are most severe and protracted in children.

Other options:

Option A: Test stool for giardiasis and give antibiotics is not preferred as the immediate next step. First, the diagnosis of HIV infection should be established before treating secondary infections.

Option C: Antibody tests for HIV are not useful in establishing the diagnosis of HIV infection. This is because all infants born to HIV-infected mothers test antibody positive at birth because of the passive transfer of maternal HIV antibodies across the placenta. It can be used as a diagnostic tool after 18 months of age.

Option D: Aerobic culture is not used for the diagnosis of HIV infection.

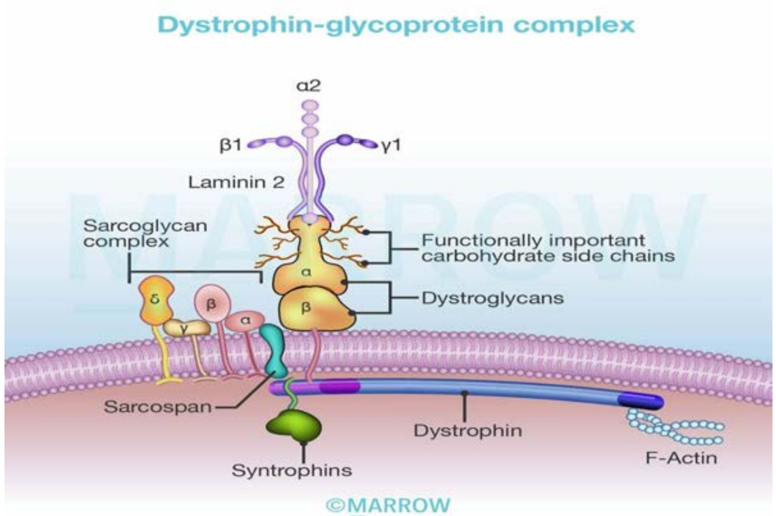
Solution to Question 7:

The clinical scenario and the muscle biopsy showing fatty replacement of muscle fibers are suggestive of Duchenne muscular dystrophy (DMD). Death in DMD occurs in the late teens and 20s (3rd decade).

DMD is the most common hereditary neuromuscular disease. It is inherited in an X-linked recessive pattern. It is caused by a mutation in the dystrophin gene on chromosome Xp21.

Normally, the dystrophin gene codes for the large protein dystrophin. Dystrophin protein forms a rod that connects the actin filaments to the transmembrane protein dystroglycan in the muscle cell membrane. This dystrophin-glycoprotein complex thereby provides strength to the muscle and connects them to the extracellular environment.

The image given below shows the dystrophin-glycoprotein complex.



Clinical features of DMD:

• Progressive weakness, developmental delay, and intellectual impairment

• In toddlers- delayed walking, falling, toe walking, and trouble running or walking upstairs

• Calf muscular pseudohypertrophy

The image given below shows pseudohypertrophy of the calf muscles.

• Gower's sign: Patient has to use their hands and arms to walk up their own body from squatting position due to the lack of hip and thigh muscle strength due to the weakness of proximal muscles, especially of the lower limbs.

• Trendelenburg gait/hip waddle

• The patient is unable to walk by the age of 12 years and death occurs by late 20 years of age, mostly due to respiratory failure during sleep.

The complications of DMD include contractures, scoliosis, dilated cardiomyopathy, and malignant hyperthermia after anesthesia.

Diagnosis of DMD:

• Elevated serum creatine kinase levels

• Polymerase chain reaction (PCR) analysis for the dystrophin gene

• If PCR is normal, but clinical suspicion is high, muscle biopsy (fatty replacement of muscle fibers) with dystrophin immunocytochemistry (absence of the normal sarcolemmal staining) is done.

Treatment of DMD includes glucocorticoids (prednisone or deflazacort), and multidisciplinary management. Eteplirsen and casimersen are the two new FDA-approved drugs for the treatment of DMD.

Note- The FDA has approved Vamorolone, a novel corticosteroid for the treatment of DMD. The drug offers enhanced anti-inflammatory properties and mineralocorticoid receptor antagonism as well. It is potentially a safer and more effective alternative to glucocorticoids.

Solution to Question 8:

A clinical scenario of a dehydrated child who is drowsy and not able to feed with skin going back very slowly on pinching is suggestive of severe dehydration (as per IMNCI).

Diarrhea is defined as a change in consistency and frequency of stools that occurs more than 3 times a day. Diarrhea, if associated with blood in stools, is called dysentery. If the episodes subside within 7 days, it is called acute diarrhea. If it persists for more than 2 weeks, it is called persistent diarrhea.

Intestinal infections are the most common cause of acute diarrhea. Rotavirus and enterotoxigenic E. coli are the major infectious causes of acute diarrhea. Other causes are certain drugs, food allergies, and systemic infections (e.g. urinary tract infection and otitis media).

The most important assessment in the evaluation of diarrhea is to look for the degree of dehydration. It is assessed as follows:

Based on the degree of dehydration after history and examination, the

estimated fluid loss is calculated as follows:

The treatment plans A,B, and C are explained in the pearl below.

Parameters	No dehydration	Some dehydration	Severe dehydration
Condition	Well alert	Restless, irritable	Lethargic orunconscious,floppy
Eyes	Normal	Sunken	Very sunken and dry
Tears	Present	Absent	Absent
Mouth and tongue	Moist	Dry	Very dry
Thirst	Drinks normallynot thirsty	Thirsty, drinks eagerly	Drinks poorly or isnot able to drink
Skin pinch	Goes back quickly	Goes back slowly	Goes back very slowly
Treatment	Plan A	Weigh the patient, if possible, and use treatment Plan B	Weigh the patient and use treatmentPlan C urgently

Degree of dehydration	Assessment of fluid loss
No dehydration	less than 50 mL/kg
Some dehydration	50-100 mL/kg
Severe dehydration	more than 100 mL/kg

Solution to Question 9:

Among the given options, babies on artificial feeds and babies of working/single parents as considered at-risk babies.

The following are considered at-risk infants:

- Birth weight <2.5 kg
- Twins
- Birth order of 5 or more
- Artificial feeding
- Weight <70% of the expected weight (i.e. II and III degrees of malnutrition)
- Failure to gain weight during 3 successive months
- Children with protein-energy malnutrition (PEM) and diarrhea
- Working mother or single parent

It is important to identify children at risk and give special intensive care. This is done because these children contribute largely to perinatal, neonatal, and infant mortality.

Solution to Question 10:

The above image is suggestive of ectopia vesicae or bladder exstrophy.

Ectopia vesicae/bladder exstrophy is a congenital abnormality that occurs when the lower abdominal wall does not form properly and the bladder is open and exposed.

The clinical features of bladder exstrophy include:

- Separation of the pubic bones with non-united symphysis pubis
- Epispadias
- In males- broad and short penis and bilateral inguinal hernia
- In females- bifid clitoris

Management involves a staged surgical reconstruction of the bladder and genitalia. Closure of the abdominal and bladder wall is done in the first year of life. This often requires an iliac osteotomy to relax the tension on the bladder and repair the abdominal wall during wound-healing.

Later, reconstruction of the bladder neck, sphincters, epispadias, and other genital defects is done.

Less satisfactorily, urinary diversion can be carried out by means of ureterosigmoid anastomosis, an ileal or colonic conduit, or continent urinary diversion.

Long-term complications include:

• Stricture at the site of the anastomosis with bilateral hydronephrosis and infection

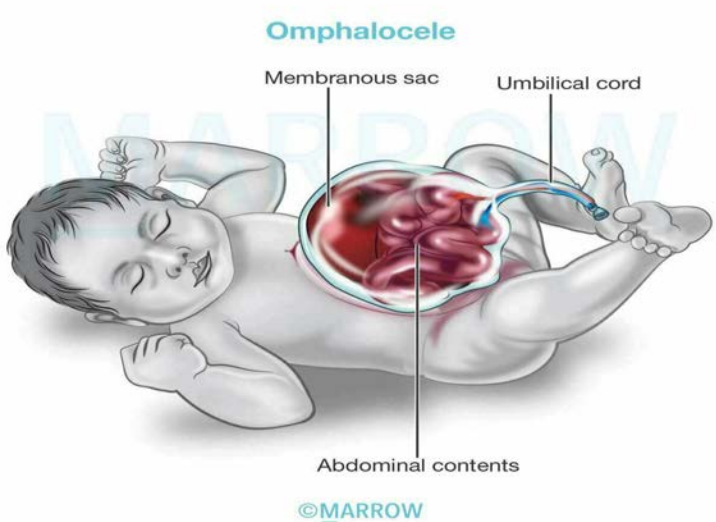
• Hyperchloremic acidosis

• An increased (20-fold) risk of tumor formation (adenoma and adenocarcinoma) at the site of a ureteroscopic anastomosis.

Other options:

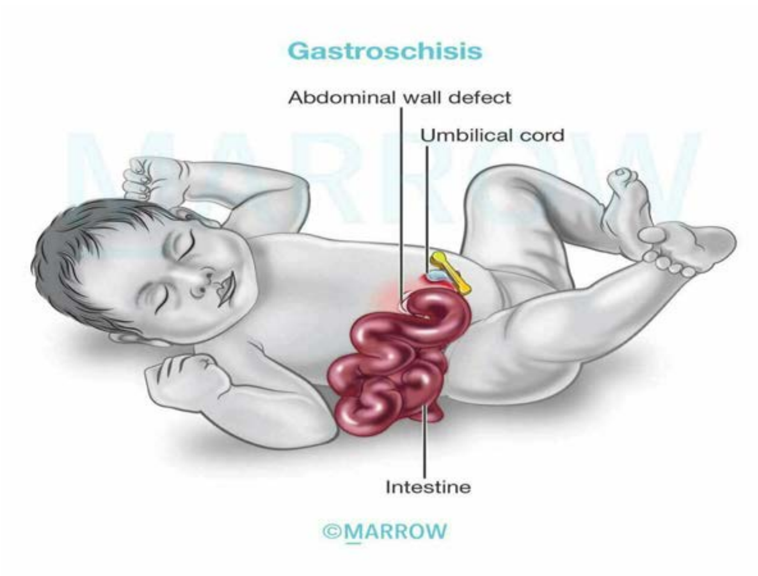
Option B: Omphalocele is a midline anterior abdominal wall defect. It occurs due to the failure of the physiological umbilical hernia to return to the abdominal cavity. The extruded abdominal contents are covered in a sac. It is associated with elevated alpha-fetoprotein levels and chromosomal abnormalities.

The image given below shows omphalocele.



Option C: In a persistent vitellointestinal duct, the vitelline duct remains patent over its entire length. This results in the formation of direct communication between the umbilicus and the intestinal tract. A fecal discharge may then be found at the umbilicus.

Option D: Gastroschisis is a condition where intestinal loops herniate through the defect in the abdominal wall. The defect usually lies to one side of the umbilicus (right side commonly). The loops of the bowel are not covered by amnion as they herniate through the abdominal wall directly into the amniotic cavity. It is associated with elevated alpha-fetoprotein levels.



Solution to Question 11:

The above scenario is suggestive of a foreign body in the airway, most likely in the bronchi. In this condition, auscultation is likely to reveal a focal area of decreased air entry due to incomplete airway obstruction

It is more commonly seen in children below 4 years. The most foreign bodies in the airway are food items. The foreign bodies can be non-irritating types (plastic, glass, or metallic foreign bodies) and irritating types (vegetables, peanuts, beans, and seeds).

The clinical presentation depends upon the site of the obstruction.

Foreign bodies in bronchi are more likely to enter the right bronchus because it is wider and more in line with the tracheal lumen.

The initial evaluation is done with a plain X-ray chest in posteroanterior and lateral views (expiratory films are useful). It may reveal:

• The radio-opaque foreign body- size, shape, and location

• In complete airway obstruction- lobar or segmental atelectasis

• In incomplete airway obstruction- the ball valve mechanism is produced when a foreign body incompletely obstructs a bronchus allowing air to enter inside but not leave. This leads to the appearance of hyperinflation or obstructive emphysema of the affected area when an expiratory X-ray film is taken as the air is not allowed to escape.

Management of foreign body airway is as follows:

• Laryngeal foreign body- Heimlich maneuver should be attempted. If it fails, cricothyrotomy or emergency tracheostomy should be done

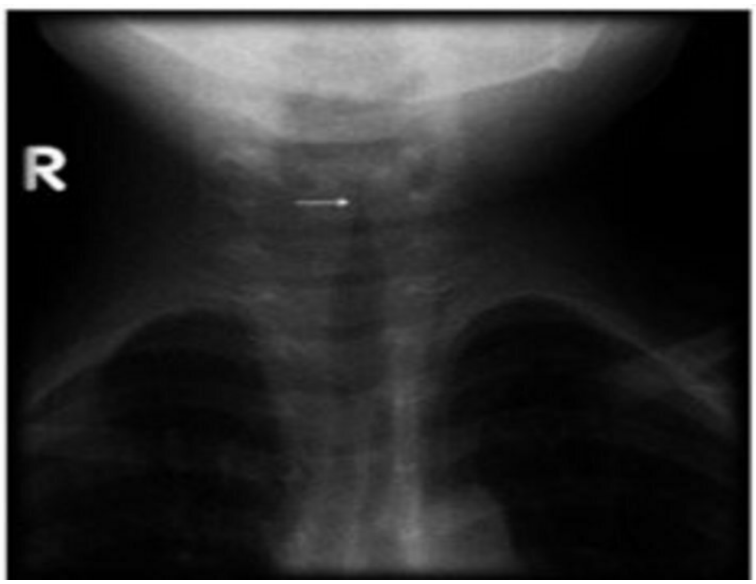
• Foreign bodies in the trachea and bronchi are removed through a rigid bronchoscope after adequate hydration and emptying of the stomach.

• Appropriate antibiotics are given to prevent secondary infection.

Option D: Laryngotracheobronchitis or croup is caused by the parainfluenza virus. It commonly affects children of age 6 months-3 years. The clinical features include barking cough, biphasic stridor, and low-grade fever. The chest X-ray shows a narrowing of the subglottic region known as the steeple sign.

The image given below shows the steeple sign seen in croup.

Site of foreign bodies	Signs and symptoms
Larynx	Complete obstruction causes asphyxiaIncomplete obstruction causes croup, hoarseness, cough, stridor, and dyspnea.
Trachea	Dysphonia, dysphagia, dry cough, and biphasic stridor.
Bronchi	May be asymptomatic.Asymmetric breath sound/air entry, coughing, and breathing.



Solution to Question 12:

The clinical scenario is suggestive of Henoch Schonlein's purpura (HSP). Glucocorticoids are used to treat HSP.

Henoch-Schönlein purpura (HSP), also known as IgA vasculitis, is a common small vessel vasculitis in childhood. It is a type III hypersensitivity reaction characterized by the deposition of IgA in the small vessels of the skin, joints, gastrointestinal tract, and kidney, resulting in vasculitis.

The exact etiology of HSP remains unknown. But, it usually occurs following an upper respiratory tract infection caused by group A Streptococcus, Staphylococcus aureus, Mycoplasma, and Adenovirus.

Children usually present with the following features:

• Palpable purpura is symmetric and occurs in the lower extremities, extensor aspect of the upper limbs, and buttocks.

• Arthralgia and arthritis- commonly affecting hip, knee, or ankle joints.

• Gastrointestinal manifestations like abdominal pain, vomiting, diarrhea, and intussusception.

• About 30% of the patients with HSP develop renal manifestations, presenting as microscopic hematuria, hypertension, and progressive glomerulonephritis.

Glucocorticoids such as prednisolone are used for the treatment of gastrointestinal manifestations in HSP. Nephritis due to IgA vasculitis should be treated with immunosuppressants like prednisolone and azathioprine.

The image given below shows palpable purpura in the lower limbs seen in HSP.

The criteria for diagnosis of childhood HSP are:

Palpable purpura with at least one of the following-

• Diffuse abdominal pain

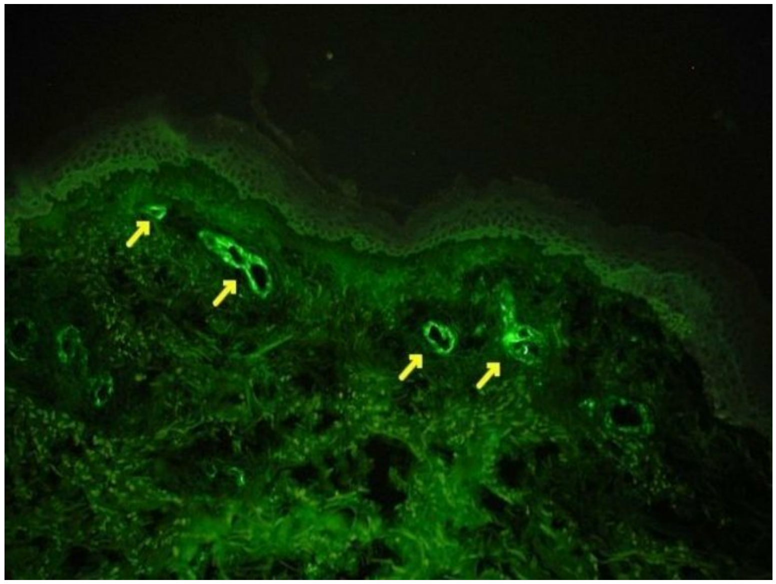
• Any biopsy showing predominant IgA deposition

• Arthritis or arthralgia

• Renal involvement (proteinuria >3 g/24 hr), hematuria, or red cell casts

Lab findings seen in HSP are nonspecific and include leukocytosis, mild anemia, and elevated erythrocyte sedimentation rate (ESR) and C-reactive protein (CRP). The platelet count is normal in HSP. A skin biopsy specimen can be useful in confirming leukocytoclastic vasculitis with IgA and C3 deposition by immunofluorescence.

The image given below shows IgA and C3 deposition in vessels seen on immunofluorescence of skin biopsy in HSP.



Other options:

Immunosuppressants like azathioprine, methotrexate and cyclosporine are used in a variety of rheumatologic disorders like vasculitis, arthritis, inflammatory bowel diseases like Crohn's disease and ulcerative colitis, autoimmune hepatitis, and lupus nephritis.

Solution to Question 13:

The true statement about minimal change disease is that it shows a good response to steroids.

Minimal change disease is the most common cause of nephrotic syndrome in children. It is more common in males and appears between 2 to 6 years of age.

The causes of MCD are mostly idiopathic. It is usually associated with Hodgkin's lymphoma (due to the increased production of cytokines), respiratory infections, prophylactic immunizations, eczema, and rhinitis.

Clinically, they mostly present with proteinuria. The child presents with a history of edema that is initially noted around the eyes and gradually progresses into generalized edema. Hypertension and hematuria are uncommon.

Levels of the following proteins are increased in nephrotic syndrome:

• Fibrinogen - leads to hypercoagulability.

• Lipoproteins - due to increased synthesis.

Levels of the following proteins are decreased in nephrotic syndrome as a consequence of selective proteinuria:

• Albumin—the protein with the largest proportion of loss in nephrotic syndrome

• Transferrin

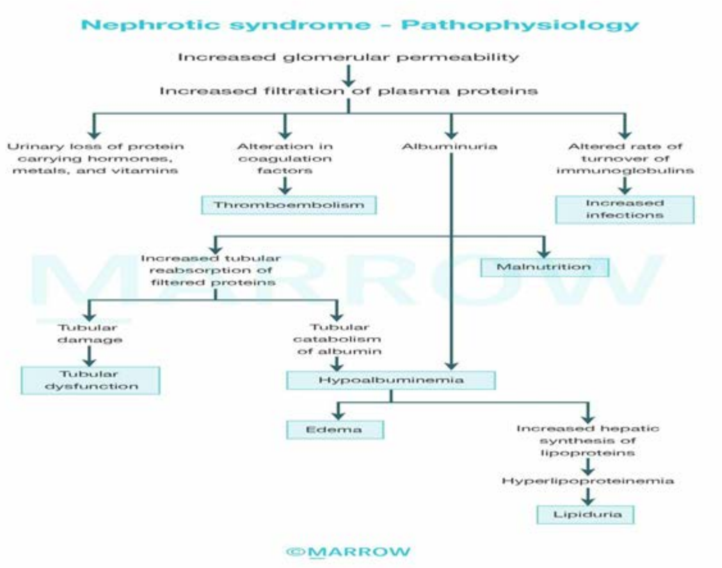
• Cholecalciferol-binding protein

• Thyroxine-binding globulin (TBG)

• Ferritin

• Coagulation factors

• Antithrombin III



Diagnosis can be made by urinalysis showing proteinuria, ranging from 3+ to 4+ protein on urine dipstick and a spot urine protein/creatinine ratio (PCR) >2. The proteinuria principally contains albumin, hence known as selective proteinuria. Serum albumin levels are <2.5 g/dL and there is an increase in serum cholesterol and triglyceride levels.

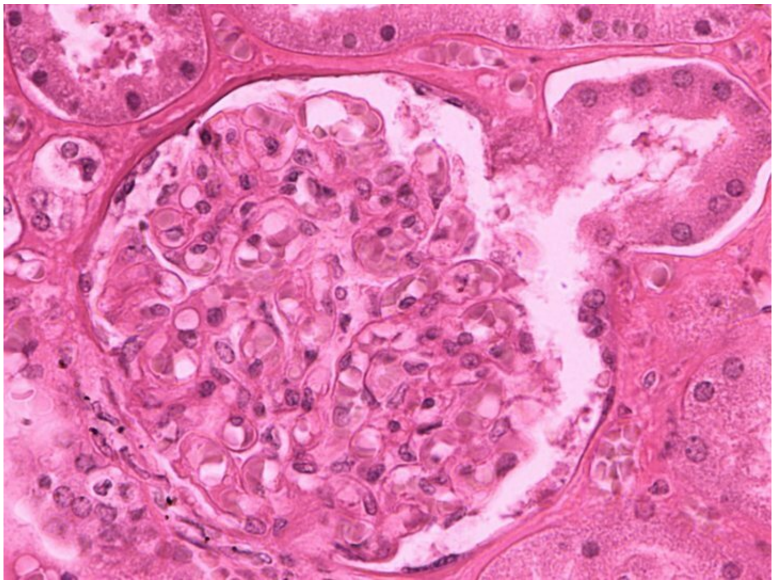
Biopsy findings include the following:

• Light microscopy- glomeruli appear normal (hence called "minimal change")

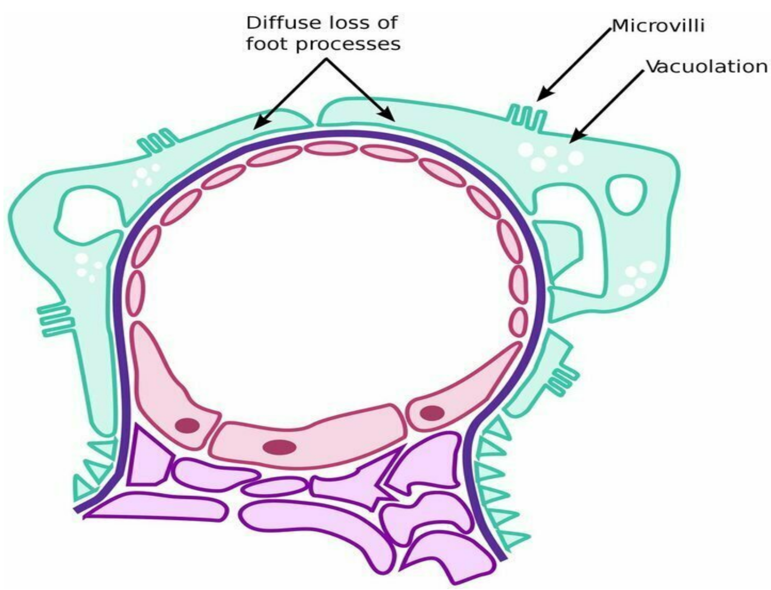
• Immunofluorescence studies- no Ig or complement deposits

• Electron microscopy- diffuse effacement of foot processes of visceral epithelial cells/podocytes

The image given below shows biopsy findings seen in minimal change disease on light microscopy.



The illustration given below shows the effacement of foot processes of podocytes as seen in MCD.



The majority of children will respond to corticosteroid (prednisone/prednisolone) treatment and will achieve complete remission (urine PCR<0.2 or <1+ protein on urine dipstick).

Solution to Question 14:

The clinical scenario of diarrhea, steatorrhea, weight loss, and growth retardation, along with positive class II HLA-DQ2, is consistent with a diagnosis of celiac disease. Patients with celiac disease are advised to avoid gluten.

The dietary recommendation for celiac disease is as follows:

Celiac disease/celiac sprue/gluten-sensitive enteropathy is an immune-mediated disorder triggered by gluten-containing foods. The presence of one of the two locus antigens HLA-DQ2 or

HLA-DQ8 is necessary, but not sufficient for the development of this condition. Detection of these is not useful for the diagnosis, however, their absence helps to exclude celiac disease.

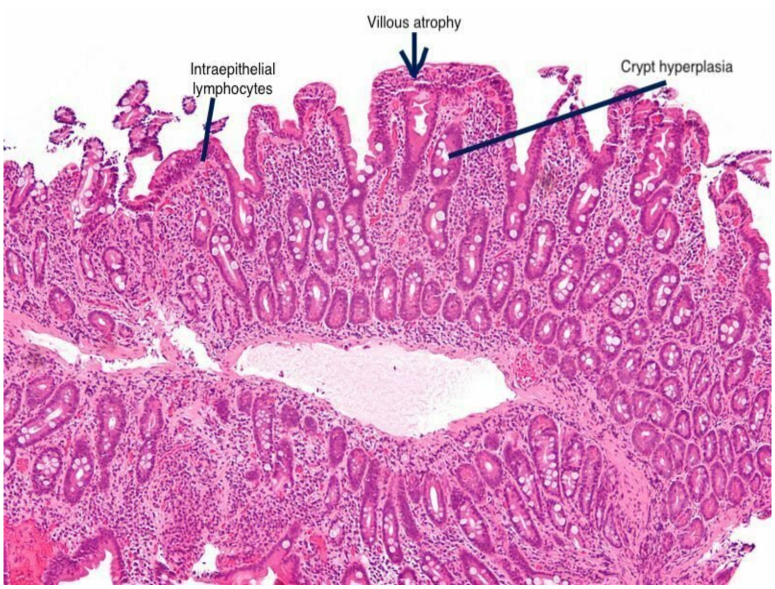
The pathogenesis involves an inflammatory response to gluten and gliadin (gluten-derived peptide). In the small intestine, gliadin is deamidated by the tissue transglutaminase (tTG), an intestinal enzyme. The deamidated gliadin is more immunogenic and is presented to helper T cells, which mediate mucosal inflammation and cause tissue damage.

Children usually present with diarrhea, weight loss, and growth failure. Other features include malabsorption, bloating, irregular bowel habits, iron deficiency anemia, osteoporosis, ataxia, and abnormal liver enzymes.

Screening for celiac disease is done by testing for serum antibodies, including tissue transglutaminase IgA, anti-endomysial, and deaminated anti-gliadin antibodies. The diagnosis in patients with positive antibodies is confirmed by biopsy of the second part of the duodenum or proximal jejunum. Microscopic findings of biopsy show intraepithelial CD8+ lymphocytes, villous atrophy, and crypt hyperplasia.

The image given below shows microscopic findings seen in celiac disease.

Diet avoided (gluten)“BRO W”	Diet Allowed“So MR P ”
Barley	Soya bean
Rye	Maize
Oats	Rice
Wheat	Potato



Other options:

Option A: Fat-free diet is advised for various disorders causing steatorrhea. These include pancreatic insufficiency, cystic fibrosis, and familial hypercholesterolemia.

Option B: Lactose-free diet is advised for lactose intolerance. It occurs due to lactase deficiency. Patients usually present with diarrhea, abdominal pain, gassiness, and bloating. A lactose tolerance test or breath hydrogen test may be used for the diagnosis.

Option C: Low-carbohydrate diet is advised for diabetes mellitus.

Paediatrics NEET 2023

Question 1:

A newborn presented with chest retractions, dyspnea, and lethargy. The pediatrician diagnosed the baby with respiratory distress syndrome. This occurs due to the deficiency of:

a) Dipalmitoyl inositol

b) Lecithin

c) Sphingomyelin

d) Dipalmitoylphosphatidylethanolamine

Question 2:

A 10-year-old child weighing 30 kg presents with a history of loose stools for 2 days. On examination, there is severe dehydration. Laboratory investigations are as follows. What is the initial management as per ISPAD guidelines?

a) Manage ABC, NS 20 mL/kg and start insulin after 1 hour

b) Manage ABC, NS 20 mL/kg along with insulin 0.1 IU/kg/hr

c) Manage ABC, NS 10 mL/kg along with insulin 0.1 IU/kg/hr

d) Manage ABC, NS 10 mL/kg and start insulin after 1 hour

Question 3:

Which of the following is the best sign to indicate adequate growth in an infant with a birth weight of 2.8 kg?

a) Increase in length of 25 centimeters in the first year

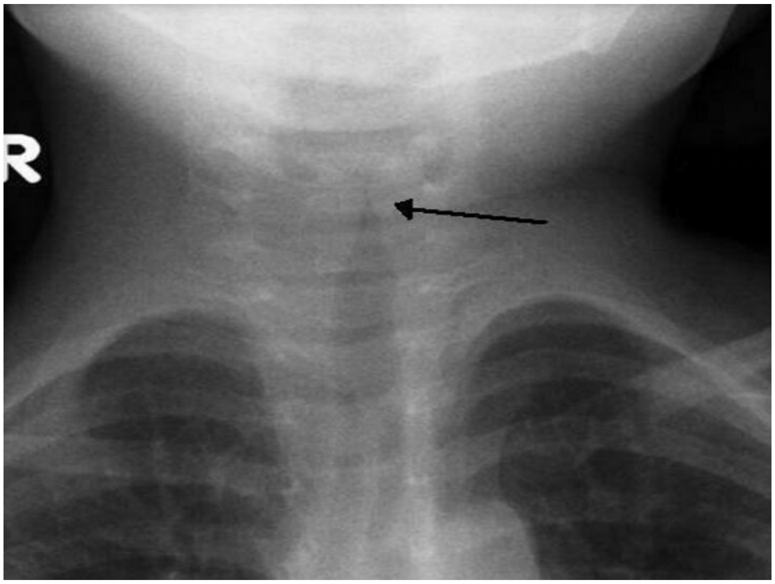
b) Weight gain of 300 grams per month till 1 year

c) Anterior fontanelle closure by 6 months of age

d) Weight under 75th percentile and height under 25th percentile

Question 4:

A child is brought to the hospital with respiratory distress and biphasic stridor. The radiograph is shown below. What is the diagnosis?



a) Acute epiglottitis

b) Acute laryngotracheobronchitis

c) Foreign body aspiration

d) Laryngomalacia

Question 5:

A 3-month-old baby comes with complaints of deafness, cataract, and patent ductus arteriosus. Which of the following is the most likely diagnosis?

a) Congenital herpes simplex virus infection

b) Congenital toxoplasmosis

c) Congenital cytomegalovirus infection

d) Congenital rubella syndrome

Question 6:

Chloride level in sweat is used in the diagnosis of which disease?

a) Phenylketonuria

b) Cystic fibrosis

c) Gaucher's disease

d) Osteogenesis imperfecta

Question 7:

A 3-month-old baby presents with jaundice and clay-colored stools. Lab investigation reveals that the baby has conjugated hyperbilirubinemia. The liver biopsy shows periductal proliferation. What is the most likely diagnosis?

a) Crigler-Najjar syndrome

b) Rotor syndrome

c) Dubin-Johnson syndrome

d) Biliary atresia

Question 8:

A child presents with fever and a rash spreading from the face, behind cheeks, and buccal mucosa to other body parts. On examination, Koplik's spot is present. What is the likely diagnosis?

a) Measles

b) Rubella

c) Varicella

d) Mumps

Question 9:

An infant presents with hepatosplenomegaly and thrombocytopenia. Neuroimaging with CT shows periventricular calcifications. What is the most likely diagnosis?

a) Congenital rubella syndrome

b) Congenital herpes simplex virus infection

c) Congenital toxoplasmosis

d) Congenital cytomegalovirus infection

Question 10:

A 3-week-old infant presents with a cough and sore throat. The mother reports that the infant develops a paroxysm of cough followed by apnea. The total leucocyte count is >50,000 cells/µL. Which of the following drugs will you prescribe for this patient?

a) Azithromycin

b) Amoxicillin

c) Cotrimoxazole

d) Clarithromycin

Answer Key

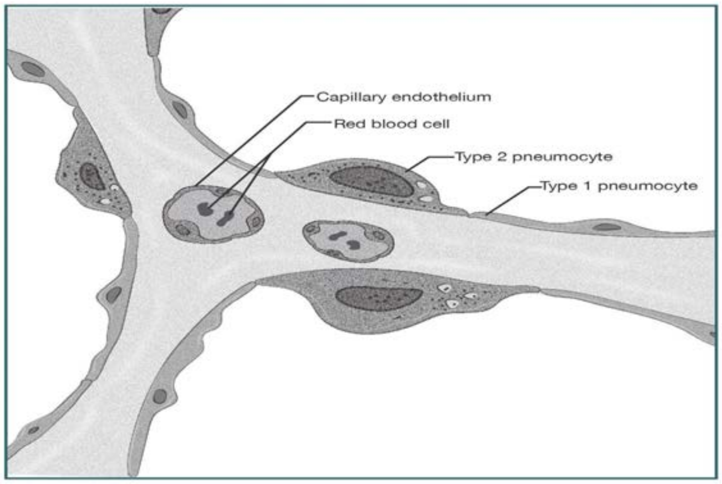
Question No.	Correct Option
1	b
2	a
3	a
4	b
5	d
6	b
7	d
8	a
9	d
10	a

Detailed Explanations

Solution to Question 1:

Respiratory distress syndrome in a neonate is likely due to the deficiency of lecithin.

Respiratory distress syndrome (RDS), also known as hyaline membrane disease, is due to the decreased production and secretion of the lung surfactant. Surfactant is a lipoprotein primarily made of phospholipids such as lecithin (otherwise known as dipalmitoylphosphatidylcholine). The other minor components are phosphatidylglycerol, apoproteins, and cholesterol. It is synthesized in type II pneumocytes/granular pneumocytes in the endoplasmic reticulum (shown below).



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RDS is commonly seen in preterm babies less than 34 weeks of gestation. Surfactant deficiency leads to an increase in the surface tension over the alveoli, causing the alveoli to collapse easily during expiration. Therefore, the baby has to put more effort to expand the lungs. Tachypnea, cyanosis, nasal flaring, prominent expiratory grunting, and intercostal and subcostal retractions are some of the common findings seen in this condition.

Chest x-ray shows features such as reticulogranular pattern, ground-glass opacity, and air bronchogram. White-out lungs can be seen on chest x-ray in severe diseases.

Management in mild to moderate RDS is with continuous positive airway pressure (CPAP), where the positive pressure keeps the alveoli open in a spontaneously breathing baby. Severe cases will require mechanical ventilation. Steroids increase the production of the surfactant, and therefore, steroids are administered antenatally to mothers with preterm labor.

Other options -

Option A: Dipalmitoyl inositol, or phosphatidylinositol, acts as a precursor for the secondary messenger system. It also helps in cell signaling and membrane trafficking.

Option C: Sphingomyelin is a dominant sphingolipid in the plasma membrane. The lecithin/sphingomyelin ratio in amniotic fluid helps in determining fetal lung maturity.

Option D: Dipalmitoyl ethanolamine is present in the cell membrane.

Solution to Question 2:

The given clinical scenario with severe dehydration, hyperglycemia, glucosuria, and metabolic acidosis is suggestive of diabetic ketoacidosis. Based on the ISPAD guidelines, the initial management includes management of the airway, breathing, and circulation (ABC) and fluid replacement with an IV bolus of 20 mL/kg NS (Normal saline). This is followed by an IV insulin infusion of 0.05 - 0.1 U/kg/hr, at least 1 hour after starting fluid replacement therapy.

The International Society for Pediatric and Adolescent Diabetes (ISPAD) guidelines provide evidence-based recommendations for the management of DKA (diabetic ketoacidosis) in children and adolescents.

Diabetic ketoacidosis is the most serious complication of type 1 diabetes mellitus. Clinical manifestations of diabetic ketoacidosis include:

• Dehydration, tachypnea with deep, sighing respiration (Kussmaul breathing)

• Nausea, vomiting, and abdominal pain that may mimic an acute abdominal condition

• Confusion, drowsiness, and coma in severe cases.

The diagnosis of DKA is based on the triad of hyperglycemia, ketosis, and metabolic acidosis. All three biochemical criteria are required to diagnose DKA

• Hyperglycemia (blood glucose >200 mg/dl)

• Venous pH < 7.3 or serum bicarbonate <18 mmol/L

• Ketonemia or ketonuria

Initial management includes management of the airway, breathing, and circulation (ABC) according to Pediatric Advanced Life Support (PALS) guidelines. Following that emergency assessment is done, which includes:

• Immediate measurement of blood glucose, blood or urine ketones, serum electrolytes and blood gases

• Assessment of level of consciousness.

• Two peripheral intravenous (IV) lines should be inserted.

The goal of therapy is to correct dehydration, correct acidosis, reverse ketosis, and gradually blood glucose concentration to near normal. Monitor for acute complications, and identify and treat any precipitating event.

Fluid replacement should always begin before starting insulin therapy. One or more boluses of 0.9% saline at 20 mL/kg are given to restore peripheral circulation. Insulin therapy with 0.05–0.1 U/kg/hr must begin at least 1 hour after starting fluid replacement therapy. Insulin of 0.05 U/kg/hr can be considered if pH is > 7.15.

Potassium replacement is given only when potassium is <5.5 mmol/L. In rare cases of hypokalemia with potassium <3.0 mmol/L, defer insulin treatment.

Bicarbonate administration is not recommended except for treatment of severe acidosis (pH < 6.9) or in cases of life-threatening hyperkalemia.

Solution to Question 3:

The best sign to indicate adequate growth in an infant is an increase in length of 25 centimeters in the first year. The growth velocity is the rate of increase in height in one year and varies with age.

The average growth velocity is 25 cm/year in the first year (birth-12 months) and an infant will be 75 cm tall at 1 year. The height velocity is 10 cm/year from 12-24 months and at 2 years, the height is 90 cm. The height velocity is 8 cm/year from 24-36 months. At 4 years, the child is 100 cm tall. After this, the child gains about 6 cm every year until 12 years.

If growth velocity is lower than expected for age, the child is likely to be suffering from a pathological cause of short stature.

Other options:

Option B: During the first 3 months of age, the weight gain is about 20-40 grams per day. From 4 months of age till 1 year of age, the weight gain is about 400 grams per month. The weight of an infant doubles his birth weight at 5 months of age and triples at 1 year of age and quadruples at 2 years of age.

Option C: The anterior fontanelle closes by 18 months of age.

Option D: According to WHO growth charts, weight under the 75th percentile and height under the 25th percentile do not indicate adequate growth.

Solution to Question 4:

The diagnosis in this child with respiratory distress and biphasic stridor is acute laryngotracheobronchitis, which is caused by the parainfluenza virus. The x-ray finding is the 'steeple sign' due to subglottic narrowing.

The age of presentation of laryngotracheobronchitis ranges from 6 months to 6 years. The clinical features include barking cough, inspiratory stridor, hoarseness, and low-grade fever.

Management:

• Racemic epinephrine nebulization leads to the rapid improvement of upper airway obstruction. Epinephrine constricts pre-capillary arterioles in the upper airway mucosa and decreases capillary hydrostatic pressure, leading to fluid resorption and improvement in airway edema.

• Dexamethasone is administered by the least invasive route (Oral > IV > IM) for all the grades of croup.

• Oxygen should be administered to children who are hypoxemic (SpO2 <92 %)

Antibiotics are not indicated in acute laryngotracheobronchitis.

Other options:

Option A: Acute epiglottitis is a medical emergency presenting in children with a rapidly progressive course of fever, sore throat, dyspnea, and respiratory obstruction, followed by a toxic appearance and drooling. The neck is hyperextended and stridor is a late finding. Radiography shows a classic thumb sign.

Option C: Foreign body aspiration initially presents with a paroxysm of cough, choking, and gagging, accompanied by new-onset wheezing and asymmetric breath sounds.

Option D: Laryngomalacia is a congenital laryngeal anomaly presenting in children with low-pitched inspiratory stridor exacerbated by exertion.

Solution to Question 5:

The most likely diagnosis is congenital rubella syndrome based on the clinical findings.

Triad of Gregg is seen in babies with congenital rubella syndrome. The features include:

• Patent ductus arteriosus (PDA)

• Cataract

• Sensorineural Hearing loss (SNHL)

Rubella or German measles is a serious infection in pregnancy that causes abortions or severe fetal defects if infection occurs in the first trimester. Transmission of the virus occurs through nasopharyngeal secretions (droplet transmission). The mother usually has a fever and generalized body rash. The rash is maculopapular in nature that begins from the face and spreads to the trunk and extremities later. The incidence and severity reduce as the pregnancy advances and is rare after 20 weeks of gestation.

There is no specific treatment for rubella virus infection in mothers. Post-exposure passive immunization can be beneficial if given within 5 days of exposure to droplets. The incidence of this disease has reduced due to the availability of MMR vaccines. MMR vaccines can be administered only to non-pregnant females as it is a live, attenuated virus. Pregnancy should be avoided for at least 1 month after vaccination.

Other options:

Option A: Herpes simplex infection can be acquired in utero, intrapartum, or during the neonatal period. The intrauterine infection causes skin vesicles, eye findings (chorioretinitis, keratoconjunctivitis), and microcephaly/hydranencephaly. Intrapartum infection presents with either:

• Skin, eye, and mouth disease

• Encephalitis

• Disseminated disease with sepsis-like presentation

Option B: Congenital toxoplasmosis occurs due to primary infection in immunologically normal pregnant women. First-trimester infection results in severe disease and third-trimester disease result in milder disease. It presents with a triad of chorioretinitis, hydrocephalus, and intracerebral calcifications.

Option C: Pregnant females infected with cytomegalovirus (CMV) present with infectious mononucleosis-like symptoms, that is fever, pharyngitis, lymphadenopathy, and polyarthritis. Transmission to the fetus increases as pregnancy advances and is maximum in the third trimester. Congenital infection includes growth restriction, microcephaly, intracranial (periventricular) calcifications, chorioretinitis, mental and motor deficits, hepatosplenomegaly, hemolytic anemia, thrombocytopenic purpura. Most infected neonates are asymptomatic at birth and later present with these symptoms. Sensorineural hearing loss is a common sequela of congenital CMV infection.

Other infections which cause congenital malformations are:

Organism	Defects
Toxoplasma	Hydro/microcephaly, convulsions, chorioretinitis, diffuse cerebral calcifications

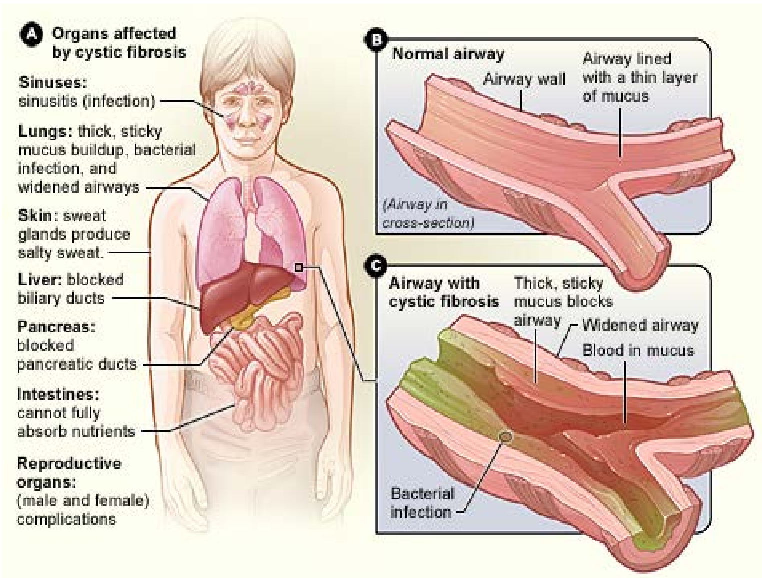
Organism	Defects
Parvovirus	Non-immune hydrops
Syphilis	Hutchinson's triad:Interstitial keratitisHutchinson's teeth Eighth nerve deafness

Solution to Question 6:

The chloride level of sweat, known as the sweat test, is used to diagnose cystic fibrosis (CF). A sweat chloride level of >60 mmol/L is diagnostic of CF.

CF is an autosomal recessive disorder with a mutation in the cystic fibrosis transmembrane regulator (CFTR) gene on chromosome 7q.

The below image depicts the organs affected by CF:



Diagnostic tests include:

1. Immunoreactive trypsinogen for newborn screening of cystic fibrosis.

2. Sweat chloride test: Increased chloride concentration >60 mmol/L is diagnostic.

3. Genetic testing for CFTR gene mutations is the gold standard and should be done for all cases.

4. Nasal transepithelial potential difference can be estimated in equivocal cases.

Diagnostic criteria:

Presence of

• Clinical symptoms affecting at least one organ system (respiratory, gastrointestinal, or genitourinary) or

• Positive newborn screening or

• A sibling with cystic fibrosis.

Evidence of dysfunction of the CFTR gene by

• Two positive sweat chloride test results obtained on separate days or

• Identification of two CFTR mutations by genetic testing or

• An abnormal measurement of nasal potential difference.

Other options:

Option A: Phenylketonuria occurs due to the deficiency of phenylalanine hydroxylase, leading to elevated phenylalanine levels. Diagnosis is confirmed by measuring plasma phenylalanine concentration.

Option C: Gaucher disease is a lysosomal storage disorder caused by a deficiency of β-glucosidase. Diagnosis is confirmed by fibroblast/leukocyte β-glucosidase activity and gene mutations.

Option D: Osteogenesis imperfecta is a genetic connective tissue disorder due to defective type-1 collagen. DNA sequencing is the first diagnostic test.

Solution to Question 7:

The most likely diagnosis in this infant with jaundice, clay-colored stools, and periductal proliferation is biliary atresia.

Biliary atresia presents with progressive jaundice, persistent acholic stools, and palpable liver with abnormal consistency. Laboratory investigations show conjugated hyperbilirubinemia. Ultrasonography reveals a triangular-cord sign (cone-shaped fibrotic mass cranial to the bifurcation of the portal vein).

HIDA test is used for screening and liver biopsy is diagnostic for biliary atresia. Biopsy shows proliferation of the bile ductule, the presence of bile plugs, and portal or perilobular edema and fibrosis, with the basic hepatic lobular architecture intact.

Other options:

Option A: Crigler-Najjar type 2 is an autosomal recessive disease with non-hemolytic unconjugated hyperbilirubinemia caused by homozygous missense mutations in UGT1A1, resulting in reduced (partial) enzymatic activity.

Option B: Rotor syndrome is an autosomal recessive disorder presenting with asymptomatic mild and fluctuating conjugated hyperbilirubinemia with normal liver histology.

Option C: Dubin-Johnson syndrome is an autosomal recessive disorder due to the absent function of multiple drug-resistant protein-2. It presents with abdominal pain, mild icterus, and conjugated hyperbilirubinemia. Biopsy shows black liver due to an accumulation of epinephrine metabolite.

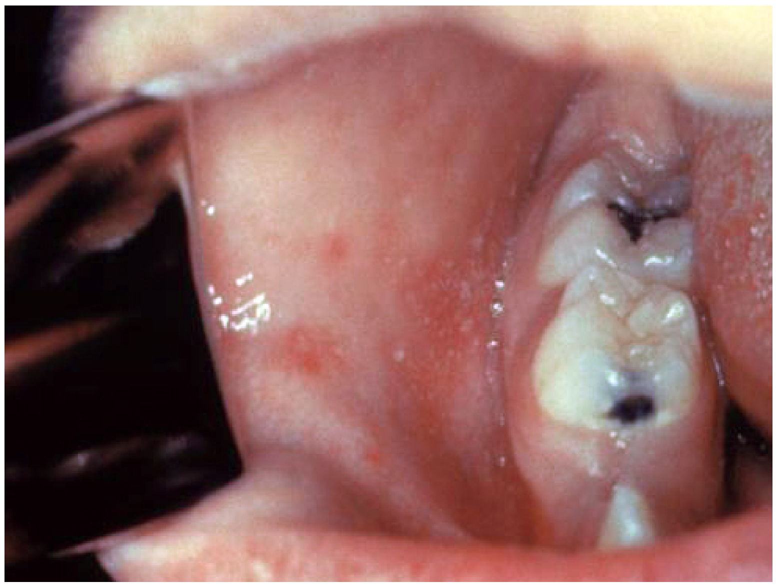
Solution to Question 8:

The likely diagnosis is measles based on the clinical symptoms and the presence of Koplik spots.

Koplik spots are discrete red lesions with central bluish-white spots seen on the buccal mucosa at the level of premolars.

The measles virus has an incubation period of 8-12 days, after which the prodrome phase begins with mild fever, conjunctivitis, upper respiratory tract infection, and pathognomic Koplik spots. This follows the appearance of maculopapular rash after 4th day of fever, which begins at the face and spreads to the torso and extremities. Lymphadenopathy may be present. Treatment is supportive. The most common complication of measles is acute otitis media.

The image below shows Koplik spots.



Other options:

Option B: Rubella presents with low-grade fever, centrifugal macular rash, and prominent lymphadenopathy (suboccipital, postauricular, and anterior cervical lymph nodes). Forchheimer spots (tiny rose-colored lesions on the soft palate) may be present.

The image below shows Forchheimer spots.

Option C: Varicella presents with fever, malaise, and anorexia, followed by a pruritic maculopapular rash, which forms vesicles with a characteristic "cropping" distribution (simultaneous presence of lesions in various stages of evolution).

Option D: Mumps presents with fever, headache, parotitis, and ipsilateral auricular pain. A morbilliform rash may occur.

Solution to Question 9:

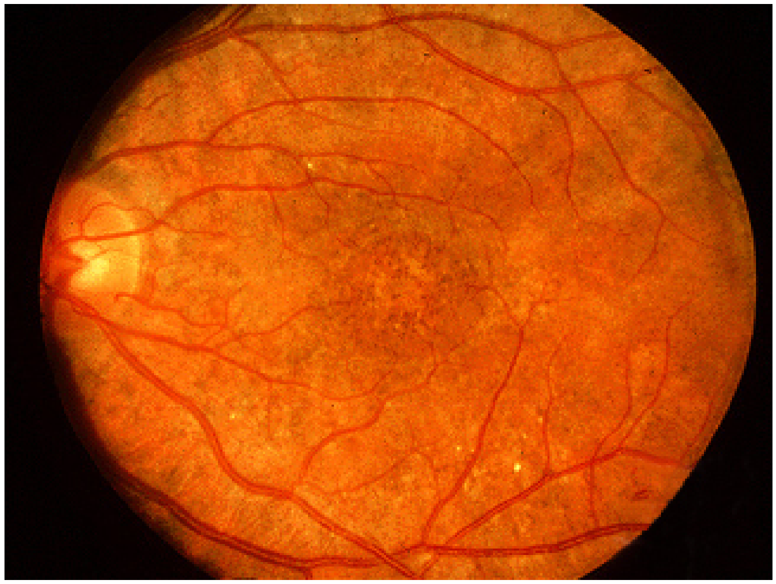
The most likely diagnosis in this infant is congenital cytomegalovirus (CMV) infection.

The clinical manifestations are hepatosplenomegaly, petechial rashes, jaundice, chorioretinitis, and microcephaly. Head neuroimaging shows intracranial periventricular calcifications. One of the long-term sequelae is sensorineural hearing loss. Diagnosis is made by isolating the virus from the urine or blood by polymerase chain reaction (PCR).

There is no definitive treatment for congenital CMV infection. Treatment with ganciclovir is indicated only in patients with progressive neurologic disease and deafness. The images below show periventricular calcifications and a blueberry muffin baby.

Other options:

Option A - Congenital rubella syndrome is due to the rubella virus. The classical triad of congenital rubella syndrome is deafness, cataract, and congenital heart disease. Patent ductus arteriosus is a common cardiac defect. Eye findings also include salt and pepper retinopathy as shown below.



Option B - Herpes simplex infection in neonates presents with skin vesicles, eye findings (chorioretinitis, keratoconjunctivitis), and microcephaly/hydranencephaly. It can also present as localized skin, eye, and mouth disease, encephalitis, or disseminated disease with a sepsis-like presentation.

Option C - Congenital toxoplasmosis is due to Toxoplasma gondii. A classic triad of chorioretinitis, hydrocephalus, and intracerebral calcifications is seen. Blueberry muffin baby appearance is also seen in this. Calcifications occur randomly throughout the brain in congenital toxoplasmosis while periventricular calcification is characteristic of congenital CMV infection.

Solution to Question 10:

The given clinical scenario of paroxysm of cough, apnea, and leukocytosis (>50,000 cells/µL) in a neonate is suggestive of pertussis. Azithromycin is the recommended drug of choice for the treatment of pertussis in neonates (< 1 month of age).

Macrolides are recommended for the treatment of pertussis. Azithromycin is the drug of choice in all age groups, for treatment or postexposure prophylaxis. Post-exposure prophylaxis is given to household close contacts and high-risk contacts (infants, immunocompromised, pregnancy).

Other options:

Option B: Amoxicillin is not used for the treatment of pertussis due to its inefficiency in the nasopharynx.

Option C: Trimethoprim-sulfamethoxazole (cotrimoxazole) is recommended as an alternative for infants >2 months old and children who are allergic to macrolides.

Option D: Clarithromycin is an alternative drug that can be used in infants aged >2 months. In children less than 2 months of age, it is contraindicated due to the risk of kernicterus.

Paediatrics INI-CET 2023

Question 1:

In a baby born with meconium-stained liquor, which of the following are recommended as per Neonatal Resuscitation guidelines?

- **a)** 1, 2, 3, 4
- **b)** 2, 4
- **c)** 3, 4
- **d)** 1, 3, 4

Question 2:

Which of the following has the least risk of perinatal transmission?

- **a)** HSV
- **b)** CMV
- **c)** Rubella
- **d)** Hepatitis B

Question 3:

Which of the following is a new drug approved by the FDA for the treatment of Rett syndrome?

- **a)** Olcegepant
- **b)** Lasmiditan
- **c)** Trofinetide
- **d)** Erenumab

Question 4:

A 6-year-old boy was brought with a history of chronic cough, and foul-smelling stools. His sibling who had similar complaints died a few years back. Sweat chloride test and CFTR gene studies were done. CFTR gene codes for proteins involved in the transport of:

a) Na+

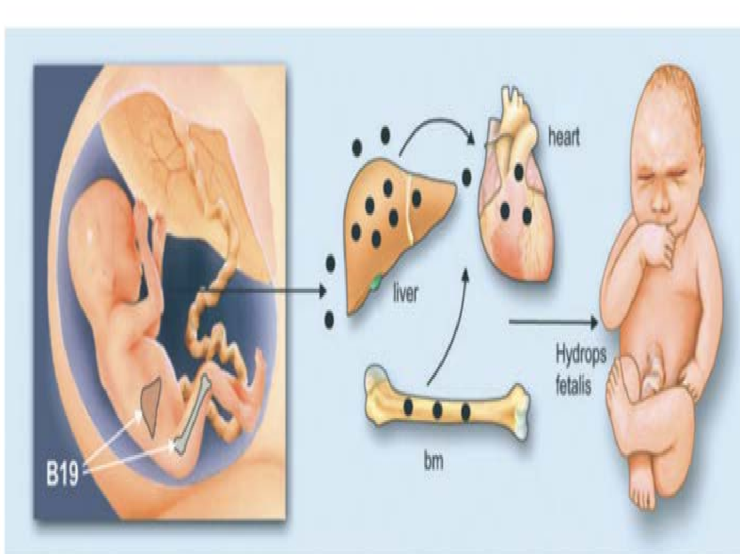
b) HCO3-

c) Cl-

d) K+

Question 5:

The perinatal transmission of infection shows the spectrum of anemia, possibly myocarditis, high output cardiac failure, and ascites. Identify the causative agent.



a) Treponema pallidum

b) CMV

c) Parvovirus B19

d) Toxoplasma gondii

Question 6:

In which of the following conditions is live vaccination not contraindicated?

a) 4 only

b) 1 only

c) 1 and 3

d) 2 and 3

Question 7:

Nada's Minor Criteria includes:

a) 3 and 4

b) 1, 2 and 3

c) 1, 2, 3 and 4

d) 2, 3 and 4

Question 8:

An 18-month-old child with a 3-day history of watery diarrhea and vomiting presented with altered sensorium. Which of the following are differential diagnoses?

a) 1, 2, 3, 4

b) 1, 2, 3

c) 1, 3, 4

d) 2, 3, 4

Question 9:

The causative agent of this condition is associated with which of the following conditions?

a) Gingivostomatitis

b) Pure red cell aplasia

c) Molluscum contagiosum

d) Kaposi sarcoma

Question 10:

A 4-year-old boy presented with a history of fever, conjunctival congestion, and erythematous desquamation of the skin as shown in the images below. A 2D echo shows a coronary aneurysm. What is the first line of treatment for his condition?

a) Aspirin

b) IVIG

c) Clopidogrel

d) LMWH

Question 11:

Which of the following is not associated with Down's syndrome?

a) Conductive hearing loss

b) Short stature

c) Hypothyroidism

d) Caudal regression

Question 12:

Which of the following is correct about congenital CMV?

- **a)** 40-50 % of the cases are symptomatic at birth
- **b)** Optimum sample for PCR is blood/ urine/ saliva sample within 2-3 weeks of birth
- **c)** If asymptomatic at birth, then there is a higher risk of developing SNHL later
- **d)** Most common congenital infection after congenital syphilis

Question 13:

A 3.5 kg term neonate born to a mother with eclampsia, did not cry at birth. Which of the following is the criteria for perinatal asphyxia?

- **a)** APGAR persists at 4-7 for > 5 minutes
- **b)** Cord pH < 7.2
- **c)** Hypocalcemia
- **d)** Hypotonia

Question 14:

Choose the correct statements for cystic fibrosis:

- **a)** i and iii
- **b)** ii and iii
- **c)** i and iv
- **d)** ii and iv

Question 15:

One-day-old term neonate presented with respiratory distress, tachypnea, SpO2-80%, and scaphoid abdomen. Which of the following cannot be used in resuscitation?

- **a)** Positive pressure ventilation
- **b)** Endotracheal intubation
- **c)** Bag and mask ventilation
- **d)** Oxygen suplementation

Question 16:

A child with type 1 DM presents with acidosis, and blood sugar 400 mg/dl. What is the further management?

a) IV insulin stat

b) Normal saline with SC insulin stat

c) Normal saline followed by insulin infusion after 1 hour

d) Normal saline and insulin infusion immediately

Question 17:

A 9-month-old child is brought to the PHC with a fever and cough. She is feeding well and has normal urine output. On examination, a respiratory rate of 48 breaths/min is found and no chest indrawing is seen. What is the next step of management?

a) Explain danger signs and send home

b) No pneumonia - Give amoxyclav + paracetamol and send home

c) Start IV ceftriaxone

d) Refer as the child has severe pneumonia

Answer Key

Question No.	Correct Option
1	c
2	c
3	c
4	c
5	c
6	a
7	a
8	c
9	b
10	b
11	d
12	b
13	d
14	c

15	c
16	c
17	a

Detailed Explanations

Solution to Question 1:

In a baby born with meconium-stained liquor, gentle suction of the nose and mouth is done in the case of a vigorous baby (Statement 3), and positive pressure ventilation (PPV) is done in the case of a non-vigorous baby after early steps (Statement 4).

A baby born through meconium-stained liquor is classified as either a vigorous baby or a non-vigorous baby. A vigorous baby has good breathing and normal tone, whereas, a non-vigorous baby has poor or no breathing with decreased tone. According to neonatal resuscitation guidelines, in a vigorous baby, gentle suction is done and the baby is kept with the mother. In a non-vigorous baby, the initial steps of resuscitation are done and is followed by PPV if the baby is not responding. Routine tracheal suctioning is not recommended (Statement 2).

The response to neonatal resuscitation at various steps is assessed based on the heart rate.

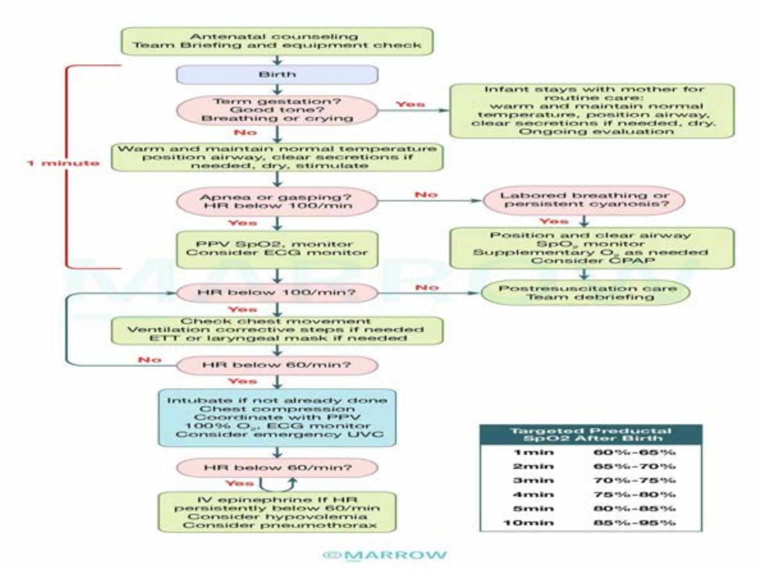
• If the heart rate is less than 100, PPV (positive pressure ventilation) is provided with bag and mask and SPO2 monitoring is done and adequate ventilation is ensured.

• If the heart rate is below 60, then chest compressions are initiated over the lower third of the sternum at a rate of 90 compressions/min.

• Recommended rate of ventilation is 30 inflations/min. (90 compressions + 30 inflation = 120 events/ min). The ratio of compressions to ventilation is 3: 1.

• If the heart rate still remains <60, despite effective compressions and ventilation, then IV epinephrine is given.

Note: For any baby, intrapartum suction of nose and mouth before delivery of shoulder is not recommended. (Statement 1)



Solution to Question 2:

Rubella has the least risk of perinatal transmission.

Rubella or German measles has an intrauterine transmission that causes serious infection in pregnancy which results in abortions or severe fetal defects if infection occurs in the first trimester. The classical triad of congenital rubella syndrome is deafness, cataract, and congenital heart disease.

Other options-

Option A: Herpes simplex infection can be acquired in utero, intrapartum/perinatally, or during the neonatal period. The intrauterine infection causes skin vesicles, eye findings (chorioretinitis, keratoconjunctivitis), and microcephaly/hydranencephaly. Intrapartum infection presents with either:

• Skin, eye, and mouth disease

• Encephalitis

• Disseminated disease with sepsis-like presentation

Option B: Cytomegalovirus, the most common congenital infection with perinatal transmission presents with hepatosplenomegaly, petechial rashes, jaundice, chorioretinitis, and microcephaly. One of the long-term sequelae is hearing loss.

Option D: Hepatitis B virus is associated with significant perinatal transmission. The most important risk factor that determines perinatal transmission is HBeAg.

Solution to Question 3:

Trofinetide is the newly FDA-approved drug for the treatment of Rett syndrome.

The mechanism by which trofinetide exerts therapeutic effects in patients with Rett syndrome is unknown. It is a synthetic analog of the amino■terminal tripeptide of insulin-like growth factor (IGF-1) designed to treat the core symptoms of Rett syndrome by potentially reducing neuroinflammation and supporting synaptic function of the protein fragment glypromate (GPE). IGF-1 is produced by both major types of brain cells – neurons and glia; IGF-1 in the brain is critical for both normal development and response to injury and disease. Trofinetide has been shown to inhibit the production of inflammatory cytokines, inhibit overactivation of microglia and astrocytes, and increase the amount of available IGF-1 that can bind to IGF-1 receptors.

It is Indicated for the treatment of Rett syndrome.

Adverse effects include diarrhea, vomiting, fever, seizure, anxiety, decreased appetite, fatigue, nasopharyngitis

Other options:

Option A: Olcegepant is a calcitonin gene-related peptide (CGRP) antagonist which was considered for the treatment of migraine, but discontinued.

Option B: Lasmiditan is a nontriptan 5HT1F receptor–selective agonist and is used to treat migraine.

Option D: Erenumab is a CGRP (calcitonin gene-related peptide) blocking antibody used in the treatment migraine.

Note: The spelling of the drug in the options has been corrected from Trofinitide to Trofinetide.

Solution to Question 4:

The given clinical scenario is suggestive of cystic fibrosis. In this condition, CFTR gene codes for proteins involved in the transport of Cl- ions.

Cystic fibrosis or mucoviscidosis is an autosomal recessive disorder with a mutation in the cystic fibrosis transmembrane regulator (CFTR) gene on chromosome 7q. The most common CFTR mutation is F508 deletion.

CFTR gene encodes for ATP-gated chloride channels in the apical membranes of epithelial cells. It regulates the flow of ions across epithelial membranes of sweat glands, respiratory airways, pancreatic ducts, etc. The defect leads to high salt concentrations in sweat and viscous luminal secretions in the respiratory and gastrointestinal tracts. Excessive loss of salt in the sweat predisposes young children to salt-depletion episodes. Therefore, these children present with hypochloremic alkalosis, resulting in increased serum bicarbonate levels.

Clinical manifestations/ complications of cystic fibrosis include:

• Sinusitis and nasal polyps

• Respiratory - copious hyper-viscous and adherent pulmonary secretions with abnormal mucociliary clearance obstruct small and medium-sized airways leading to respiratory compromise. The disease can be further complicated by bronchiectasis and allergic bronchopulmonary aspergillosis (ABPA).

• Gastrointestinal -

• Thick and adherent exocrine secretions obstruct pancreatic ducts and impair the production and flow of digestive enzymes

• Tissue destruction of the exocrine pancreas, with fibrotic scarring, cyst proliferation, and loss of acinar tissue - fat malabsorption, steatorrhea, fat-soluble vitamin deficiencies

• Obstruction of intrahepatic bile ducts and parenchymal fibrosis leading to hepatic insufficiency and focal biliary cirrhosis

• Distal intestinal obstructive syndrome in adults and meconium ileus in neonates

• Reproductive - males have infertility due to the congenital absence of bilateral vas deferens and defective sperm transport, resulting in obstructive azoospermia. Females have infertility due to abnormalities of the reproductive tract secretions.

The diagnosis is confirmed if both of the following criteria are met:

• Presence of

• Clinical symptoms affecting at least one organ system (respiratory, gastrointestinal, or genitourinary) or

• Positive newborn screening or

• A sibling with cystic fibrosis

• Evidence of dysfunction of the CFTR gene by

• Two positive sweat chloride test results were obtained on separate days or

• Identification of two CFTR mutations by genetic testing or

• An abnormal measurement of nasal potential difference

Cystic fibrosis requires a multidisciplinary approach to management. The correction of the defect can be via:

• Lumacaftor, an FDA-approved drug that corrects the misfolding and reverts the Phe508del mutation defect

• Ivacaftor potentiates CFTR channel opening and stimulates ion transport

End-stage pulmonary failure requires lung transplantation.

Solution to Question 5:

The given clinical vignette with anemia, myocarditis, high output cardiac failure, ascites and the image showing hydrops fetalis point to a diagnosis of erythema infectiosum (Fifth disease) caused by parvovirus B19.

Parvovirus B19 infects fetal erythroid progenitor cells and causes anemia. Severe anemia leads to high-output cardiac failure and hydrops fetalis. In addition to bone marrow, the liver is also affected causing ascites, because it is involved in erythropoiesis in the early fetal life.

Erythema infectiosum presents as a prodrome of low-grade fever, headache, and a mild upper respiratory tract infection after an incubation period of 4-28 days. A characteristic rash occurs in 3 stages:

• Erythematous facial flushing causing a slapped cheek appearance in the 1st stage

• Diffuse macular erythema spreads to the trunk and proximal extremities in the 2nd stage.

• Central clearing of the macular lesions, giving the rash a lacy, reticulated appearance in the 3rd stage.

The rash is more prominent on extensor surfaces and spares the palms and soles. At this stage, affected children are afebrile and do not appear ill. Children with erythema infectiosum are not likely to be infectious at presentation.

Complications of the disease include:

• Arthralgias or arthritis

• Transient Aplastic Crisis: A transient arrest of erythropoiesis and absolute reticulocytopenia in serum hemoglobin in individuals with chronic hemolytic conditions like sickle cell anemia

• Thrombocytopenic purpura

• Myocarditis

• A papular-purpuric "gloves-and-socks" syndrome

• Neurological: Aseptic meningitis, encephalitis, and peripheral neuropathy, stroke in children with sickle cell disease

• Infection-associated hemophagocytic syndrome in immunocompromised patients

• Maternal infection is associated with nonimmune fetal hydrops and intrauterine fetal demise

There is no data to support the use of IVIG for postexposure prophylaxis, however, it has been used with some success to treat B19-related episodes of anemia and bone marrow failure in immunocompromised children. There is no specific antiviral therapy for B19 infection.

Solution to Question 6:

Live vaccination is not contraindicated in complement deficiency. In DiGeorge syndrome, Wiskott-Aldrich Syndrome, and Ataxia telangiectasia, live vaccines are contraindicated.

In patients with DiGeorge syndrome, Wiskott-Aldrich Syndrome, and Ataxia telangiectasia, there is a risk of severe or fatal infections due to live vaccines because of their impaired T-cell immunity. In contrast, a patient with a complement deficiency has an intact T-cell and B-cell immunity, which are crucial for building an immune response against live vaccines. These patients have an increased susceptibility to certain bacterial infections, especially those caused by Neisseria species, due to defects in the complement system, which is part of the innate immune system. But their adaptive immune system, which is primarily responsible for responding to vaccinations, is generally intact. Therefore, live vaccinations are not typically contraindicated in these patients.

DiGeorge syndrome is due to a developmental defect involving the endodermal derivatives of the 3rd and 4th pharyngeal pouches. Cell-mediated immunity is defective due to decreased T-cells. Affected individuals also have depressed graft rejection, hypocalcemia, and cardiac defects. For severe T cell deficiencies (e.g. severe combined immunodeficiency disease [SCID], complete DiGeorge syndrome), no live viral or live bacterial vaccines should be given.

Wiskott-Aldrich syndrome(WAS) is an X-linked recessive immunodeficiency disorder. It is caused by mutations in the Wiskott-Aldrich syndrome protein (WASp) that primarily affects the T lymphocytes. In patients with partial T cell deficiencies (e.g., partial DiGeorge syndrome, Wiskott-Aldrich syndrome), all live viral vaccines are avoided.

Ataxia telangiectasia is caused due to mutation of the ATM gene. This gene is involved in the repair of double-strand DNA breaks. This defect in DNA repair leads to impaired Ig isotype switching and the formation of antigen receptors. As a result, the patient is immunodeficient with partial T cell deficiency, and hence live vaccines are avoided.

Solution to Question 7:

Nada's Minor Criteria includes abnormal second heart sound and abnormal BP.

Nada's criteria is used to diagnose congenital heart disease in children. The presence of 1 major or 2 minor criteria indicates congenital heart disease.

Major criteria	Minor criteria
Systolic murmur grade 3 or more	Systolic murmur < grade 3
Any diastolic murmur	Abnormal second sound (S2)
Cyanosis	Abnormal ECG
Congestive cardiac failure	Abnormal chest X-ray finding s
	Abnormal BP

Solution to Question 8:

The differential diagnoses in a child with a history of watery diarrhea and altered sensorium include severe dehydration, cerebral venous thrombosis, and hyponatremia.

The lack of dysentery(bloody diarrhoea) and the short period of interval between diarrhoeal episodes and the symptom complex, help to rule out Haemolytic Uremic Syndrome (HUS). HUS is a form of microangiopathic hemolytic anemia usually occurring in children less than 5 years of age. There are 2 forms of HUS:

• Typical HUS which is usually followed by a gastrointestinal infection most commonly by E.coli. The infection in turn affects the gastric mucosa and alters the endothelial cells which promotes platelet activation and aggravation.

• Atypical HUS which occurs due to deficiency in certain proteins like complement factor H, factor I, or CD46.

Clinical features of the condition include bloody diarrhea which occurs 5-10 days before the onset of hemolytic uremic syndrome, abdominal pain, fatigue, cramping, and vomiting. The most dangerous complication is renal failure requiring dialysis or renal transplantation. Treatment is mainly supportive and the nature of the illness is usually self-limiting.

Differential diagnoses in a child presenting with altered sensorium/ seizures with diarrhea:

• Hypo- or hypernatremia

• Hypoglycemia

• Encephalitis or meningitis

• Febrile seizures

• Cerebral or saggital sinus thrombosis

Solution to Question 9:

The given image showing the classical slapped cheek appearance is strongly suggestive of erythema infectiosum. It is caused by Parvovirus. Parvovirus B19 is associated with pure red cell aplasia.

Clinically, parvovirus B-19 infection may have several presentations-

• Erythema infectiosum- Also known as the fifth disease. The incubation period after infection is typically around 1-2 weeks. The hallmark of erythema infectiosum is the characteristic rash, which appears initially as erythematous facial flushing, often described as a “slapped cheek” appearance. Later the rash spreads rapidly or concurrently to the trunk and proximal extremities as diffuse macular erythema. The child may be febrile.

• Transient aplastic crisis- It may be seen in patients with underlying chronic hemolytic anemia like sickle-cell disease, thalassemia, hereditary spherocytosis, and RBC enzyme deficiencies like pyruvate kinase deficiency. Patients with aplastic crisis appear ill with fever, malaise, and pale with tachycardia and tachypnea due to profound anemia. The rash may or may not be present.

• Pure red cell aplasia- Erythrocyte P blood group antigen is the primary cell receptor for the virus. Persistent Parvovirus B19 infection in immunocompromised patients can cause chronic suppression of bone marrow and chronic anemia leading to pure red cell aplasia.

• Maternal infection is associated with non-immune fetal hydrops and intrauterine fetal demise.

• Only about 7% of children with erythema infectiosum have, whereas 80% of adults have joint involvement.

• Infection-associated hemophagocytic syndrome in immunocompromised patients.

The diagnosis of erythema infectiosum is made clinically and does not require investigations routinely. Detection of viral DNA in the blood by PCR or nucleic acid hybridization may be done for immunocompromised cases or for prenatal diagnosis of infection.

It is self-limited and resolves on its own. IVIG may be given to immunocompromised children.

Solution to Question 10:

This child with a history of fever, conjunctival congestion, strawberry tongue, and erythematous desquamation of the skin is likely to have Kawasaki's disease. The first line of treatment in Kawasaki disease is IVIG (intravenous immunoglobulin).

Kawasaki's disease is also known as mucocutaneous lymph node syndrome. It is a vasculitis condition of unknown etiology. Clinical features include a high spiking fever that is unresponsive to antipyretics. The fever can last between 5 days to 3-4 weeks. The five principal clinical criteria for Kawasaki's disease are:

• Bilateral nonexudative conjunctival infection with limbal sparing

• Erythema of the oral and pharyngeal mucosa, strawberry tongue, and red, cracked lips

• Edema and erythema of the hands and feet

• Various forms of rash- maculopapular, erythema multiforme, scarlatiniform

• Nonsuppurative cervical lymphadenopathy

Diagnosis requires fever and at least four of the five principal criteria. Myocarditis develops frequently in acute Kawasaki disease. It presents as tachycardia and reduced ventricular systolic function. The tachycardia will be disproportionate to the fever. Patients may develop coronary artery aneurysms in the 2nd to 3rd week of illness. They are generally asymptomatic and diagnosed on echocardiography. However, if large it can lead to death. Thus, echocardiography should be performed at diagnosis and after 1-2 weeks.

Other symptoms include periungual desquamation, gastrointestinal symptoms (vomiting, abdominal pain, diarrhea), cough, rhinorrhea, and arthritis. Laboratory evaluation will reveal elevated leukocytes, platelets, erythrocyte sedimentation rate, and C-reactive protein levels.

Treatment involves 2g/kg of intravenous immunoglobulin as an infusion over 10-12 hours within 10 days of onset. Moderate to high-dose aspirin should be administered until the patient is afebrile.

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Solution to Question 11:

Among the choices provided, Caudal regression is not associated with Down’s syndrome. It occurs more frequently in infants of diabetic mothers (both preconceptionally and early in pregnancy) than in other infants although the exact mechanism is unclear.

Down’s syndrome is an autosomal disorder. It was described by Langdon Down in 1866. It is also called trisomy 21 or mongolism. The clinical features of Down's syndrome seen in older children are:

• Short stature

• Small round head

• Narrow tilted eye slits

• Mal-formed ears

• Short broad hands

• Lax limbs

• Mental retardation

Conditions most commonly associated with Down’s syndrome:

• Congenital heart disease (in about 35% of cases)

• Gastrointestinal anomalies (in about 10%)

• Chronic serous otitis media (in >50%)

• Conductive hearing loss

• Hypothyroidism

• Alzheimer’s disease (in the 30's and 40’s)

• Epilepsy (in about 10%)

• Ocular disorders

• Reduced fertility

• Reduced life span (Often due to complications like infections)

The frequency of Down's syndrome increases with rising maternal age. The risk of a woman who is 20 is 1 in 3000 but that of a woman who is 45 is 1 in 50.

It seems to be unaffected by the age of the father.

Note: Genetic counseling: It can either be prospective or retrospective. Prospective genetic counseling refers to the prevention of disease by the prevention or reduction of heterozygous marriage like in sickle cell anemia or thalassemia. Retrospective genetic counseling means that the hereditary disorder has already occurred in the family and methods like contraception, pregnancy termination or sterilization are to be used to avoid any serious health disorder in the unborn baby.

Methods of early diagnosis and treatment are early detection of genetic carriers, prenatal diagnosis, screening of newborn infants, and recognizing pre-clinical cases.

Solution to Question 12:

The correct statement regarding CMV infection is optimum sample for PCR is blood/ urine/ saliva sample within 2-3 weeks of birth

CMV is the most common congenital infection (Option D). The clinical manifestations are hepatosplenomegaly, petechial rashes, jaundice, chorioretinitis, and microcephaly. One of the long-term sequelae is sensorineural hearing loss. Neuroimaging shows intracranial periventricular calcifications. Other manifestations include IUGR, hydrops, microcephaly, pneumonitis, chorioretinitis, seizures, abnormal dentition, and hypocalcified enamel.

Only 10% of the infected babies are symptomatic (Option A). Asymptomatic neonates have lesser chances of having late sequelae i.e. sensorineural hearing loss. About half of the symptomatic infants develop long-term sequelae which include sensorineural hearing loss (SNHL), vision loss, seizure disorders, mental retardation, and developmental delay. However, in asymptomatic infants, only about 13.5% develop long-term sequelae which is commonly SNHL (option C).

Diagnosis can be made by the presence of IgM antibodies in the fetal blood. Isolation of the virus from the urine or blood by PCR is more sensitive. The investigation of choice is PCR or viral culture. It should be done within the first three weeks of life. After this period, it is difficult to distinguish between a congenital and an acquired infection. Since not all affected infants are viremic at birth, urine and/or saliva are more reliable samples than blood (Option B).

There is no definitive treatment for congenital CMV infection. Treatment with ganciclovir is indicated only in patients with progressive neurologic disease and deafness. Below is the image showing the blueberry muffin baby.

Solution to Question 13:

Among the given options, the essential criterion for perinatal asphyxia is hypotonia.

Perinatal asphyxia is an insult to the fetus or newborn due to a lack of oxygen (hypoxia) and/or a lack of perfusion (ischemia) to the various organs. It is often associated with tissue lactic acidosis and hypercarbia.

Essential criteria for perinatal asphyxia:

• Umbilical arterial blood sample indicating prolonged metabolic or mixed acidemia (pH < 7.O).

• Apgar score of 0-3 persisting for >5 minutes.

• Neurological manifestations appearing in the immediate neonatal period Ex:- seizures, coma, hypotonia, or hypoxic ischemic encephalopathy (HIE).

• Neonatal multiorgan dysfunction.

Solution to Question 14:

In cystic fibrosis, there are increased chloride levels in sweat and decreased Chloride levels in pancreatic juice. The chloride level of sweat, known as the sweat test, is used to diagnose cystic fibrosis (CF). A sweat chloride level of >60 mmol/L is diagnostic of CF.

Cystic fibrosis or mucoviscidosis is an autosomal recessive disorder with a mutation in the cystic fibrosis transmembrane regulator (CFTR) gene on chromosome 7q. The most common CFTR mutation is F508 deletion.

CFTR gene encodes for ATP-gated chloride channels in the apical membranes of epithelial cells. The function of the CFTR gene on chloride channels is organ-specific. In sweat glands, the normal function of the CFTR gene is to reabsorb chloride ions. But in the pancreatic ducts and respiratory airways, the normal function of the CFTR gene is to secrete chloride ions. So, in cystic fibrosis, due to defect in the chloride channels, chloride levels in the sweat increase, and chloride levels in the pancreatic juices decreases. Excessive loss of salt in the sweat predisposes young children to salt-depletion episodes. Therefore, these children present with hypochloremic alkalosis, resulting in increased serum bicarbonate levels.

Clinical manifestations/ complications of cystic fibrosis include:

• Sinusitis and nasal polyps

• Respiratory - copious hyper-viscous and adherent pulmonary secretions with abnormal mucociliary clearance obstruct small and medium-sized airways leading to respiratory compromise. The disease can be further complicated by bronchiectasis and allergic bronchopulmonary aspergillosis (ABPA).

• Gastrointestinal -

• Thick and adherent exocrine secretions obstruct pancreatic ducts and impair the production and flow of digestive enzymes

• Tissue destruction of the exocrine pancreas, with fibrotic scarring, cyst proliferation, and loss of acinar tissue - fat malabsorption, steatorrhea, fat-soluble vitamin deficiencies

• Obstruction of intrahepatic bile ducts and parenchymal fibrosis leading to hepatic insufficiency and focal biliary cirrhosis

• Distal intestinal obstructive syndrome in adults and meconium ileus in neonates

• Reproductive - males have infertility due to the congenital absence of bilateral vas deferens and defective sperm transport, resulting in obstructive azoospermia. Females have infertility due to abnormalities of the reproductive tract secretions.

The diagnosis is confirmed if both of the following criteria are met:

• Presence of

• Clinical symptoms affecting at least one organ system (respiratory, gastrointestinal, or genitourinary) or

• Positive newborn screening or

• A sibling with cystic fibrosis

• Evidence of dysfunction of the CFTR gene by

• Two positive sweat chloride test results were obtained on separate days or

• Identification of two CFTR mutations by genetic testing or

• An abnormal measurement of nasal potential difference

Cystic fibrosis requires a multidisciplinary approach to management. The correction of the defect can be via:

• Lumacaftor, an FDA-approved drug that corrects the misfolding and reverts the Phe508del mutation defect

• Ivacaftor potentiates CFTR channel opening and stimulates ion transport

End-stage pulmonary failure requires lung transplantation.

Solution to Question 15:

The given clinical scenario of a one-day term-old neonate presented with respiratory distress, tachypnea, SpO2-80%, and scaphoid abdomen is suggestive of congenital diaphragmatic hernia. Bag and mask ventilation is to be avoided in the management of this baby. Bag and mask ventilation may increase abdominal distension and limit the expansion of the hypoplastic lungs.

Congenital diaphragmatic hernia (CDH) is a birth defect of the diaphragm. Malformation of the diaphragm allows the abdominal organs to push into the chest cavity, hindering proper lung formation and resulting in pulmonary hypoplasia. The most common type of CDH is a Bochdalek hernia and in most cases involves the left side. CDH is a life-threatening pathology in infants and a major cause of morbidity and mortality due to two complications: pulmonary hypoplasia and pulmonary hypertension.

Babies with CDH present with respiratory distress characterized by tachypnea, grunting, use of accessory muscles, and cyanosis. They may also have a scaphoid abdomen and increased chest wall diameter. Auscultation of the chest may reveal bowel sounds with reduced breath sounds.

Diagnosis: A chest radiograph and nasogastric tube passage can confirm the diagnosis of CDH. Prenatal diagnosis can be made using ultrasonography between 16 to 24 weeks of gestation. High-speed fetal magnetic resonance imaging (MRI) can further define the lesion.

Management of CDH:

The immediate step in the management is the naso- or orogastric tube placement for decompression. The airway is to be secured with endotracheal intubation and respiratory distress is stabilised.

Inhaled nitric oxide is used widely for its pulmonary vasodilatory effect. Conventional mechanical ventilation, high-frequency oscillatory ventilation (HFOV), and extracorporeal membrane oxygenation (ECMO) are the 3 main strategies to support respiratory failure in newborn with CDH.

Surgical repair of the defect is to be considered at least 48 hours after stabilization and resolution of pulmonary hypertension. During surgery, the viscera are reduced into the abdominal cavity, and the posterolateral defect in the diaphragm is closed using interrupted, nonabsorbable sutures.

Solution to Question 16:

A given clinical scenario of a child with type 1 diabetes mellitus (DM) presenting with acidosis and a high blood sugar level (hyperglycemia) describes a scenario suggestive of diabetic ketoacidosis (DKA), a serious complication of uncontrolled diabetes. the further management would be Normal saline followed by insulin infusion after 1 hour.

The International Society for Pediatric and Adolescent Diabetes (ISPAD) guidelines provide evidence-based recommendations for the management of DKA (diabetic ketoacidosis) in children and adolescents.

Diabetic ketoacidosis is the most serious complication of type 1 diabetes mellitus. Clinical manifestations of diabetic ketoacidosis include:

• Dehydration, tachypnea with deep, sighing respiration (Kussmaul breathing)

• Nausea, vomiting, and abdominal pain that may mimic an acute abdominal condition

• Confusion, drowsiness, and coma in severe cases.

The diagnosis of DKA is based on the triad of hyperglycemia, ketosis, and metabolic acidosis. All three biochemical criteria are required to diagnose DKA

• Hyperglycemia (blood glucose >200 mg/dl)

• Venous pH < 7.3 or serum bicarbonate <18 mmol/L

• Ketonemia or ketonuria

Initial management includes management of the airway, breathing, and circulation (ABC) according to Pediatric Advanced Life Support (PALS) guidelines. Following that emergency assessment is done, which includes:

• Immediate measurement of blood glucose, blood or urine ketones, serum electrolytes and blood gases

• Assessment of level of consciousness.

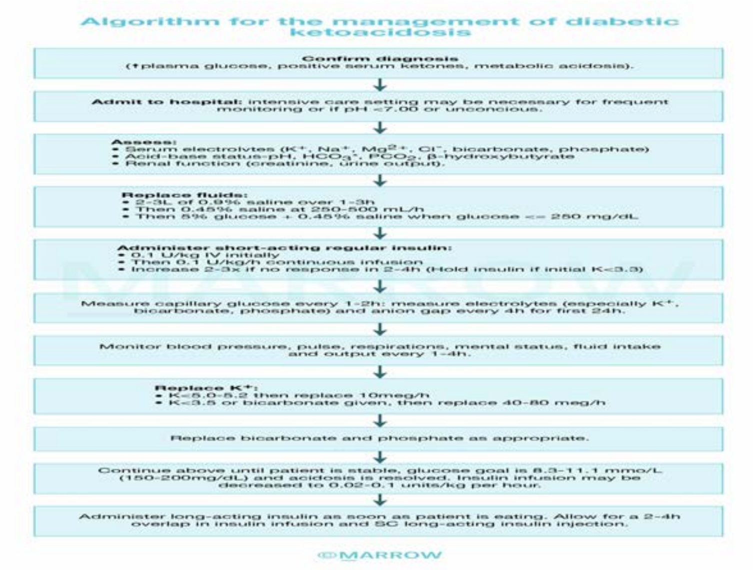
• Two peripheral intravenous (IV) lines should be inserted.

The goal of therapy is to correct dehydration, correct acidosis, reverse ketosis, and gradually blood glucose concentration to near normal. Monitor for acute complications, and identify and treat any precipitating event.

Fluid replacement should always begin before starting insulin therapy. One or more boluses of 0.9% saline at 10-20 mL/kg are given to restore peripheral circulation. Insulin therapy with 0.05–0.1 U/kg/hr must begin at least 1 hour after starting fluid replacement therapy. Insulin of 0.05 U/kg/hr can be considered if pH is > 7.15.

Potassium replacement is given only when potassium is <5.5 mmol/L. In rare cases of hypokalemia with potassium <3.0 mmol/L, defer insulin treatment.

Bicarbonate administration is not recommended except for treatment of severe acidosis (pH < 6.9) or in cases of life-threatening hyperkalemia.



Solution to Question 17:

In the given clinical scenario, a 9-month-old child with normal breathing (> 50/min RR for infants 2 months to 12 months is considered as fast breathing) and no chest indrawing is seen. So, the next step in management would be to explain danger signs and send them home.

According to the IMNCI, three key clinical signs are used to assess a sick child with suspected pneumonia:

• Respiratory rate (RR): distinguishes children who have pneumonia from those who do not. Children are classified under pneumonia if:

• > 60 breaths per minute if age is < 2 months

• > 50/min RR for infants 2 months to 12 months

• > 40/min RR for children 12 months to 5 years of age

• Lower chest wall indrawing refers to the indrawing of the lower chest wall with respect to the child's breathing and indicates pneumonia

• Stridor indicates those with severe pneumonia requiring hospital admission.

The revised classification of acute respiratory infections is simplified to include only two categories of pneumonia:

• Cough and cold, no pneumonia - requires home remedy

• Pneumonia - with fast breathing and/or chest indrawing, which requires home therapy with oral amoxicillin.

• Severe pneumonia - pneumonia with any general danger sign,
which requires a referral and injectable therapy.

Danger signs of severe pneumonia are:

• Not able to drink

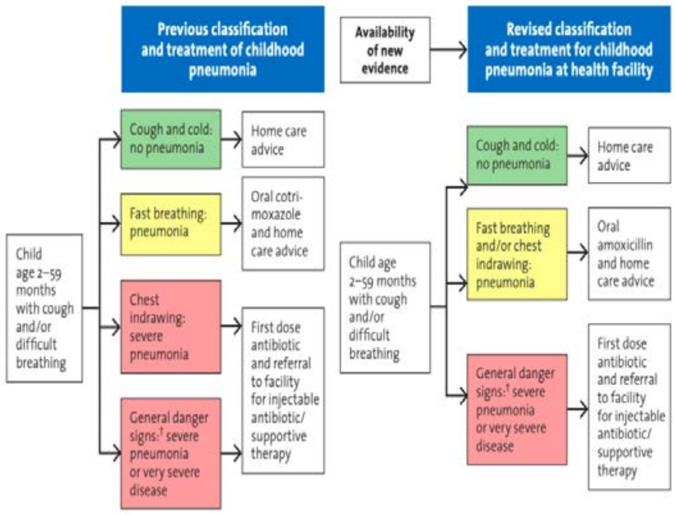
• Persistent vomiting

• Convulsions

• Lethargic or unconscious

• Stridor in a calm child

• Severe malnutrition



† Not able to drink, persistent vomiting, convulsions, lethargic or unconscious, stridor in a calm child or severe malnutrition.

Paediatrics INI-CET 2024

Question 1:

Which of the following is not seen in Kawasaki disease?

- **a)** Purulent conjunctivitis
- **b)** Strawberry tongue
- **c)** Rash
- **d)** Cervical lymphadenopathy

Question 2:

What is the ideal maintenance fluid of choice in children?

- **a)** 5% dextrose and N/2 saline
- **b)** 5% dextrose and N saline
- **c)** 10% dextrose and N/2 saline
- **d)** 5% dextrose and N/4 saline

Question 3:

Which of the following is not seen in nutritional rickets?

- **a)** Low phosphorus
- **b)** Normal calcium
- **c)** Urine calcium: creatinine ratio > 15
- **d)** High PTH

Question 4:

Which of the following is CORRECT regarding febrile seizure

- **a)** A,B,C
- **b)** A,C,D
- **c)** A,B,C,D

d) A,B,D

Question 5:

What is the rate of increase in head circumference in the first 3 months of life?

a) 1 cm/month

b) 2 cm/month

c) 4 cm/month

d) 0.5 cm/month

Question 6:

Which of the following holds true for "some dehydration" in a child:

a) 1, 3, 5 only

b) 1, 2, 4 only

c) 1, 3, 4, 5 only

d) 1, 2, 3, 4

Question 7:

Which of the following is not a suitable method for free flow oxygen delivery to a newborn baby?

a) Oxygen tube kept close to the baby's nose and mouth

b) Oxygen mask placed firmly over the baby's mouth and nose

c) Self-inflating bag attached to an oxygen mask placed over baby's mouth and nose

d) A reservoir bag attached to oxygen mask kept over the baby's nose and mouth

Question 8:

All are true about renal tubular acidosis in children except

a) Type 4 is due to defective ammoniagenesis and aldosterone resistance

b) Type 2 is Fanconi and hypophosphatemic rickets

c) Type 4 with paradoxical aciduria and hypokalemia

d) Type 2 - abnormal loss of HCO_3

Question 9:

Which of the following is a reliable sign of "some dehydration" in severe acute malnutrition?

a) Increased thirst

b) Sunken eyes

c) Skin pinch that goes back slowly

d) Altered sensorium

Question 10:

A 5-year-old has the following anthropometry values. According to WHO criteria for malnutrition, what is the diagnosis?

a) Severe acute malnutrition

b) Chronic malnutrition

c) Moderate acute malnutrition

d) Acute on chronic malnutrition

Question 11:

Downe's score includes which all of the following?

a) 1,2,3

b) 2,3

c) 1,2,4

d) 2, 4

Question 12:

A 15 year old boy presents with short stature. On examination, his upper segment/lower segment ratio is 0.8:1. Which of the following is the likely diagnosis?

a) Spondyloepiphyseal dysplasia

b) Achondroplasia

c) Growth hormone deficiency

d) Hypothyroidism

Question 13:

Which of the following MRI findings will be seen in a child with spastic quadriplegia?

a) Periventricular leukomalacia

b) Multicystic encephalomalacia

c) Cerebellar degeneration

d) Basal ganglia calcification

Question 14:

A 10 year boy presents to you with sexual maturity rating 5, height for age > chronological age, hyperpigmentation and short stature. His blood pressure measured 180/110mmHg. Which of the following enzymes deficiency is responsible for this condition?

a) 11 beta hydroxylase

b) 21 alpha hydroxylase

c) 3 beta dehydrogenase

d) Cholesterol side chain cleavage enzyme

Question 15:

INSURE is associated with which of the following conditions?

a) Respiratory distress in new born

b) Birth Asphyxia

c) Neonatal jaundice

d) Transient tachypnea of newborn

Question 16:

All of the following are included under diagnostic criteria of severe acute malnutrition except

a) Height for age

b) Weight for height

c) Pedal edema

d) MUAC

Question 17:

In children diagnosed with congenital heart disease, NADA's criteria include all the following, except?

a) Diastolic murmur

b) Increased troponin-T

c) Abnormal X-ray

d) Congestive cardiac failure

Question 18:

Which of the following is false regarding neonatal cholestasis?

a) Kasai procedure should be done within 90 days

b) HIDA scan is an investigation of choice

c) Biliary atresia is the most common cause

d) Liver biopsy rules out other causes

Question 19:

What causes the closure of ductus arteriosus?

a) Increase in Oxygen

b) Decrease in CO_2

c) Increase in pulmonary vascular resistance

d) Fall in concentration of catecholamines

Question 20:

A 4-year-old child was diagnosed with croup. Which of the following is the most common causative organism?

a) Hemophilus influenza

b) Streptococcus pneumonia

c) Influenza virus

d) Diphtheria

Question 21:

What would be the age of a child who copies crosses, tell stories, and goes to the toilet alone?

a) 3 years

b) 4 years

c) 5 years

d) 2 years

Question 22:

Which of the following is not true about breastmilk jaundice?

a) Breastfeeding need not be stopped for the treatment

b) Most of the children require phototherapy

c) Exclusive breastfeeding with a total bilirubin of more than 10mg/dl after 3-4 weeks of birth

d) Unconjugated hyperbilirubinemia

Question 23:

Which of the following is not a feature of cystic fibrosis?

a) Nasal polyposis

b) Biliary atresia

c) Meconium ileus

d) Bleeding diathesis

Question 24:

A 2-week-old infant presents with a history of epistaxis and recurrent rectal bleeding. The parents also reported a history of continuous bleeding from the umbilicus since birth. Laboratory investigations revealed normal prothrombin time, normal thrombin time, normal partial thromboplastin time, normal platelet count, and normal function. The urea clot solubility test was positive. What is the most likely deficient factor in this patient?

a) Factor XIII

b) Factor XI

c) Factor X

d) Factor XII

Question 25:

Which of the following indicates the primary failure of recovery in an admitted patient with severe acute malnutrition?

a) i,ii,iii

b) i,ii,iv

c) i,iii,iv

d) ii,iii,iv

Answer Key

Question No.	Correct Option
1	a
2	b
3	c
4	b
5	b
6	c
7	c
8	c
9	a
10	b
11	c
12	a
13	b
14	a

15	a
16	a
17	b
18	a
19	a
20	c
21	b
22	b
23	b
24	a
25	a

Detailed Explanations

Solution to Question 1:

Purulent conjunctivitis is not seen in Kawasaki's disease. Non-purulent conjunctivitis is a feature seen.

Kawasaki disease is also known as mucocutaneous lymph node syndrome. It is a vasculitis condition of unknown etiology. Clinical features include a high spiking fever that is unresponsive to antipyretics. The fever can last between 5 days to 3-4 weeks. The five principal clinical criteria for Kawasaki's disease are:

• Bilateral non-purulent conjunctival infection with limbal sparing

• Erythema of the oral and pharyngeal mucosa, strawberry tongue (option B), and red, cracked lips

• Edema and erythema of the hands and feet

• Various forms of rash- maculopapular, erythema multiforme, scarlatiniform (option C)

• Nonsuppurative cervical lymphadenopathy (option D)

The image below shows bilateral non-purulent conjunctival redness, strawberry tongue, and edema and erythema of the hands:

Diagnosis requires fever and at least four of the five principal criteria. Myocarditis develops frequently in acute Kawasaki disease. It presents as tachycardia and reduced ventricular systolic function. The tachycardia will be disproportionate to the fever. Patients may develop coronary artery aneurysms in the 2nd to 3rd week of illness. They are generally asymptomatic and diagnosed on echocardiography. However, if large it can lead to death. Thus, echocardiography should be performed at diagnosis and after 1-2 weeks.

Treatment involves 2g/kg of intravenous immunoglobulin as an infusion over 10-12 hours within 10 days of onset. Moderate to high-dose aspirin should be administered until the patient is afebrile.

Solution to Question 2:

5% dextrose and N saline is the ideal maintenance fluid in children.

5% dextrose provides 17 calories/100 mL and nearly 20% of the daily caloric needs. This level is enough to prevent ketone production and helps minimise protein degradation. However, the child will lose weight on this regimen and therefore it is necessary to start enteral feeds sooner or if not possible, total parenteral nutrition is recommended.

Other options:

Hypotonic fluids like N/2 or N/4 saline can increase the risk of hyponatraemia and free water movement inside RBCs, leading to haemolysis. Hence, isotonic fluids with 5% dextrose are recommended as standard maintenance fluids in children.

Solution to Question 3:

Urine calcium: creatinine ratio > 15 is not seen in a case of nutritional rickets. Instead low urine calcium and high urinary phosphate excretion are seen due to stimulation of PTH(parathormone) in vitamin D deficiency(nutritional rickets).(Option C)

Initially, low calcium levels in the blood stimulate increased PTH secretion from the parathyroid glands that results in bone resorption and increased reabsorption of calcium and increased excretion of phosphorus in the kidneys.

Hence blood calcium levels can be low to normal with low phosphorus levels and high ALP due to increased bone resorption activity.

Note - High urinary calcium excretion is associated with hereditary hypophosphatemic rickets with hypercalciuria.

Nutritional rickets is a skeletal disorder primarily caused by a deficiency in vitamin D, calcium, or phosphorus, or a combination of these nutrients. Rickets typically affect children during periods of rapid growth and bone development.

25-hydroxy vitamin D level should be measured to diagnose vitamin D deficiency rickets. The initial lab tests in this child should also include serum calcium, phosphorous, alkaline phosphatase (ALP), and parathyroid hormone (PTH). Lab findings reveal decreased calcium and phosphate levels, increased PTH, and ALP.

Treatment with oral vitamin D is preferred. Patients who show no response should be evaluated for other disorders, such as hypophosphatemic rickets, vitamin D-dependent rickets, renal tubular acidosis-associated rickets, and chronic kidney disease-associated rickets.

Solution to Question 4:

Chance of recurrence after one episode of febrile seizure is not 54%

Febrile seizures are seizures that-

• Occur between the age of 6 months and 5 years

• Occur at a temperature of 38°C (100.4°F) or higher

• Are not due to any CNS infection or any metabolic imbalance

• Occur without a history of prior afebrile seizures

• Occur with no post-ictal neurological deficit.

Types of febrile seizures:

Investigations:

• Lumbar puncture should be performed in the first episode of seizures with fever, for all infants <6 months age (to exclude meningitis).

• EEG and neuroimaging are advised for children with atypical febrile seizures.

Treatment:

Antipyretics and proper hydration. Midazolam or diazepam (rectal) can be used if seizures present.

Prophylaxis:

• Intermittent prophylaxis: If the febrile seizures are 3 or more in 6 months, or 4 or more seizures in 1 year, oral clobazam (1mg/kg/day) for 3 days during the fever episode.

• Continuous prophylaxis: If intermittent prophylaxis fails or in recurrent atypical febrile seizures, sodium valproate and phenobarbitone can be used.

Simple febrile seizures	Complex (atypical) febrile sei zures
Occurs within 24 hours of th e onset of fever	Can occur at any time during a fever
Generalized seizures	Focal seizures
Duration is <15 min	Prolonged
No recurrence within 24 hrs of seizures	Recurrent
Chances of going into epileps y 1%	Chances of going into epileps y 6%

Solution to Question 5:

Rate of increase of head circumference in the first 3 months of life is 2 cm/month.

The maximum circumference of the head from the occipital protuberance to the supraorbital ridges on the forehead is recorded using the cross-tape method.

Head circumference increases at the following rates:

• 2 cm/month in first 3 months

• 1 cm/month in the next 3 months

• 0.5 cm/month in the next 6 months

Head circumference is equal to chest circumference at 1 year of age.

Head circumference is more than chest circumference by 2.5 cm at birth.

• At birth: Head circumference > Chest circumference

• At 9-12 months: Head circumference = Chest circumference

• After 1 year: Head circumference < Chest circumference

Age	Head circumfere nce
At bir th	33 to 35cm
3 m	40 cm
6 m	43 cm
1 year	46 cm

Age	Head circumference
2 years	48 cm
3 years	49 cm
4 years	50 cm
12 years	52 cm (adult size)

Solution to Question 6:

Increased thirst, delayed skin pinch, sunken eyes, and reduced urine output indicate some dehydration. Lethargy indicates severe dehydration.

The degree of dehydration is assessed as follows:

Based on the degree of dehydration after history and examination, the estimated fluid loss is calculated as follows:

Parameters	No dehydration	Some dehydration	Severe dehydration
Condition	Well alert	Restless, irritable	Lethargic or unconscious,floppy
Eyes	Normal	Sunken	Very sunken and dry
Tears	Present	Absent	Absent
Mouth and tongue	Moist	Dry	Very dry
Thirst	Drinks normallynot thirsty	Thirsty, drinks eagerly	Drinks poorly or isnot able to drink
Skin pinch	Goes back quickly	Goes back slowly	Goes back very slowly
Urine output	Normal	Decreased	Oliguria, anuria
Capillary refill	2-3 seconds	3-4 seconds	>4 seconds
Treatment	Plan A	Weigh the patient, if possible, and use treatment Plan B	Weigh the patient and use treatmentPlan C urgently

Degree of dehydration	Assessment of fluid loss
No dehydration	less than 50 mL/kg
Some dehydration	50-100 mL/kg
Severe dehydration	more than 100 mL/kg

Solution to Question 7:

Self-inflating bag is not used for free flow oxygen delivery. It is used to provide positive pressure ventilation. Self-inflating bag and mask is used in neonatal resuscitation.

Bag and mask ventilation is used as the first step in airway management to provide positive pressure ventilation. It serves as a rescue technique to provide oxygenation and ventilation in patients with difficult airways until tracheal intubation is achieved.

The bag in this apparatus is a self-inflating bag. This allows for it to be used without a pressurized gas supply and is delivered to the patient when the bag is squeezed (compressed). A patent airway and a good seal are essential to facilitate bag and mask ventilation.

The rescuer should be positioned at the patient’s head. A one-person technique involves the "E-C seal" in which your first and second digits form a "C" over the mask with your thumb pressing down by the nasal bridge, your second digit over the bottom of the mask by the mouth, and your third through fifth digits forming an "E" and applying pressure to the mandible to hold the mask tight

Effective ventilation can be confirmed by:

• Chest rise on squeezing the bag

• Condensation in the clear mask

• Pulse oximetry shows improved saturation

• Capnography shows exhaled carbon dioxide

Mask ventilation is usually contraindicated in patients with an increased risk of regurgitation. It should be used with caution in patients with severe facial trauma.

The following image shows the parts of a self-inflating bag and mask apparatus with a reservoir bag.

Solution to Question 8:

Hyperkalemia and metabolic acidosis are features of type 4 renal tubular acidosis

Type IV (hyperkalemic) renal tubular acidosis: Hyperkalemia with distal RTA occurring due to aldosterone resistance or deficiency is termed type 4 RT A. Aldosterone directly stimulates the proton pump, increases Na+ absorption resulting in negative intrahibular potential and increases urinary K+ losses and stimulates basolateral Na+ /K+ ATPase. Hence, aldosterone deficiency or resistance is expected to cause hyperkalemia and acidosis.

The clinical features seen in Type IV RTA are as follows:

• Hyperkalemia: Hypoaldosteronism results in an increase in potassium concentration.

• Metabolic acidosis: The important buffers in urine are ammonia and phosphate. Hyperkalemia impairs ammonia production in the proximal tubule, which reduces the buffering of hydrogen ions. This results in decreased hydrogen ion excretion.

• Urinary pH<5.5: Due to the decreased production of ammonia in the proximal tubule, excess protons are excreted in the renal tubules, thus decreasing the urinary pH.

RTA type I (distal RTA) is seen in autoimmune diseases like Sjogren's syndrome, systemic lupus erythematosus, systemic sclerosis, medullary sponge kidney disease, cirrhosis, drugs like amphotericin B, lithium, etc. The patients develop hypokalemia and metabolic acidosis with alkaline urine (urinary pH >5.5).

RTA type II (proximal RTA) is seen in Fanconi syndrome, Wilson's disease, heavy metals ingestion, etc., The patients develop hypokalemia and metabolic acidosis with acidic urine (urinary pH <5.5). It occurs due to generalized proximal tubule dysfunction. Defective HCO3- resorption is the primary dysfunctional mechanism causing metabolic acidosis and is also associated with glycosuria, aminoaciduria, and phosphaturia. Rickets is rare in isolated proximal RTA but common in Fanconi syndrome.

RTA type III (mixed RTA) is seen in inherited mutations of carbonic anhydrase II. A combined pathology of type I and type II RTA is seen in this type. It results in hypokalemia and metabolic acidosis.

Solution to Question 9:

Increased thirst is the best indicator of 'some dehydration' in severe acute malnutrition.

Increased thirst is the most reliable indicator because impaired skin pinch and sunken eyes can also be seen due to loss of subcutaneous fat in a child with acute malnutrition. Altered sensorium is a feature of severe dehydration.

The degree of dehydration is assessed as follows:

Based on the degree of dehydration after history and examination, the estimated fluid loss is calculated as follows:

Parameters	No dehydration	Some dehydration	Severe dehydration
Condition	Well alert	Restless, irritable	Lethargic orunconscious,flop py
Eyes	Normal	Sunken	Very sunken and dry
Tears	Present	Absent	Absent
Mouth and ton gue	Moist	Dry	Very dry
Thirst	Drinks normallynot thirst y	Thirsty, drinks eagerly	Drinks poorly or isnot able to drink
Skin pinch	Goes back quickly	Goes back slowly	Goes back very slowly
Urine output	Normal	Decreased	Oliguria, anuria
Capillary refill	2-3 seconds	3-4 seconds	>4 seconds
Treatment	Plan A	Weigh the patient, if possible, and use treatment Plan B	Weigh the patient and use tre atmentPlan C urgently

Degree of dehydrati on	Assessment of fluid loss
No dehydration	less than 50 mL/kg
Some dehydration	50-100 mL/kg
Severe dehydration	more than 100 mL/kg

Solution to Question 10:

According to WHO criteria for malnutrition, the diagnosis of this child is chronic malnutrition. (Option B)

Height for age less than -2 SD (-2.5 in this case) indicates moderate stunting, pointing towards a diagnosis of chronic malnutrition.

Weight for age less than -2 SD (-3.2 in this case) indicates underweight but it cannot differentiate between wasting and stunting.

Weight for height:

• less than - 3SD suggests severe acute malnutrition (SAM)

• between -2 to -3 SD suggests moderate acute malnutrition

Weight for height in this case is -1.7 SD. Hence it is not a case of SAM or moderate acute malnutrition.

Solution to Question 11:

Respiratory rate, cyanosis and grunting are a part of Downe's score. Feeding is not a part of it.

Downes’ Score is a comprehensive scoring system and can be applied to any gestational age and condition. Scoring is to be done every half an hour and charting is carried out to monitor progress.

Downe’s score of >6 means impending respiratory failure.

Score	0	1	2
Respiratory rate	<60/min	60–80/min	>80/min
Air entry	Normal	Mild decrease	Marked decrease
Cyanosis	Nil	In-room air	On >40% O_2
Grunting	None	Only on auscultation	Audible with the naked ear
Retraction	Nil	Mild	Moderate

Solution to Question 12:

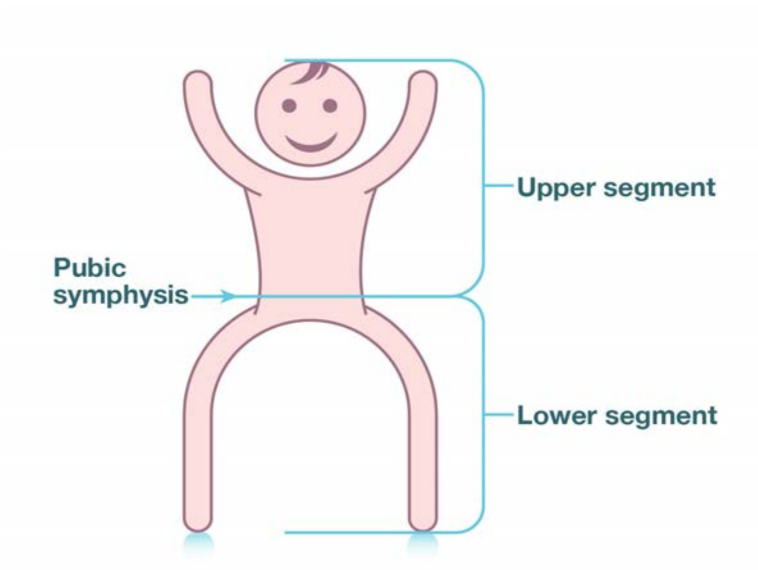
Upper segment to lower segment ratio (U/L) less than 1 indicates spondyloepiphyseal dysplasia.

The upper segment to lower segment (U/L) ratio in this child is 0.8. In normal development, the U/L ratio is 1 in children more than 7 years.

The lower segment(L) is defined as the length from the symphysis pubis to the floor. The upper segment(U) is measured by subtracting the lower segment from the height.

The U/L ratio is

- 1.7 at birth
- 1.3 at 3 years
- 1 after 7 years
- 0.9 in adults



High U/L ratio:

• Rickets

• Achondroplasia (short limb dwarfism)

• Turner syndrome

• Untreated congenital hypothyroidism

Low U/L ratio:

• Spondyloepiphyseal dysplasia

• Marfan syndrome

• Morquio syndrome

• Vertebral anomalies

Solution to Question 13:

Multicystic encephalomalacia is the MRI finding seen in children with spastic quadriplegia.

Cerebral palsy is classified into several types based on its clinical features.

Spastic quadriplegia is the most common subtype seen in India. It is often caused by perinatal asphyxia and neonatal illness where all four limbs are affected. MRI would reveal multicystic encephalomalacia. It is associated with a poor outcome.

Spastic diplegia is commonly seen in premature infants with a history of birth asphyxia. It is second most common type of cerebral palsy in India after spastic quadriplegia. Both the lower limbs are involved, with spasticity in the hip adductors. This leads to a characteristic commando crawling in infants and a scissoring gait in older children. Brain MRI shows scarring and shrinkage in the periventricular white matter with compensatory enlargement of the cerebral ventricles, called periventricular leukomalacia (option A). IQ of these children is usually normal and has a good prognosis.

Spastic hemiplegia affects one-half of the body. It is seen due to a vascular insult leading to a perinatal stroke event. Early hand preference is seen (before 1 year), where one side of the body is always preferred as the other side is weak. MRI shows focal changes and porencephalic cysts due to infarcts.

Dyskinetic/ Extrapyramidal palsy is associated with birth asphyxia and kernicterus in the early neonatal period. Unlike spastic diplegia, the upper limbs are affected more than the lower limb. Brain MRI reveals lesions in the basal ganglia and thalamus (option D).

Cerebellar degenerations are associated with ataxic cerebral palsy (option C).

Note: Periventricular leukomalacia can also be seen in spastic quadriplegia but multicystic encephalomalacia is more common.

Solution to Question 14:

11 beta hydroxylase deficiency is the enzyme deficiency responsible for this condition. It is one of the variants of congenital adrenal hyperplasia(CAH) where there is increased aldosterone and testosterone activity with reduced cortisol levels.

The given clinical scenario of precocious puberty(SMR grading of 5 in a 10 year old boy) with height age> chronological age with short stature and increased blood pressure 180/110 suggests 11 beta hydroxylase deficient variant of CAH.

Solution to Question 15:

INSURE is associated with respiratory distress in newborns.

INSURE is abbreviated as intubate-intratracheal surfactant administration-extubation. It is a treatment modality performed in neonates with respiratory distress especially in preterm where surfactant production is reduced. It is indicated in neonates who are not maintaining oxygen saturation >90% while breathing 40-70% FiO2 in nasal CPAP(continuous positive airway pressure).

Extubation back to nasal CPAP is done immediately once the infant is stable, usually within minutes to <1 hr of surfactant administration. The amount of nasal CPAP required usually decreases after approximately 72 hr of age and most infants can be weaned from nasal CPAP.

Variations of the INSURE method are MIST (minimally invasive surfactant therapy) and LISA (less invasive surfactant administration), in which a small feeding tube, rather than an endotracheal tube (ETT), is used to deliver intratracheal surfactant to a spontaneously breathing infant on nasal CPAP.

The combination of early rescue surfactant by these methods along with nasal CPAP has been associated with the reduced need for mechanical ventilation and prevention of bronchopulmonary dysplasia.

Solution to Question 16:

Height for age is not a diagnostic criterion of severe acute malnutrition.

Criteria for severe acute malnutrition (SAM) in a 6-month-old child are mid-upper arm circumference, symmetrical pedal edema , and weight for height. Height for age is not the criteria for the diagnosis of severe acute malnutrition.

The presence of any of the following confers severe acute malnutrition in children 6 to 59 months of age:

• Weight-for-height below -3 standard deviation (<-3SD) on the WHO Growth Standard (Option B)

• Presence of bipedal edema (Option C)

• Mid-upper arm circumference (MUAC) below 11.5 cm (Option D)

Once the child is diagnosed with severe acute malnutrition, the doctor must assess for the following complications:

• Severe edema (+++)

• Low appetite (failed appetite test)

• Medical complications (severe anemia, pneumonia, diarrhea, dehydration, cerebral palsy, tuberculosis, HIV, heart disease, etc.)

One or more danger signs as per IMNCI

If any of the complications are present, the child is diagnosed with complicated SAM and is managed inpatient in the facility.

If the above-mentioned signs are absent, the child is diagnosed with uncomplicated SAM and supervised management at home is given.

Solution to Question 17:

Increased troponin-T is not included in NADA's criteria.

Nada's criteria is used to diagnose congenital heart disease in children. The presence of 1 major or 2 minor criteria indicates congenital heart disease.

Major criteria	Minor criteria
Systolic murmur grade 3 or moreAny diastolic murmur (Option A)CyanosisCongestive cardiac failure (Option D)	Systolic murmur < grade 3Abnormal second sound (S2)Abnormal ECGAbnormal chest X-ray findings (Option C)Abnormal BP

Solution to Question 18:

Kasai procedure should be done within 60 days of age, and not 90 days. Delay beyond 60 days significantly reduces success rates and increases the risk of requiring liver transplantation.

Neonatal cholestasis refers to prolonged jaundice in newborns caused by impaired bile flow.

Option B: HIDA scan is the investigation of choice. The hepatobiliary iminodiacetic acid (HIDA) scan is used to evaluate bile flow from the liver into the intestine and can help differentiate biliary atresia from other causes of neonatal cholestasis.

Option C: Biliary atresia is the most common cause of neonatal cholestasis, this condition requires surgical intervention.

Option D: Liver biopsy is a key diagnostic tool in neonatal cholestasis as it can confirm biliary atresia and rule out other causes like metabolic diseases or neonatal hepatitis.

Solution to Question 19:

An increase in oxygen causes the closure of ductus arteriosus.

Mechanism of closure of the ductus arteriosus:

1. Functional Closure (Minutes to Hours): Increased oxygen tension (PaO■) after birth causes smooth muscle contraction in the ductus. Decreased prostaglandin E2 (PGE2) levels (due to loss of placental production and increased pulmonary clearance) also contribute.

2. Anatomical Closure (Days to Weeks): Smooth muscle proliferation and fibrosis lead to permanent closure, forming the ligamentum arteriosum.

Thus, oxygen, reduced PGE2, and smooth muscle constriction drive ductus closure.

Changes in vascular resistance at birth:

Pulmonary vascular resistance (PVR): Before birth, the lungs are fluid-filled, and PVR is high. At birth, with the first breath, the lungs expand, oxygen tension rises, and pulmonary blood flow increases. This causes a decrease in PVR.

Systemic vascular resistance (SVR): At birth, clamping of the umbilical cord eliminates the low-resistance placental circulation. This leads to an increase in SVR.

Solution to Question 20:

Among the options, the Influenza virus is the most common causative organism of croup.

Croup, or laryngotracheobronchitis, is a viral infection affecting the upper airway, primarily in children aged 6 months to 6 years. It causes inflammation of the larynx, trachea, and bronchi, leading to characteristic symptoms such as:

- Barking cough (seal-like cough)
- Inspiratory stridor
- Hoarseness
- Respiratory distress in severe cases

The most common causative organism for croup is the parainfluenza virus. While other viruses like influenza, respiratory syncytial virus (RSV), and adenovirus can also cause croup, parainfluenza is the predominant cause.

Management:

- Racemic epinephrine nebulization leads to the rapid improvement of upper airway obstruction. Epinephrine constricts pre-capillary arterioles in the upper airway mucosa and decreases capillary hydrostatic pressure, leading to fluid resorption and improvement in airway edema.
- Dexamethasone-by the least invasive route (Oral>IV>IM). It is indicated in all the grades of croup.
- Oxygen should be administered to children who are hypoxemic (SpO2 <92 %)

Antibiotics are not indicated in laryngotracheobronchitis.

Other options:

Option A: Hemophilus influenza is associated with epiglottitis.

Option B: Streptococcus pneumonia is associated with pneumonia, otitis media, or meningitis, not croup.

Option D: Diphtheria is associated with pseudomembrane in the throat and airway obstruction

Solution to Question 21:

The combination of copying a cross, telling stories, and independent toileting aligns with the developmental milestones of a 4-year-old child.

Developmental milestones

At 1 year

- Motor: Walks with support
- Adaptive: Mature pincer grasp
- Language: Speaks a few words
- Social: Plays simple ball game
- Domestic mimicry at 18 months

At 2 years (Option D)

- Motor: Walks up and downstairs, one step at a time
- Adaptive: Makes a tower of 6 blocks
- Language: Simple sentences, a vocabulary of 50-100 words, uses pronouns at 2.5 yrs
- Social: Asks for food and starts toilet training, listens to stories

At 3 years (Option A)

- Motor: Rides tricycle
- Adaptive: Makes a tower of 9 blocks, handedness is established
- Language: Knows full name, age, gender
- Social: Plays simple games and engages in parallel play with other children

At 4 years

- Motor: Hops
- Adaptive: Draws square and constructs a bridge with blocks
- Language: Tells stories
- Social: Goes to the toilet alone

At 5 years (Option C)

- Motor: Skips
- Adaptive: Draws a triangle
- Language: Asks the meaning of words, names 4 colors
- Social: Dresses and undresses

Solution to Question 22:

Most of the children suffering from breastmilk jaundice do not require phototherapy.

Breastmilk jaundice is due to significant unconjugated hyperbilirubinemia (Option D), after the 7th day and may last for 2–3 weeks. This may be caused by the breast milk that might contain β-glucuronidase, which deconjugates intestinal bilirubin and promotes its absorption and other inhibitors of glucuronyl transferase. Persistently high bilirubin levels may require changing from breast milk to infant formula for 24–48 h and/or treatment with phototherapy, without cessation of breastfeeding. (Option A)

Most cases of breast milk jaundice are mild and self-limiting, and phototherapy is required only in rare severe cases where bilirubin levels are significantly elevated. (Option B)

Breast milk jaundice is typically observed in exclusively breastfed infants, often with elevated bilirubin (>10mg/dl) levels persisting beyond the neonatal period (after 3-4 weeks of birth). (Option C)

Solution to Question 23:

Biliary atresia is not a feature of cystic fibrosis.

Cystic Fibrosis (CF) is a genetic disorder that primarily affects the lungs, digestive system, and other organs. It is caused by mutations in the CFTR gene, which leads to the production of thick and sticky mucus. The CFTR gene is on chromosome 7q. The most common CFTR mutation is F508 deletion. This mucus blocks ducts and passages in various organs, causing problems.

Features of CF:

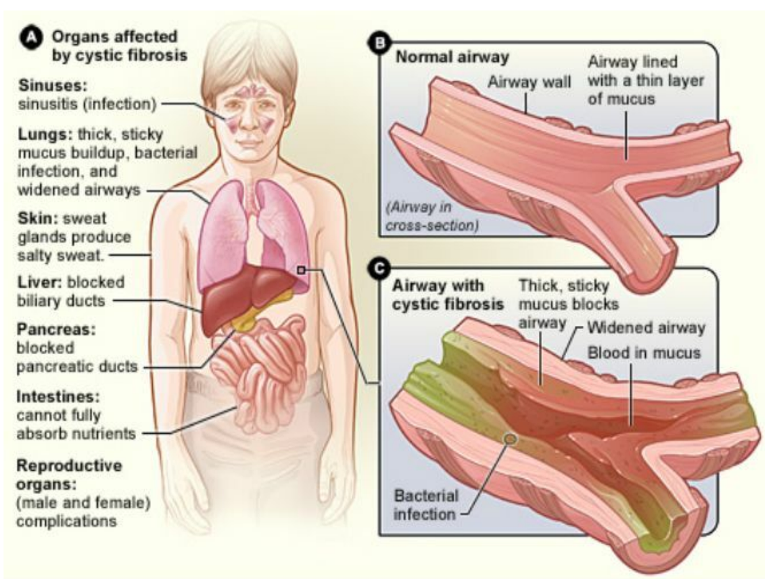
• Respiratory Symptoms: Chronic cough, recurrent lung infections, bronchiectasis, and nasal polyps. (Option A)

• Digestive Issues: Pancreatic insufficiency, malabsorption, meconium ileus (Option C), and fat-soluble vitamin deficiencies which causes bleeding diathesis (Option D).

• Sweat Gland Abnormalities: Elevated sweat chloride levels.

• Reproductive: Infertility in males (due to absent vas deferens) and reduced fertility in females.

• Other Complications: CF-related diabetes (CFRD), osteoporosis, and liver disease.



The diagnosis is confirmed if both of the following criteria are met:

1. Presence of

• Clinical symptoms affecting at least one organ system (respiratory, gastrointestinal, or genitourinary) or

• Positive newborn screening or

• A sibling with cystic fibrosis.

2. Evidence of dysfunction of the CFTR gene by

• Two positive sweat chloride test results were obtained on separate days or

• Identification of two CFTR mutations by genetic testing or

• An abnormal measurement of nasal potential difference.

Cystic fibrosis requires a multidisciplinary approach to management. The correction of the defect can be done via:

• Lumacaftor, corrects the misfolding and reverts the mutation defect.

• Ivacaftor potentiates CFTR channel opening and stimulates ion transport.

End-stage pulmonary failure requires lung transplantation.

Solution to Question 24:

The infant's history of umbilical bleeding, recurrent bleeding episodes, normal coagulation studies, and positive urea clot solubility test strongly suggest Factor XIII deficiency.

FXIII cross-links fibrin polymers and strengthens the clot, making it resistant to premature breakdown. Its deficiency leads to delayed wound healing, spontaneous bleeding, and in severe cases, life-threatening hemorrhages.

Presents with recurrent umbilical bleeding since birth and Epistaxis and rectal bleeding

Laboratory Findings: Normal PT, PTT, TT, platelet count and function: These indicate that the coagulation cascade and platelets are functioning normally up to fibrin formation. Positive urea clot solubility test: This is a classic diagnostic test for Factor XIII deficiency. Clots deficient in Factor XIII dissolve in 5M urea because they lack cross-linked fibrin

The preferred treatment is FXIII concentrates. Fresh frozen plasma (FFP) and cryoprecipitate are alternatives. Antifibrinolytics like tranexamic acid can be adjunctive for minor bleeding. Regular monitoring and prophylaxis are crucial for long-term management and preventing complications.

Other Options:

Option B: Factor XI: Deficiency typically causes mild to moderate bleeding, mainly in mucosal sites, and does not present with umbilical bleeding.

PT and PTT may be prolonged, which is not the case here.

Option C: Factor X: Deficiency causes severe bleeding and affects PT and PTT (both would be prolonged), unlike in this case where all tests are normal.

Option D: Factor XII: Deficiency does not cause bleeding; it is often asymptomatic. It can lead to prolonged PTT, which is not observed here.

Solution to Question 25:

Failure to regain appetite by day 4, failure to start to loose edema by day 4 and presence of edema by day 10 indicate primary failure of recovery in admitted patient with severe acute malnutrition.

For assessing primary failure of recovery in a patient with severe acute malnutrition (SAM), key clinical benchmarks guide the evaluation.

Difference between primary and secondary failure of malnutrition treatment:

Inpatient cases

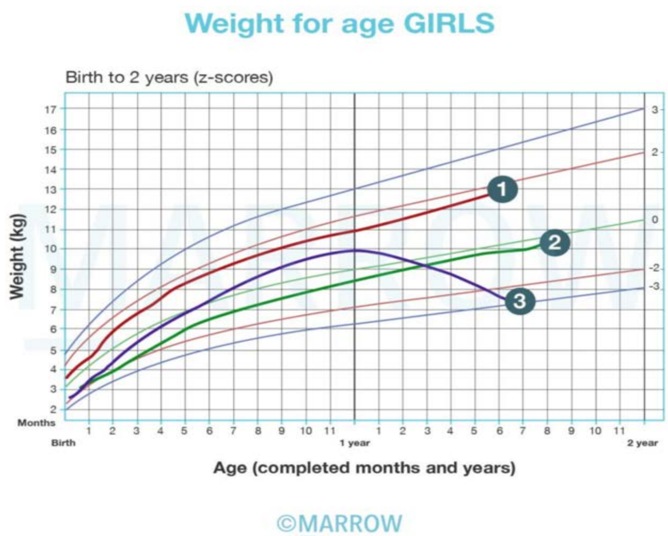
Day of admission is considered as day 0.

Primary Failure	Secondary failure
Failure to regain appetite by day 4 (i)Failure to start losing edema by day 4 (ii)Presence of edema on day 10 (iii)Failure to gain at least 5 g/kg/day by day 10	Failure to gain 5 g/kg/day for 3 consecutive days during rehabilitationReappearance of edema after 4 weeks

Paediatrics NEET 2024

Question 1:

Three female children graphs were plotted on an IAP growth chart, which of them signals growth failure?



a) Only 2 and 3

b) Only 3

c) 1, 2 and 3

d) Only 1

Question 2:

A neonate has bilateral chorioretinitis with intraparenchymal calcifications. She also has hydrocephalus. What is the causative organism?

a) Trypanosoma

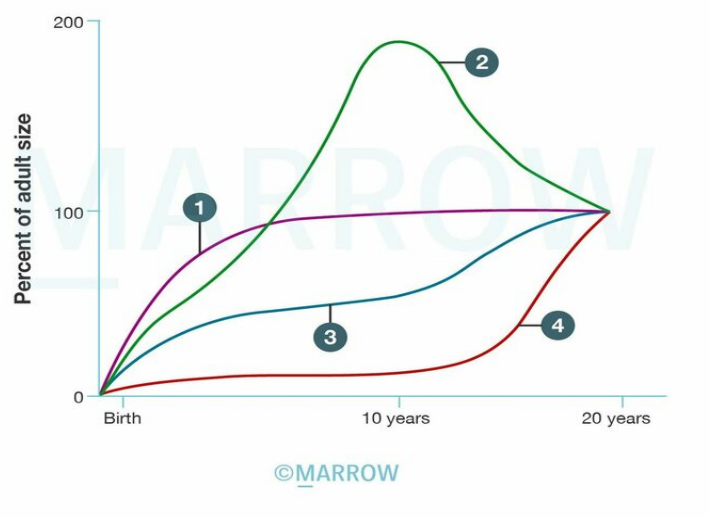
b) Toxoplasma gondi

c) Taenia solium

d) Cytomegalovirus

Question 3:

What is represented by each of the following growth curves?



a) 1 - Liver, 2 - Spleen, 3- Ovary, 4 - Testis

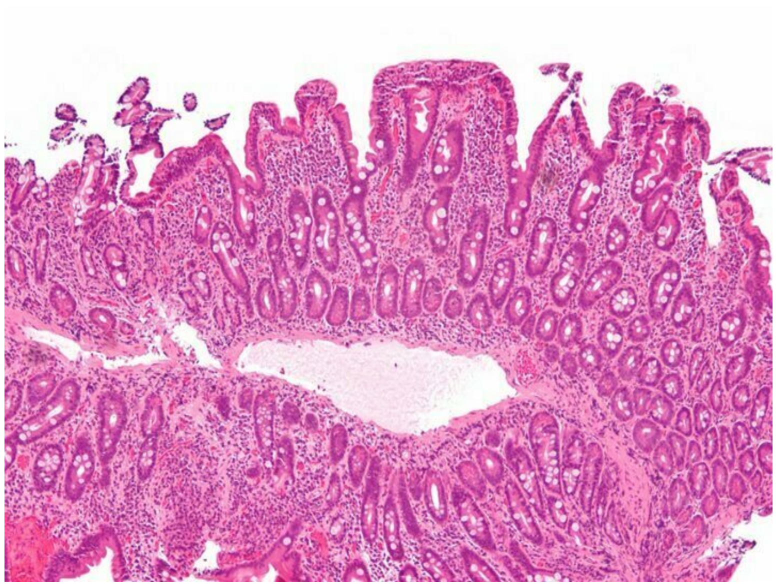
b) 1 - Brain, 2- Tonsils, 3 - BMI, 4 - Genital Organs

c) 1 - Brain, 2 - somatic growth, 3- Ovary, 4 - Height

d) 1- Brain, 2- Waldeyer's ring, 3- somatic growth, 4- Testicular growth

Question 4:

A mother brought her 2-year-old child to the OPD with complaints of lack of weight gain, abdominal pain on eating, and poor eating habits. The child was advised gluten-free diet after which there was improvement in the general condition and weight gain. What is the probable diagnosis?



a) Topical sprue

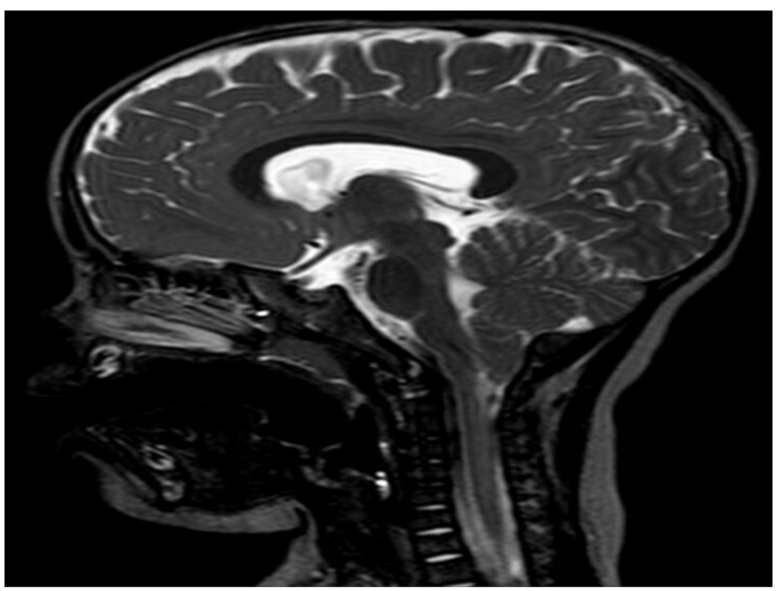
b) Celiac disease

c) Whipple's disease

d) Intestinal malabsorption

Question 5:

Identify the condition shown in the image:



a) Chiari type 1

b) Dandy-Walker malformation

c) Vein of Galen malformation

d) Corpus callosum agenesis

Question 6:

A newborn baby presented with a difficulty in breathing. Auscultation of the chest revealed gurgling sounds and heart sounds shifted to the right. The chest radiograph confirmed a congenital diaphragmatic hernia. Which among the following is the most common location for this condition?

a) Esophageal opening

b) Left posterolateral

c) Between sternum and coastal cartlage

d) Central tendon

Question 7:

A child presented with failure to thrive and recurrent infections. On examination, tonsils are absent. Further evaluation confirms the deficiency of adenosine deaminase. What is the likely diagnosis?

a) Lesch-Nyhan syndrome

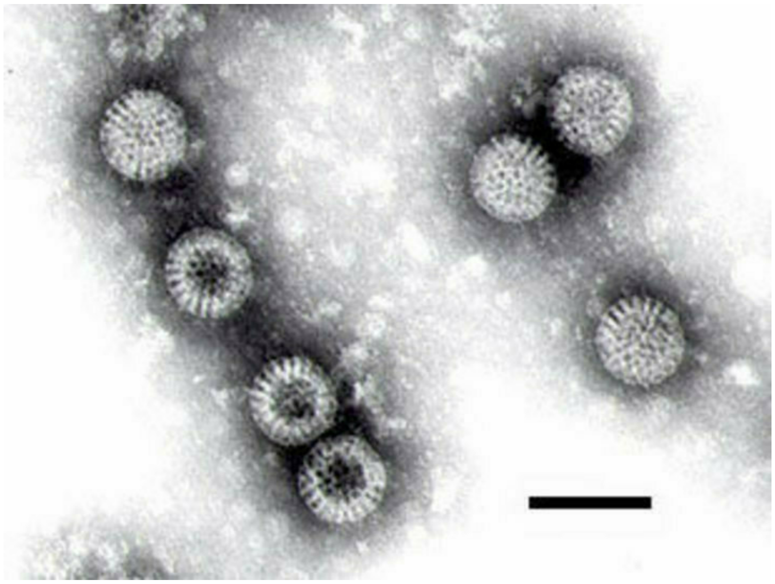
b) Ataxia telengeictasia

c) Severe Combined Immuno Deficiency

d) Friedrich's ataxia

Question 8:

A child presented with diarrhea and dehydration. There was no blood in the stools. A stool examination revealed a spoke wheel pattern. Identify the causative organism.



a) Adenovirus

b) E.coli

c) Norwalk virus

d) Rota virus

Answer Key

Question No.	Correct Option
1	b
2	b
3	d
4	b
5	a
6	b
7	c
8	d

Detailed Explanations

Solution to Question 1:

Out of the growth curves (weight-for-age), only growth curve 3 shows growth failure. This pattern shows fluctuation in growth percentiles, which might indicate a history of growth issues or inconsistent growth. The drop to below -2SD is significant and suggests growth failure as it indicates the child is significantly below the normal range.

The nutritional status of a child is assessed in terms of anthropometry:

The World Health Organization (WHO) has established international standards for normal child growth from birth to 5 years of age. The normalization of the anthropometric measures is done in terms of z-scores (standard deviation scores). The anthropometric indicators are marked in a growth chart which is compared with a reference curve.

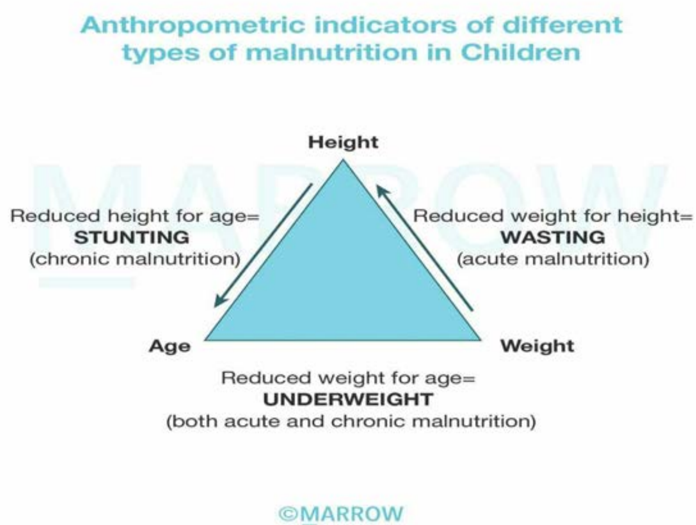
The anthropometric measures used are weight for age, height for age, weight for height, BMI, and mid-upper arm circumference.

• Low weight for height, or wasting, indicates acute malnutrition. It is also mentioned as wasting.

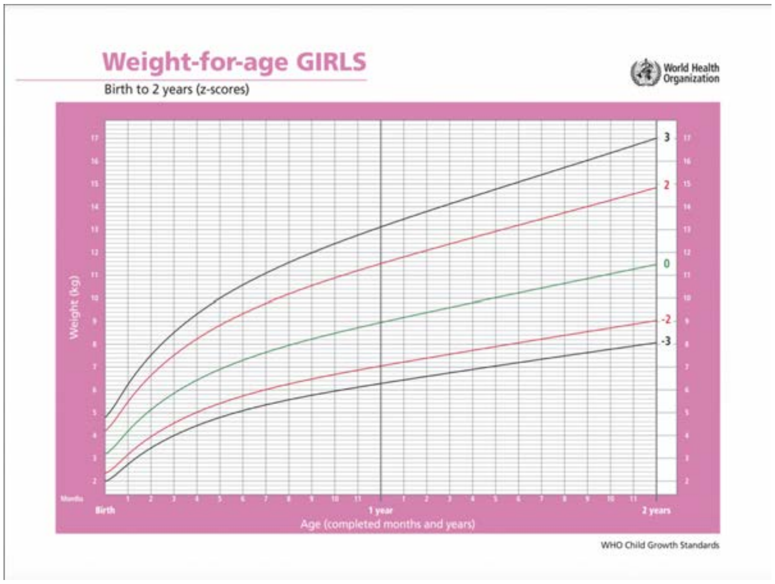
• High weight-for-height indicates overweight.

• Low height-for-age, or stunting, indicates chronic malnutrition. This is the best indicator of long-term nutritional status in the pediatric population.

• Low weight-for-age indicates acute or chronic malnutrition. It does not differentiate between wasting and stunting. Hence, it has limited clinical significance.



The below image shows the WHO growth chart for girls (6 months-2years). Boys have similar charting in blue color.



Solution to Question 2:

The presence of bilateral chorioretinitis with intraparenchymal calcifications and hydrocephalus in a neonate is suggestive of congenital toxoplasmosis.

The characteristic triad of congenital toxoplasmosis is chorioretinitis, hydrocephalus, and diffuse intracranial calcifications. Clinical manifestations apart from the triad include hydrops fetalis, intrauterine growth retardation, prematurity, peripheral retinal scars, persistent jaundice, and mild thrombocytopenia.

Congenital toxoplasmosis occurs when a pregnant woman acquires the infection during gestation. The organism may cross to the fetus transplacentally or at the time of vaginal delivery. The severity of the disease in the neonate is greater if the woman gets infected early in her pregnancy. The causative agent is Toxoplasma gondii, an obligate intracellular protozoan.

The diagnosis is made by the detection of IgM anti-toxoplasma antibodies. IgG can cross the placenta is hence, cannot confirm infection in the infant. IgM immunosorbent agglutination assay (ISAGA) and IgA ELISA are the most useful diagnostic tests.

Spiramycin can be administered for cases of acute infection in early pregnancy to reduce vertical transmission. It does not cross the placenta and therefore cannot be used to treat an infected fetus. Pyrimethamine-sulfonamide (to be avoided in the first trimester) is used when a fetal infection is confirmed or in maternal infections after 18 weeks.

Other options:

Option A: Trypanosoma is a genus of protozoan parasites responsible for diseases like African trypanosomiasis (sleeping sickness) and American trypanosomiasis (Chagas disease). These diseases present with different clinical features, such as fever, swollen lymph nodes, and neurological symptoms in African trypanosomiasis, and fever, swelling at the infection site, and cardiomyopathy in Chagas disease.

Option C: Taenia solium is a parasitic tapeworm transmitted through undercooked pork, causing intestinal infection (taeniasis) or more severe cyst formation in tissues, including the brain (cysticercosis). It's prevalent in regions with poor sanitation and where pork is consumed.

Option D: Congenital cytomegalovirus presents with hepatosplenomegaly, petechial rashes, jaundice, chorioretinitis, and microcephaly. Head neuroimaging shows intracranial periventricular calcifications. One of the long-term sequelae is sensorineural hearing loss. Diagnosis is made by isolating the virus from the urine or blood by polymerase chain reaction (PCR)

Solution to Question 3:

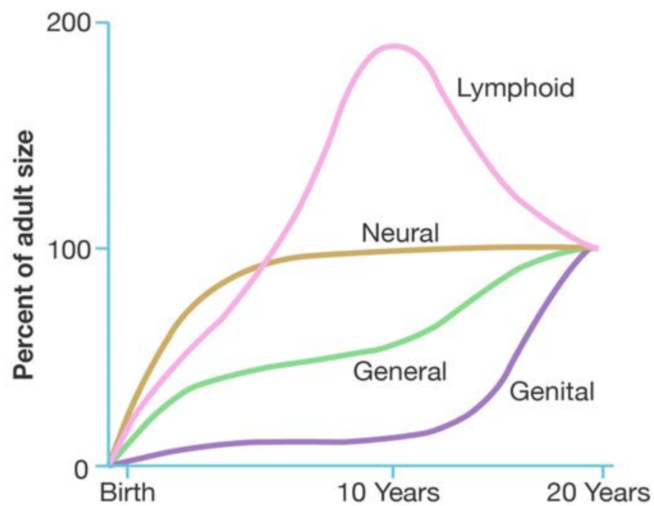
The correct answer is 1- Brain, 2- Waldeyer's ring, 3- somatic growth, 4- Testicular growth.

Curve 1: The neural growth(brain) attains 90% of the growth by the age of 2 years.

Curve 2: The tonsils are collections of lymphatic tissue located within the pharynx. They collectively form a ringed arrangement, known as Waldeyer's ring. Tonsils/lymphatic growth in children occur between 4-8 years of age and then shrink down in adult age.

Curve 3: Somatic growth shows a sigmoid curve, rapid growth in the initial years, and then during puberty.

Curve 4: Testicular growth/Gonadal growth starts during puberty, before that the curve is flat.



Solution to Question 4:

The clinical presentation and the histopathological findings are suggestive of celiac disease.

Celiac disease is an autoimmune gluten-induced enteropathy where HLA DQ2 and DQ8 are expressed. In the small intestine, peptides of gluten are presented to helper T cells which mediate mucosal inflammation. Antiendomysial and anti-tTG antibodies are found in serological studies.

Clinical Features:

• Gastrointestinal Symptoms: Chronic diarrhoea, abdominal pain, vomiting, weight loss, and failure to thrive.

• Extraintestinal Symptoms: Anemia, fatigue, elevated liver enzymes, neurologic disorders, short stature, dental enamel defects, arthralgia, aphthous stomatitis.

Investigation:

• Serologic Tests: Initial testing for anti-TG2 IgA antibodies and total IgA. If anti-TG2 is positive and IgA levels are normal, CD is likely.

• Genetic Testing: For HLA-DQ2 or DQ8 haplotypes.

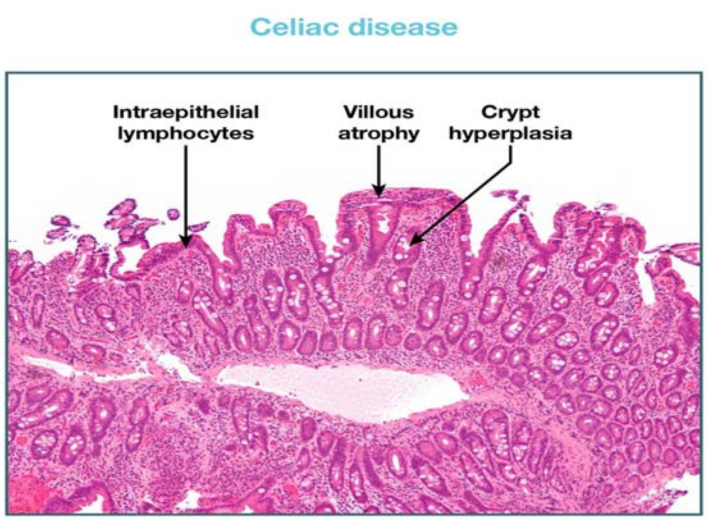
• Endoscopy and Biopsy: Multiple biopsies from the small intestine to check for villous atrophy.

• Biopsy Findings: villous atrophy: flattening of the villi in the small intestine, crypt hyperplasia: enlargement of the crypts, intraepithelial lymphocytosis: Increased number of lymphocytes in the epithelium.

Treatment:

Gluten-Free Diet: To manage celiac disease, patients must strictly avoid all foods containing gluten. Safe alternatives include rice, corn, maize, buckwheat, millet, amaranth, quinoa, sorghum, potato, soybean, tapioca, and teff, along with bean and nut flours.

Biopsy findings can be seen below:



Other options:

Option A: Tropical sprue is another malabsorption disorder seen as diarrhoea, steatorrhea, nutritional deficiencies, and weight loss. It is prevalent in some tropical areas hence the name. It is mostly attributed to infectious agents considering its response to antibiotics. It is diagnosed by a small intestinal biopsy which shows a close resemblance to that seen in celiac disease but it’s not specifically triggered by gluten and is more common in adults.

Option C: Whipple Disease is a multisystemic disease caused by T.whipplei. Small intestinal biopsy shows PAS-positive macrophages.

Option D: Intestinal Malabsorption is a general term used for conditions that cause poor nutrient absorption, which could include celiac disease but is not specific enough to explain the clear response to a gluten-free diet.

Solution to Question 5:

The MRI image in the question shows a downward displacement of the cerebellar tonsils through the foramen magnum, which is characteristic of Chiari type 1 malformation.

This condition involves the herniation of the cerebellar tonsils into the spinal canal and is often associated with headaches, neck pain, and sometimes neurological deficits. It may be identified on imaging, such as MRI, where the cerebellar tonsillar descent is clearly visualized.

Arnold Chiari malformation is the most common malformation of the posterior fossa and hindbrain. There are two types of Chiari malformations-

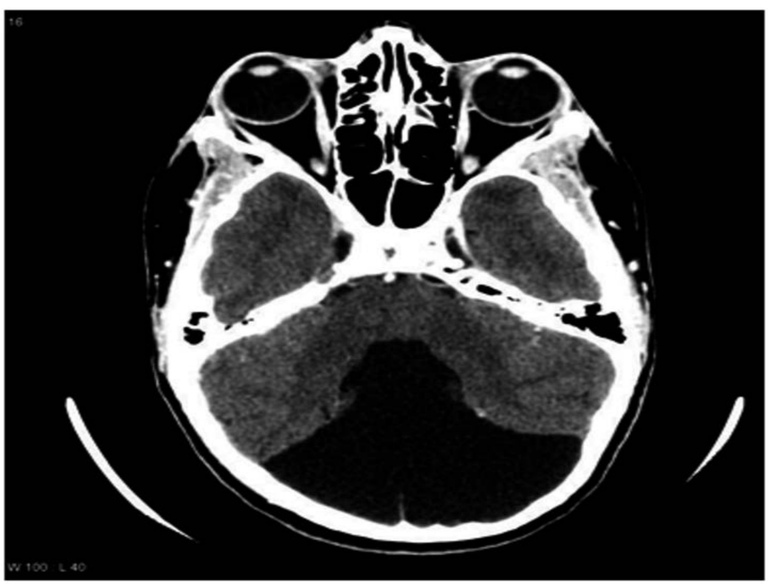
• Chiari I malformation- It is the more common type. There is downward displacement of the cerebellar tonsils through the foramen magnum. It may be associated with syringomyelia.

• Chiari II malformation: Displacement of the medulla, the fourth ventricle, and cerebellar vermis through the foramen magnum. It is almost always associated with a myelomeningocele.

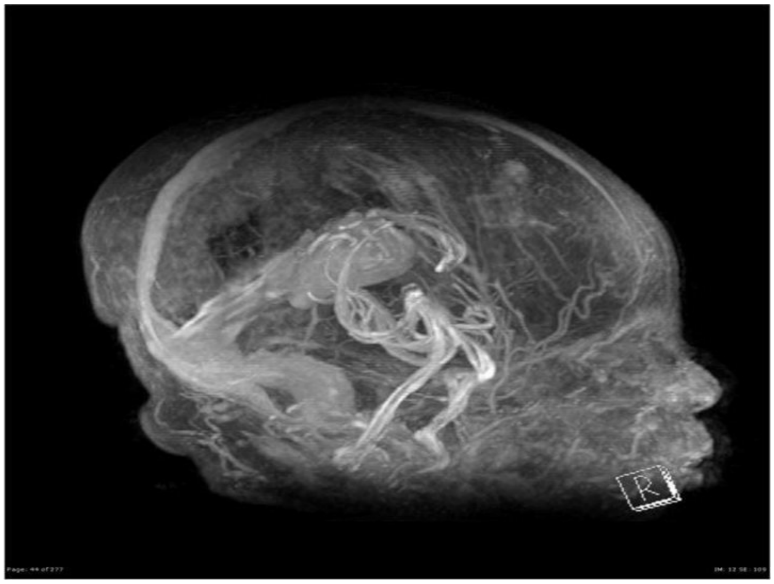
Other options:

Option B: Dandy walker malformation is a developmental anomaly characterized by a cystic dilation of the fourth ventricle, hypoplasia of the cerebellar vermis, an enlarged posterior fossa, and an upward displacement of the cerebellum. The atresia of the foramina of Magendie and Luschka leads to obstructed CSF flow, resulting in non-communicating hydrocephalus, causing symptoms such as vomiting, seizures, bulging fontanelles, and developmental delays. Diagnostic imaging includes ultrasound, CT, or MRI.

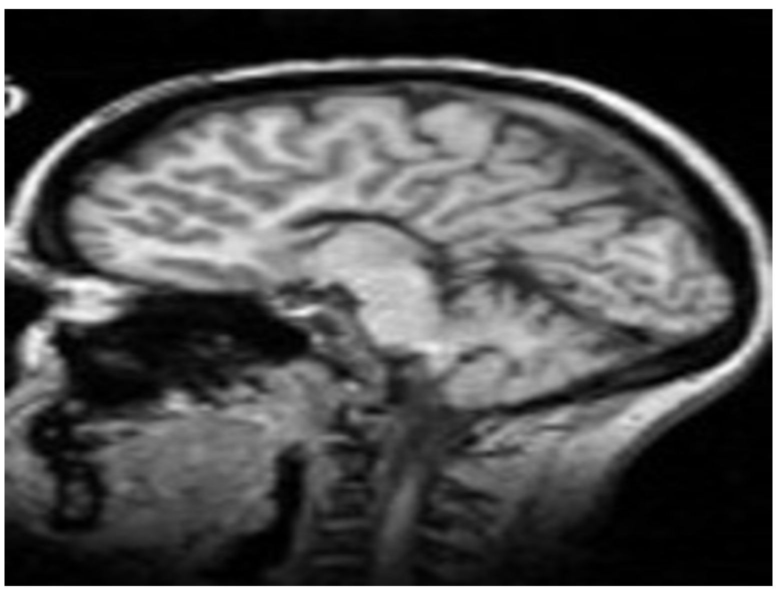
Given below is an image of Dandy walker malformation-



Option C: The MR angiography in the vein of Galen malformation shows an enlarged midline venous structure, with multiple arteriovenous communications leading to aneurysmal dilatation. The image below shows the malformation:



Option D: Agenesis of the corpus callosum is one of the most common congenital cerebral malformations which is morphologically the complete or partial absence of corpus callosum and not defined by functional or behavioral abnormalities. The given image shows the MRI findings:



Solution to Question 6:

Congenital diaphragmatic hernia with the features given in the clinical vignette occurs most commonly in a left posterolateral defect.

Bochdalek hernia is the most common congenital diaphragmatic hernia which occurs on the left posterolateral side due to a defect in the pleuroperitoneal membrane. The abdominal contents in the thorax compress the developing lungs causing pulmonary hypoplasia which manifests as respiratory distress in most infants. Chest X-ray shows an indistinct diaphragm with radiolucent para cardiac shadow and absent bowel loops in the abdomen. Opacities can be seen in the thorax if solid organs or omentum also herniate. Surgical repair is indicated in all cases either by a thoracic or abdominal approach.

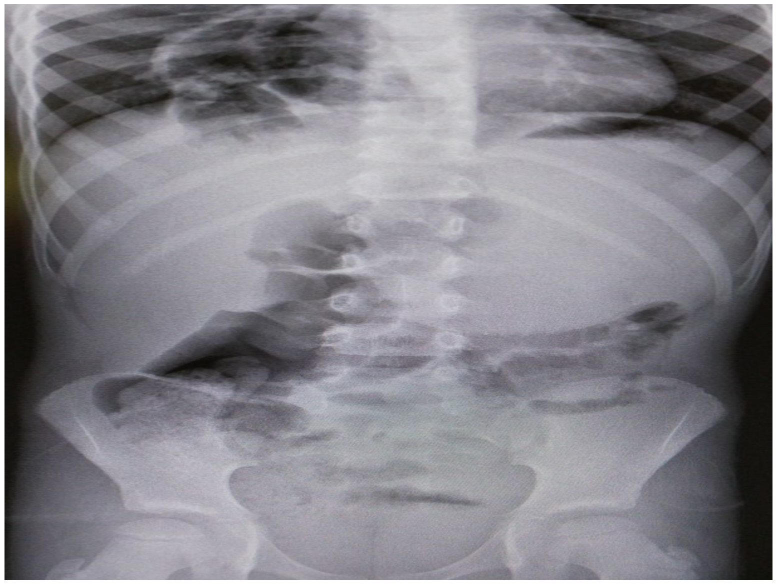
Given below is an X-ray showing Bochdalek's hernia

Other options:

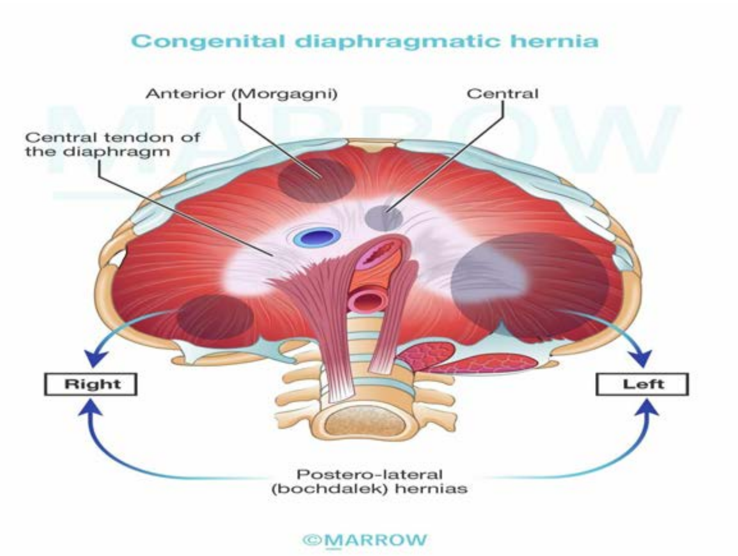
Option A- Sliding and rolling hernias occur via the oesophagal opening(hiatal hernia). Sliding hernia (more common) occurs when the gastroesophageal junction(GEJ) and a part of the stomach slide up into the thorax via the opening. A rolling hernia occurs when the GEJ stays below the diaphragm, but the stomach goes up, lying parallel to the oesophagus in the thorax. Hiatal hernias occur in adults.

Option C- The space between the sternum and costal cartilage is called the space of Larry or space of Morgagni. A defect in this space causes Morgagni hernia, which is the second most common hernia in children. It occurs on the right anteromedial aspect and is often less symptomatic.

The following is an X-ray showing Morgagni's Hernia :



Option D- A defect in the central tendon causes a central congenital diaphragmatic hernia.



Solution to Question 7:

The given clinical scenario is suggestive of severe combined immunodeficiency (SCID) due to a deficiency of adenosine deaminase.

Affected children present with prominent thrush (oral candidiasis), extensive diaper rash, and failure to thrive. Some patients develop a morbilliform rash shortly after birth because maternal T cells are transferred across the placenta and attack the fetus, causing graft-versus-host disease (GVHD).

Persons with SCID are extremely susceptible to recurrent, severe infections by a wide range of pathogens, including:

• Candida albicans

• Pneumocystis jiroveci

• Pseudomonas

• Cytomegalovirus

• Varicella

The thymus biopsy shows remnants of Hassall's corpuscles, indicating severe combined immunodeficiency (SCID) due to adenosine deaminase (ADA) deficiency. SCID affects both humoral and cell-mediated immune responses, primarily impacting T cells and resulting in the absence or severe underdevelopment of lymphoid tissues like lymph nodes and tonsils.

There are two main types of SCID:

• X-linked SCID: The most common form, caused by a mutation in the common γ-chain (γc) subunit of cytokine receptors. This leads to reduced T-cell numbers and impaired antibody synthesis despite normal B-cell counts. The thymus shows undifferentiated epithelial cells.

• Autosomal recessive SCID: Often due to ADA deficiency, causing toxic metabolite accumulation that damages T cells, resulting in greater T-cell reduction.

In both forms, the thymus is small and lacks lymphoid cells, but ADA deficiency uniquely features remnants of Hassall’s corpuscles.

Note: X-linked SCID is the first human disease in which gene therapy has been successful.

Other options:

Option A: Lesch-Nyhan syndrome is due to a deficiency of the enzyme Hypoxanthine-guanine phosphoribosyltransferase(HGPRT). It is an X-linked recessive disease. The clinical features include intellectual disability, self-mutilation, aggression, hyperuricemia, and gout.

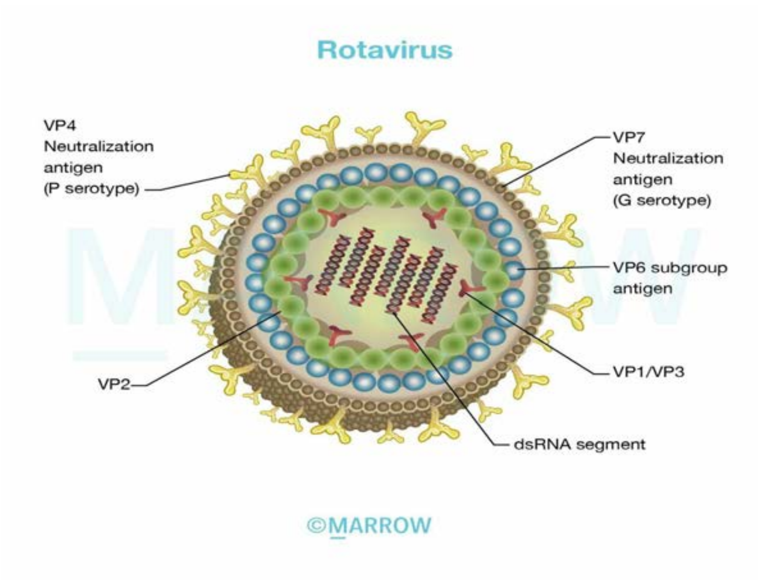
Option B: Ataxia Telangiectasia is a progressive neurodegenerative genetic disease causing poor coordination and small dilated blood vessels. It is autosomal recessive in inheritance due to ATM gene mutation on Ch 11q22-23.

Option D: Friedrich's ataxia involves the posterior column, pyramidal, and spinocerebellar tracts. Clinical features include cardiomyopathy, optic atrophy, and diabetes mellitus. It should be differentiated from other forms of ataxia as Tabes dorsalis and SACD(Subacute Combined Degeneration).

Solution to Question 8:

The given clinical scenario along with the electron micrograph image of the stool sample showing a spoke wheel pattern with complaints of non-bloody diarrhea and dehydration in a child is suggestive of rotavirus.

Rotavirus is a nonenveloped, double-stranded (ds) RNA virus. Rotaviruses appear as cartwheel-shaped structures with short spokes radiating from them. It has 11 segments of double-stranded RNA. They exhibit genetic reassortment due to the presence of this segmented RNA.



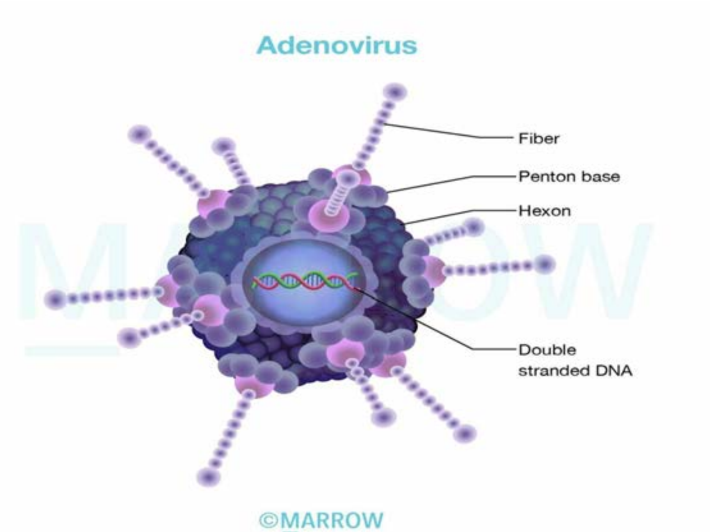
Rotavirus is the most common identifiable viral cause of diarrhea in children below the age of 5 years and is most frequent between 6-24 months of age. The incubation period is 1-3 days. Most cases are asymptomatic or with mild symptoms like watery diarrhea, vomiting, low-grade fever, and temporary lactose intolerance. Diagnosis can be made by rapid antigen detection of rotavirus in stool specimens.

Rotavirus vaccine is associated with a potential risk of intussusception, especially when the first dose of these vaccines is given to infants aged >12 weeks. Because of this risk, the rotavirus vaccine is not a part of catch-up immunization programs.

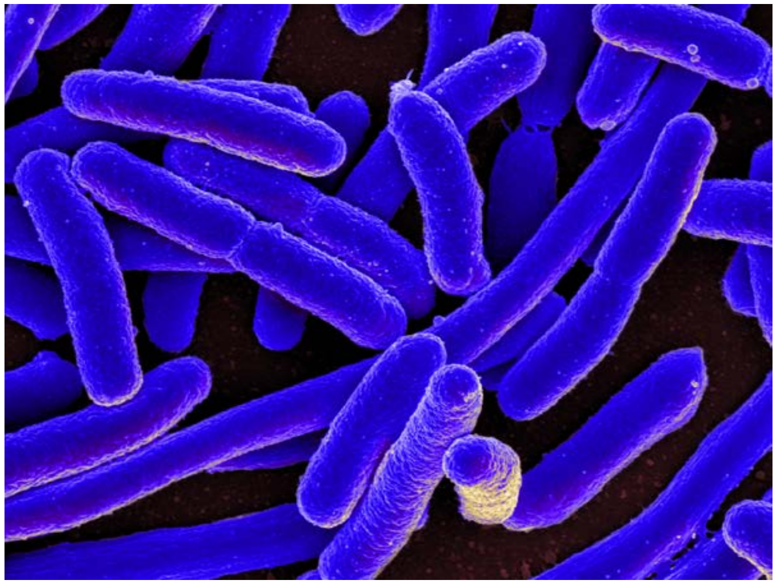
Symptoms associated with different levels of dehydration are classified into 3 types no/minimal dehydration, mild to moderate dehydration, and severe dehydration.

Other options:

Option A: Adenovirus contains double-stranded DNA. They have an icosahedral symmetry with structures known as fibers extruding from the vertices. They have a space vehicle-shaped appearance.



Option B: E coli are motile, facultative anaerobes in GIT. It is a gram-negative, flagellated rod. It is the most frequent cause of acute complicated urinary tract infections. It is a lactose fermenter and hence, grows on MacConkey medium to produce pink colonies.



Option C: Norwalk virus is a small, round, non-enveloped virion with a single-stranded positive-sense RNA genome with icosahedral symmetry. When viewed under an electron microscope, it has a cup-shaped morphology. Clinical features include nausea, vomiting, and diarrhoea.